

A proof of the Erdős–Gallai cycle decomposition conjecture

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Abstract

The Erdős–Gallai conjecture (Erdős Problem #184) asserts that the edge set of every n -vertex simple graph can be partitioned into $O(n)$ cycles and single edges. We prove this conjecture: every n -vertex graph can be decomposed into $O(n)$ cycles and edges. The best previous bound, $O(n \log^* n)$, is due to Bucić–Montgomery [2].

Our argument builds on the method of [2] and has three new ingredients. First, τ -rules placed inside the recursion of Bucić–Montgomery’s Lemma 14, together with a cap on the degrees of guest vertices, produce a hierarchy $\text{HB}^{*\tau+}$ in which hub concentration is negligible for every run. Second, *standalone parts are chained* into cycles through junction paths that run inside lent random colour classes of ancestors at least two rounds back (a transparent vortex, TPV, together with a junction synchronisation that splits every large class into cherries, JS-LC); only parity leftovers, single edges, remain. Third, *each end of a leftover single edge receives a private junction edge*, which joins it to a junction vertex (possibly shared with other ends) in a sparse random pool inside its ancestor’s expander; the resulting simple quotient graphs have at most a vanishing fraction of n vertices in total (the fraction tends to 0 as the degree threshold D_* grows), every decomposition of a quotient lifts back, and the quotients are handled by induction on the number of vertices.

The constants are galactic and are not computed: the degree threshold D_* is only required to be sufficiently large (each condition on it holds for all sufficiently large D_*), and it is of tower type ($\log_2 \log_2 D_* \geq 2^8$ is one of these conditions). The bound $f(n) = O(n)$ has also been formally verified in the Lean 4 proof assistant.

1 Introduction, conventions, cited inputs, constants, dependency structure

For a graph G let $f(G)$ be the least number of cycles and single edges that partition $E(G)$, and let $f(n)$ be the maximum of $f(G)$ over graphs G on n vertices (Definition 1.4). Erdős and Gallai conjectured that $f(n) = O(n)$ (see [6]). Repeatedly removing a longest cycle gives $f(n) = O(n \log n)$ [6]; Conlon, Fox and Sudakov [4] proved $f(n) = O(n \log \log n)$; and Bucić–Montgomery [2] proved $f(n) = O(n \log^* n)$, which was the best previously known bound. Lovász [11] proved that every n -vertex graph decomposes into at most $n/2$ paths and cycles.

In this paper we prove that $f(n) = O(n)$, confirming the conjecture of Erdős and Gallai. The precise statement is Theorem 1.1, which is proved in Section 7. The proof builds on the method of Bucić–Montgomery [2]: it keeps the round structure of their argument, and it uses, and in several places modifies, their lemmas on expanders and path connectivity. Remark 1.2 says where the new ingredients lie, and the overview that follows it describes the architecture of the argument. Section 1 contains the overview, the conventions, every cited result (each published result of [2] is quoted in our notation, with its number in [2]), one elementary decomposition bound and one standard

concentration inequality that are proved in full instead of being cited (Fact 1.6 and Lemma 1.27), the constants and the single list of galactic conditions, the symbol table, and the dependency table of all numbered statements. The bound $f(n) = O(n)$ has also been formally verified in Lean 4 with Mathlib; Section 8 describes the formalization.

Theorem 1.1 (Main Theorem). *Let $\varepsilon, \sigma, C', A, N_0, D_*$ be as in Definition 1.31, with D_* satisfying (Γ1)–(Γ4) of Condition 1.33. Put*

$$C_0 := \frac{D_*}{2} + 1085 + \varepsilon_1(D_*) \quad \text{and} \quad c_{\text{EG}} := \max\left\{\frac{C_0}{1 - 2\theta_Q}, \frac{N_0}{2}\right\},$$

where $\varepsilon_1(D_*)$ and $\theta_Q = \varepsilon_2(D_*)$ are the explicit functions of D_* defined in Proposition 7.18 and Lemma 7.16. Then every graph G on n vertices satisfies

$$f(G) \leq c_{\text{EG}} n.$$

In particular $f(n) = O(n)$.

Dependencies: 7.22 (forward pointer; in Table 3 the proof is the node `s7:thmMainProof`).

The proof of Theorem 1.1 is given in Section 7, as Corollary 7.22. Theorem 1.1 is the only statement of this paper that is stated before the statements it depends on.

Remark 1.2 (organization of the proof; the new ingredients). (i) Compared with [2], the new ingredients of the argument are concentrated in the following places:

- (a) the routing engine inside JS-LC (Lemmas 6.6, 6.5, 6.3 and 6.4 as used for cherry systems at junction multiplicity $2M_l - 2$ in Lemma 6.13);
 - (b) Lemma 14^τ (Lemma 2.6) and part (iv) of Theorem 6.10: together they modify the core Lemma 14 of [2] instead of using it as a black box;
 - (c) the composition JV^{+*} of Section 7; within it, the most delicate points are property (J1) of Lemma 6.15, aggregated over all MIX-parts, on which Lemma 7.7 rests; the conditioning in Lemma 7.10 and in the ultra-hub bound; the lift, including the cross-round use of junction edges through the Lent sets; and the existence of one joint outcome;
 - (d) every place where a per-vertex multiplicity feeds a path-connectivity parameter t : property (P4) of the vortex runs of Section 4, Step 6 of Lemma 5.7, and the joint routing of JS-LC at junction multiplicity $2M_l - 2$. In particular, Lemma 4.3(c) bounds the number of arc ends at a vertex by its degree, so that a vertex lies in at most $M_l - 1$ pairs, and the U-lent classes are path connected for the multiplicity parameter $t_Y = \lceil \lambda_r^{1.6} \rceil \geq M_l$ (Definition 3.10; rows 4 and 7 of Table 5).
- (ii) The argument relies in particular on the following properties, each proved where indicated:
- (1) the hub concentration is negligible: for every valid $\text{HB}^{*\tau+}$ run (Definition 2.7) and every designation, $\sum_{(Y,l)} m_{Y,l} \leq \varepsilon_{\text{CONC}}(D_*) n$ (Theorem 6.10(iv));
 - (2) in every round, JS-LC emits at most one J^{hub} -edge of any given class at any given hub (property (J1) of Lemma 6.15);
 - (3) the lift of every cycle of a decomposition of a quotient is a cycle of G , and the lifted objects are pairwise edge-disjoint (Lemma 7.9(ii)); every edge of G is covered exactly once, and in particular no junction edge is used both by a lifted cycle and by the device of its owner (Lemma 7.9(iii) and Theorem 6.19(b));

- (4) every per-round cost term is of one of the three kinds listed in the paragraph “Per-round costs here” of [the overview](#); in particular, no per-round cost term of order $\Theta(n)$ occurs other than a once-per-vertex charge.

Dependencies: none.

Overview: the architecture, and where the $\log^*$ disappears

This overview is prose; it contains no statement. Every claim it makes about the present argument is proved at the place it points to; the paragraph on [\[2\]](#) describes their published proof.

The chain. The argument is a chain of five links:

$$\text{UNI-MIX} \longrightarrow \text{MIX-C on } \text{HB}^{*\tau+} \text{ with no VX-parts} \longrightarrow \text{JV}^{+*} \longrightarrow \text{UH}^* \longrightarrow \text{layered HI}''.$$

UNI-MIX and MIX-C decompose all edges except a set of parity leftovers; JV^{+*} joins the ends of the leftovers by private junction edges to pooled junction vertices, and so routes them into simple quotient graphs; UH^* controls the size of these quotients (round-split pools and sub-layered ultra hubs); the layered induction HI'' closes the argument. Section [2](#) builds the hierarchy $\text{HB}^{*\tau+}$ (Lemma [14^{\tau}](#), Lemma SEP, the overlap bounds, rule GC, ORIGIN^τ , the tower facts); Section [3](#) proves the random splitting and linking tools (Lemma [15⁺](#), Theorem [16^{*}](#), Lemma HB) and defines the lending colouring COL-JV; Section [4](#) proves the vortex tools (TPV, the four-phase P-vortex, Theorem VX^+); Section [5](#) treats light parts (zones, the child and parent sides of the Theorem-U machinery, Lemma K-RED); Section [6](#) treats standalone parts (GATE, MED, EQ-LPT, PAR, HCC-P, Theorems CONC and CONC-L, Lemma JS-LC, Lemma J^+ and the MIX-C assembly for an abstract J-consumer); Section [7](#) constructs the junction-vertex quotient, bounds its size and the cost, and proves Theorem JV^{+*} , Theorem HI'' and the Main Theorem.

Where Bucić–Montgomery lose the factor $\log^* n$. The proof of Theorem 2 of [\[2\]](#) iterates their Lemma 26 about $\log^* n$ times: each iteration removes long cycles, splits the remaining edges into expanders by Lemma 14 of [\[2\]](#) (with $s = 0$), and decomposes each (necessarily small) expander into cycles and few edges; the average degree of the leftover drops from d to a polylogarithm of d . Each iteration costs $\Theta(n)$ objects, charged to all vertices that are active in it, and there are $\log^* n$ iterations.

Per-round costs here. The hierarchy $\text{HB}^{*\tau+}$ has the same round structure (rounds $l = 1, \dots, R$, with $R = O(\log^* n)$). In the present argument every per-round cost term is of one of three kinds.

- (a) *Decaying in the round index.* It is bounded by a constant multiple of one of n/M_l , $n/\log P_l$, $n\lambda_r^{-1/2}$, n/P_r , $nM_l^3 \log \lambda_{l-2}/\lambda_{l-2}^{95}$ (the parameters are those of Definition [2.7](#)) and $n\eta(\lambda_{l-2})$, where $\eta(x) = \log_2(2A \log_2(A \log_2 x))/(\log_2 x)^2$; the last bounds the number cap_l of visit-capping pieces of the Theorem-U chaining (Lemma [5.7\(ii\)](#), Step 4 of its proof). The hub charge is of this kind: a vertex can be a hub in many rounds, and the round- l term is $\sum_{Z \in \text{Std}_l} |A_Z| \leq 15.2 \varepsilon n / \log P_l$ (Proposition [2.13](#), (K2)), which sums over all rounds to at most $\varepsilon_A n$ (Lemma [2.17\(e\)](#)). Because the round parameters grow like a tower towards earlier rounds, each of these sums over all rounds to at most a quantity that depends only on D_* , tends to 0 as $D_* \rightarrow \infty$, and is uniform in R (tower-lacunary sums, Lemma [2.16](#); the terms $nM_l^3 \log \lambda_{l-2}/\lambda_{l-2}^{95}$ sum to at most $2F(\log_2 D_*) n$, with $F(x) = (3184 + 30400 \log_2 x) x^{-90.2}$, by Lemma [7.16\(iv\)](#), and the terms cap_l are part of $\varepsilon_{\text{ch}}(D_*) n$ in Lemma [5.7\(ii\)](#)).

- (b) *A once-per-vertex global charge.* Fresh ports (a vertex is fresh in at most two rounds, and TPV pays 169 per fresh port: $169 \cdot 2n = 338n$); parentless light incidences (at most two per vertex, each paying 369: $369 \cdot 2n$); unpaired fresh legs (at most $2n$); long cycles (at most n); and the final leftover E_0 (fewer than $D_*n/2$ edges).
- (c) *Per-ancestor terms with a factor $R - r$.* Some terms are charged to an ancestor Y of round r once in every later round, so that in total they carry a factor $R - r$: the τ -, d^* - and guest-cap terms of the concentration bound (Theorems 6.9(iii) and 6.10(iv)), and the lost-parent mass $\text{lp} = \sum_{Y \in \text{Bad}} |Y| (R - r(Y))$ (Definition 5.3). The factor satisfies $R - r \leq 2 \log^* d_r + 2 \leq \log \lambda_r$ (Lemma 2.17(a),(d)), so it depends on d_r alone, and it is outweighed by the decay of the rest of the term in λ_r (for lp , by $\mathbb{P}(Y \in \text{Bad}) \leq |Y|^{-2}$, Lemma 5.4(c)). Hence these terms, too, are at most (for lp : in expectation) n times a function of D_* that tends to 0 as $D_* \rightarrow \infty$ and is uniform in R (Theorem 6.10(iv), Lemma 5.5). This is why $\log^* D_*$, and not $\log^* n$, appears in $\varepsilon_{\text{CONC}}$.

The three mechanisms.

- (a) *No hub concentration.* The τ -rules inside the recursion of Lemma 14 of [2], together with rule GC, give $\sum_{(Y,l)} m_{Y,l} \leq \varepsilon_{\text{CONC}}(D_*)n$ for every valid run and every designation (Theorem 6.10(iv)). The chaining cost of JS-LC is at most $1.5 \sum_{(Y,l)} m_{Y,l}$ plus a decaying term, hence a vanishing fraction of n .
- (b) *TPV and JS-LC.* Classed ports are never retired by the vortex of their own part; their edges become beads, which are chained into cycles through junction paths inside lent classes of an ancestor of round at most $l - 2$. Only parity leftovers remain; they form the J-set J_l of Lemma 6.15.
- (c) *Private junction edges.* Each end of a J-item receives a private junction edge, which joins it to a junction vertex in a pool of density M_l^{-2} inside the expander of its ancestor, which has minimum degree larger than $s_r/2 \gg M_l$. A junction vertex may serve several ends, but every junction edge serves at most one (Lemma 7.9(iii)). The quotient Q_l is simple, every decomposition of Q_l lifts to a decomposition of the corresponding edges of G (Lemma 7.9), and $\sum_l |V(Q_l)| \leq \theta_Q n \leq n/4$ (Lemma 7.16). The Erdős–Gallai bound for the smaller graphs Q_l is supplied by minimality in the layered induction HI'' (Theorem 7.20).

The cost line. On the outcome chosen in Section 7 (Proposition 7.18),

$$f(G) \leq \left(\frac{D_*}{2} + \underbrace{745}_{\text{K-RED}} + \underbrace{338}_{\text{TPV, fresh}} + \underbrace{2}_{\text{fresh legs}} + \varepsilon_1(D_*) \right) n + 2 \sum_l f(Q_l),$$

with $745 + 338 + 2 = 1085$. Together with $\sum_l |V(Q_l)| \leq \theta_Q n$ and $\theta_Q \leq 1/4$, the induction of Theorem 7.20 gives $f(G) \leq C_0 n + 2c_{\text{EG}} \theta_Q n \leq c_{\text{EG}} n$.

1.1 Conventions

Convention 1.3 (graphs, logarithms, neighbourhoods, balls). (a) Graphs are finite and simple, unless they are explicitly called multigraphs (auxiliary multigraphs occur only in Sections 5 and 7). G always denotes the input graph and $n := |V(G)|$. For any graph H we write $|H| := |V(H)|$, as in [2]. For $k \in \mathbb{N}$, $[k] := \{1, \dots, k\}$.

- (b) $\log = \log_2$ throughout, as in [2]. We put $\log^{[0]} x := x$ and $\log^{[k]} x := \log(\log^{[k-1]} x)$, and $\log^* x$ is the least integer $k \geq 0$ with $\log^{[k]} x \leq 1$, as in [2].
- (c) For a vertex v of a graph H , $N_H(v)$ is its set of neighbours and $d_H(v) = \deg_H(v)$ its degree; $\delta(H)$ and $\Delta(H)$ are the minimum and maximum degree. For $U \subseteq V(H)$, $N_H(U)$ is the set of vertices of $V(H) \setminus U$ that have a neighbour in U . $H[U]$ is the induced subgraph; $H - U$ and $H \setminus U$ both denote $H[V(H) \setminus U]$; for $F \subseteq E(H)$, $H - F$ is obtained by deleting the edges of F (keeping all vertices). For disjoint $A, B \subseteq V(H)$, $E_H(A, B)$ is the set of edges with one end in A and the other in B , and $e_H(A, B) := |E_H(A, B)|$; for an edge set F and a vertex v , $\deg_F(v)$ is the number of edges of F at v .
- (d) A *path through V* is a path all of whose interior vertices lie in V ; its ends are unrestricted. For $U, V \subseteq V(H)$ and an integer $i \geq 0$, $B_H^i(U, V)$ is the set of vertices of V that can be reached by a path through V of length at most i starting at a vertex of U (the starting vertex need not lie in V); thus $B_H^i(U, V) \subseteq V$ (Cited result 1.7).
- (e) Families of pairs of vertices are *multisets*: a pair may occur several times, and “every vertex lies in at most t pairs” counts occurrences (Cited result 1.9).

Dependencies: none.

Definition 1.4 (objects and f). An *object* is a cycle (of length at least 3) or a single edge. A *decomposition* of an edge set F is a partition of F into the edge sets of objects; a decomposition of a graph H is a decomposition of $E(H)$. $f(F)$ is the least number of objects in a decomposition of F (so $f(\emptyset) = 0$); $f(H) := f(E(H))$; and $f(n) := \max\{f(H) : |V(H)| = n\}$.

Dependencies: none.

Fact 1.5 (elementary properties of f). (a) $f(F) \leq |F|$ for every edge set F .

(b) If F_1, F_2 are disjoint edge sets, then $f(F_1 \cup F_2) \leq f(F_1) + f(F_2)$.

(c) If H is the vertex-disjoint union of graphs H_1 and H_2 , then $f(H) = f(H_1) + f(H_2)$.

(d) Adding or deleting isolated vertices does not change f . In particular $f(n)$ is non-decreasing in n .

Dependencies: 1.4.

Proof. (a) The single edges of F form a decomposition of F into $|F|$ objects. (b) The union of a decomposition of F_1 and a decomposition of F_2 is a decomposition of $F_1 \cup F_2$, because $F_1 \cap F_2 = \emptyset$. (c) By (b), $f(H) \leq f(H_1) + f(H_2)$. Conversely, the edge set of every object induces a connected graph, so every object of a decomposition of $E(H)$ has all its edges in $E(H_1)$ or all in $E(H_2)$; hence every decomposition of $E(H)$ splits into a decomposition of $E(H_1)$ and one of $E(H_2)$, and $f(H) \geq f(H_1) + f(H_2)$. (d) $f(H)$ depends only on $E(H)$, which does not change. Adding an isolated vertex to an n -vertex graph gives an $(n+1)$ -vertex graph with the same f , so $f(n+1) \geq f(n)$. $\square$

Fact 1.6 (the long-cycle bound). (a) Every graph H with $h \geq 1$ vertices has a decomposition into at most $h(\log h + 1)$ objects, at most $h - 1$ of which are single edges. In particular $f(H) \leq h(\log h + 1)$.

- (b) Let $N > 1$ and $L := \log N$, and let α and β be real numbers with $0 \leq \alpha \leq N$, $\beta > 0$ and $4\beta \leq L$. If F is the edge set of a (simple) graph, so that F has no loops and no parallel edges, and every edge of F has both ends in a set W with $|W| \leq \beta\alpha/L$, then F has a decomposition into at most $\beta\alpha$ objects, at most $|W|$ of which are single edges.

The argument, iterated removal of a long cycle, is the elementary $O(n \log n)$ bound (see [6] and [2]); it is proved here in full. It uses that graphs are simple (Convention 1.3(a)) in two places: in the claim, where a vertex of degree at least d has at least d neighbours, and through $|E(H)| \leq \binom{h}{2}$. Part (b) is the form in which the vortex runs of Section 4 are finished.

Dependencies: 1.3, 1.4.

Proof. (a) Put $m := |E(H)|$, so that $m \leq \binom{h}{2} \leq h^2/2$.

Claim. If a graph H' with vertex set $V(H)$ has $m' \geq h$ edges, then H' contains a cycle of length at least m'/h . Put $d := \max\{2, \lceil m'/h \rceil\}$. Then $d - 1 \leq m'/h$: if $\lceil m'/h \rceil \geq 2$ then $d - 1 = \lceil m'/h \rceil - 1 < m'/h$, and otherwise $d - 1 = 1 \leq m'/h$ because $m' \geq h$. Delete from H' , one at a time, a vertex whose degree in the current graph is at most $d - 1$, as long as such a vertex exists. Every edge of H' disappears exactly once, together with the first of its two ends to be deleted. Suppose that all h vertices are deleted. The last of them has degree 0 when it is deleted, and each of the others has degree at most $d - 1$ when it is deleted; hence $m' \leq (h - 1)(d - 1) \leq (h - 1)m'/h < m'$, since $m' \geq h \geq 1$, a contradiction. So the deletions stop at a subgraph H^* of H' with at least one vertex and minimum degree at least $d \geq 2$. Let $v_0 v_1 \cdots v_r$ be a longest path in H^* . Every neighbour of v_0 in H^* is one of $v_1, \dots, v_r$, since otherwise the path could be extended. As v_0 has at least d neighbours in H^* , the largest index i with $v_0 v_i \in E(H^*)$ satisfies $i \geq d \geq 2$, so $v_0 v_1 \cdots v_i v_0$ is a cycle of H' of length $i + 1 > d \geq \lceil m'/h \rceil \geq m'/h$. This proves the claim.

Removal of long cycles. Put $H_0 := H$ and $m_i := |E(H_i)|$. For $i = 0, 1, \dots$, as long as $m_i \geq h$, choose by the claim (applied to $H' := H_i$) a cycle C_i of H_i of length at least m_i/h , and put $H_{i+1} := H_i - E(C_i)$. As $m_{i+1} < m_i$, this stops, at some $t \geq 0$ with $m_t \leq h - 1$. The cycles $C_0, \dots, C_{t-1}$ together with the m_t edges of H_t , taken as single edges, form a decomposition of $E(H)$ into $t + m_t$ objects, of which $m_t \leq h - 1$ are single edges. For $i < t$ we have $m_{i+1} = m_i - |E(C_i)| \leq (1 - 1/h)m_i$; hence $m_i \leq (1 - 1/h)^i m$ for $0 \leq i \leq t$.

We use $(1 - 1/h)^h \leq 1/2$. For $h = 1$ the left-hand side is 0. For $h \geq 2$ put $x := 1/h$; then $(1 - x)^h(1 + x)^h = (1 - x^2)^h \leq 1$ and, by the binomial theorem, $(1 + x)^h \geq 1 + hx = 2$, so $(1 - x)^h \leq 1/2$.

If $t = 0$, the decomposition has $m_0 \leq h - 1 \leq h(\log h + 1)$ objects, as $\log h \geq 0$. Let $t \geq 1$, and write $t - 1 = qh + r'$ with integers $q \geq 0$ and $0 \leq r' \leq h - 1$. Since C_{t-1} was chosen, $m_{t-1} \geq h$; as $t - 1 \geq qh$ and $0 \leq 1 - 1/h \leq 1$,

$$h \leq m_{t-1} \leq (1 - 1/h)^{t-1} m \leq ((1 - 1/h)^h)^q m \leq 2^{-q} m \leq 2^{-q} h^2/2.$$

Hence $2^{q+1} \leq h$, i.e. $q + 1 \leq \log h$, and $t = qh + r' + 1 \leq (q + 1)h \leq h \log h$. So the decomposition has $t + m_t \leq h \log h + h - 1 \leq h(\log h + 1)$ objects.

(b) Note that $L > 0$ as $N > 1$. If $W = \emptyset$, then $F = \emptyset$, and the empty decomposition has $0 \leq \beta\alpha$ objects. Otherwise $h := |W| \geq 1$, and $h \leq \beta\alpha/L \leq \beta N/L \leq N/4$ because $\alpha \leq N$, $\beta > 0$ and $4\beta \leq L$; hence $\log h \leq \log N - 2 = L - 2$. By (a), applied to the graph with vertex set W and edge set F , the set F has a decomposition into at most $h(\log h + 1) \leq h(L - 1) \leq hL \leq \beta\alpha$ objects, at most $h - 1 \leq |W|$ of which are single edges. $\square$

1.2 Cited results of Bucić–Montgomery

Every published result of [2] used in this paper is stated below in our notation, with its number in [2]; numbers refer to the arXiv version arXiv:2211.07689v2. Where this paper modifies a proof

of [2] (Lemma 14, Propositions 12 and 13, Lemma 17, Proposition 18, Lemma 19, Lemma 9) or extracts an explicit bound from a proof (Lemma 25), the relevant steps of the proof are summarised as well; the modified proofs themselves are written out in full where they are used. All items of this subsection are published results.

Cited result 1.7 ([2, Section 2.1, Section 2.6]). [2] uses the following notation, which is also the notation of this paper (Convention 1.3). For a graph G and $v \in V(G)$, $d_G(v)$ is the degree and $N_G(v)$ the neighbourhood of v ; $\Delta(G)$ is the maximum degree; for $U \subseteq V(G)$, $N_G(U)$ is the set of vertices of $V(G) \setminus U$ with a neighbour in U ; $G[V]$ is the induced subgraph, $G \setminus V := G[V(G) \setminus V]$, and $G - F$ is obtained by deleting the edge set F ; $|G|$ is the number of vertices. All logarithms have base 2; $\log^{[0]} n = n$, $\log^{[k]} n$ is the k -fold iterated logarithm, and $\log^* n$ is the least $k \geq 0$ with $\log^{[k]} n \leq 1$. A path through V is a path with all internal vertices in V , and $B_G^i(U, V)$ is the set of vertices of V that can be reached by a path through V of length at most i starting at a vertex of U (the starting vertex is not required to lie in V), so that $B_G^1(U, V) \subseteq V$.

Dependencies: none (published).

Cited result 1.8 ([2, Theorem 4] (Chernoff)). Let n be an integer, $0 \leq \delta, p \leq 1$, $X \sim \text{Bin}(n, p)$ and $\mu := \mathbb{E}X = np$. Then

$$\mathbb{P}(X > (1 + \delta)\mu) \leq e^{-\delta^2\mu/3} \quad \text{and} \quad \mathbb{P}(X < (1 - \delta)\mu) \leq e^{-\delta^2\mu/2}.$$

Dependencies: none (published).

Cited result 1.9 ([2, Definition 7]). A graph G is (ℓ, t) -path connected through a vertex set $V \subseteq V(G)$ if, for every collection $\mathcal{P} \subseteq \binom{V(G)}{2}$ in which every vertex lies in at most t pairs, there are edge-disjoint paths $P_{x,y}$, $\{x, y\} \in \mathcal{P}$, such that each $P_{x,y}$ is an xy -path through V of length at most ℓ .

Multiset clause (quoted in substance from [2]): here $\binom{V(G)}{2}$ denotes the multiset of pairs of distinct vertices of G ; in particular the same pair may occur several times in $\mathcal{P}$. Thus a pair occurring k times receives k pairwise edge-disjoint paths, and the bound t counts occurrences. The ends x, y are unrestricted (they need not lie in V); only interior vertices must lie in V .

Dependencies: none (published).

Cited result 1.10 ([2, Proposition 8]). Let $1 \leq \ell, t \leq n$, let G be an n -vertex graph and let $V \subseteq V(G)$ satisfy $|V| \geq 4t - 2$ and $|B_G^\ell(U, V)| > |V|/2$ for every $U \subseteq V(G)$ with $|U| = t$. If $x_1, \dots, x_{2t-1}, y_1, \dots, y_{2t-1}$ are distinct vertices of G , then for some $j \in [2t-1]$ there is an $x_j y_j$ -path in G through V of length at most $4\ell \log n$.

Dependencies: none (published).

Cited result 1.11 ([2, Lemma 9]). Let $n \geq 2$ and $1 \leq \ell, k \leq n$. Let G be an n -vertex graph and $V \subseteq V(G)$ with $|V| \geq n/8 + 1$, and suppose that for every non-empty $U \subseteq V(G)$ and every $F \subseteq E(G)$ with $|F| \leq 2^9 k |U| (\ell \log n)^2$ we have $|B_{G-F}^\ell(U, V)| > |V|/2$. Then G is $(4\ell \log n, k)$ -path connected through V .

Structure of the proof in [2] (modified in Lemma 3.7, which uses Haxell's theorem, Cited result 1.29, in place of Theorem 6). Let $\mathcal{P}$ be a collection of pairs in which every vertex lies in at most k pairs, let $r := |\mathcal{P}|$, and index the pairs by $i \in [r]$. Let H_i be the hypergraph on $E(G)$ whose edges are the edge sets of the paths through V of length at most $4\ell \log n$ joining the i -th pair. For $I \subseteq [r]$ let M_I be a maximal matching of $\bigcup_{i \in I} H_i$; the claim is $|M_I| \geq 4\ell \log n \cdot |I|$. If not, the

set F of edges covered by M_I has $|F| < (4\ell \log n)^2 |I|$. Let $I' \subseteq I$ be maximal such that no vertex lies in two pairs of I' ; then $2k|I'| \geq |I|$ and $|I'| \leq n/2$. With $t := \lceil |I'|/16 \rceil$ one has $2t - 1 \leq |I'|$, $4t - 2 \leq \lceil n/8 \rceil + 1 \leq |V|$ and $|F| < 8kt(8\ell \log n)^2$, so every t -set U satisfies $|B_{G-F}^\ell(U, V)| > |V|/2$; Proposition 8 (Cited result 1.10), applied to $G - F$ and $2t - 1$ pairs of I' , gives a path in $G - F$ joining a pair of I' , i.e. an edge of some H_j disjoint from M_I , contradicting maximality. Theorem 6 of [2] (Aharoni–Haxell [1]) then gives the paths. Since the pairs are indexed by $i \in [r]$, and the inequality $2k|I'| \geq |I|$ counts the pairs of I with multiplicity (every pair of I meets a vertex of a pair of I' , and each of the $2|I'|$ vertices of pairs of I' lies in at most k pairs of I), the argument applies verbatim to multisets $\mathcal{P}$.

Dependencies: none (published).

Cited result 1.12 ([2, Definition 11]). An n -vertex graph G is an (ε, s) -expander if for every $U \subseteq V(G)$ and every $F \subseteq E(G)$ with $1 \leq |U| \leq 2n/3$ and $|F| \leq s|U|$ we have

$$|N_{G-F}(U)| \geq \frac{\varepsilon|U|}{\log^2 n}.$$

Remark (from [2]; hypotheses made explicit): if $n \geq 2$ and $\varepsilon > 0$ then, for every $v \in V(G)$, the pair $U = \{v\}$, $F = \{\text{edges at } v\}$ violates this inequality, so necessarily $|F| > s$; hence $\delta(G) > s$ for every (ε, s) -expander G with $\varepsilon > 0$ on at least two vertices. (For $\varepsilon = 0$ every graph satisfies the inequality; wherever this remark is used below, $\varepsilon \geq 2^{-7}$.)

Dependencies: none (published).

Cited result 1.13 ([2, Proposition 12]). For $U \subseteq V(G)$ and $d > 0$ let $N_{G,d}(U)$ be the set of vertices of $V(G) \setminus U$ with at least d neighbours in U . Let G be an n -vertex (ε, s) -expander, $U \subseteq V(G)$ with $1 \leq |U| \leq 2n/3$, and F a set of at most $s|U|/2$ edges. Then for every $0 < d \leq s$, either (a) $|N_{G-F}(U)| \geq s|U|/(2d)$, or (b) $|N_{G-F,d}(U)| \geq \varepsilon|U|/\log^2 n$.

Proof in [2] (its counting is modified in Proposition 3.4). If (a) fails, the set $X := N_{G-F}(U) \setminus N_{G-F,d}(U)$ has fewer than $s|U|/(2d)$ vertices, and the set F' of edges of $G - F$ between U and X has $|F'| \leq d|X| \leq s|U|/2$. Then $|F \cup F'| \leq s|U|$ and $N_{G-F-F'}(U) = N_{G-F,d}(U)$, so the expansion of G gives (b).

Dependencies: none (published).

Cited result 1.14 ([2, Proposition 13]). There is n_0 such that the following holds for $n \geq n_0$, $1 \geq \varepsilon \geq 2^{-9}$ and $s \geq 8 \log^{13} n$. Let G be an n -vertex (ε, s) -expander, $U \subseteq V(G)$ with $|U| \leq 2n/3$, and F a set of at most $s|U|/4$ edges. Then $G - F$ contains either

- (a) $|U|/\log^7 n$ vertex-disjoint stars, each with $\log^9 n$ leaves, with centre in U and all leaves in $V(G) \setminus U$, or
- (b) a bipartite subgraph H with classes U and $X \subseteq V(G) \setminus U$ such that $|X| \geq \varepsilon|U|/(2 \log^2 n)$, every vertex of X has degree at least $\log^4 n$ in H , and every vertex of U has degree at most $2 \log^9 n$ in H .

Structure of the proof in [2] (modified in Proposition 3.4). Take a maximal family of vertex-disjoint stars as in (a), with centre set $C \subseteq U$ and leaf set L . If (a) fails, then $|C| \leq |U|/\log^7 n$, $|L| \leq |U| \log^2 n$, and by maximality no vertex of $U \setminus C$ has $\log^9 n$ neighbours in $G - F$ outside $U \cup L$, whence $|N_{G-F}(U \setminus C)| < 2|U| \log^9 n$. With $d := \log^4 n$ and $\Delta := 2 \log^9 n$, go through the vertices v of $V(G) \setminus U$ in turn and add a star with centre v and d leaves in U whenever the degrees

in U stay at most Δ ; X is the set of centres added and H the union of the stars. The saturated set U' (vertices of $U \setminus C$ of H -degree exactly Δ) satisfies $|U'| \leq 2d|L|/\Delta \leq \varepsilon|U|/(8\log^2 n)$, because a saturated vertex has at least $\Delta/2$ of its H -neighbours in L . With $B := C \cup U'$, Proposition 12 applied to $U \setminus B$ excludes option (a) there, and every vertex of $N_{G-F,d}(U \setminus B) \setminus B$ lies in X , giving $|X| \geq \varepsilon|U|/(2\log^2 n)$.

Dependencies: none (published).

Cited result 1.15 ([2, Lemma 14]). Given an n -vertex graph G , a non-negative integer s and $\varepsilon \leq 2^{-5}$, one can delete at most $4sn \log n$ edges of G so that the remaining edges can be partitioned into graphs $G_1, \dots, G_r$ with $\sum_{i=1}^r |G_i| \leq 2n$, each G_i an (ε, s) -expander. (For $s = 0$ nothing is deleted.)

Structure of the proof in [2] (modified in Lemma 2.6). The proof is by induction on n under the stronger bound $\sum_i |G_i| \leq 2n - 2n/(2 + \log n)$, which is at least n ; the case $n = 1$ is trivial, and if G is an (ε, s) -expander one takes $G_1 = G$. Otherwise there is a witness (U, F) : $1 \leq |U| \leq 2n/3$, $F \subseteq E(G)$, $|F| \leq s|U|$ and $|N_{G-F}(U)| < \varepsilon|U|/\log^2 n$. Put

$$G_1 := G[U \cup N_{G-F}(U)] - F, \quad G_2 := G \setminus U - E(G_1) - F,$$

which edge-partition $G - F$, and $n_1 := |G_1| = |U \cup N_{G-F}(U)|$, $n_2 := |G_2| = n - |U|$. Then

$$(6) \quad n_1 + n_2 = n + |N_{G-F}(U)| < n + \varepsilon n_1 / \log^2 n;$$

$$(7) \quad n_2 < n \quad \text{and} \quad n_1 \leq |U| + \varepsilon|U|/\log^2 n \leq 2n/3 + \varepsilon n \leq 3n/4 < n.$$

The induction hypothesis gives sets $E_i \subseteq E(G_i)$ with $|E_i| \leq 4sn_i \log n_i$ and partitions of $G_i - E_i$ into (ε, s) -expanders $G_{i,j}$ with $\sum_j |G_{i,j}| \leq 2n_i - 2n_i/(2 + \log n_i)$. Using $\log n_1 \leq \log(3n/4) < \log n - 2/5$:

$$(8) \quad |F| + |E_1| + |E_2| \leq s(|U| + 4n_1 \log n_1 + 4n_2 \log n_2) \leq s(4(n_1 + n_2) \log n - \frac{3}{5}n_1) \leq 4sn \log n,$$

$$(9) \quad \frac{2n_1}{2 + \log n_1} > \frac{2n_1}{2 + \log n} + \frac{n_1}{10 \log^2 n},$$

where the last step of (8) uses (6). For $n \geq 3$, (9) follows from $\log n_1 < \log n - 2/5$ and $n_1 < n$: then $\frac{2}{2 + \log n_1} - \frac{2}{2 + \log n} > \frac{0.8}{(2 + \log n)^2} \geq \frac{1}{10 \log^2 n}$, where the last step holds because $\log n \geq \log 3 > 2/(2\sqrt{2} - 1)$. For $n = 2$ we have $n_1 = 1$ by (7), and (9) reads $1 > \frac{2}{3} + \frac{1}{10}$. Combining (9) with (6) gives $\sum_{i,j} |G_{i,j}| \leq (n_1 + n_2)(2 - \frac{2}{2 + \log n}) - \frac{n_1}{10 \log^2 n} \leq 2n - \frac{2n}{2 + \log n}$. The deleted edges are $F \cup E_1 \cup E_2$.

Dependencies: none (published).

Cited result 1.16 ([2, Lemma 15]). Let $n, s \in \mathbb{N}$, $k \geq 1$ an integer, and $0 < \varepsilon \leq 1$, and let G be an n -vertex (ε, s) -expander with $s \geq 2^{12}\varepsilon^{-1}k^2 \log^4 n$. Then there are edge-disjoint graphs $G_1, \dots, G_k$ with $E(G) = \bigcup_i E(G_i)$, each an $(\varepsilon/4, \sqrt{s\varepsilon}/(8k \log n))$ -expander with vertex set $V(G)$.

This result is quoted for comparison only and is not used: it is replaced by Lemma 3.2 (Lemma 15⁺), which is proved directly and loses neither the factor 4 in ε nor the square root in s .

Dependencies: none (published).

Cited result 1.17 ([2, Theorem 16]). Let G be an n -vertex (ε, s) -expander with $1 \geq \varepsilon \geq 2^{-7}$ and $s \geq \log^{135} n$, and let $V \subseteq V(G)$ contain each vertex independently with probability $1/3$. Then with high probability G is $(4 \log^5 n, 2^8 \log^5 n)$ -path connected through V .

This theorem is not used as a black box. Its proof (split $E(G)$ by Lemma 15 into $k = 2^{17} \log^{42} n$ expanders; apply Lemma 19 in each; for a given U and F with $|F| \leq 2^{17}|U| \log^{15} n$ some class contains at most $|U|/\log^{27} n$ edges of F ; conclude by Lemma 9 with $k = 2^8 \log^5 n$ and $\ell = \log^4 n$) is modified in Theorem 3.8.

Dependencies: none (published).

Cited result 1.18 ([2, Lemma 17]). Let $n \geq 2$ and let G be an n -vertex (ε, s) -expander with $2^{-9} \leq \varepsilon \leq 1$ and $s \geq 8 \log^{13} n$. Let $U \subseteq V(G)$ satisfy $|N_G(U)| \geq |U| \log^{24} n$, let $F \subseteq E(G)$ satisfy $|F| \leq |U|$, and let $V \subseteq V(G)$ contain each vertex independently with probability $1/3$. Then with probability $1 - e^{-\Omega(|U| \log^2 n)}$ we have $|B_{G-F}^{\log^4 n}(U, V)| > |V|/2$.

Proof parameters and structure in [2] (modified in Lemma 3.5). $\ell := \log^4 n$, $q := 3/11$, and p is defined by $(1-p)^{\ell-1} = 11/12$, so that $p \geq 1/(15 \log^4 n)$; V is realised as $V_1 \cup \dots \cup V_\ell$ with independent V_i , each vertex in V_i with probability p for $i < \ell$ and q for $i = \ell$. B_i is the set of vertices reachable from U by a path of $G - F$ of length at most i whose interior lies in $V_1 \cup \dots \cup V_{i-1}$; it is determined by $U, V_1, \dots, V_{i-1}$, every vertex of $N_{G-F}(B_i)$ with a $G - F$ -neighbour in $B_i \cap V_i$ lies in B_{i+1} , and $B_\ell \cap V_\ell \subseteq B_{G-F}^\ell(U, V)$. The growth of B_i is shown via Proposition 13: in case (a) by Chernoff's bound on the star centres, in case (b) by a bounded-differences inequality for the number of vertices of X with an H -neighbour in V_i ; the final stage uses Chernoff's bound with density q .

Dependencies: none (published).

Cited result 1.19 ([2, Proposition 18]). Let $n \geq 2$, $0 < \varepsilon \leq 1$ and $s \geq \log^{24} n$. Let G be an n -vertex (ε, s) -expander and $U \subseteq V(G)$ with $|U| \leq 2n/3$. Then there is $U' \subseteq U$ with $|N_G(U')| \geq |U'| \log^{24} n$ and $|U'| \geq \varepsilon|U|/(3 \log^{26} n)$.

Proof in [2] (modified in Lemma 3.6). Put $\theta := \log^{24} n$ and let $U' \subseteq U$ be maximal with $|N_G(U')| \geq \theta|U'|$. Let F be the set of edges from $U \setminus U'$ to $V(G) \setminus (U' \cup N_G(U'))$. Expansion applied to U and F gives $\varepsilon|U|/\log^2 n \leq |N_{G-F}(U)| \leq |N_G(U')| < (|U'| + 1)\theta + 1 \leq 3|U'|\theta$.

Remark. The maximality step gives, for every $v \in U \setminus U'$, fewer than $\theta + 1$ neighbours outside $U' \cup N_G(U')$ (adding v to U' removes at most the one vertex v from the neighbourhood); [2] write “at most θ ”, which is exact only when θ is an integer. Hence $|F| \leq (\theta + 1)|U \setminus U'| \leq (\theta + 1)|U|$, and the proof needs $s \geq \theta + 1$ rather than $s \geq \theta$. Lemma 3.6 is formulated with $s_1 \geq \theta_* + 1$ accordingly.

Dependencies: none (published).

Cited result 1.20 ([2, Lemma 19]). Let G be an n -vertex (ε, s) -expander with $2^{-9} \leq \varepsilon \leq 1$ and $s \geq 2 \log^{24} n$, and let $V \subseteq V(G)$ contain each vertex independently with probability $1/3$. Then with probability $1 - o(1/n)$, for every non-empty $U \subseteq V(G)$ and every $F \subseteq E(G)$ with $|F| \leq |U|/\log^{27} n$, we have $|B_{G-F}^{\log^4 n}(U, V)| > |V|/2$. (The statement in [2] says “every U ”; the empty set is excluded implicitly.)

Structure of the proof in [2] (modified in Lemma 3.6). Call U' well-expanding if $|N_G(U')| \geq |U'| \log^{24} n$. A union bound of the failure probability of Lemma 17 over all pairs (U', F) with U' well-expanding and $|F| \leq |U'|$ gives failure probability $o(1/n)$. For $|U| \leq 2n/3$, Proposition 18 gives a well-expanding $U' \subseteq U$ with $|U'| \geq \varepsilon|U|/(3 \log^{26} n) \geq |F|$, and the ball is monotone in U ; for $|U| > 2n/3$ one passes to a subset $\bar{U}$ with $n/2 \leq |\bar{U}| \leq 2n/3$.

Dependencies: none (published).

Cited result 1.21 ([2, Theorem 21] (Lovász [11])). Every n -vertex graph can be decomposed into at most $n/2$ paths and cycles.

Dependencies: none (published).

Cited result 1.22 ([2, Corollary 22]). Every graph can be decomposed into paths such that each vertex is an end of at most two of the paths. (The proof in [2] adds a new vertex joined to every vertex of even degree, applies Theorem 21, and deletes the new vertex.)

Consequence used in this paper. If every edge of a graph H has both ends in a set W , then $E(H)$ decomposes into at most $|W|$ paths. Indeed, apply the corollary to the graph $(W, E(H))$:

every path has two distinct ends in W and every vertex of W is an end of at most two paths, so twice the number of paths is at most $2|W|$.

Dependencies: none (published).

Cited result 1.23 ([2, Lemma 25], explicit form). *Let $\varepsilon \geq 2^{-5}$ and $m \geq \max(2, 2^{30}/\varepsilon^2)$ (so $m \geq 2^{40}$ suffices at $\varepsilon = 2^{-5}$). Every m -vertex $(\varepsilon, 0)$ -expander contains a cycle of length at least $\varepsilon^2 m / (18 \log^4 m)$. (The bound $m \geq 2$, which excludes the one-vertex graph, follows from $m \geq 2^{30}/\varepsilon^2$ unless $\varepsilon^2 > 2^{29}$; without it the statement would be false for such ε at $m = 1$.)*

Explicit form from the proof in [2]. The statement of [2] says “a cycle of length $\Omega(m/\log^4 m)$ ”. The proof runs a depth-first search, finds a moment at which the unexplored set U and the processed set R have equal size, so that the current path P contains $N_G(U)$ and has at least $\varepsilon m / (3 \log^2 m)$ vertices; it splits P into consecutive segments X, Y, Z with $|X|, |Z| \geq |P|/3$ and $\varepsilon^2 m / (18 \log^4 m) \leq |Y| < \varepsilon^2 m / (9 \log^4 m)$. If X and Z are joined by a path avoiding Y , a shortest such path together with a segment of P is a cycle containing Y , of length at least $|Y| \geq \varepsilon^2 m / (18 \log^4 m)$; otherwise expansion of the side of size at most $m/2$ gives a contradiction. This explicit bound is the form used in Lemma 2.9.

Dependencies: none (published).

Remark 1.24 (where the results of [2] are used).

Result of [2]	Used in
Definition 7 (1.9)	Sections 3, 4, 5, 6
Lemma 9, Proposition 8 (1.11, 1.10)	Lemma 3.7
Theorem 6 (Aharoni–Haxell)	not used (Lemma 3.7 uses Haxell’s theorem [8], Cited result 1.29, instead)
Definition 11 (1.12)	everywhere
Propositions 12–13, Lemma 17, Proposition 18, Lemma 19	Section 3 (proof of Theorem 16*)
Lemma 14 and its proof (1.15)	Section 2
Corollary 22 (1.22)	Section 4
Lemma 25 (1.23)	Lemma 2.9
Theorem 4 (1.8)	Sections 3, 4
Lemma 15, Theorem 16, Theorem 21	not used directly (Lemma 15 is replaced by Lemma 3.2; Theorem 16 by Theorem 3.8; Theorem 21 enters only through Corollary 22)

Theorem 2 of [2] (the bound $O(n \log^* n)$) is not used, and neither are Theorem 24 and Lemma 26 of [2], which are used only in its proof: the vortex runs of Section 4 are finished with the elementary long-cycle bound of Fact 1.6. The three results of the row “not used directly” are not dependencies of this remark.

Dependencies: 1.7, 1.8, 1.9, 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.18, 1.19, 1.20, 1.22, 1.23, 1.6.

1.3 Other cited and standard results

The following standard results are used. Where a short derivation of the exact form used here is needed, it is written out. One of them, Lemma BBD below (a Bernstein inequality for functions of independent indicators with bounded differences), is not cited but proved in full. Its probabilistic part uses only finite sums. Its analytic part uses only elementary properties of the exponential function (such as $e^v \geq 1 + v$ for real v) and the fact that a differentiable function whose derivative is non-negative on an interval is non-decreasing there; the one elementary one-variable inequality that it needs is proved there.

Cited result 1.25 (Chernoff bounds for sums of independent indicators; [9, Theorems 2.1 and 2.8]). Let $X = \sum_{i=1}^m I_i$ be a sum of independent indicator variables with $\mathbb{P}(I_i = 1) = p_i$ (a Poisson-binomial variable), and let $0 \leq \delta \leq 1$.

(a) If $\mu \geq \mathbb{E}X$ then $\mathbb{P}(X \geq (1 + \delta)\mu) \leq e^{-\delta^2\mu/3}$; if $0 \leq \mu \leq \mathbb{E}X$ then $\mathbb{P}(X \leq (1 - \delta)\mu) \leq e^{-\delta^2\mu/2}$. In particular the two bounds of Cited result 1.8 hold verbatim with $\mu = \mathbb{E}X$.

(b) If $\mathbb{E}X \leq \mu$ then for every integer $j \geq 1$,

$$\mathbb{P}(X \geq j) \leq \frac{\mu^j}{j!} \leq \left(\frac{e\mu}{j}\right)^j.$$

Dependencies: none (standard).

Derivation of the forms used. (a) By [9, Theorems 2.1 and 2.8], with $\lambda := \mathbb{E}X$ and $t \geq 0$, $\mathbb{P}(X \geq \lambda + t) \leq \exp(-t^2/(2(\lambda + t/3)))$ and $\mathbb{P}(X \leq \lambda - t) \leq \exp(-t^2/(2\lambda))$. For the upper tail put $t := (1 + \delta)\mu - \lambda \geq \delta\mu$; the exponent $t^2/(2(\lambda + t/3))$ is increasing in t and, for fixed μ , decreasing in λ (as λ decreases, t increases and $\lambda + t/3 = \frac{2}{3}\lambda + \frac{1}{3}(1 + \delta)\mu$ decreases), so it is at least its value at $\lambda = \mu$, namely $\delta^2\mu/(2(1 + \delta/3)) \geq \delta^2\mu/3$ since $\delta \leq 1$. For the lower tail write $\lambda = \mu + a$ with $a \geq 0$ and put $t := \lambda - (1 - \delta)\mu = \delta\mu + a \geq 0$. Then $t^2 = \delta^2\mu^2 + 2\delta\mu a + a^2 \geq \delta^2\mu^2 + \delta^2\mu a = \delta^2\mu\lambda$ (as $\delta \leq 2$), so $t^2/(2\lambda) \geq \delta^2\mu/2$ (the case $\lambda = 0$ is trivial). (b) If $X \geq j$ then some j -set $S \subseteq [m]$ has $I_i = 1$ for all $i \in S$, so $\mathbb{P}(X \geq j) \leq \sum_{|S|=j} \prod_{i \in S} p_i \leq (\sum_i p_i)^j / j! \leq \mu^j / j!$, because every product $\prod_{i \in S} p_i$ occurs $j!$ times in the expansion of $(\sum_i p_i)^j$ and all terms are non-negative. Finally $j! \geq (j/e)^j$. $\square$

Cited result 1.26 (Markov's inequality and its two uses). If $X \geq 0$ and $a > 0$ then $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Consequently:

(a) for $X \geq 0$, $\mathbb{P}(X \leq 3\mathbb{E}X) \geq 2/3$;

(b) for $X_1, X_2 \geq 0$, $\mathbb{P}(X_1 \leq 4\mathbb{E}X_1 \text{ and } X_2 \leq 4\mathbb{E}X_2) \geq 1/2$;

(c) (a) and (b) hold with $\mathbb{P}$ and $\mathbb{E}$ replaced by the conditional probability and expectation given any event of positive probability, or given any fixed outcome of variables that are independent of the variables being drawn.

Dependencies: none (standard).

Derivation. (a) If $\mathbb{E}X = 0$ then $X = 0$ almost surely. Otherwise $\mathbb{P}(X > 3\mathbb{E}X) \leq \mathbb{P}(X \geq 3\mathbb{E}X) \leq 1/3$. (b) Similarly $\mathbb{P}(X_i > 4\mathbb{E}X_i) \leq 1/4$ for $i = 1, 2$, and a union bound gives failure probability at most $1/2$. (c) The same argument in the conditional probability space; if the conditioning fixes variables independent of those being drawn, the conditional law of the latter is their unconditional law. $\square$

Lemma 1.27 (Lemma BBD: Bernstein inequality for bounded differences). Let $m \geq 0$ be an integer, let $p_1, \dots, p_m \in [0, 1]$, and let $I_1, \dots, I_m$ be independent random variables with $\mathbb{P}(I_k = 1) = p_k$ and $\mathbb{P}(I_k = 0) = 1 - p_k$; write $I := (I_1, \dots, I_m)$. Let $\Psi: \{0, 1\}^m \rightarrow \mathbb{R}$, and let $b \geq 0$ and $c_1, \dots, c_m \in [0, b]$ be such that $|\Psi(x) - \Psi(x')| \leq c_k$ whenever $x, x' \in \{0, 1\}^m$ differ only in the k -th coordinate. Let $\beta > 0$ with $\beta \geq \sum_{k=1}^m p_k(1 - p_k)c_k^2$. Then for every $a \geq 0$,

$$\mathbb{P}(\Psi(I) \geq \mathbb{E}\Psi(I) + a) \leq \exp\left(-\frac{a^2}{2(\beta + ba/3)}\right) \quad \text{and} \quad \mathbb{P}(\Psi(I) \leq \mathbb{E}\Psi(I) - a) \leq \exp\left(-\frac{a^2}{2(\beta + ba/3)}\right).$$

Dependencies: none (standard).

Proof. This is a standard Bernstein-type inequality; we prove it by the exponential-moment method. All sums in this proof are finite. For $0 \leq k \leq m$, $x = (x_1, \dots, x_k) \in \{0, 1\}^k$ and $z = (z_{k+1}, \dots, z_m) \in \{0, 1\}^{m-k}$ put

$$\text{wt}_k(x) := \prod_{j=1}^k (p_j x_j + (1 - p_j)(1 - x_j)), \quad \text{wt}'_k(z) := \prod_{j=k+1}^m (p_j z_j + (1 - p_j)(1 - z_j))$$

(empty products are 1), and write $(x, z) \in \{0, 1\}^m$ for the concatenation. These weights are non-negative, $\sum_{x \in \{0, 1\}^k} \text{wt}_k(x) = \prod_{j \leq k} (p_j + (1 - p_j)) = 1$, and likewise $\sum_{z \in \{0, 1\}^{m-k}} \text{wt}'_k(z) = 1$. By independence, $\mathbb{P}(I = x) = \text{wt}_m(x)$ for $x \in \{0, 1\}^m$. Hence $\mathbb{P}(I \in S) = \sum_{x \in S} \text{wt}_m(x)$ for $S \subseteq \{0, 1\}^m$, and $\mathbb{E}g(I) = \sum_x \text{wt}_m(x)g(x)$ for $g: \{0, 1\}^m \rightarrow \mathbb{R}$.

Step 1: partial averages. For $0 \leq k \leq m$ and $x \in \{0, 1\}^k$ put

$$\Psi_k(x) := \sum_{z \in \{0, 1\}^{m-k}} \text{wt}'_k(z) \Psi(x, z).$$

Then $\Psi_m = \Psi$, and, as $\text{wt}'_0 = \text{wt}_m$, Ψ_0 is the constant $\sum_x \text{wt}_m(x) \Psi(x) = \mathbb{E}\Psi(I)$. Let $1 \leq k \leq m$ and $y \in \{0, 1\}^{k-1}$. Split the sum defining $\Psi_{k-1}(y)$ according to the first coordinate z_k of $z = (z_k, z')$, and use $\text{wt}'_{k-1}(z_k, z') = (p_k z_k + (1 - p_k)(1 - z_k)) \text{wt}'_k(z')$. This gives

$$\Psi_{k-1}(y) = p_k \Psi_k(y, 1) + (1 - p_k) \Psi_k(y, 0). \quad (1.1)$$

Put $\Xi_k(y) := \Psi_k(y, 1) - \Psi_k(y, 0) = \sum_z \text{wt}'_k(z) (\Psi(y, 1, z) - \Psi(y, 0, z))$. The points $(y, 1, z)$ and $(y, 0, z)$ differ only in the k -th coordinate, so $|\Xi_k(y)| \leq \sum_z \text{wt}'_k(z) c_k = c_k$. By (1.1),

$$\Psi_k(y, x_k) - \Psi_{k-1}(y) = (x_k - p_k) \Xi_k(y) \quad (x_k \in \{0, 1\}). \quad (1.2)$$

Step 2: an elementary inequality. For every real u ,

$$\left(1 - \frac{u}{3}\right) e^u \leq 1 + \frac{2u}{3} + \frac{u^2}{6}. \quad (1.3)$$

Indeed, let $\chi(u)$ be the right side minus the left side. Then $\chi(0) = 0$ and $\chi'(u) = \frac{2}{3} + \frac{u}{3} - \left(\frac{2}{3} - \frac{u}{3}\right) e^u$, so $\chi'(0) = 0$; and $\chi''(u) = \frac{1}{3}(1 - (1 - u)e^u) \geq 0$, because $1 - u \leq e^{-u}$. Hence χ' is non-decreasing, so $\chi' \leq 0$ on $(-\infty, 0]$ and $\chi' \geq 0$ on $[0, \infty)$. Thus χ is non-increasing on $(-\infty, 0]$ and non-decreasing on $[0, \infty)$, and $\chi \geq \chi(0) = 0$. Now let $0 \leq \zeta < 3$ and $u \leq \zeta$. Then $1 - u/3 \geq 1 - \zeta/3 > 0$ and $1 + \frac{2u}{3} + \frac{u^2}{6} = \left(1 - \frac{u}{3}\right)(1 + u) + \frac{u^2}{2}$. Dividing (1.3) by $1 - u/3$ gives

$$e^u \leq 1 + u + \frac{u^2}{2(1 - u/3)} \leq 1 + u + \frac{u^2}{2(1 - \zeta/3)}. \quad (1.4)$$

Step 3: one coordinate. Let $\eta \geq 0$ with $\eta b < 3$ (it is chosen in Step 5), and put $\zeta := \eta b$. Let $1 \leq k \leq m$ and $y \in \{0, 1\}^{k-1}$, and write $p := p_k$ and $\Xi := \Xi_k(y)$. The numbers $\eta(1 - p)\Xi$ and $-\eta p \Xi$ are at most $\eta|\Xi| \leq \eta c_k \leq \eta b = \zeta$. Apply (1.4) to both, multiply by $p \geq 0$ and $1 - p \geq 0$ respectively, and add. Since $p(1 - p) - (1 - p)p = 0$ and $p(1 - p)^2 + (1 - p)p^2 = p(1 - p)$, this gives

$$p e^{\eta(1-p)\Xi} + (1 - p) e^{-\eta p \Xi} \leq 1 + \frac{\eta^2 p(1 - p)\Xi^2}{2(1 - \zeta/3)} \leq \exp\left(\frac{\eta^2 p_k(1 - p_k)c_k^2}{2(1 - \zeta/3)}\right), \quad (1.5)$$

where the last step uses $\Xi^2 \leq c_k^2$ and $1 + v \leq e^v$.

Step 4: induction over the coordinates. For $0 \leq k \leq m$ put

$$\mathcal{E}_k := \sum_{x \in \{0,1\}^k} \text{wt}_k(x) \exp(\eta(\Psi_k(x) - \Psi_0)).$$

Then $\mathcal{E}_0 = 1$, and $\mathcal{E}_m = \mathbb{E} \exp(\eta(\Psi(I) - \mathbb{E}\Psi(I)))$. Let $1 \leq k \leq m$, and write $x \in \{0,1\}^k$ as (y, x_k) with $y \in \{0,1\}^{k-1}$; then $\text{wt}_k(y, 1) = p_k \text{wt}_{k-1}(y)$ and $\text{wt}_k(y, 0) = (1 - p_k) \text{wt}_{k-1}(y)$. By (1.2), and then by (1.5) and the non-negativity of all weights,

$$\begin{aligned} \mathcal{E}_k &= \sum_{y \in \{0,1\}^{k-1}} \text{wt}_{k-1}(y) e^{\eta(\Psi_{k-1}(y) - \Psi_0)} \left(p_k e^{\eta(1-p_k)\Xi_k(y)} + (1-p_k) e^{-\eta p_k \Xi_k(y)} \right) \\ &\leq \mathcal{E}_{k-1} \exp\left(\frac{\eta^2 p_k (1-p_k) c_k^2}{2(1-\zeta/3)}\right). \end{aligned}$$

By induction on k , and since $\sum_k p_k (1-p_k) c_k^2 \leq \beta$, $\mathcal{E}_m \leq \exp(\eta^2 \beta / (2(1-\zeta/3)))$.

Step 5: the upper tail. Let $S := \{x \in \{0,1\}^m : \Psi(x) \geq \Psi_0 + a\}$. For $x \in S$ we have $\exp(\eta(\Psi(x) - \Psi_0 - a)) \geq 1$, as $\eta \geq 0$, and all terms below are non-negative. Hence

$$\begin{aligned} \mathbb{P}(\Psi(I) \geq \mathbb{E}\Psi(I) + a) &= \sum_{x \in S} \text{wt}_m(x) \leq e^{-\eta a} \sum_{x \in \{0,1\}^m} \text{wt}_m(x) e^{\eta(\Psi(x) - \Psi_0)} \\ &= e^{-\eta a} \mathcal{E}_m \leq \exp\left(-\eta a + \frac{\eta^2 \beta}{2(1-\eta b/3)}\right). \end{aligned}$$

Now choose $\eta := a/(\beta + ba/3) \geq 0$. Since $\beta > 0$, we have $\eta b/3 = (ba/3)/(\beta + ba/3) < 1$, so $\eta b < 3$ as required in Step 3, and $1 - \eta b/3 = \beta/(\beta + ba/3)$. Therefore

$$-\eta a + \frac{\eta^2 \beta}{2(1-\eta b/3)} = -\frac{a^2}{\beta + ba/3} + \frac{a^2}{(\beta + ba/3)^2} \cdot \frac{\beta(\beta + ba/3)}{2\beta} = -\frac{a^2}{2(\beta + ba/3)},$$

which is the first bound.

Step 6: the lower tail. The function $-\Psi$ satisfies the hypotheses with the same $b, c_1, \dots, c_m$ and β , and $\mathbb{E}(-\Psi(I)) = -\mathbb{E}\Psi(I)$. So the first bound for $-\Psi$ is the second bound for Ψ . $\square$

Cited result 1.28 (Hall's theorem [7]). *Let $\mathcal{J}$ be a finite set and, for every $a \in \mathcal{J}$, let $\mathcal{T}(a)$ be a finite set. If $|\bigcup_{a \in \mathcal{J}'} \mathcal{T}(a)| \geq |\mathcal{J}'|$ for every $\mathcal{J}' \subseteq \mathcal{J}$, then there is an injective map χ from $\mathcal{J}$ to $\bigcup_{a \in \mathcal{J}} \mathcal{T}(a)$ with $\chi(a) \in \mathcal{T}(a)$ for every $a \in \mathcal{J}$ (a system of distinct representatives).*

Dependencies: none (standard).

Cited result 1.29 (Haxell's condition for matchability [8]). *Let $q \geq 1$ be an integer, and let $\mathcal{H}$ be a hypergraph (a set of subsets, called edges, of a finite vertex set) whose vertex set is the disjoint union of two sets $\mathcal{X}$ and $\mathcal{Y}$, such that every edge e satisfies $|e \cap \mathcal{X}| = 1$ and $|e \cap \mathcal{Y}| \leq q$. Suppose that for every $\mathcal{X}' \subseteq \mathcal{X}$ and every $Z \subseteq \mathcal{Y}$ with $|Z| \leq (2q-1)(|\mathcal{X}'|-1)$ there is an edge e of $\mathcal{H}$ with $e \cap \mathcal{X} \subseteq \mathcal{X}'$ and $e \cap Z = \emptyset$. Then $\mathcal{H}$ has an $\mathcal{X}$ -saturating matching, i.e. a set of pairwise disjoint edges such that every vertex of $\mathcal{X}$ lies in one of them.*

For $\mathcal{X}' = \emptyset$ there is no such Z , because $(2q-1)(0-1) < 0$. For $q = 1$ (edges with at most two vertices) the result is Hall's theorem; a vertex a with $\{a\} \in \mathcal{H}$ is matched by that edge. In the later literature the result of [8] is quoted for hypergraphs whose edges have at most (in some sources: exactly) r vertices, one of them in $\mathcal{X}$, with the bound $(2r-3)(|\mathcal{X}'|-1)$. With $r = q+1$, the maximum edge size, this bound is $(2(q+1)-3)(|\mathcal{X}'|-1) = (2q-1)(|\mathcal{X}'|-1)$: the stated form agrees

with Haxell's theorem as quoted there, and the case $q = 1$ (Hall's theorem) confirms this reading of the parameters. The only application, in the proof of Lemma 3.7, verifies the hypothesis for every Z with $|Z| < q^2|\mathcal{X}'|$; so that application stays valid if $2q - 1$ is replaced by any positive factor at most q^2 . Unlike the other cited items of this subsection, which are standard textbook results, this is a published research result. The derivation below shows that the stated form follows from the special case in which every edge meets $\mathcal{Y}$ in exactly q vertices; so the stated form holds if [8] proves either of the two quoted forms. After the derivation, a complete proof of the stated form is written out (Haxell's alternating-tree argument), so that the application does not depend on the precise form in which the result is stated in [8].

Dependencies: none (published).

Derivation from the case $|e \cap \mathcal{Y}| = q$. Let $\mathcal{H}$ be as stated, and let $\mathcal{X}_0$ be the set of $a \in \mathcal{X}$ such that $\{a\}$ is an edge. Every edge e with $e \cap \mathcal{X} \not\subseteq \mathcal{X}_0$ has $e \cap \mathcal{Y} \neq \emptyset$ (otherwise $e = \{a\}$ with $a \in \mathcal{X}_0$). For each such e take a set N_e of $q - |e \cap \mathcal{Y}|$ new vertices, the sets N_e pairwise disjoint and disjoint from $\mathcal{X} \cup \mathcal{Y}$, and put $e^+ := e \cup N_e$. Let $\mathcal{H}^+$ be the hypergraph on $(\mathcal{X} \setminus \mathcal{X}_0) \sqcup \mathcal{Y}^+$, where $\mathcal{Y}^+ := \mathcal{Y} \cup \bigcup_e N_e$, whose edges are these sets e^+ . Every edge of $\mathcal{H}^+$ meets $\mathcal{X} \setminus \mathcal{X}_0$ in exactly one vertex and $\mathcal{Y}^+$ in exactly q vertices.

We check the hypothesis for $\mathcal{H}^+$. Let $\mathcal{X}' \subseteq \mathcal{X} \setminus \mathcal{X}_0$ and $Z^+ \subseteq \mathcal{Y}^+$ with $|Z^+| \leq (2q - 1)(|\mathcal{X}'| - 1)$. For every $z \in Z^+ \setminus \mathcal{Y}$ let f_z be the unique edge of $\mathcal{H}$ with $z \in N_{f_z}$, and choose a vertex $b_z \in f_z \cap \mathcal{Y}$. Put $Z := (Z^+ \cap \mathcal{Y}) \cup \{b_z : z \in Z^+ \setminus \mathcal{Y}\} \subseteq \mathcal{Y}$. Then $|Z| \leq |Z^+|$, so the hypothesis for $\mathcal{H}$ gives an edge e with $e \cap \mathcal{X} \subseteq \mathcal{X}'$ and $e \cap Z = \emptyset$. As $|e \cap \mathcal{X}| = 1$, the set $e \cap \mathcal{X}$ is a single vertex of $\mathcal{X}' \subseteq \mathcal{X} \setminus \mathcal{X}_0$; so $e \cap \mathcal{X} \not\subseteq \mathcal{X}_0$, the set e^+ is an edge of $\mathcal{H}^+$, and $e^+ \cap (\mathcal{X} \setminus \mathcal{X}_0) = e \cap \mathcal{X} \subseteq \mathcal{X}'$. Moreover $e^+ \cap Z^+ = \emptyset$: a vertex of $e^+ \cap Z^+$ lies either in $e \cap \mathcal{Y}$, and then in $Z \cap e$; or it is some $z \in N_e$, and then $f_z = e$ and $b_z \in e \cap Z$. Both contradict $e \cap Z = \emptyset$.

By the case $|e \cap \mathcal{Y}| = q$, $\mathcal{H}^+$ has an $(\mathcal{X} \setminus \mathcal{X}_0)$ -saturating matching $\{e_1^+, \dots, e_k^+\}$. The edges $e_1, \dots, e_k$ of $\mathcal{H}$ are pairwise disjoint, as $e_i \subseteq e_i^+$, and they cover $\mathcal{X} \setminus \mathcal{X}_0$, as $e_i \cap \mathcal{X} = e_i^+ \cap (\mathcal{X} \setminus \mathcal{X}_0)$. The edges $\{a\}$, $a \in \mathcal{X}_0$, are disjoint from each other and from every e_i , since $e_i \cap \mathcal{X} \subseteq \mathcal{X} \setminus \mathcal{X}_0$. So $\{e_1, \dots, e_k\} \cup \{\{a\} : a \in \mathcal{X}_0\}$ is an $\mathcal{X}$ -saturating matching of $\mathcal{H}$. $\square$

Proof of the stated form. This is Haxell's alternating-tree argument. For an edge e let $a(e)$ denote the vertex of $e \cap \mathcal{X}$. We use induction on $|\mathcal{X}|$. If $\mathcal{X} = \emptyset$, the empty set of edges is an $\mathcal{X}$ -saturating matching. Let $|\mathcal{X}| \geq 1$ and fix $a_0 \in \mathcal{X}$. The edges of $\mathcal{H}$ that do not contain a_0 form a hypergraph on $(\mathcal{X} \setminus \{a_0\}) \sqcup \mathcal{Y}$ that satisfies the hypothesis with $\mathcal{X} \setminus \{a_0\}$ in place of $\mathcal{X}$: for $\mathcal{X}' \subseteq \mathcal{X} \setminus \{a_0\}$, the edge e given by the hypothesis for $\mathcal{H}$ has $e \cap \mathcal{X} \subseteq \mathcal{X}'$, so $a_0 \notin e$. By the induction hypothesis this hypergraph has an $(\mathcal{X} \setminus \{a_0\})$ -saturating matching.

Call a set $\mathcal{M}$ of pairwise disjoint edges of $\mathcal{H}$ *good* if no edge of $\mathcal{M}$ contains a_0 and every vertex of $\mathcal{X} \setminus \{a_0\}$ lies in an edge of $\mathcal{M}$. Since every edge meets $\mathcal{X}$ in exactly one vertex, the map $m \mapsto a(m)$ is then a bijection from $\mathcal{M}$ onto $\mathcal{X} \setminus \{a_0\}$. We have just seen that a good set exists. It suffices to find a good $\mathcal{M}$ and an edge x with $a(x) = a_0$ and $x \cap m \cap \mathcal{Y} = \emptyset$ for every $m \in \mathcal{M}$: then x is disjoint from every $m \in \mathcal{M}$ (as $a(m) \neq a_0$), and $\mathcal{M} \cup \{x\}$ is an $\mathcal{X}$ -saturating matching.

Configurations. Let $\mathcal{M}$ be good. A *configuration* for $\mathcal{M}$ is a sequence $(x_1, \mathcal{S}_1, \dots, x_k, \mathcal{S}_k)$ with $k \geq 0$, of edges x_i of $\mathcal{H}$ and sets $\mathcal{S}_i \subseteq \mathcal{M}$, with the following properties. Put $\mathcal{X}^{(0)} := \{a_0\}$, $Z^{(0)} := \emptyset$ and, for $1 \leq i \leq k$,

$$\mathcal{X}^{(i)} := \{a_0\} \cup \{a(m) : m \in \mathcal{S}_1 \cup \dots \cup \mathcal{S}_i\}, \quad Z^{(i)} := \mathcal{Y} \cap \left(x_1 \cup \dots \cup x_i \cup \bigcup_{m \in \mathcal{S}_1 \cup \dots \cup \mathcal{S}_i} m \right).$$

The properties are, for every $1 \leq i \leq k$: (C1) $a(x_i) \in \mathcal{X}^{(i-1)}$ and $x_i \cap Z^{(i-1)} = \emptyset$; (C2) $\mathcal{S}_i = \{m \in$

$\mathcal{M} : m \cap x_i \cap \mathcal{Y} \neq \emptyset\}$ and $\mathcal{S}_i \neq \emptyset$. The *signature* of the configuration is the sequence $(|\mathcal{S}_1|, \dots, |\mathcal{S}_k|)$ of positive integers.

The sets $\mathcal{S}_i$ are pairwise disjoint: if $j < i$ and $m \in \mathcal{S}_j$, then $m \cap \mathcal{Y} \subseteq Z^{(i-1)}$, which x_i avoids by (C1), so $m \notin \mathcal{S}_i$. As $m \mapsto a(m)$ is injective on $\mathcal{M}$ and $a(m) \neq a_0$, we get $|\mathcal{X}^{(k)}| - 1 = \sum_{i \leq k} |\mathcal{S}_i| \leq |\mathcal{M}| = |\mathcal{X}| - 1$. Each $m \in \mathcal{S}_i$ shares a vertex of $\mathcal{Y}$ with x_i , so $|(m \cap \mathcal{Y}) \setminus x_i| \leq q - 1$. Hence, using $|\mathcal{S}_i| \geq 1$,

$$\begin{aligned} |Z^{(k)}| &\leq \sum_{i \leq k} \left(|x_i \cap \mathcal{Y}| + \sum_{m \in \mathcal{S}_i} |(m \cap \mathcal{Y}) \setminus x_i| \right) \leq \sum_{i \leq k} (q + (q - 1)|\mathcal{S}_i|) \\ &\leq (2q - 1) \sum_{i \leq k} |\mathcal{S}_i| = (2q - 1)(|\mathcal{X}^{(k)}| - 1). \end{aligned}$$

So the hypothesis, with $\mathcal{X}' := \mathcal{X}^{(k)}$ and $Z := Z^{(k)}$, gives an edge x with $a(x) \in \mathcal{X}^{(k)}$ and $x \cap Z^{(k)} = \emptyset$.
(*)

Swap step. Suppose that $\mathcal{M}$ is good, that $(x_1, \mathcal{S}_1, \dots, x_j, \mathcal{S}_j)$ is a configuration for $\mathcal{M}$, and that y is an edge with $a(y) \in \mathcal{X}^{(j)}$, $y \cap Z^{(j)} = \emptyset$ and $y \cap m \cap \mathcal{Y} = \emptyset$ for every $m \in \mathcal{M}$. If $a(y) = a_0$, then $\mathcal{M}$ and $x := y$ are as required, and the swap step ends. Otherwise $a(y) = a(m)$ for exactly one $m \in \mathcal{M}$, and $m \in \mathcal{S}_i$ for exactly one $i \leq j$. Put $\mathcal{M}' := (\mathcal{M} \setminus \{m\}) \cup \{y\}$. It is good: y shares no vertex of $\mathcal{Y}$ with an edge of $\mathcal{M}$, the vertex $a(y) = a(m)$ lies in no edge of $\mathcal{M} \setminus \{m\}$, $a_0 \notin y$, and $a(m)$ is covered by y . For $i' \leq j$ we have $x_{i'} \cap \mathcal{Y} \subseteq Z^{(j)}$, so $y \cap x_{i'} \cap \mathcal{Y} = \emptyset$; and $m \in \mathcal{S}_{i'}$ only for $i' = i$. Hence $\{m' \in \mathcal{M}' : m' \cap x_{i'} \cap \mathcal{Y} \neq \emptyset\}$ equals $\mathcal{S}_{i'}$ for $i' < i$ and $\mathcal{S}_i \setminus \{m\}$ for $i' = i$, and the sets $\mathcal{X}^{(i')}$ and $Z^{(i')}$ with $i' < i$ are the same for $\mathcal{M}'$ as for $\mathcal{M}$. So, if $\mathcal{S}_i \neq \{m\}$, then $(x_1, \mathcal{S}_1, \dots, x_{i-1}, \mathcal{S}_{i-1}, x_i, \mathcal{S}_i \setminus \{m\})$ is a configuration for $\mathcal{M}'$, and the swap step ends. If $\mathcal{S}_i = \{m\}$, then $(x_1, \mathcal{S}_1, \dots, x_{i-1}, \mathcal{S}_{i-1})$ is a configuration for $\mathcal{M}'$, the edge x_i satisfies $a(x_i) \in \mathcal{X}^{(i-1)}$ and $x_i \cap Z^{(i-1)} = \emptyset$ by (C1), and x_i shares no vertex of $\mathcal{Y}$ with an edge of $\mathcal{M}'$; we repeat the swap step with $\mathcal{M}'$, this configuration and $y := x_i$. As $i - 1 < j$, and as $\mathcal{X}^{(0)} = \{a_0\}$, the swap step ends after at most $j + 1$ rounds.

One move. Let $\mathcal{M}$ be good, with a configuration of signature $(s_1, \dots, s_k)$, and let x be as in (*). If x shares a vertex of $\mathcal{Y}$ with some edge of $\mathcal{M}$, append x and $\mathcal{S}_{k+1} := \{m \in \mathcal{M} : m \cap x \cap \mathcal{Y} \neq \emptyset\}$; by (*) this is a configuration for $\mathcal{M}$, of signature $(s_1, \dots, s_k, s_{k+1})$. Otherwise apply the swap step with $j := k$ and $y := x$. Either it finds $\mathcal{M}$ and x as required, or it ends with a good $\mathcal{M}'$ and a configuration for $\mathcal{M}'$ of signature $(s_1, \dots, s_{i-1}, s_i - 1)$ for some $i \leq k$ with $s_i \geq 2$.

Termination. For finite sequences σ, τ of positive integers write $\sigma \prec \tau$ if the word $\sigma\infty$ precedes the word $\tau\infty$ in the lexicographic order on words over $\{1, 2, \dots\} \cup \{\infty\}$. As ∞ occurs only as the last letter, neither of two distinct such words is a prefix of the other, so $\prec$ is a strict total order. A move that does not find $\mathcal{M}$ and x as required replaces the signature σ by some $\sigma' \prec \sigma$: an extension σ' of σ has a finite entry where $\sigma\infty$ has ∞ , and $(s_1, \dots, s_{i-1}, s_i - 1)$ is smaller at position i . Every signature has positive entries with sum at most $|\mathcal{X}| - 1$, so there are finitely many signatures, and a sequence that is strictly decreasing for $\prec$ has no repetition; hence it is finite. Start with a good $\mathcal{M}$ and the configuration with $k = 0$, and make moves as long as possible. After finitely many moves a swap step finds a good $\mathcal{M}$ and an edge x as required. $\square$

Cited result 1.30 (*T*-joins and Euler circuits). (a) Let H be a connected graph and $T \subseteq V(H)$ with $|T|$ even. Every spanning tree S of H contains a *T*-join, i.e. a set $J \subseteq E(S)$ such that the vertices of odd J -degree are exactly the vertices of T .

(b) A connected graph (or multigraph, possibly with loops) with at least one edge in which every vertex has even degree has a closed trail using every edge exactly once.

- (c) A graph in which every vertex has even degree is an edge-disjoint union of cycles; a directed graph without loops in which every vertex has in-degree equal to out-degree is an arc-disjoint union of directed cycles.

The Lovász path decomposition is used only through Cited result 1.22.

Dependencies: none (standard).

Derivation of (a) and (c). (a) Root S at a vertex ρ , and put the edge from a vertex $v \neq \rho$ to its parent into J if and only if the subtree of S rooted at v contains an odd number of vertices of T . For $v \neq \rho$ the J -degree of v is congruent modulo 2 to the number of T -vertices in its subtree minus those in the subtrees of its children, i.e. to $\mathbf{1}[v \in T]$; for ρ it is congruent to $|T| - \mathbf{1}[\rho \in T] \equiv \mathbf{1}[\rho \in T]$, since $|T|$ is even. (b) is Euler's theorem. (c) If some edge (arc) remains, start at one of its ends and walk along unused edges (leaving along unused out-arcs); since every vertex has even degree (balanced degrees), the walk can always continue until it revisits a vertex, which closes a cycle (directed cycle); delete it and repeat. $\square$

1.4 Constants, their order, and the galactic conditions

Definition 1.31 (the constants). (i) $\varepsilon := 2^{-5}$, $\sigma := 100$, $C' := 103$ and $A := 105$.

- (ii) N_0 is an absolute constant, i.e. it depends only on σ , C' and A , with $N_0 \geq 2^{40}$ (redundant: size condition (i), $L \geq 2^{10}$, already forces every $N \geq N_0$ to satisfy $N \geq 2^{1024}$, i.e. $\lceil N_0 \rceil \geq 2^{1024}$) and N_0 at least each of the absolute size thresholds required in the proofs of Lemma 4.2, Lemma 4.3 and Theorem 4.4. Every requirement on N_0 in this item is an *eventuality*: it holds for every sufficiently large value of N_0 . Indeed $N_0 \geq 2^{40}$ is a lower bound, and every other requirement asks that an explicit inequality in N , which holds for all sufficiently large N , hold for every $N \geq N_0$. So such an N_0 exists (finitely many eventualities have a common threshold), and only its existence is used; no numerical value of N_0 enters any proof. (Each of these three statements carries the hypothesis that its vertex set has at least N_0 vertices, and its proof records the size conditions it needs. The lemmas of Section 5 carry the hypothesis $n \geq N_0$ through the setting paragraph of that section and use N_0 only to apply these three statements. Lemma 3.2 and Theorem 3.8 need no lower bound on the number of vertices.)
- (iii) D_* is a constant satisfying (T1)–(T4) of Condition 1.33. Each of these conditions is an eventuality in D_* : once N_0 is fixed, every sufficiently large D_* satisfies all of them (Lemma 7.21). So such a constant exists; only its existence is used, and no numerical value of D_* enters any proof.
- (iv) $\varepsilon_1(D_*)$ and $\varepsilon_2(D_*)$ are the explicit functions of D_* defined in Proposition 7.18 and Lemma 7.16 respectively, and $\theta_Q := \varepsilon_2(D_*)$. Each is a finite sum of explicit terms depending only on D_* , ε , σ , C' and A .
- (v) $C_0 := D_*/2 + 1085 + \varepsilon_1(D_*)$; the function ε_1 is counted exactly once. By (T4), $C_0 \leq D_*/2 + 1091$.
- (vi) $c_{\text{EG}} := \max\{C_0/(1 - 2\theta_Q), N_0/2\}$.
- (vii) (Generic notation, proofs only; permitted, but not used in the present text.) Inside a proof, $\varepsilon(D_*)$ may be used as shorthand for an expression written out explicitly in the same proof that depends only on D_* , ε , σ , C' , A and tends to 0 as $D_* \rightarrow \infty$. It never appears in the

final bound of a statement; every named small quantity ($\varepsilon_A, \varepsilon_U, \varepsilon_K = \varepsilon_{\text{ch}}, \varepsilon_{\text{CONC}}, \varepsilon_M, \varepsilon_X, \varepsilon_1, \varepsilon_2$) is given by an explicit formula in its defining statement.

Dependencies: none.

(The references of this definition to later statements are not dependencies; see the paragraph on forward references in Section 1.6. For instance, ε_1 and ε_2 are defined in Proposition 7.18 and Lemma 7.16.)

Remark 1.32 (order of the constants). The constants are fixed in the following order:

1. $\sigma, C', A, \varepsilon$;
2. N_0 ;
3. D_* ;
4. $\varepsilon_1(D_*), \varepsilon_2(D_*) = \theta_Q$ and C_0 ;
5. c_{EG} .

Nothing earlier in this list depends on anything later. In particular C_0 and θ_Q depend on D_* only (given the fixed $\sigma, C', A, \varepsilon$): they do not depend on G , on n , on the run of the hierarchy, on the designation, or on c_{EG} . The constant c_{EG} enters only as the constant c of the induction, Theorem 7.20, which is applied last; this is why that induction is not circular. The constants of steps 2 and 3 are only required to be sufficiently large: every condition on N_0 (Definition 1.31(ii)) and on D_* (Condition 1.33) is an eventuality in that constant, given the constants of the earlier steps. Table 1, after Condition 1.33, lists every such condition with the variable in which it is eventual, the constants it may depend on, and the values at which it is used.

Dependencies: 1.31.

Condition 1.33 (the galactic conditions on D_*). The constant D_* satisfies the following conditions. This is the single list of galactic conditions of the paper; every use cites the specific item. Every condition is an *eventuality* in D_* : once N_0 is fixed, it holds for all sufficiently large D_* (Lemma 7.21). Only the existence of a suitable D_* is used: no proof evaluates any of the inequalities below at a particular numerical value.

($\Gamma 1$) (*Tower condition*.) $D_* > 2$, so that $\log_2 \log_2 D_*$ is defined and positive; and for every real $\mu \geq \log_2 \log_2 D_*$, with $\lambda := 2^\mu$, the following hold:

- (a) $\mu \geq 2^8$;
- (b) $2^\mu \geq 2^{14} A \mu^3$;
- (c) $2A \log_2(A\mu) \leq 1.6 \mu$;
- (d) $2 \log_2 \mu + 8 \leq \mu$ (this follows from (a), and is listed separately for direct citation);
- (e) (the condition (G^*)) $\lambda^{36} \geq 2^{240} (A\mu)^{46A}$;
- (f) for every row of the COL-JV table (Table 5), every inequality in column 3 of that row holds at λ , where $\bar{M} := (A\mu)^{2A}$ and $\bar{k} := 192\lambda^3 + \frac{4}{3}\bar{M}^2$ as in the caption of that table. (Row 13 is (G^*), and the inequality of row 12 is item (c). This item points forward to explicit formulas, as ($\Gamma 4$) does for ε_1 and ε_2 : column 3 consists of 18 explicit inequalities, two in row 1, five in row 8 and one in each other row, and they involve only $\mu, \lambda = 2^\mu, \bar{M}$ and $\bar{k}$.)

Each item consists of explicit inequalities in the single real variable μ , and each of them holds for all sufficiently large μ (Lemma 7.21); ($\Gamma 1$) asks that all of them hold on the whole ray $\mu \geq \log_2 \log_2 D_*$. So ($\Gamma 1$) is upward closed in D_* . (The requirement $D_* > 2$ only makes the ray well defined: whenever $\log_2 \log_2 D_*$ is defined, (a) at $\mu = \log_2 \log_2 D_*$ already gives $D_* \geq 2^{2^{256}}$.)

Every round $r \leq R$ of the hierarchy has average degree $d_r \geq D_*$ (Definition 2.7), so $\lambda_r := \log_2 d_r$ satisfies $\log_2 \lambda_r \geq \log_2 \log_2 D_*$, and (a)–(f) hold at $\mu = \log_2 \lambda_r$. They also hold at $\mu = \log_2 \log_2 d$ for every $d \geq D_*$, and at $\mu = \log_2 D_*$ (as $t \geq \log_2 t$ for $t > 0$). By (a) at $\mu = \log_2 \log_2 D_*$, $\log_2 D_* \geq 2^{256}$. The consequences used later, each proved where it is used, are: every requirement (column 2) and every column-3 inequality of the COL-JV table (Lemma 3.11(ii)); the tower facts of Proposition 2.14, Lemma 2.16 and Lemma 2.17 (among them $\theta_r^{\text{GC}} > \tau_r$ and $\lambda_r \geq 2^{2 \log^* d_r + 2}$); and Proposition 2.11 and elementary estimates in Sections 5–7, each of which cites the item it uses.

- ($\Gamma 2$) (a) $D_* \geq 2^{117}$; hence $\log_2(t_l^{\text{HB}} d_l) \geq 117$ in every round l , the fixed-point requirement of the Lemma-25 size cap (Lemma 2.9);
- (b) $P_{l-2}/2 \geq M_l \log^4 M_l$ for every $3 \leq l \leq R$, including rounds with $M_l = 2^{40}$ (light parts two rounds back are admissible parents, Proposition 2.20);
- (c) every per-ancestor failure probability of Lemma 3.13 and of Lemma 5.4 is at most $|V(Y)|^{-2}$ (the failure-union lines of Table 5).

Each of (a)–(c) is implied by ($\Gamma 1$): (a) is immediate from ($\Gamma 1$); (b) is proved in Lemma 2.17(b); and the bounds of (c) are proved in Lemma 3.13 (with rows 6 and 7 of Table 5) and in Lemma 5.4(b),(c). None of these proofs, and none of the results they use, uses (b) or (c). They are listed separately for traceability.

- ($\Gamma 3$) $(\log_2 D_*)^{C'} \geq 2N_0$.

Hence every pre-part has at least $P_R \geq (\log D_*)^{C'} \geq 2N_0$ vertices, every light part at least $P_R/2 \geq N_0$, every ancestor, part and TPV, P-vortex or VX^+ run meets the absolute size thresholds of its tools, and every $o(1)$ term in a per-part success probability is at most $1/100$ (each proved where used).

- ($\Gamma 4$) $\varepsilon_1(D_*) \leq 6$ and $\theta_Q = \varepsilon_2(D_*) \leq 1/4$, where ε_1 and ε_2 are the explicit expressions of Proposition 7.18 and Lemma 7.16.

These expressions collect the following terms, with the coefficients stated; here ε_1 contains $3\varepsilon_X$ and ε_2 contains $12\varepsilon_X$, where $\varepsilon_X = \varepsilon_U + 10.3/D_* + 2.1(6 \log_2 D_* + 12)/D_*$ (Lemma 7.15):

- the concentration term $1.5\varepsilon_{\text{CONC}}(D_*)$ of ε_1 (it does not occur in ε_2), where

$$\varepsilon_{\text{CONC}}(D_*) = 2^{\sigma+15} \frac{\log^* D_*}{\log D_*} + 200\varepsilon \frac{\log^* D_*}{C' \log \log D_*} + \frac{4(2 \log^* D_* + 2)}{(\log D_*)^{1/2}}$$

(Theorem 6.10(iv); its first two terms bound the τ - and d^* -terms of all ancestors as in Theorem 6.9(iii), with a spare factor 2, and its last term bounds the guest-cap terms $\theta_r^{\text{GC}}(Y^0)$ of the light parts);

- $\varepsilon_A = 31\varepsilon/(C' \log \log D_*)$ (defined in Lemma 2.17(e)), with coefficient 169 in ε_1 and 12 in ε_2 ;
- ε_U , the demotion and lost-parent expectations (Lemma 5.5), with coefficient 3 in ε_1 and 12 in ε_2 (through ε_X);

- ε_K , the U-chaining term (Lemma 5.9), with coefficient 1 in ε_1 (it does not occur in ε_2);
- the pool term $\sum_l (6 \log_2 M_l + 12)/M_l \leq 2.1(6 \log_2 D_* + 12)/D_*$, whose bound enters with coefficient 3 in ε_1 and 12 in ε_2 (through ε_X);
- in ε_2 only, the quotient term $2F(\log_2 D_*)$, with coefficient 4 (so ε_2 contains $8F(\log_2 D_*)$), where $F(x) = (3184 + 30400 \log_2 x) x^{-90.2}$ (Lemma 7.16(iv));
- terms $O(1/D_*)$: $(252 + 9 + 3 \cdot 10.3)/D_* = 291.9/D_*$ in ε_1 and $(12 \cdot 10.3 + 4 \cdot 60)/D_* = 363.6/D_*$ in ε_2 .

Remarks. (1) *Eventualities.* Each of (Γ1)–(Γ4) holds for all sufficiently large D_* , once N_0 is fixed: (Γ1) because each of its finitely many items holds for all sufficiently large μ (and its range $\mu \geq \log_2 \log_2 D_*$ shrinks as D_* grows); (Γ2)(a) and (Γ3) because they are lower bounds on D_* , and (Γ2)(b),(c) because they follow from (Γ1); and (Γ4) because $\varepsilon_1(D_*) \rightarrow 0$ and $\varepsilon_2(D_*) \rightarrow 0$ as $D_* \rightarrow \infty$. This is proved in Lemma 7.21. Hence every sufficiently large D_* satisfies all four conditions, and only the existence of such a D_* is used.

(2) *Order of choice; no circularity.* Table 1 lists the constants in the order in which they are chosen (Remark 1.32), the conditions on them, the variable in which each condition is an eventuality, and the constants it may depend on. A condition on the constant of a step depends only on constants of earlier steps: the size conditions, (Γ1), (Γ2)(a) and (Γ4) depend on absolute constants only, and (Γ3) depends on N_0 , which is chosen before D_* . Every condition is used only at values of its variable that are at least the chosen threshold: the size conditions at $N = |Z|$ for the vertex set Z of a vortex run, which is a pre-part or a light part and so has $N \geq P_R/2 \geq N_0$ by (Γ3); the items of (Γ1) at values $\mu \geq \log_2 \log_2 D_*$, as described there; and (Γ2)(a), (Γ3) and (Γ4) at D_* itself. The quantities of a run $(d_r, \lambda_r, |V(Y)|, L_Y)$ are not chosen: they are universally quantified, and they exceed the thresholds because $d_r \geq D_*$ in every round $r \leq R$ and by (Γ3). The hypotheses $n \geq N_0$ and $d_1 \geq D_*$ on the graph are not consequences of these conditions: a graph with $n < N_0$ or with $d_1 < D_*$ is handled directly in steps (2) and (3) of the proof of Theorem 7.20, because $c_{EG} \geq N_0/2$ and $c_{EG} \geq C_0 \geq D_*/2$. Finally $\varepsilon_1, \varepsilon_2 = \theta_Q$ and C_0 are functions of D_* , and c_{EG} is defined last; it is used only as the constant c of Theorem 7.20, and no condition on N_0 or D_* involves it. So no choice is circular.

Dependencies: 1.31, 1.32.

1.5 The symbol table

Every symbol that is used outside a single proof has exactly one meaning, given in Table 2. A symbol listed there is not reused locally. Symbols local to one proof or one run are introduced there and have no meaning elsewhere: the sans-serif symbols of a vortex run (Convention 4.1); the starred parameters of the proof of Theorem 3.8; the node notation of Lemma 14^r and Lemma SEP (Section 2); the cluster, layer, demand, padding and potential notation of Lemmas 6.3 and 6.6; and the letters S_w, μ, μ^*, T, m_w of Lemmas 7.10 and 7.11. The letter η is deliberately local as well and is not in the table: it denotes the failure terms η_{TPV}, η_{PV} and η_{VX} of the three vortex statements of Section 4; the bijections η_h of Definition 7.4 and Construction 7.5; the functions $\eta(x)$ of Lemma 2.16(iv) and of the proof of Lemma 5.7(ii) (the latter also in the overview); and numbers inside single proofs (for instance those of Lemmas 1.27 and 6.13). The column “macro” gives the L^AT_EX macro of the preamble (“–” if the symbol has none), and the column “defined in” points to the defining statement (forward pointers are allowed in this table).

Table 2. The symbol table.

Step	Constant; conditions on it	Eventual in	Depends on	Used at
1	$\varepsilon = 2^{-5}$, $\sigma = 100$, $C' = 103$, $A = 105$	explicit	–	everywhere
2	N_0 : $N_0 \geq 2^{40}$ (redundant: size condition (i), $L \geq 2^{10}$, already forces every $N \geq N_0$ to satisfy $N \geq 2^{1024}$, i.e. $\lceil N_0 \rceil \geq 2^{1024}$), and the size conditions of Lemma 4.2, Lemma 4.3 and Theorem 4.4 hold for every $N \geq N_0$	N (so every requirement is upward closed in N_0)	absolute constants only	$N = Z $ for the vertex set Z of a vortex run; $N \geq N_0$ by (Γ3)
3	D_* : (Γ1), i.e. $D_* > 2$ and (a)–(f), including (G*) and the column-3 inequalities of Table 5	μ , on the ray $\mu \geq \log_2 \log_2 D_*$; hence D_*	absolute constants only	$\mu = \log_2 \lambda_r$ ($r \leq R$), $\log_2 \log_2 d$ ($d \geq D_*$; e.g. $\log_2 x$ for $x \geq \log_2 D_*$), $\log_2 D_*$
	(Γ2)(a): $D_* \geq 2^{117}$	D_*	absolute constants only	$d_l \geq D_*$ (Lemma 2.9)
	(Γ2)(b),(c)	consequences of (Γ1)	absolute constants only; not a condition	every round, every ancestor
	(Γ3): $(\log_2 D_*)^{C'} \geq 2N_0$	D_*	N_0 (step 2)	sizes of parts, $P_R/2 \geq N_0$
	(Γ4): $\varepsilon_1(D_*) \leq 6$ and $\varepsilon_2(D_*) \leq 1/4$	D_*	absolute constants only	D_* (cost line, Section 7; Definition 1.31(v); Remark 5.10(b))
4	$\varepsilon_1(D_*)$, $\theta_Q = \varepsilon_2(D_*)$, C_0	defined	D_*	Theorem 7.20
5	$c_{\text{EG}} = \max\{C_0/(1 - 2\theta_Q), N_0/2\}$	defined	C_0, θ_Q, N_0	Theorem 7.20, as its constant c

Table 1: The order in which the constants are chosen, and the conditions on them (Condition 1.33, Remark (2)). “Eventual in x ” means: holds for all sufficiently large x once the constants of the earlier steps are fixed. A condition on the constant of a step depends only on earlier steps. For steps 2 and 3, the last column gives the values at which a condition is used; each of them is at least the chosen threshold.

Symbol	Macro	Meaning	Defined in
<i>General</i>			
$f(F), f(G), f(n)$	<code>\f</code>	least number of objects (cycles, single edges) partitioning $F, E(G)$; maximum over n -vertex G	1.4
n	–	number of vertices of the input graph G	1.3
$\log, \log^*$	<code>\log, \logs</code>	base 2; iterated logarithm	1.3
$N_G(U), B_G^i(U, V)$	<code>\Nbr, \Ball</code>	outside neighbourhood; ball through V	1.7
$\mathbb{P}, \mathbb{E}, \mathbf{1}[\cdot], \text{Bin}$	<code>\Prob, \Ex, \ind, \Bin</code>	probability, expectation, indicator, binomial	–
<i>Constants</i>			
$\varepsilon = 2^{-5}; \sigma = 100; C' = 103; A = 105$	<code>\eps, -, \Cp, \Aexp</code>	fixed parameters	1.31
$N_0, D_*, C_0, c_{\text{EG}}$	<code>\Nzero, \Dstar, \Czero, \cEG</code>	size threshold; galactic degree threshold; main constants	1.31
$\varepsilon_1, \varepsilon_2 = \theta_Q$	<code>\epsOne, \epsTwo, \thetaQ</code>	cost and quotient-size functions of D_*	7.18, 7.16
$\varepsilon_A, \varepsilon_U, \varepsilon_K = \varepsilon_{\text{ch}}, \varepsilon_{\text{CONC}}, \varepsilon_M, \varepsilon_X$	<code>\epsA, \epsU, \epsK, \epsChain, \epsCONC, -, \epsX</code>	named small functions of D_*	2.17, 5.5, 5.7/5.9, 6.10, 6.19, 7.15
$\varepsilon(D_*)$	<code>\epsD</code>	generic, inside proofs only	1.31(vii)
$(\Gamma 1) - (\Gamma 4), (G^*)$	<code>\Gc, \Gstar</code>	galactic conditions	1.33
$c_{\text{OV}} = \log_2(4/3)$	<code>\cOV</code>	overlap constant	2.5
$\psi(x) = (6 \log_2 x + 12)/x$	–	tower sum weight	2.17

(Table 2, continued.) *Hierarchy*

$\text{HB}^{*\tau+}$ (HB^* , $\text{HB}^{*\tau}$ only in comparisons)	$\backslash\text{HBtp}$ ($\backslash\text{HBs}$, $\backslash\text{HBt}$)	the hierarchy	2.7
G_l , G'_l , d_l , $\lambda_l = \log d_l$, $-$, $\backslash\text{Cyc}$ R , E_0 , Cyc_l		round- l graph, the same after removing long cycles; average degree; its logarithm; last round; leftover; long cycles	2.7
$t_l^{\text{HB}} = \log^2 d_l$; $M_l \in \mathbb{N}$ (a ceiling, (R2)); $\Lambda_l = \log M_l$; $s_l = \lceil \Lambda_l^\sigma \rceil$; $P_l = \lceil \lambda_l^{C'} \rceil$; $\tau_l = \lceil 128s_l \log^2 M_l \rceil$	$\backslash\text{tHB}$	round parameters	2.7
$\mathcal{Q}$	$\backslash\text{piece}$	piece of the $s = 0$ run of Lemma 14	2.7
Z^0 , X_Z^0 , S_Z , $\text{home}(v)$, D_l	$\backslash\text{home}$	pre-part, its leaf expander, guests, home, duplicated vertices	2.7
light part $Z = Z^0 \setminus S_Z$; $X_Z = X_Z^0 - S_Z$; Std_l ; GC-part; $\theta_l^{\text{GC}}(Z^0) = \lceil Z^0 \lambda_l^{-1/2} \rceil$	$\backslash\text{Std}$, $\backslash\text{thGC}$	light/standalone split; guest cap	2.7
$E_l(Z)$	$-$	edges assigned at round l to Z	2.7
witness (U, F) , N , F_0 , H_{out} , U' , N' , F_1 , H_{in} , N'' , F'' , G_1 , G_2	$\backslash\text{Hout}$, $\backslash\text{Hin}$	τ -rules (local to Section 2)	2.1, 2.2
leaf $\mathfrak{L}$, Dup , Dup_l^*	$\backslash\text{Leaf}$, $\backslash\text{Dup}$, $\backslash\text{Dups}$	leaf of a recursion; vertices in at least two leaves (of one recursion; of round l 's two-level recursion)	2.3, 2.7
ancestor Y ; $V(Y)$; H_Y ; ε_Y ; s_Y ; $L_Y = \log V(Y) $	$-$	ancestor of round r and its expander	2.8
$\text{anc}_l(x)$; A_Z ; U_Z ; F_Z ; Q_Z	$\backslash\text{anc}$	ancestors; hubs; ports; fresh ports; classed ports	2.8
$\mu_r(w)$, dup_r , $\text{mult}_r(w)$	$\backslash\text{dup}$, $\backslash\text{mult}$	number of round- r pre-parts containing w ; duplication; number of round- r ancestors containing w	2.8
$j_0(x)$, $j_1(x)$	$-$	first round in which x lies in a pre-part; in a light part	2.8
ν_l	$-$	number of ancestors of rounds $\leq l-2$ (at most $2.74n/P_{l-2}$)	2.13 (bound: 2.17(b))
(α) , (β)	$-$	ORIGIN edge types	2.19

(Table 2, continued.) *Colouring, lending, zones, light parts*

Own_Y , $k_{\text{lend}}(Y)$, k_{own}	Lend_Y , $\backslash\text{Own}$, $\backslash\text{kend}$, $\backslash\text{kown}$	$\backslash\text{Lend}$, $\backslash\text{IU}$, $\backslash\text{IJS}$, $\backslash\text{IJV}$	two-stage colouring lent index families	3.10 3.10
$\mathcal{I}_U$, $\mathcal{I}_{JS}$, $\mathcal{I}_{JV}$	$\mathcal{L}_{Y,l,c,\varsigma}^U$, $\mathcal{L}_{Y,l,j}^{JS}$, $\mathcal{L}_{Y,l}^{JV}$	$\backslash\text{LU}$, $\backslash\text{LJS}$, $\backslash\text{LJV}$	U-, JS- and JV-lent classes	3.10
$K_l^{JS} = M_l^2$; $t_l^{JS} = 2M_l + 2$; $\rho_l = M_l^{-4}$; $\text{lab}_{Y,l}(y)$; $T_j(Y, l)$	$\backslash\text{KJS}$, $\backslash\text{tJS}$, $\backslash\text{lab}$		JS layers, multiplicity, label probability, labels, junction sets	3.10
$p_Y = 1/(2k_{\text{lend}}(Y))$	–		probability that an edge of H_Y lies in $\mathcal{L}_{Y,l}^{JV}$	3.10
$T_Y^{\text{sl}} = \lceil L_Y^2 \rceil$	$\backslash\text{Tslot}$		number of U-slots	3.10
$t_Y = \lceil \lambda_r^{1.6} \rceil$	–		U-multiplicity parameter of a light ancestor Y of round r	3.10
$\mathcal{A}(w)$	$\backslash\text{Aw}$		candidate sets of Lemma HB	3.9
$\text{choice}(v)$, $\varpi_Y = L_Y^{-2}$, sublabel (l, c, ς) , $\text{Zone}_{Y,l,c,\varsigma}$, ρ_Y	$\backslash\text{zp}$, $\backslash\text{uslot}$, $\backslash\text{Zone}$		Theorem-U zones	5.1, 5.2
(E1), demoted, parent-bad, good parent, Bad, $\text{Rt}(Z)$, $\text{Pl}(Z)$, $\text{par}(v)$, dem, lp	$\backslash\text{Bad}$, $\backslash\text{Rt}$, $\backslash\text{Pl}$, $\backslash\text{parU}$		stage-2 statuses of light parts	5.3
φ	$\backslash\text{extph}$		external phase map: phase of a vertex's round- l zone, or *	4.3, 5.6
U-bundle quotient $\mathcal{M}_{l,c,i}$	$\text{Bdl}_{l,c}$; U-multigraph	$\backslash\text{Bdl}$, $\backslash\text{UQ}$	Theorem-U chaining	5.7
$\text{Lent}_U^{\text{U}}(Y)$, $\text{Lent}^{\text{ext}}(Y)$, K^{std}	$\backslash\text{LentU}$, $\backslash\text{Lentext}$, $\backslash\text{Kstd}$		used U-lent edges; externally used lent edges; standalone edges	5.9

(Table 2, continued.) *Standalone parts and chaining*

designation	δ , class	–	class of a classed port	6.8
$Y(u)$				
$d_{Y,l}(h)$, $m_{Y,l}$, $d_{Y,l}^*$, $\alpha_{Y,l}$		–	class degree, its maximum, duplicated class ports, guest core degree	6.8
$\mathfrak{B}_{Y,l}$, $\gamma_l = \lfloor P_{l-2}/M_l \rfloor$, $\backslash\text{Bead}$			deterministic bead graph; giant threshold ((Y,l) is giant iff a component has more than $2\gamma_l$ edges)	6.8
giant				
$c_{h,l}^{\text{agg}}$, $c_x(Z)$, $c_{\text{pp}}(u)$		$\backslash\text{cagg}$, $\backslash\text{cpp}$	class counts at hubs, fresh centres, ports	6.8
cluster	$\mathfrak{K}$	$= \backslash\text{clu}$, $\backslash\text{exc}$, $\backslash\text{MED}$, $\backslash\text{pot}$	clusters	6.2, 6.3
$(A_{\mathfrak{K}}, U_{\mathfrak{K}}, B_{\mathfrak{K}})$, $b(\mathfrak{K})$, $\text{exc}(u)$, $\Phi(\mathfrak{K})$, $\text{MED}(<)$, pot				
$\mathcal{X}_j$, $\text{pad}(u)$, $\text{dm}^\pm(u)$, Φ_j , $\mathcal{P}_j$, F_j		$\backslash\text{LayP}$, $\backslash\text{pad}$, $\backslash\text{dem}$	HCC-P data	6.6
lend-bad ; Lost_Z , Lost_l ; $\text{Ret}_Z = A_Z \cup F_Z \cup \text{Lost}_Z$; $Q_Z^* = Q_Z \setminus \text{Lost}_Z$; O_Z ; X_U		$\backslash\text{Lost}$, $\backslash\text{Ret}$, $\backslash\text{Qs}$, $\backslash\text{XU}$	stage-2 lending data and the functional X_U	6.11
$\text{sc}_{Y,l}$, $g_{Y,l}$, $S_0(Y,l)$		$\backslash\text{sco}$	cherry split cost, number of cherry classes, non-giant system (local to the proof)	6.13 (proof)
$J_l = J_l^{\text{lost}} + J_l^{\text{hub}} + J_l^{\text{fr}} + J_l^{\text{par}}$; (J1)–(J3)		$\backslash\text{Jlost}$, $\backslash\text{Jhub}$, $\backslash\text{Jfr}$, $\backslash\text{Jpar}$, $\backslash\text{Jp}$	the J-set	6.15
J-consumer, $\mathcal{O}_l$, $\text{Lent}_l^{\text{JV}}$, (JC1)–(JC3)		$\backslash\text{Obj}$, $\backslash\text{LentJV}$, $\backslash\text{JC}$	consumer interface	6.16
$\text{Lent}(Y)$, $\text{Lent}^{\text{U}}(Y)$, $\text{Lent}^{\text{JS}}(Y)$, $\text{Lent}^{\text{JV}}(Y)$, $E^{\text{rem}}(Y)$, $H_0(Y)$		$\backslash\text{Lent}$, $\backslash\text{LentU}$, $\backslash\text{LentJS}$, $\backslash\text{LentJV}$, $\backslash\text{Erem}$	used lent edges and remainders (defined recursively along the processing order)	6.17, 6.18
$E^{\text{V}}(Z)$, $E^{\text{Q}}(Z)$, B_Z		$= \backslash\text{EV}$, $\backslash\text{EQ}$	TPV output	4.2, 6.17
$E^{\text{Q}}(Z)$				

(Table 2, continued.) *Junction-vertex quotient*

$\text{pl}(v), q_l = M_l^{-2}, \pi_{l,r} = 2^{-(l-1-r)}, \text{Pool}_{l,r}, \text{Pool}_l, r(w)$	$\backslash \text{plab}, \backslash \text{Pool}$	round-split pool	7.1
$\text{Cand}_l(u), H_l^{\text{cd}} = \lambda_{l-2}^{95}/(8M_l^2), \text{JV-bad}$	$\backslash \text{Cand}, \backslash \text{Hcd}$	candidates	7.2
$\xi_l, \mathfrak{F}_l^-$	$\backslash \text{Past}$	round- l randomness; the fixed past	7.4
items, live items, Live, $c_h^{\text{live}}, \theta_l^{\text{ult}} = \lfloor M_l H_l^{\text{cd}}/7 \rfloor$, ultra, $k_h, K_l^{\text{HUB}} = 4M_l$	$\backslash \text{Live}, \backslash \text{clive}, \backslash \text{thult}, \backslash \text{KHUB}$	round-step data	7.5
PAR object, cherry, un-paired fresh leg	–	PAR data	7.5
$\text{Used}(u), \text{Used}_\kappa(h)$	$\backslash \text{Used}$	junctions already used	7.5
$B_\kappa^P, B_\kappa^H, B^{(\iota)}, Q_l, \mathcal{D}_l$	$\backslash \text{BP}, \backslash \text{BH}, \backslash \text{Dec}$	layers, sub-layers, quotient, its decomposition	7.5
$\text{copies}_l, \text{pay}_l^{\text{rd}}$	$\backslash \text{copies}, \backslash \text{payrd}$	round- l quotient size and random payments	7.5
$m_\kappa(h, w), m_\kappa(w, w')$	–	multiplicities in HUB and PAR layers	7.7, 7.10
$t_l^{\text{CC}} = \lceil 2 \log_2 M_l \rceil$	$\backslash \text{tCC}$	parameter of Lemma CC	7.10
$X_{V,l}, X_V$	$\backslash \text{XV}$	junction-copy weight	7.12
$\omega_l(v)$	–	pooled-vertex weight ($c_{v,l}^{\text{agg}}$ for $v \in D_l, M_l$ otherwise)	7.13
$X_{\text{pool}}, X' = X_U + X_{\text{pool}} + X_V$	$\backslash \text{Xpool}, \backslash \text{Xp}$	stage-1 functional	7.14
$\det_l$	$\backslash \text{dett}$	deterministic part of the quotient size	7.16

1.6 The dependency table

Table 3 has one row for every numbered statement of Sections 1–7, in the order in which the statements appear (sectioning units such as the overview are not statements and have no row). The columns are: the label; a short name and the number of the statement; its section; and the statements it depends on (this column copies the “Dependencies” line printed at the end of the statement).

What an entry means. The entries of the row of a statement A are its logical inputs: the statements that the statement or the proof of A uses through a reference, by name (for instance “by Hall’s theorem” or “by Markov’s inequality”), or through a named item such as $(\Gamma 3)$, $(K1)$, $(J2)$, $\text{COL}(g)$ or $(R5)$. A fact that is proved in the running text between two statements (for instance (5.1), (5.2), (6.1), the observations of Section 4.1, or the bounds on the numbers of PAR objects and hub items in the opening paragraph of the subsection on quotient size and payments in Section 7) has no row; a statement that uses such a fact lists the statements from which the fact is proved. Convention 1.3 (graphs are finite and simple, unless they are called multigraphs) holds throughout; it is listed only in the rows of the statements that cite it, directly or through an observation of Section 4.1. Definition 1.31 fixes the constants used throughout. It is listed in the row of every later statement that cites it or invokes a requirement that it places on a constant (for instance “by the choice of N_0 ”); every other statement that uses one of these constants reaches it through the entries of its row. The row of a remark lists the earlier statements that the remark is about. A few rows also list the result of [2] whose proof the statement modifies and writes out in full. Theorem 3.8 does not list [2, Theorem 16]: its proof follows that of [2, Theorem 16], but it is written out in full in Section 3 and uses neither that theorem nor a step of its proof. A few rows list

a statement that is used only where the statement of the row is applied, or only through another entry of the same row: for instance $(\Gamma 3)$ in the rows of Lemma 4.2, Lemma 4.3 and Theorem 4.4, and $(\Gamma 2)$ in the row of Lemma 2.17. These entries are harmless over-declarations. The following are not entries: pointers that only say where a result is proved, used or modified, or that it is not used (except in the row of Theorem 1.1, whose one entry is its proof, Corollary 7.22; see below); comparisons with [2]; and the mere use of notation introduced elsewhere. The items whose name is marked “not used” (or “not used directly”) are entries of no row.

Forward references. Apart from pointers (see above) and the reference of Theorem 1.1 to its proof, Corollary 7.22, four kinds of forward reference occur in the text. They are not entries, and none of them is circular. (1) Constants. Theorem 1.1, Definition 1.31, Remark 1.32 and Condition 1.33 use explicit functions of D_* ($\varepsilon_1, \varepsilon_2$ and the terms they collect), explicit size thresholds and, in $(\Gamma 1)(f)$, the explicit inequalities of column 3 of Table 5; these are written out in later statements, proofs and tables. These are explicit formulas; no lemma of the paper is needed to state them. What is used about them is that finitely many explicit inequalities hold eventually: in N for N_0 (Definition 1.31(ii) and the size conditions of Section 4), and in D_* for D_* (Lemma 7.21). The existence of N_0 and of such a D_* is used only in Corollary 7.22; every other statement uses only the properties that Definition 1.31 requires of the fixed constants. Remark 1.32 also says where constants enter later (for instance, that c_{EG} enters only through the induction of Theorem 7.20); this is a description, verified in those statements. (2) Announced consequences. Condition 1.33 records consequences of its items that are proved later, where they are used. For instance, $(\Gamma 2)(b)$ is proved in Lemma 2.17(b), and the bounds of $(\Gamma 2)(c)$ in Lemma 3.13 (with rows 6 and 7 of Table 5) and in Lemma 5.4(b),(c). None of these lemmas uses $(\Gamma 2)(b)$ or (c) , and $(\Gamma 2)(a)$ enters them only through Lemma 2.9; the only statement that uses $(\Gamma 2)(b),(c)$ as such, Lemma 7.21, lists all three lemmas. Definition 7.2 notes that every JV-good port u has $|\text{Cand}_l(u)| \geq H_l^{\text{cd}}$. This is Lemma 7.3(iv), which is proved from the definition of JV-good and from Lemma 7.3(ii), not from the note. Definition 3.10 notes, in its paragraph “Consumers”, that for $r \geq R - 1$ the set $\text{Lent}(Y)$ is empty, so that $E^{\text{rem}}(Y) = E_r(Y)$ or $H_0(Y) = E_r(Y)$ (Construction 6.17); and Definition 5.3 notes that for $r \geq R - 1$ Lemma 5.4(c) reduces to (b). None of these conditions and definitions uses these consequences. (3) Constructions. That Construction 6.17 applies every device within its hypotheses is shown in Lemma 6.18 and Theorem 6.19(a); that Construction 7.5 is well defined is Lemma 7.6(i), (ii), (iv) and (v). These results are proved about the constructions. Besides earlier statements, each part uses only the definitions of the steps it concerns and of the steps before them (for Construction 6.17, the steps at the rounds already processed), and not the well-definedness of the step that it proves. (4) Definition 7.4 describes the choice of the stage-1 outcome through the functional X' of Definition 7.14, and the choice of each ξ_l through step (f) of Construction 7.5. These choices are made in Lemma 7.17, whose row lists Construction 7.5 and Lemma 7.15 (the latter rests on Definition 7.14). Moreover, X' is a function of the run, δ and stage 1 only (Definition 7.14), so the selection (1e) of Definition 7.4 depends neither on the later stages of the schedule nor on the round step.

Table 3. Dependency table of all numbered statements.

Label	Name	S.	Depends on
s1:thmMain	Main Theorem (1.1)	1	7.22
s1:remStatus	organization of the proof (1.2)	1	–
s1:convGraphs	graph conventions (1.3)	1	–
s1:defObject	objects; f (1.4)	1	–
s1:factAdd	properties of f (1.5)	1	1.4
s1:factEG0	long-cycle bound (1.6)	1	1.3, 1.4
s1:citNotation	B–M notation (1.7)	1	–
s1:citChernoff	B–M Thm 4 (Chernoff) (1.8)	1	–
s1:citDef7	B–M Def 7 (1.9)	1	–
s1:citProp8	B–M Prop 8 (1.10)	1	–
s1:citLem9	B–M Lemma 9 (1.11)	1	–
s1:citDef11	B–M Def 11 (1.12)	1	–
s1:citProp12	B–M Prop 12 (1.13)	1	–
s1:citProp13	B–M Prop 13 (1.14)	1	–

(Table 3, continued.)

Label	Name	S.	Depends on
s1:citLem14	B–M Lemma 14 and its proof (1.15)	1	–
s1:citLem15	B–M Lemma 15 (not used) (1.16)	1	–
s1:citThm16	B–M Thm 16 (not used; see Thm 16*) (1.17)	1	–

(Table 3, continued.)

Label	Name	S.	Depends on
s1:citLem17	B–M Lemma 17 (1.18)	1	–
s1:citProp18	B–M Prop 18 (1.19)	1	–
s1:citLem19	B–M Lemma 19 (1.20)	1	–
s1:citThm21	B–M Thm 21 (not used directly) (1.21)	1	–
s1:citCor22	B–M Cor 22 (1.22)	1	–
s1:citLem25	B–M Lemma 25 (1.23)	1	–
s1:remBMused	where B–M is used (1.24)	1	1.7, 1.8, 1.9, 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.18, 1.19, 1.20, 1.22, 1.23, 1.6

(Table 3, continued.)

Label	Name	S.	Depends on
s1:citChernoffGen	Chernoff (Poisson-binomial) (1.25)	1	–
s1:citMarkov	Markov (1.26)	1	–
s1:lemBBD	Lemma BBD (Bernstein, bounded differences) (1.27)	1	–
s1:citHall	Hall’s theorem (1.28)	1	–
s1:citHaxell	Haxell 1995 (1.29)	1	–

(Table 3, continued.)

Label	Name	S.	Depends on
s1:citEuler	T -joins, Euler (1.30)	1	–
s1:defConstants	the constants (1.31)	1	–
s1:remOrder	order of constants (1.32)	1	1.31
s1:condGamma	galactic conditions (1.33)	1	1.31, 1.32

(Table 3, continued.)

Label	Name	S.	Depends on
s2:defWitness	witness (2.1)	2	1.12, 1.31
s2:defTauRules	τ -rules and split (2.2)	2	2.1
s2:lemSEP	Lemma SEP (2.3)	2	2.2
s2:lemThinCut	thin cut (2.4)	2	2.3, 2.2
s2:lemOVgeneric	Lemma OV (generic overlap) (2.5)	2	2.3, 2.1, 1.15
s2:lem14tau	Lemma 14 τ (2.6)	2	1.15, 1.12, 2.2, 2.4, 2.5, 2.3, 2.1
s2:defHBtp	the hierarchy HB $^{*\tau+}$ (2.7)	2	1.15, 2.6, 1.31, 2.5, 2.3

(Table 3, continued.)

Label	Name	S.	Depends on
s2:defAncestors	ancestors, typing (2.8)	2	2.7
s2:lemCap	Lemma-25 size cap (2.9)	2	1.23, (Γ_2), 2.7, 2.6
s2:lemHS	Lemma HS (2.10)	2	1.12
s2:propStructure	structure of a run (2.11)	2	2.7, 2.8, 2.6, 2.9, 2.10, 1.12, 1.15, 2.3, (Γ_1)
s2:lemEL	edge laminarity EL (2.12)	2	2.7, 2.8
s2:propOV	overlap constants (2.13)	2	2.5, 2.6, 2.11, 2.8, 2.7, 2.3
s2:propDegRec	degree recursion (2.14)	2	2.6, 2.9, 2.13, 2.7, 2.3, (Γ_1), 2.5

(Table 3, continued.)

Label	Name	S.	Depends on
s2:propExists	existence, termination (2.15)	2	2.6, 2.14, 2.7, 2.9, 2.5, 2.1, 2.13
s2:lemLacunary	tower-lacunary sums (2.16)	2	2.14, (Γ_1)
s2:lemTower	tower facts (2.17)	2	2.14, 2.16, 2.7, (Γ_1), (Γ_2), 2.11, 2.9, 2.13, 2.8
s2:lemGC	rule GC (2.18)	2	2.7, 2.17, 2.12, 2.8
s2:propOrigin	ORIGIN $^\tau$ (2.19)	2	2.4, 2.3, 2.9, 2.13, 2.12, 2.7, 2.18, 2.5, 2.11
s2:propParentless	fresh, parentless mass (2.20)	2	2.11, 2.17, 2.9, 2.8, 2.7
s2:remTauConstants	$\tau+$ constants (2.21)	2	2.13, 2.6, 2.17, 2.7, 2.5

(Table 3, continued.)

Label	Name	S.	Depends on
s3:lemMonotone	monotonicity (3.1)	3	1.9, 1.12
s3:lemL15p	Lemma 15 $^+$ (3.2)	3	1.12, 1.8
s3:remMultiset	multiset remark (3.3)	3	1.9, 1.11
s3:propP13s	Prop 13* (3.4)	3	1.13, 1.14, 1.12
s3:lemL17s	Lemma 17* (3.5)	3	3.4, 1.18, 1.27, 1.8, 1.25
s3:lemP18s	Prop 18*, Lemma 19* (3.6)	3	1.19, 1.20, 3.5, 1.12

(Table 3, continued.)

Label	Name	S.	Depends on
s3:lemL9rho	Lemma 9 $_\rho$ (3.7)	3	1.11, 1.10, 1.29, 1.9, 3.3
s3:thmT16s	Theorem 16* (3.8)	3	3.2, 3.4, 3.5, 3.6, 3.7, 3.1, 1.12, 1.8
s3:lemHB	Lemma HB (3.9)	3	1.28, 1.12
s3:defCOL	lending data COL-JV (3.10)	3	2.8, 2.11, 2.17
s3:lemCOLJV	COL-JV table (3.11)	3	(Γ_1), (G^*), 2.17, 3.10, 2.8, 2.11, 3.2, 3.8

(Table 3, continued.)

Label	Name	S.	Depends on
s3:lemCOLJVe	eventual form of COL-JV (3.12)	3	–
s3:lemCOL	Lemma COL (3.13)	3	3.10, 3.2, 3.8, 3.1, 3.11, 2.17, 2.11, 2.8, (Γ_1)

(Table 3, continued.)

Label	Name	S.	Depends on
s4:convVortex	vortex notation (4.1)	4	–
s4:lemTPV	Lemma TPV (4.2)	4	3.2, 3.8, 3.9, 3.1, 1.22, 1.6, 1.8, 1.26, (Γ 3), 4.1, 1.3, 1.31

(Table 3, continued.)

Label	Name	S.	Depends on
s4:lemPV	Lemma PV (P-vortex) (4.3)	4	3.8, 3.9, 3.1, 1.22, 1.6, 1.8, 1.26, (Γ 3), 4.1, 4.2, 1.3, 1.31
s4:thmVXp	Theorem VX ⁺ (4.4)	4	3.2, 3.8, 3.9, 3.1, 1.22, 1.6, 1.8, (Γ 3), 4.1, 1.3, 1.31
s4:remTPVVX	TPV versus VX ⁺ (not used) (4.5)	4	4.2, 4.4

(Table 3, continued.)

Label	Name	S.	Depends on
s5:defZones	zones (5.1)	5	2.8, 3.10, 2.11, 2.17, (Γ 1)
s5:lemZones	zone lemma (5.2)	5	5.1, 2.11, 2.17, (Γ 1), 3.10
s5:defStages	stage-2 statuses (5.3)	5	5.1, 3.10, 3.9, 2.20, 2.8, 2.11, 2.17, 1.9
s5:lemE1	bad probabilities (5.4)	5	3.13, 3.9, 5.2, 5.3, 1.25, 3.10, 3.2, 3.11, 2.11, 2.17, (Γ 1)
s5:lemExpect	expected bad mass (5.5)	5	5.4, 5.3, 2.13, 2.17, 2.16, (Γ 1), 2.11
s5:lemChild	child side (5.6)	5	4.3, 5.3, 5.2, 3.13, 2.11, 3.10, (Γ 3)
s5:lemParent	parent side (5.7)	5	5.6, 5.3, 5.2, 3.10, 3.11, 3.1, 1.30, 2.13, 2.17, 2.11, (Γ 1), 1.9
s5:lemDemoted	demoted parts (5.8)	5	4.4, 2.11, 3.11, 2.17, (Γ 1), (Γ 3), 5.3, 5.7
s5:lemKRED	Lemma K-RED (5.9)	5	5.6, 5.7, 5.8, 2.11, 2.20, 2.12, 3.10, 5.3
s5:remConstants	369 and 745 (5.10)	5	4.3, 5.9, 2.11, 2.20, 2.13, 2.17, 4.4, (Γ 4), 1.6

(Table 3, continued.)

Label	Name	S.	Depends on
s6:lemGATE	Lemma GATE (6.1)	6	1.4
s6:defCluster	clusters (6.2)	6	–
s6:lemMED	Lemma MED (6.3)	6	6.2
s6:lemEQLPT	Lemma EQ-LPT (6.4)	6	–
s6:lemPAR	Lemma PAR (6.5)	6	1.30
s6:lemHCCP	Lemma HCC-P (6.6)	6	6.1, 6.3, 6.2
s6:lemHCCglob	union of systems (6.7)	6	1.5, 6.6
s6:defDesign	designation, classes (6.8)	6	2.8, 2.7
s6:thmCONC	Theorem CONC (6.9)	6	2.19, 2.17, 2.13, 2.16, 6.8, 2.7, (Γ 1)
s6:thmCONCL	Theorem CONC-L (6.10)	6	2.19, 2.3, 2.4, 2.18, 2.12, 2.17, 2.16, 6.9, 2.13, 2.7, 6.8, (Γ 1)

(Table 3, continued.)

Label	Name	S.	Depends on
s6:defLending	lending statuses (6.11)	6	3.10, 3.13, 5.3, 6.8, 5.1
s6:lemLost	the sets Lost (6.12)	6	6.11, 3.13, 5.4, 5.5, 2.11, 2.17, 3.10
s6:lemJSLC	Lemma JS-LC (6.13)	6	6.5, 6.3, 6.4, 6.6, 6.7, 6.8, 6.11, 3.13, 3.1, 2.12, 2.9, 2.13, 2.17, 2.7, 2.11, 3.10, 1.9
s6:remStar	star split (not used) (6.14)	6	6.13
s6:lemJplus	Lemma J ⁺ (6.15)	6	6.13, 6.5, 6.8, 2.11, 2.12, 2.9, 3.10, 2.7, 6.6, 6.11
s6:defJconsumer	J-consumer (6.16)	6	3.10, 6.15
s6:consOrder	processing order (6.17)	6	4.2, 5.6, 5.7, 5.8, 6.13, 6.16, 6.11, 5.3, 4.4, 3.10
s6:lemLent	Lent sets (6.18)	6	6.17, 6.11, 6.16, 6.13, 5.7, 5.9, 3.10, 6.15, 5.3, 4.2, 5.6, 5.8, 4.4
s6:thmMIXC	Theorem MIX-C (6.19)	6	6.17, 4.2, 5.9, 5.8, 6.13, 6.15, 6.16, 6.18, 6.10, 2.11, 2.20, 2.13, 2.17, 3.10, 3.13, 4.3, 5.6, 4.4, 2.7, (Γ 3), 6.8, 6.11

(Table 3, continued.)

Label	Name	S.	Depends on
s7:defPool	pool labels (7.1)	7	2.17, 3.10
s7:defCand	candidates (7.2)	7	7.1, 3.10, 6.8
s7:lemCand	candidate counts (7.3)	7	7.2, 7.1, 3.10, 3.11, 2.11, 2.17, 1.25
s7:defSchedule	randomness schedule (7.4)	7	3.10, 5.1, 7.1, 6.11, 5.3, 7.2, 4.2, 4.3, 4.4, 5.6, 5.8
s7:consRound	JV ⁺ round step (7.5)	7	6.15, 6.16, 7.2, 7.1, 7.4, 1.26
s7:lemWellDef	round step well defined (7.6)	7	7.5, 6.15, 7.3, 3.11, 1.28, 2.7
s7:lemMULT	Lemma MULT (7.7)	7	7.5, 6.15, 7.2, 7.1, 2.13
s7:lemSimple	Q_l is simple (7.8)	7	7.5

(Table 3, continued.)

Label	Name	S.	Depends on
s7:lemLift	Lemma JV-L (lift) (7.9)	7	7.5, 7.8, 7.6, 6.15, 6.16, 3.10, 2.11, 6.11
s7:lemCC	Lemma CC (7.10)	7	7.5, 7.6, 1.25, 6.15
s7:lemUltra	ultra-hub copies (7.11)	7	7.5, 7.6, 6.15, 1.25, 7.7, 7.3
s7:lemVstar	the weight X_V (7.12)	7	7.1, 2.13, 2.17
s7:lemPay	payments (7.13)	7	7.5, 7.6, 6.15, 2.20, 2.17
s7:defXprime	the functional X' (7.14)	7	6.11, 7.12, 7.13
s7:lemEXprime	expectation of X' (7.15)	7	7.14, 6.12, 7.3, 7.12, 2.13, 2.11, 2.17, 5.5, 2.9
s7:lemUHsplit	Lemma UH*-split (7.16)	7	7.10, 7.7, 7.11, 7.12, 7.15, 7.4, 2.13, 2.17, 2.16, (Γ4), (Γ1), 7.5, 7.6, 2.7, 6.15

(Table 3, continued.)

Label	Name	S.	Depends on
s7:lemOneOutcome	one joint outcome (7.17)	7	7.4, 7.5, 6.17, 4.2, 4.3, 4.4, 5.6, 5.8, 7.15, 7.13, 7.16, 1.26, (Γ3)
s7:propCost	cost (7.18)	7	6.19, 6.10, 7.9, 7.13, 7.15, 7.17, 5.9, 2.20, 2.13, 7.5, 2.17, (Γ4), 6.11, 7.14
s7:thmJVps	Theorem JV ⁺ (7.19)	7	7.18, 7.16, 7.8, 7.9, 7.17, 6.19, 2.15, 7.13, 7.11, 1.33
s7:thmHI	Theorem HI'' (7.20)	7	1.5, 1.4, 1.31
s7:lemGammaSat	satisfiability of the conditions (7.21)	7	1.33, 7.18, 7.16, 5.5, 5.9, 2.17, 7.15, 6.10, 1.31, 3.13, 5.4, 3.12
s7:thmMainProof	proof of the Main Theorem (7.22)	7	7.20, 7.19, 2.15, 7.21, 6.8, 1.33, 1.31
s7:remNonCirc	non-circularity (7.23)	7	7.20, 7.19, 7.18, 7.16, 7.10, 7.11, 7.13, 7.8, 7.5, 4.2, 4.3, 4.4, (Γ4), 1.6

Every edge of this table points from a statement to one appearing earlier in the paper; the only exception is the Main Theorem, stated in Section 1 and proved in Section 7.

2 The hierarchy $HB^{*\tau+}$

This section defines the hierarchy $HB^{*\tau+}$ on which the whole construction runs, and proves the structural facts about it that later sections use. The hierarchy follows the round structure of the proof of [2, Theorem 2]: in each round, long cycles are removed, the remaining graph is cut by the recursion of [2, Lemma 14] with $s = 0$ into pieces, and each large piece is cut again into robust expanders. There are two changes inside a round. First, the robust recursion is replaced by Lemma 14^τ below: at every split, vertices with many deleted edges across the cut are moved into the separator (the τ -rules), which caps the number of deleted edges from any vertex into the non-duplicated vertices of a leaf (the thin-cut lemma). Second, a *guest cap* (rule (GC)) is added to the definition of light pre-parts. The leaves of this recursion are still certified expanders, and the overlap between leaves stays small (Lemma OV).

All constants in this section are the constants of $\text{HB}^{*\tau+}$. The hierarchy without the τ -rules and without (GC) (“plain HB^* ”), and the hierarchy with the τ -rules but without (GC) (“ $\text{HB}^{*\tau}$ ”), appear only in the final remark of this section, for comparison. Round parameters are written t_l^{HB} , τ_r and θ_r^{GC} .

Standing assumption. Throughout this section, as everywhere in this text, D_* denotes the constant of Definition 1.31(iii): it satisfies $(\Gamma 1)$ – $(\Gamma 4)$ (Condition 1.33). The proofs use these conditions either directly or through the results they apply; for instance the termination part of Proposition 2.15 uses Proposition 2.14, and hence $(\Gamma 1)$.

Local notation. In the first two subsections of this section the letter H denotes a node graph with $m = |H|$ vertices, and (U, F) , N , F_0 , F_1 , F'' , H_{out} , H_{in} , U' , N' , N'' , G_1 , G_2 , n_1 , n_2 , ν^* are local to these subsections, as are S (a total leaf size), M (a size threshold) and c (a constant) in Lemma OV. Throughout, $\varepsilon = 2^{-5}$, $\log = \log_2$, and $|H| := |V(H)|$ (Convention 1.3).

2.1 Witnesses, the τ -rules and separation

Definition 2.1 (witness and its minimal part). Let $s \geq 0$ and let H be a graph on $m = |H|$ vertices that is not an (ε, s) -expander (Cited result 1.12). Then $m \geq 2$, since a graph on one vertex has no set U with $1 \leq |U| \leq 2m/3$. A *witness* at H is a pair (U, F) with $U \subseteq V(H)$ and $F \subseteq E(H)$ such that

$$1 \leq |U| \leq \frac{2}{3}m, \quad |F| \leq s|U|, \quad |N_{H-F}(U)| < \frac{\varepsilon|U|}{\log^2 m}.$$

Since H is not an (ε, s) -expander, a witness exists. For a witness (U, F) put

$$N := N_{H-F}(U), \quad F_0 := E_H(U, V(H) \setminus (U \cup N)).$$

Fact. $F_0 \subseteq F$ and $N_{H-F_0}(U) = N$. In particular (U, F_0) is again a witness at H , with the same set N .

Dependencies: 1.12, 1.31

Proof of the Fact. Let $uv \in F_0$ with $u \in U$ and $v \notin U \cup N$. If $uv \notin F$, then uv is an edge of $H - F$, so $v \in N_{H-F}(U) = N$, a contradiction. Hence $F_0 \subseteq F$. Consequently $H - F \subseteq H - F_0$, so $N \subseteq N_{H-F_0}(U)$. Conversely, let $v \notin U$ have a neighbour $u \in U$ with $uv \notin F_0$. If $v \notin N$ then $uv \in E_H(U, V(H) \setminus (U \cup N)) = F_0$, a contradiction; so $v \in N$. Thus $N_{H-F_0}(U) = N$. Finally $|F_0| \leq |F| \leq s|U|$, so (U, F_0) is a witness. $\square$

Definition 2.2 (the τ -rules and the split). Let H , m and s be as in Definition 2.1, let (U, F) be a witness at H with N and F_0 as there, write $V := V(H)$, and let $\tau > s$ be real. The τ -rules are:

- (1) $H_{\text{out}} := \{v \in U : \deg_{F_0}(v) \geq \tau\}$, $U' := U \setminus H_{\text{out}}$, $N' := N \cup H_{\text{out}}$;
- (2) $F_1 := E_H(U', V \setminus (U' \cup N'))$, $H_{\text{in}} := \{x \in V \setminus (U' \cup N') : \deg_{F_1}(x) \geq \tau\}$, $N'' := N' \cup H_{\text{in}}$, $F'' := E_H(U', V \setminus (U' \cup N''))$.

The *split of H by the τ -rules* has the two children $G_1 := H[U' \cup N''] - F''$ and $G_2 := H - U' - E(G_1) - F''$, and its *deleted set* is F'' . Every edge of F'' has one end in U' and the other outside $U' \cup N''$, so no edge of F'' lies inside $U' \cup N''$, and every edge of F'' meets U' . Hence

$$G_1 = H[U' \cup N''], \quad G_2 = H[V \setminus U'] - E(H[N'']). \quad (2.1)$$

Facts.

- (a) $U' \cap N'' = \emptyset$, $U' \cup N' = U \cup N$, and $F'' \subseteq F_1 \subseteq F_0 \subseteq F$.
- (b) $E(H) = E(G_1) \sqcup E(G_2) \sqcup F''$ (disjoint union), $V(G_1) = U' \cup N''$ and $V(G_2) = V \setminus U'$. A vertex of U' lies only in G_1 , a vertex of N'' lies in both children, and a vertex of $V \setminus (U' \cup N'')$ lies only in G_2 .
- (c) Given the pair (U, N) , the objects $H_{\text{out}}, U', N', F_1, H_{\text{in}}, N'', F'', G_1, G_2$ depend only on (U, N) , because they are defined from $F_0 = E_H(U, V \setminus (U \cup N))$. The choice of F matters only through $N = N_{H-F}(U)$: every witness (U, F) yields the same split as its minimal part (U, F_0) . (It is not true that the choice of F is irrelevant, since F determines N .)

Dependencies: 2.1

Proof of the Facts. (a) $U' \subseteq U$ and $N \cap U = \emptyset$, so $U' \cap N = \emptyset$; $H_{\text{out}} \cap U' = \emptyset$ by definition; and $H_{\text{in}} \cap (U' \cup N') = \emptyset$ by definition. Hence $U' \cap N'' = \emptyset$. As $H_{\text{out}} \subseteq U$, we get $U' \cup N' = (U \setminus H_{\text{out}}) \cup N \cup H_{\text{out}} = U \cup N$. Therefore $F_1 = E_H(U', V \setminus (U \cup N)) \subseteq E_H(U, V \setminus (U \cup N)) = F_0$, and since $N' \subseteq N''$, $F'' = E_H(U', V \setminus (U' \cup N'')) \subseteq E_H(U', V \setminus (U' \cup N')) = F_1$. Finally $F_0 \subseteq F$ by Definition 2.1.

(b) The vertex sets follow from (2.1), and the statement about where vertices go follows from $U' \cap N'' = \emptyset$. Let $e = xy \in E(H)$. If $x, y \in U' \cup N''$ then $e \in E(G_1)$, and $e \notin F''$. Otherwise one end, say y , lies outside $U' \cup N''$. If $x \in U'$ then $e \in F''$. If $x \notin U'$ then both ends lie in $V \setminus U'$ and not both lie in N'' , so $e \in E(G_2)$; and $e \notin F''$ because e does not meet U' . An edge in $E(G_1) \cap E(G_2)$ would have both ends in $(U' \cup N'') \cap (V \setminus U') = N''$, which (2.1) excludes from G_2 . So the three sets are disjoint and cover $E(H)$.

(c) F_0 is determined by (U, N) , and every later object in (1)–(2) and in (2.1) is defined from U , N and F_0 alone. By the Fact of Definition 2.1, (U, F_0) is a witness with the same N . $\square$

Lemma 2.3 (Lemma SEP). *A split recursion on a graph H_0 is a finite rooted binary tree whose nodes ν carry graphs H_ν , with $H_\nu = H_0$ at the root, such that at every non-leaf node ν there are disjoint sets $U'_\nu, N''_\nu \subseteq V(H_\nu)$ for which the two children ν_1, ν_2 of ν carry*

$$H_{\nu_1} = H_\nu[U'_\nu \cup N''_\nu], \quad H_{\nu_2} = H_\nu[V(H_\nu) \setminus U'_\nu] - E(H_\nu[N''_\nu]),$$

and the deleted set at ν is $F''_\nu := E_{H_\nu}(U'_\nu, V(H_\nu) \setminus (U'_\nu \cup N''_\nu))$. (By (2.1), every split by the τ -rules has this form.) We write $|\nu| := |H_\nu|$, say that a vertex v lies in ν if $v \in V(H_\nu)$, and call the leaf nodes leaves (distinct leaf nodes are distinct leaves even if their vertex sets coincide). A deleted edge is an edge of some F''_ν . Let Dup be the set of vertices lying in at least two leaves. Then:

- (0) *at every non-leaf ν , $E(H_\nu) = E(H_{\nu_1}) \sqcup E(H_{\nu_2}) \sqcup F''_\nu$; a vertex of U'_ν lies only in ν_1 , a vertex of N''_ν lies in both children, every other vertex of ν lies only in ν_2 , and $|\nu_1| + |\nu_2| = |\nu| + |N''_\nu|$. Every vertex of a node lies in at least one leaf below it, and the total size of the leaves is $|H_0| + \sum_\nu |N''_\nu|$, the sum over the non-leaf nodes;*
- (i) *every edge xy of H_0 either lies in exactly one leaf, which contains x and y , or is deleted at exactly one node ν . In the second case x and y lie in ν and go to different children of ν , each to only one child, and ν is the first node on the path of nodes having xy as an edge at which this happens;*
- (ii) *if $u \notin \text{Dup}$ and $u \in V(H_0)$, then the nodes in which u lies are exactly the ancestors of the unique leaf $\mathfrak{L}_u$ containing u ($\mathfrak{L}_u$ included), and $u \notin N''_\nu$ at every node ν on that path;*

- (iii) if $u \notin \text{Dup}$, $u, h \in V(H_0)$, $h \notin V(\mathfrak{L}_u)$, and ν^* is the deepest ancestor of $\mathfrak{L}_u$ in which h lies, then every deleted edge hu' with $u' \in V(\mathfrak{L}_u) \setminus \text{Dup}$ is deleted at ν^* .

Dependencies: 2.2

Proof. (0) Fix a non-leaf ν and write $H = H_\nu$, $V = V(H)$, $U' = U'_\nu$, $N'' = N''_\nu$. The proof of Fact (b) of Definition 2.2 used only that $U' \cap N'' = \emptyset$ and the formulas (2.1), so it gives $E(H) = E(H_{\nu_1}) \sqcup E(H_{\nu_2}) \sqcup F''_\nu$ and the statement about where vertices go. As $U' \cap N'' = \emptyset$, $|\nu_1| + |\nu_2| = |U'| + |N''| + |V| - |U'| = |\nu| + |N''|$. Every vertex of ν lies in a child of ν , so by induction on the height of the subtree every vertex of a node lies in some leaf below it. Summing $|\nu_1| + |\nu_2| - |\nu| = |N''|$ over all non-leaf nodes gives the total leaf size.

(i) Let Ω_{xy} be the set of nodes having xy as an edge; the root is in Ω_{xy} . Every edge of a child is an edge of its parent, so Ω_{xy} is closed under taking parents. By (0), at a non-leaf $\nu \in \Omega_{xy}$ the edge xy lies in exactly one of $E(H_{\nu_1})$, $E(H_{\nu_2})$ and F''_ν ; so at most one child of ν is in Ω_{xy} , and none is exactly when $xy \in F''_\nu$. Hence Ω_{xy} is a path from the root ending at a node $\nu_\circ$. If $\nu_\circ$ is a leaf, then xy is an edge of exactly one leaf, namely $\nu_\circ$ (other leaves are not in Ω_{xy}), and xy is deleted nowhere (at nodes of $\Omega_{xy} \setminus \{\nu_\circ\}$ it passes to a child; at nodes outside Ω_{xy} it is not an edge). If $\nu_\circ$ is not a leaf, then $xy \in F''_{\nu_\circ}$, xy lies in no leaf, and it is deleted at no other node. At $\nu_\circ$, one end lies in $U'_{\nu_\circ}$ and goes only to the first child, and the other lies outside $U'_{\nu_\circ} \cup N''_{\nu_\circ}$ and goes only to the second. At every earlier node of Ω_{xy} the edge xy passes to a child, which contains both x and y .

(ii) By (0), u lies in at least one leaf, hence in exactly one, $\mathfrak{L}_u$. The nodes in which u lies form a set closed under taking parents (vertex sets shrink downwards), and at each non-leaf node of this set at least one child contains u . If some node ν had both children containing u , then u would lie in a leaf below each child, i.e. in two distinct leaves. So exactly one child of every such non-leaf node contains u : the nodes containing u form a path from the root to a leaf, which is $\mathfrak{L}_u$. A vertex of N''_ν lies in both children of ν , so $u \notin N''_\nu$ on this path.

(iii) As $h \in V(H_0)$, the root is an ancestor of $\mathfrak{L}_u$ in which h lies, so ν^* exists. Let $u' \in V(\mathfrak{L}_u) \setminus \text{Dup}$. By (ii) applied to u' , whose unique leaf is $\mathfrak{L}_u$, the nodes containing u' are exactly the ancestors of $\mathfrak{L}_u$. Suppose hu' is deleted at ν . Then h and u' lie in ν , so ν is an ancestor of $\mathfrak{L}_u$ in which h lies, and ν is not below ν^* . By (i), at ν the vertices h and u' go to different children, each to only one. The vertex u' goes to the child of ν on the path to $\mathfrak{L}_u$, so h does not lie in that child. Hence ν is the deepest ancestor of $\mathfrak{L}_u$ containing h , that is, $\nu = \nu^*$. $\square$

Lemma 2.4 (thin cut). *Let $\tau > 0$ and consider a split recursion (Lemma 2.3) in which every split is produced by the τ -rules at threshold τ : at each non-leaf node ν there are $s_\nu < \tau$ and a witness at H_ν (with parameter s_ν) whose τ -rules (Definition 2.2) give U'_ν and N''_ν . Then for every leaf $\mathfrak{L}$ and every vertex h ,*

$$\#\{\text{deleted edges } hu : u \in V(\mathfrak{L}) \setminus \text{Dup}\} \leq \lceil \tau \rceil - 1,$$

which is $\tau - 1$ when τ is an integer (as in all applications), and this number is 0 if $h \in V(\mathfrak{L})$. Moreover, if $u \notin \text{Dup}$ then every edge at u is either deleted or an edge of the unique leaf containing u .

Dependencies: 2.3, 2.2

Proof. The condition $s_\nu < \tau$ only ensures that the τ -rules are defined (Definition 2.2 asks for $\tau > s$); the argument below uses only the structure of the τ -rules. Fix $\mathfrak{L}$ and h ; we may assume h lies in the root (otherwise no edge of H_0 meets h) and that some deleted edge hu with $u \in V(\mathfrak{L}) \setminus \text{Dup}$ exists. For such u we have $\mathfrak{L}_u = \mathfrak{L}$, so by Lemma 2.3(ii) the nodes containing u are the ancestors of $\mathfrak{L}$.

Case $h \in V(\mathfrak{L})$. Let hu be deleted at ν . Then ν is an ancestor of $\mathfrak{L}$, and by Lemma 2.3(i) the vertices h and u go to different children of ν , each to only one. But the child of ν on the path to $\mathfrak{L}$ contains $V(\mathfrak{L}) \ni h, u$. This is a contradiction, so the number is 0.

Case $h \notin V(\mathfrak{L})$. Let ν^* be the deepest ancestor of $\mathfrak{L}$ in which h lies. By Lemma 2.3(iii), every deleted edge hu with $u \in V(\mathfrak{L}) \setminus \text{Dup}$ is deleted at ν^* , i.e. lies in $F'' = F''_{\nu^*}$. Use the notation of Definition 2.2 at ν^* . The vertex h does not lie in the child of ν^* on the path to $\mathfrak{L}$ (the *path child*), so $h \notin N''$, since vertices of N'' lie in both children.

Case A: the path child is G_1 . Then $h \notin V(G_1) = U' \cup N''$; in particular $h \notin U' \cup N'$ and $h \notin H_{\text{in}}$, so by the definition of H_{in} we have $\deg_{F_1}(h) < \tau$, i.e. $\deg_{F_1}(h) \leq \lceil \tau \rceil - 1$. As $F'' \subseteq F_1$, at most $\lceil \tau \rceil - 1$ edges of F'' meet h .

Case B: the path child is G_2 . Then $h \notin V(G_2) = V \setminus U'$, so $h \in U' = U \setminus H_{\text{out}}$, and by the definition of H_{out} , $\deg_{F_0}(h) < \tau$. As $F'' \subseteq F_0$, at most $\lceil \tau \rceil - 1$ edges of F'' meet h .

In both cases the count is at most $\lceil \tau \rceil - 1$. The last assertion is Lemma 2.3(i): an edge uy that is not deleted lies in exactly one leaf, which contains u and is therefore the unique leaf $\mathfrak{L}_u$. $\square$

2.2 Overlap, and Lemma 14 $^\tau$

Lemma 2.5 (generic overlap bound, Lemma OV). *Let $c > 0$ with $3.42c\varepsilon < 1$, and put $c_{\text{OV}} := \log_2(4/3)$. Consider a split recursion (Lemma 2.3) on a root graph with $n_0 \geq 1$ vertices such that at every non-leaf node ν , with $m := |\nu|$,*

$$m \geq 2, \quad |\nu_1| \leq \frac{3}{4}m, \quad |N''_\nu| \leq \frac{c\varepsilon |U'_\nu|}{\log^2 m}.$$

(Recall that vertices of U'_ν go only to the first child ν_1 .) Let S be the total size of the leaves. For $M \geq 2$ let $\Delta_{\geq M} := \sum_\nu |N''_\nu|$, the sum over the non-leaf nodes ν with $|\nu| \geq M$ (the number of duplication events at nodes of size at least M ; this is a number, not to be confused with the vertex set Dup of Lemma 2.3), and for a vertex v let $\text{dup}_{\geq M}(v)$ be the number of non-leaf nodes ν with $|\nu| \geq M$ and $v \in N''_\nu$ (so $\sum_v \text{dup}_{\geq M}(v) = \Delta_{\geq M}$). Then:

- (a) $\Delta_{\geq M} \leq c\varepsilon \left(\frac{1}{\log^2 M} + \frac{1}{c_{\text{OV}} \log M} \right) S \leq \frac{3.42c\varepsilon S}{\log M};$
- (b) for every vertex v , the number of leaves $\mathfrak{L}$ with $|\mathfrak{L}| \geq M$ containing v is at most $1 + \text{dup}_{\geq M}(v)$; consequently $\sum_{\mathfrak{L}: |\mathfrak{L}| \geq M} |\mathfrak{L}| - \left| \bigcup_{\mathfrak{L}: |\mathfrak{L}| \geq M} V(\mathfrak{L}) \right| \leq \Delta_{\geq M};$
- (c) $S \leq n_0 / (1 - 3.42c\varepsilon).$

Instance $c = 1$ (the $s = 0$ recursion). Call a split recursion an $s = 0$ recursion if at every non-leaf node ν there is a witness (U_ν, F_ν) at H_ν with parameter $s = 0$ (Definition 2.1) and $U'_\nu = U_\nu$, $N''_\nu = N_{H_\nu}(U_\nu)$. This is exactly the recursion in the proof of [2, Lemma 14] with $s = 0$ (Cited result 1.15): there $F_\nu = \emptyset$, the children are $H_\nu[U_\nu \cup N_\nu]$ and $H_\nu \setminus U_\nu - E(H_\nu[U_\nu \cup N_\nu]) = H_\nu[V(H_\nu) \setminus U_\nu] - E(H_\nu[N_\nu])$, and the deleted set $E_{H_\nu}(U_\nu, V(H_\nu) \setminus (U_\nu \cup N_\nu))$ is empty. (It is also the split by the τ -rules at $s = 0$ for any $\tau > 0$: then $F_0 = \emptyset$ and $H_{\text{out}} = H_{\text{in}} = \emptyset$.) Every $s = 0$ recursion satisfies the hypotheses with $c = 1$; hence its total leaf size is at most $n_0 / (1 - 3.42\varepsilon) \leq 1.12n_0$. The lemma is applied with $c = 1.6$ to the recursion of Lemma 14 $^\tau$ below, and with $c = 1.6$ to the two-level recursion of a round of $\text{HB}^{*\tau+}$.

Dependencies: 2.3, 2.1, 1.15

Proof. (a) If no non-leaf node has size at least M there is nothing to prove. For every non-leaf node ν with $|\nu| \geq M$ and every $u \in U'_\nu$, give the pair (ν, u) the charge $c\varepsilon/\log^2 |\nu|$. The total charge at ν is $c\varepsilon|U'_\nu|/\log^2 |\nu| \geq |N''_\nu|$, so $\Delta_{\geq M}$ is at most the total charge. A *leaf copy* is a pair $(\mathfrak{L}, v)$ with $\mathfrak{L}$ a leaf and $v \in V(\mathfrak{L})$; there are exactly S leaf copies. For each charged pair (ν, u) , the vertex u lies in ν_1 , hence (Lemma 2.3(0)) in some leaf $\mathfrak{L}$ below ν_1 ; fix one and map (ν, u) to the leaf copy $(\mathfrak{L}, u)$. The pair (ν, u) is recovered from u and ν , and ν is an ancestor of $\mathfrak{L}$; so each leaf copy $(\mathfrak{L}, u)$ receives the charges of pairs (ν, u) for a set of ancestors ν of $\mathfrak{L}$ with $|\nu| \geq M$, $u \in U'_\nu$ and $\mathfrak{L}$ below ν_1 . List these ancestors as $\nu^{(1)}, \nu^{(2)}, \dots, \nu^{(k)}$ from $\mathfrak{L}$ upwards. For $i < k$, the node $\nu^{(i)}$ lies strictly below $\nu^{(i+1)}$ on the path from $\nu^{(i+1)}$ to $\mathfrak{L}$, which passes through the first child of $\nu^{(i+1)}$; vertex sets shrink downwards, so $|\nu^{(i)}| \leq |\nu_1^{(i+1)}| \leq \frac{3}{4}|\nu^{(i+1)}|$. Hence $|\nu^{(i)}| \geq (4/3)^{i-1}|\nu^{(1)}| \geq (4/3)^{i-1}M$, i.e. $\log |\nu^{(i)}| \geq \log M + (i-1)c_{\text{OV}}$. The charge received by $(\mathfrak{L}, u)$ is therefore at most

$$c\varepsilon \sum_{j \geq 0} \frac{1}{(\log M + jc_{\text{OV}})^2} \leq c\varepsilon \left(\frac{1}{\log^2 M} + \int_0^\infty \frac{dx}{(\log M + xc_{\text{OV}})^2} \right) = c\varepsilon \left(\frac{1}{\log^2 M} + \frac{1}{c_{\text{OV}} \log M} \right),$$

because the summand is decreasing in j . Summing over the S leaf copies gives the first bound. For $M \geq 2$ we have $\log M \geq 1$, so $1/\log^2 M \leq 1/\log M$, and $1 + 1/c_{\text{OV}} = 3.4094\dots \leq 3.42$.

(b) Let $\mathcal{C}_v$ be the set of nodes of size at least M in which v lies. It is closed under taking parents (vertex sets, hence sizes, shrink downwards), so if nonempty it is a rooted subtree containing the root. Every leaf of the recursion of size at least M that contains v belongs to $\mathcal{C}_v$ and has no children, so it is a leaf of $\mathcal{C}_v$. In a finite rooted tree the number of leaves is $1 + \sum_\nu (\kappa_\nu - 1)$, the sum over the nodes with $\kappa_\nu \geq 1$ children; here $\kappa_\nu \leq 2$, and $\kappa_\nu = 2$ means that v lies in both children of ν , i.e. $v \in N''_\nu$ (Lemma 2.3(0)), with $|\nu| \geq M$. So the number of such leaves is at most $1 + \text{dup}_{\geq M}(v)$. Summing $(\#\{\mathfrak{L} \ni v : |\mathfrak{L}| \geq M\} - 1)^+ \leq \text{dup}_{\geq M}(v)$ over v gives the consequence.

(c) All non-leaf nodes have size at least 2, so by Lemma 2.3(0), $S = n_0 + \sum_\nu |N''_\nu| = n_0 + \Delta_{\geq 2} \leq n_0 + 3.42c\varepsilon S$ by (a) with $M = 2$. Rearranging gives (c).

Instance $c = 1$. By Definition 2.1 with $s = 0$ we have $F_\nu = \emptyset$, so $N_\nu = N_{H_\nu}(U_\nu)$, and every neighbour of U_ν outside U_ν lies in N_ν ; this gives the three formulas stated. The node has $m \geq 2$ vertices (Definition 2.1), and $|N''_\nu| = |N_\nu| < \varepsilon|U_\nu|/\log^2 m$, which is the hypothesis with $c = 1$; and $|\nu_1| = |U_\nu| + |N_\nu| < (1 + \varepsilon)\frac{2}{3}m < \frac{3}{4}m$, using $\log m \geq 1$. Finally $1/(1 - 3.42/32) = 1.1196\dots \leq 1.12$. $\square$

Lemma 2.6 (Lemma 14 $^\tau$). *Let H_0 be a graph on n_0 vertices, $s \geq 1$ an integer, $\varepsilon = 2^{-5}$, and $\tau \geq 128s \log^2 n_0$. The τ -run of H_0 (with parameters s, τ) is the recursion that, at a node H , stops if H is an (ε, s) -expander (the node is then a leaf), and otherwise picks any witness (U, F) at H with parameter s (Definition 2.1) and splits H by the τ -rules at threshold τ (Definition 2.2; note $\tau \geq 128s > s$ whenever $n_0 \geq 2$). Let Dup be the set of vertices lying in at least two leaves. Then, for every choice of witnesses:*

(a) *at every split, at a node of size m ,*

$$|H_{\text{out}}| + |H_{\text{in}}| \leq \frac{|F_0|}{\tau} \leq \frac{s|U|}{\tau} \leq \frac{|U|}{128 \log^2 m}, \quad |N''| < \frac{1.25 \varepsilon |U|}{\log^2 m} < \frac{1.5 \varepsilon |U|}{\log^2 m},$$

$|U'| \geq \frac{127}{128}|U| > 0$, $U \subseteq U' \cup N''$, $n_1 := |U' \cup N''| \leq 0.698m < \frac{3}{4}m$ and $n_2 := m - |U'| < m$. The recursion terminates; it is a split recursion; at most $4sn_0 \log n_0$ edges are deleted; the remaining edges are partitioned into the leaves; every leaf is an (ε, s) -expander; and the total size of the leaves is at most $2n_0 - 2n_0/(2 + \log n_0)$;

- (b) (*OV* at 1.6ε) at every split, $|N''| \leq 1.6\varepsilon|U'|/\log^2 m$, the vertices of U' go only to G_1 , and $|G_1| \leq \frac{3}{4}m$. Hence, by Lemma 2.5 with $c = 1.6$, the total leaf size S satisfies $S \leq n_0/(1 - 5.5\varepsilon) \leq 1.21n_0$, and, by the first bound of Lemma 2.5(a) with $c = 1.6$, for every $M \geq 2$ the duplication at nodes of size at least M is $\Delta_{\geq M} \leq 5.46\varepsilon S/\log M$;
- (c) for every leaf $\mathfrak{L}$ and every vertex h , if $\tau > 0$ then $\#\{\text{deleted edges } hu : u \in V(\mathfrak{L}) \setminus \text{Dup}\} \leq \lceil \tau \rceil - 1$ ($= \tau - 1$ for integer τ); the number is 0 if $h \in V(\mathfrak{L})$;
- (d) if $u \notin \text{Dup}$, every edge at u is deleted or lies in the unique leaf containing u .

Dependencies: 1.15, 1.12, 2.2, 2.4, 2.5, 2.3, 2.1

Proof. If $n_0 = 1$ the root is an (ε, s) -expander, no edge is deleted, and (a), (b), (d) and the last clause of (c) are trivial (the bound in (c) assumes $\tau > 0$); let $n_0 \geq 2$. Every node of the recursion is a subgraph of H_0 on $m \leq n_0$ vertices, so $\tau \geq 128s \log^2 m$ at every node. The argument follows the proof of [2, Lemma 14] (Cited result 1.15) with the split of Definition 2.2 in place of the split of [2].

Counting at one split. Fix a non-leaf node H on $m \geq 2$ vertices, with witness (U, F) , and use the notation of Definition 2.2; put $V := V(H)$. Every edge of F_0 has exactly one end in U , so $\sum_{v \in U} \deg_{F_0}(v) = |F_0|$. Let $a := \sum_{v \in H_{\text{out}}} \deg_{F_0}(v)$; then $\tau|H_{\text{out}}| \leq a$. By Fact (a) of Definition 2.2, $F_1 = E_H(U', V \setminus (U \cup N))$ is F_0 minus the edges at H_{out} , so $|F_1| = |F_0| - a$. Every edge of F_1 has exactly one end outside $U' \cup N' = U \cup N$, so $\tau|H_{\text{in}}| \leq \sum_{x \notin U \cup N} \deg_{F_1}(x) = |F_1|$. Adding, $\tau(|H_{\text{out}}| + |H_{\text{in}}|) \leq |F_0| \leq |F| \leq s|U|$. With $\tau \geq 128s \log^2 m$ this gives $|H_{\text{out}}| + |H_{\text{in}}| \leq |U|/(128 \log^2 m)$, and $1/128 = \varepsilon/4$. Therefore

$$|N''| = |N| + |H_{\text{out}}| + |H_{\text{in}}| < \frac{\varepsilon|U|}{\log^2 m} + \frac{\varepsilon|U|}{4\log^2 m} = \frac{1.25\varepsilon|U|}{\log^2 m}.$$

Since $|H_{\text{out}}| \leq |U|/128$, $|U'| \geq \frac{127}{128}|U| > 0$. As $H_{\text{out}} \subseteq N' \subseteq N''$, $U = U' \cup H_{\text{out}} \subseteq U' \cup N''$, so $|U| \leq n_1$. Also $n_1 = |U'| + |N''| \leq |U| + |N''| < (1 + 1.5\varepsilon)|U| \leq \frac{2}{3}(1 + \frac{3}{64})m \leq 0.698m$, using $\log m \geq 1$; and $n_2 = |V \setminus U'| = m - |U'| < m$. By Fact (b) of Definition 2.2, $n_1 + n_2 = m + |N''|$. Finally $\log n_1 \leq \log m + \log 0.698 < \log m - \frac{2}{5}$, since $\log_2 0.698 = -0.518 \dots$

Termination. Both children of every split have fewer vertices than their parent, so every root-to-node path has at most n_0 nodes, and the binary tree is finite. By (2.1) the recursion is a split recursion. Its leaves are (ε, s) -expanders by the stopping rule, and by Lemma 2.3(i) every edge of H_0 is either deleted (exactly once) or lies in exactly one leaf.

The induction. We show by induction on the height of the subtree below a node ν , with $m = |\nu|$, that the subtree deletes at most $4sm \log m$ edges and its leaves have total size at most $2m - 2m/(2 + \log m)$. For a leaf this holds, since nothing is deleted and $m \leq 2m - 2m/(2 + \log m)$ (as $2 + \log m \geq 2$). Let ν be a non-leaf node, with n_1, n_2 as above ($1 \leq n_1, n_2 < m$), and let E_i^{del} be the set of edges deleted in the subtree below the i -th child; by induction $|E_i^{\text{del}}| \leq 4sn_i \log n_i$. The edges deleted in the subtree below ν are $F'' \cup E_1^{\text{del}} \cup E_2^{\text{del}}$, and $|F''| \leq |F_0| \leq s|U| \leq sn_1$. As $|U| \leq n_1$, we have $|N''| < 1.5\varepsilon n_1 / \log^2 m$. Hence, as in [2, inequality (8)],

$$\begin{aligned} \frac{|F''| + |E_1^{\text{del}}| + |E_2^{\text{del}}|}{s} &\leq n_1 + 4n_1 \log n_1 + 4n_2 \log n_2 \leq n_1 + 4n_1 \left(\log m - \frac{2}{5} \right) + 4n_2 \log m \\ &= 4(n_1 + n_2) \log m - \frac{3}{5}n_1 = 4(m + |N''|) \log m - \frac{3}{5}n_1 \\ &< 4m \log m + n_1 \left(\frac{6\varepsilon}{\log m} - \frac{3}{5} \right) \leq 4m \log m, \end{aligned}$$

since $6\varepsilon/\log m \leq 6/32 < 3/5$. For the leaf sizes put $L := \log m \geq 1$. As $n_2 < m$, $2n_2/(2 + \log n_2) \geq 2n_2/(2 + L)$. As $\log n_1 < L - \frac{2}{5}$,

$$\frac{2n_1}{2 + \log n_1} > \frac{2n_1}{\frac{8}{5} + L} = \frac{2n_1}{2 + L} + \frac{\frac{4}{5}n_1}{(\frac{8}{5} + L)(2 + L)} \geq \frac{2n_1}{2 + L} + \frac{n_1}{10L^2},$$

where the last step is $8L^2 \geq (\frac{8}{5} + L)(2 + L)$, i.e. $7L^2 - 3.6L - 3.2 \geq 0$, which holds for $L \geq 1$ (the left side is 0.2 at $L = 1$ and increasing for $L \geq 1$). By induction the total leaf size below ν is therefore

$$\begin{aligned} \sum_{i=1}^2 \left(2n_i - \frac{2n_i}{2 + \log n_i} \right) &< (n_1 + n_2) \left(2 - \frac{2}{2 + L} \right) - \frac{n_1}{10L^2} = (m + |N''|) \left(2 - \frac{2}{2 + L} \right) - \frac{n_1}{10L^2} \\ &\leq 2m - \frac{2m}{2 + L} + 2|N''| - \frac{n_1}{10L^2} \\ &< 2m - \frac{2m}{2 + L} + \left(3\varepsilon - \frac{1}{10} \right) \frac{n_1}{L^2} \leq 2m - \frac{2m}{2 + L}, \end{aligned}$$

as in [2, inequality (9)], because $3\varepsilon = 3/32 < 1/10$. At the root this gives (a).

(b) By (a), $|N''| < 1.5\varepsilon|U|/\log^2 m \leq 1.5 \cdot \frac{128}{127}\varepsilon|U'|/\log^2 m < 1.52\varepsilon|U'|/\log^2 m$. (The first bound of (a) even gives $|N''| < 1.25 \cdot \frac{128}{127}\varepsilon|U'|/\log^2 m < 1.26\varepsilon|U'|/\log^2 m$; the looser constant 1.6 suffices.) Vertices of U' go only to G_1 and $|G_1| = n_1 \leq \frac{3}{4}m$ (Fact (b) of Definition 2.2 and (a)); every non-leaf node has $m \geq 2$. As $3.42 \cdot 1.6\varepsilon = 0.171 < 1$, Lemma 2.5 applies with $c = 1.6$: by its part (c), $S \leq n_0/(1 - 5.472\varepsilon) \leq n_0/(1 - 5.5\varepsilon) = 1.2075 \dots n_0 \leq 1.21 n_0$; by its part (a), $\Delta_{\geq M} \leq 1.6\varepsilon(1 + 1/c_{OV})S/\log M \leq 5.46\varepsilon S/\log M$, since $1.6(1 + 1/c_{OV}) = 5.455 \dots$

(c), (d) Every split is produced by the τ -rules at threshold τ from a witness with parameter $s < \tau$, so Lemma 2.4 applies. $\square$

2.3 The hierarchy

Definition 2.7 (the hierarchy $\text{HB}^{*\tau+}$). Let G be a graph on n vertices, and let $\varepsilon = 2^{-5}$, $\sigma = 100$, $C' = 103$, $A = 105$ and D_* be as in Definition 1.31. Put $G_1 := G$. All graphs G_l , G'_l below have vertex set $V(G)$. For $l = 1, 2, \dots$:

- (R0) $d_l := 2|E(G_l)|/n$. If $d_l < D_*$, stop, and put $E_0 := E(G_l)$ and $R := l - 1$. Otherwise put $\lambda_l := \log d_l$.
- (R1) $t_l^{\text{HB}} := \log^2 d_l$. As long as the current graph contains a cycle of length at least $t_l^{\text{HB}} d_l$, delete one (the lexicographically first, for a fixed order of $V(G)$). The deleted cycles form Cyc_l ; the remaining graph is G'_l . Thus G'_l has no cycle of length at least $t_l^{\text{HB}} d_l$.
- (R2) $M_l := \lceil \max(2^{40}, 2^{16} t_l^{\text{HB}} d_l \log^4(t_l^{\text{HB}} d_l)) \rceil \in \mathbb{N}$, $\Lambda_l := \log M_l$, $s_l := \lceil \Lambda_l^\sigma \rceil$, $P_l := \lceil \lambda_l^{C'} \rceil$, $\tau_l := \lceil 128 s_l \log^2 M_l \rceil$. The ceiling makes M_l a natural number, as its later uses require (it counts classes and colours, for instance $K_l^{\text{JS}} = M_l^2$ in Definition 3.10 and the palettes $[3M_l]$, $[4M_l]$ of Section 7); Λ_l is the logarithm of this integer. Besides integrality, only two facts about the ceiling are used: $M_l \geq \max(2^{40}, 2^{16} t_l^{\text{HB}} d_l \log^4(t_l^{\text{HB}} d_l))$, and $M_l \leq \max(2^{40}, 2^{16} t_l^{\text{HB}} d_l \log^4(t_l^{\text{HB}} d_l)) + 1$.
- (R3) Run an $s = 0$ recursion (Lemma 2.5) on G'_l that stops exactly at $(\varepsilon, 0)$ -expanders: a node is a leaf iff its graph is an $(\varepsilon, 0)$ -expander, and otherwise any witness with parameter 0 is used. This is the recursion of the proof of [2, Lemma 14] with $s = 0$ (Cited result 1.15). Its leaves are the $s = 0$ pieces $\mathcal{Q}$ of round l . Inside every piece with $|\mathcal{Q}| \geq P_l$, run the τ -run of $\mathcal{Q}$ with

$s = s_l$ and $\tau = \tau_l$ (Lemma 2.6), with any witnesses. The *two-level recursion* of round l is the split recursion obtained from the $s = 0$ recursion by attaching, at every piece $\mathcal{Q}$ with $|\mathcal{Q}| \geq P_l$, the tree of its τ -run. Its leaves are the pieces of size less than P_l and the leaves of the τ -runs. The *round- l pre-parts* are the leaves of the τ -runs with at least P_l vertices; equivalently, the leaves of the two-level recursion with at least P_l vertices. For a pre-part named Z , Z^0 denotes its vertex set and X_Z^0 its leaf graph, a graph on Z^0 . Finally $\text{Dup}_l^* :=$ the set of vertices lying in at least two leaves (of any size, singletons included) of the two-level recursion of round l .

- (R4) $D_l :=$ the set of vertices lying in at least two round- l pre-parts. Fix an order of the round- l pre-parts (the *home order*). For a vertex v lying in some pre-part, $\text{home}(v) = \text{home}_l(v) :=$ the first pre-part containing v . The *guests* of a pre-part Z are $S_Z := \{v \in Z^0 \cap D_l : \text{home}(v) \neq Z\}$. The pre-part Z is *light* iff

$$(L1) \quad |S_Z| \leq |Z^0|/2;$$

$$(L2) \quad \text{every } u \in Z^0 \text{ has at most } s_l/2 \text{ neighbours in } S_Z \text{ in the graph } G'_l[Z^0];$$

$$(GC) \quad \text{no } x \in S_Z \text{ has at least}$$

$$\theta_l^{\text{GC}}(Z^0) := \lceil |Z^0| \lambda_l^{-1/2} \rceil$$

neighbours in $Z^0 \setminus S_Z$ in the graph X_Z^0 (the *guest cap*).

Otherwise Z is *standalone*. A pre-part satisfying (L1) and (L2) but failing (GC) is a *GC-part*; it is standalone. A light pre-part Z gives the *light part* with vertex set $Z^0 \setminus S_Z$ and graph $X_Z := X_Z^0 - S_Z$. Std_l is the set of round- l standalone pre-parts; it contains the GC-parts and the pre-parts failing (L1) or (L2), and it is a deterministic function of the run.

- (R5) The edges of G'_l are assigned in this order. (1) For every light pre-part Z the edges of X_Z are assigned to the light part Z ; for every standalone pre-part Z the edges of X_Z^0 are assigned to Z . (The graphs X_Z^0 of distinct pre-parts are edge-disjoint, being distinct leaves of the two-level recursion, Lemma 2.3(i), and $X_Z \subseteq X_Z^0$; so no edge is assigned twice in this step.) (2) Every edge not yet assigned whose ends both lie in some light part is assigned to the first such light part (in the home order). (3) Every edge not yet assigned whose ends both lie in Z^0 for some standalone pre-part Z is assigned to the first such pre-part (in the home order). (4) Every other edge of G'_l passes down: $E(G_{l+1})$ is the set of these edges. $E_l(Z)$ denotes the set of edges assigned at round l to Z (a light part or a standalone pre-part).

Naming. Pre-parts are leaves, so distinct pre-parts are counted separately even if their vertex sets coincide. If a pre-part Z is light, the letter Z also denotes the vertex set $Z^0 \setminus S_Z$ of its light part. A *round- l part* is a light part or a standalone pre-part of round l . We write home_l , S_Z and so on with the round when it is not clear from the context.

A *valid* $\text{HB}^{*\tau+}$ run is an execution of this procedure with any witnesses in both levels of (R3), any vertex order in (R1), and any home orders in (R4). At $s = 0$ the τ -rules are idle (Lemma 2.5), so the first level of (R3) is the recursion of [2] unchanged; the τ -rules act only inside the pieces.

Dependencies: 1.15, 2.6, 1.31, 2.5, 2.3

Definition 2.8 (ancestors, typing, multiplicities). Fix a valid $\text{HB}^{*\tau+}$ run. An *ancestor of round r* is either

- a light part Y of round r ; then $V(Y) := Y$ (the vertex set $Y^0 \setminus S_Y$), $H_Y := X_Y$, $\varepsilon_Y := 2^{-6}$ and $s_Y := s_r/2$; or

- a standalone pre-part $Y \in \text{Std}_r$, GC-parts included; then $V(Y) := Y^0$, $H_Y := X_Y^0$, $\varepsilon_Y := 2^{-5}$ and $s_Y := s_r$.

We write $r(Y)$ for the round of Y and $L_Y := \log |V(Y)|$. Every ancestor corresponds to a distinct pre-part $Y^0 \supseteq V(Y)$ of its round. For $l \geq 3$ and a vertex x ,

$$\text{anc}_l(x) := \{Y : Y \text{ an ancestor of a round } r \leq l - 2 \text{ with } x \in V(Y)\},$$

and $\text{anc}_l(x) := \emptyset$ for $l \leq 2$. For $Z \in \text{Std}_l$:

- the *hubs* of Z are $A_Z := Z^0 \cap D_l$;
- the *ports* of Z are $U_Z := Z^0 \setminus D_l$;
- the *fresh ports* are $F_Z := \{x \in U_Z : \text{anc}_l(x) = \emptyset\}$ (so all ports are fresh for $l \leq 2$);
- the *classed ports* are $Q_Z := U_Z \setminus F_Z$.

For a round r and a vertex w : $\mu_r(w)$ is the number of round- r pre-parts containing w (so $D_r = \{w : \mu_r(w) \geq 2\}$); $\text{dup}_r := \sum_w (\mu_r(w) - 1)^+$; and $\text{mult}_r(w)$ is the number of ancestors Y of round r with $w \in V(Y)$. Finally $j_0(x)$ is the first round in which x lies in a pre-part, and $j_1(x)$ the first round in which x lies in a light part (∞ if there is none).

Dependencies: 2.7

2.4 Basic properties of valid runs

Lemma 2.9 (Lemma-25 size cap). (i) If $m \geq 2^{40}$, $T \geq 2^{117}$ and $m < 18432 T \log^4 m$, then $m \leq 2^{16} T \log^4 T$.

(ii) In every round $l \leq R$ of a valid $\text{HB}^{*\tau+}$ run, every $s = 0$ piece $\mathcal{Q}$ satisfies $|\mathcal{Q}| \leq M_l$. Hence every round- l pre-part has $|Z^0| \leq M_l$; for every round- l part Z (light or standalone) every vertex is incident with at most $M_l - 1$ edges of $E_l(Z)$; and $\tau_l \geq 128 s_l \log^2 |\mathcal{Q}|$, so Lemma 2.6 applies to the τ -run of every piece with $|\mathcal{Q}| \geq P_l$.

Remark (not used). Part (i) is false for small T : for $T = 2^{20}$ and $m = 2^{55}$ we have $m \geq 2^{40}$ and $18432 T \log^4 m = 2^{57.29\dots} > m$, but $2^{16} T \log^4 T = 2^{53.28\dots} < m$. The uniform form $m \leq \max(2^{40}, 2^{16} T (\log T + 40)^4)$ holds for all $T \geq 1$.

Dependencies: 1.23, (F2), 2.7, 2.6

Proof. (i) Put $\chi(y) := y / \log^4 y$; since $\frac{d}{dy} \ln \chi(y) = \frac{1}{y} (1 - \frac{4}{\ln y})$, χ is increasing for $y > e^4$. The hypothesis says $\chi(m) < 18432 T$. Put $x := \log T \geq 117$ and $B := 2^{16} T \log^4 T = 2^{16} T x^4$, so $\log B = 16 + x + 4 \log x$. We have $(2^{16}/18432)^{1/4} = (32/9)^{1/4} = 1.37317\dots$, and $0.37317 x \geq 16 + 4 \log x$ for $x \geq 117$: at $x = 117$ the two sides are $43.66\dots$ and $43.48\dots$, and for $x \geq 117$ the derivative of the left side ($0.373\dots$) exceeds that of the right side ($4/(x \ln 2) < 0.05$). Hence $(32/9)^{1/4} x \geq 16 + x + 4 \log x = \log B$, i.e. $2^{16} x^4 \geq 18432 (\log B)^4$, i.e. $\chi(B) \geq 18432 T$. If $m > B$, then, as $B > e^4$, $\chi(m) > \chi(B) \geq 18432 T$, contradicting the hypothesis. So $m \leq B$.

Remark. The numbers in the counterexample are direct computations. For the uniform form put $B' := 2^{16} T (x+40)^4$ with $x = \log T \geq 0$; then $\log B' = 16 + x + 4 \log(x+40)$ and $\chi(B') \geq 18432 T$ is equivalent to $(32/9)^{1/4} (x+40) \geq 16 + x + 4 \log(x+40)$, which follows from $0.37317 x + 38.9 \geq 4 \log(x+40)$; this holds at $x = 0$ ($38.9 \geq 21.3$) and the left side grows faster (0.373 against at most $4/(40 \ln 2) < 0.15$). As $B' > e^4$, $m > \max(2^{40}, B')$ would give $\chi(m) > \chi(B') \geq 18432 T$.

(ii) Let $l \leq R$ and $T := t_l^{\text{HB}} d_l = d_l \log^2 d_l$. By $(\Gamma 2)(a)$ (Condition [\(T2\)](#)), $d_l \geq D_* \geq 2^{117}$, so $\log^2 d_l \geq 1$ and $T \geq 2^{117}$; and $M_l \geq \max(2^{40}, 2^{16} T \log^4 T)$ by (R2). Let $\mathcal{Q}$ be an $s = 0$ piece and $m := |\mathcal{Q}|$. If $m < 2^{40}$ then $m \leq M_l$. Otherwise the graph of $\mathcal{Q}$ is an $(\varepsilon, 0)$ -expander (the stopping rule in (R3)) with $\varepsilon = 2^{-5}$ on $m \geq 2^{40} = 2^{30}/\varepsilon^2$ vertices, so by Cited result [1.23](#) it contains a cycle of length at least $\varepsilon^2 m / (18 \log^4 m) = m / (18432 \log^4 m)$. This cycle lies in G'_l , which has no cycle of length at least T by (R1). Hence $m < 18432 T \log^4 m$, and (i) gives $m \leq 2^{16} T \log^4 T \leq M_l$. Every pre-part is a leaf of the τ -run of some piece, so $Z^0 \subseteq V(\mathcal{Q})$ and $|Z^0| \leq M_l$. By (R5) every edge of $E_l(Z)$ has both ends in Z^0 (in the light part $Z \subseteq Z^0$ if Z is light), so every vertex meets at most $|Z^0| - 1 \leq M_l - 1$ of them. Finally $\tau_l \geq 128 s_l \log^2 M_l \geq 128 s_l \log^2 |\mathcal{Q}|$. $\square$

Lemma 2.10 (Lemma HS). *Let $\varepsilon' > 0$ and $s \geq 0$, let X be an (ε', s) -expander on a vertex set of size $z \geq 11$, let $W \subseteq V(X)$ with $|W| \leq z/2$, and suppose every $u \in V(X) \setminus W$ has at most $s_W \leq s$ neighbours in W in X . Then $X - W$ is an $(\varepsilon'(\log z'/\log z)^2, s - s_W)$ -expander on $z' := z - |W|$ vertices, and $\varepsilon'(\log z'/\log z)^2 \geq \varepsilon'/2$. In particular, if X is a $(2^{-5}, s)$ -expander and $s_W = s/2$, then $X - W$ is a spanning $(2^{-6}, s/2)$ -expander of $V(X) \setminus W$.*

Dependencies: [1.12](#)

Proof. Note $z' \geq z/2 \geq 5.5$. Let $U \subseteq V(X) \setminus W$ with $1 \leq |U| \leq 2z'/3$ and $F' \subseteq E(X - W)$ with $|F'| \leq (s - s_W)|U|$. Put $F := F' \cup E_X(U, W)$. Then $|E_X(U, W)| \leq s_W |U|$, so $|F| \leq s|U|$, and $|U| \leq 2z'/3 \leq 2z/3$. By Cited result [1.12](#) applied to X , $|N_{X-F}(U)| \geq \varepsilon' |U| / \log^2 z$. No vertex of W lies in $N_{X-F}(U)$, because all edges between U and W are in F . For $v \notin U \cup W$ and $u \in U$, the edge uv (if present) lies in $X - W$ and not in $E_X(U, W)$, so $uv \notin F$ iff $uv \notin F'$. Hence $N_{X-F}(U) = N_{(X-W)-F'}(U)$, and

$$|N_{(X-W)-F'}(U)| \geq \frac{\varepsilon' |U|}{\log^2 z} = \varepsilon' \left(\frac{\log z'}{\log z} \right)^2 \frac{|U|}{\log^2 z'}.$$

This is the expansion claimed. Next, $(\log z'/\log z)^2 \geq \frac{1}{2}$ iff $z' \geq z^{1/\sqrt{2}}$. Since $z' \geq z/2$, it suffices that $z^{1-1/\sqrt{2}} \geq 2$, i.e. $\log z \geq 1/(1 - 1/\sqrt{2}) = 3.414\dots$, i.e. $z \geq 10.67$; this holds as $z \geq 11$. The last sentence follows because an (ε_a, s) -expander is an (ε_b, s) -expander whenever $\varepsilon_b \leq \varepsilon_a$. $\square$

Proposition 2.11 (structure of a valid run). *For every valid $\text{HB}^{*\tau+}$ run on a graph G with $n \geq N_0$ and $d_1 \geq D_*$:*

- (i) *each X_Z^0 is a $(2^{-5}, s_l)$ -expander on Z^0 ; each light X_Z is a $(2^{-6}, s_l/2)$ -expander on the light part Z ; every H_Y has minimum degree greater than s_Y ; and $s_r \geq \lambda_r^{100}$ for every $r \leq R$;*
- (ii) *the conclusions of Lemma [2.9\(ii\)](#) hold;*
- (iii) *$E(G_{l+1}) \subseteq E(G'_l) \subseteq E(G_l)$ for every $l \leq R$, so $d_{l+1} \leq d_l$; and*

$$E(G) = \bigsqcup_{l \leq R} E(\text{Cyc}_l) \sqcup E_0 \sqcup \bigsqcup_{Y \text{ light}} E_{r(Y)}(Y) \sqcup \bigsqcup_{l \leq R} \bigsqcup_{Z \in \text{Std}_l} E_l(Z)$$

(disjoint unions); $E(H_Y) \subseteq E_{r(Y)}(Y)$ for every ancestor Y ; every edge of $E_{r(Y)}(Y)$ has both ends in $V(Y)$; the graphs H_Y of all ancestors Y of all rounds are pairwise edge-disjoint, and the graphs X_Z^0 of the pre-parts of one round are pairwise edge-disjoint; $\sum_{l \leq R} |\text{Cyc}_l| \leq n$ (the number of long cycles); and $|E_0| < D_ n/2$;*

(iv) *light parts of one round are pairwise vertex-disjoint, and $|Y| \geq |Y^0|/2 \geq P_r/2$ for every light part Y of round r ; every vertex outside D_l lies in at most one round- l pre-part; the port sets U_Z of distinct $Z \in \text{Std}_l$ are pairwise disjoint; and each vertex lies in at most one light part per round, namely in the light part of $\text{home}_l(v)$ if that pre-part is light.*

Dependencies: 2.7, 2.8, 2.6, 2.9, 2.10, 1.12, 1.15, 2.3, (T1)

Proof. By (T1)(a) (Condition (T1)) at $\mu = \log \log D_*$, we have $\log \log D_* \geq 2^8$, so $\lambda_l \geq \log D_* \geq 2^{256}$ for every $l \leq R$; we use only much weaker consequences, such as $P_l \geq 11$ and $D_* \geq 8$.

(iv) Let v lie in the light part of a light pre-part Z of round l , i.e. $v \in Z^0 \setminus S_Z$. If $v \in D_l$ then, as $v \notin S_Z$, $\text{home}_l(v) = Z$. If $v \notin D_l$ then Z is the only round- l pre-part containing v , so again $\text{home}_l(v) = Z$. Hence v lies only in the light part of $\text{home}_l(v)$ (if that pre-part is light), which proves the first and last assertions. $|Y| = |Y^0| - |S_Y| \geq |Y^0|/2$ by (L1), and $|Y^0| \geq P_r$. By definition of D_l , a vertex outside D_l lies in at most one round- l pre-part, so the sets $U_Z = Z^0 \setminus D_l$ are pairwise disjoint.

(i) Each X_Z^0 is a leaf of a τ -run, hence a $(2^{-5}, s_l)$ -expander by the stopping rule (Lemma 2.6(a)). For a light pre-part Z apply Lemma 2.10 to $X = X_Z^0$, $z = |Z^0| \geq P_l \geq 11$, $W = S_Z$ and $s_W = s_l/2$: $|S_Z| \leq |Z^0|/2$ by (L1), and X_Z^0 is a subgraph of $G'_l[Z^0]$ (node graphs of a split recursion are subgraphs of the root), so by (L2) every vertex has at most $s_l/2$ neighbours in S_Z in X_Z^0 . So $X_Z = X_Z^0 - S_Z$ is a $(2^{-6}, s_l/2)$ -expander on $Z^0 \setminus S_Z$. For an ancestor Y , H_Y is an (ε_Y, s_Y) -expander on $|V(Y)| \geq P_r/2 \geq 2$ vertices (by (iv)), so its minimum degree exceeds s_Y by the remark in Cited result 1.12. Finally $M_r \geq 2^{16}T \log^4 T \geq T = d_r \log^2 d_r \geq d_r$ (with $T = t_r^{\text{HB}} d_r$), so $\Lambda_r \geq \lambda_r$ and $s_r \geq \Lambda_r^{100} \geq \lambda_r^{100}$.

(ii) is Lemma 2.9(ii).

(iii) By (R1), $E(G_l) = E(\text{Cyc}_l) \sqcup E(G'_l)$. By Lemma 2.3(i) applied to the two-level recursion of round l , every edge of G'_l is deleted exactly once or lies in exactly one leaf. By (R5), every edge of G'_l is assigned to exactly one light part or standalone pre-part, or passes to G_{l+1} : step (1) assigns each edge at most once (the X_Z^0 are edge-disjoint and $X_Z \subseteq X_Z^0$), and steps (2)–(4) treat each remaining edge once. So $E(G'_l) = \bigsqcup_{Z \text{ light}} E_l(Z) \sqcup \bigsqcup_{Z \in \text{Std}_l} E_l(Z) \sqcup E(G_{l+1})$; in particular $E(G_{l+1}) \subseteq E(G'_l) \subseteq E(G_l)$. By (R0), $E(G_{R+1}) = E_0$, and iterating from $l = 1$ to $l = R$ gives the displayed partition. By (R5)(1), $E(X_Y)$ is assigned to a light Y and $E(X_Y^0)$ to a standalone Y , so $E(H_Y) \subseteq E_{r(Y)}(Y)$; and each step of (R5) assigns an edge to Y only if both its ends lie in $V(Y)$. Two pre-parts of the same round are distinct leaves of the two-level recursion, so their leaf graphs are edge-disjoint by Lemma 2.3(i). For distinct ancestors $Y \neq Y'$ (of any rounds), $E(H_Y) \subseteq E_{r(Y)}(Y)$ and $E(H_{Y'}) \subseteq E_{r(Y')}(Y')$, and these assigned sets are disjoint by the partition above. (The graphs X_Y^0 of light pre-parts of different rounds need not be edge-disjoint: an edge of X_Y^0 between a guest $x \in S_Y$ and the light part Y may be assigned by no step of (R5) at round $r(Y)$, pass down, and lie in a leaf graph of a later round. Only the statement above is claimed.)

Long cycles. For $l \leq R$ put $y_l := d_l \geq D_*$; by the first part of (iii), $y_{l+1} \leq y_l$. The cycles of Cyc_l are edge-disjoint, each has length at least $t_l^{\text{HB}} d_l = y_l \log^2 y_l$, and together they have at most $|E(G_l)| - |E(G'_l)| \leq |E(G_l)| - |E(G_{l+1})| = \frac{n}{2}(y_l - y_{l+1})$ edges. Hence

$$|\text{Cyc}_l| \leq \frac{n}{2} \cdot \frac{y_l - y_{l+1}}{y_l \log^2 y_l}.$$

Group the rounds $l \leq R$ by $k := \lfloor \log y_l \rfloor \geq k_0 := \lfloor \log D_* \rfloor$. The rounds with a given k are consecutive, and for them $y_l \geq 2^k$ and $\log^2 y_l \geq k^2$, while the sum of $y_l - y_{l+1}$ over them is at most the first y_l of the group, which is less than 2^{k+1} . So these rounds contribute at most $\frac{n}{2} \cdot \frac{2^{k+1}}{2^k k^2} = n/k^2$, and $\sum_{l \leq R} |\text{Cyc}_l| \leq n \sum_{k \geq k_0} k^{-2} \leq n/(k_0 - 1) \leq n$, as $k_0 \geq 3$.

Finally $E_0 = E(G_{R+1})$ and $d_{R+1} < D_*$ by (R0), so $|E_0| = nd_{R+1}/2 < D_* n/2$. $\square$

Lemma 2.12 (edge laminarity, EL). *For every ancestor Y of round r and every round $l > r$, no edge of G_l has both ends in $V(Y)$.*

Dependencies: 2.7, 2.8

Proof. By (R1) and (R5)(4), $E(G_{j+1}) \subseteq E(G'_j) \subseteq E(G_j)$ for every round j , so $E(G_l) \subseteq E(G_{r+1}) \subseteq E(G'_r)$. Let $e \in E(G_{r+1})$; then e is an edge of G'_r that was not assigned at round r . If Y is light and both ends of e lie in $V(Y) = Y$, then e , if not assigned in step (1) of (R5), is assigned in step (2) to the first light part containing both ends; a contradiction. If Y is standalone (a GC-part or not) and both ends lie in $V(Y) = Y^0$, then e , if not assigned in steps (1) or (2), is assigned in step (3) to the first standalone pre-part whose vertex set contains both ends; again a contradiction. $\square$

Proposition 2.13 (overlap constants on $\text{HB}^{*\tau+}$). *Fix a valid $\text{HB}^{*\tau+}$ run and a round $r \leq R$. Let S_r be the total size of the leaves of the two-level recursion of round r , and S_r^Q the total size of the $s = 0$ pieces. Then $S_r^Q \leq 1.12n$ and $S_r \leq 1.21n$, and:*

- (K1) $\sum |Z^0| \leq 1.37n$, the sum over the round- r pre-parts; $|\text{Std}_r| \leq 1.37n/P_r$; the number of ancestors of round r is at most $1.37n/P_r$ (each ancestor corresponds to a distinct pre-part $Y^0 \supseteq V(Y)$); hence the number ν_l of ancestors of rounds at most $l - 2$ satisfies $\nu_l \leq 1.37n \sum_{r \leq l-2} P_r^{-1}$;
- (K2) $|D_r| \leq \text{dup}_r \leq 7.6\epsilon n / \log P_r$, and $\sum_{Z \in \text{Std}_r} |AZ| \leq \sum |Z^0 \cap D_r| \leq 2\text{dup}_r \leq 15.2\epsilon n / \log P_r$, the middle sum over all round- r pre-parts;
- (K3) the total size $\sum |Z^0|$ of the round- r pre-parts failing (L1) is at most $30.4\epsilon n / \log P_r$, and $\sum_Y |Y^0 \cap \text{Dup}_r^*| \leq 16\epsilon n / \log P_r$, the sum over all round- r pre-parts Y ;
- (F11) $\sum_w \text{mult}_r(w) = \sum_Y |V(Y)| \leq 1.37n$, the second sum over the ancestors Y of round r , and $\text{mult}_r(w) \leq \mu_r(w)$ for every vertex w .

Dependencies: 2.5, 2.6, 2.11, 2.8, 2.7, 2.3

Proof. The composite tree. The two-level recursion of round r is a split recursion on G'_r , whose vertex set $V(G)$ has n elements. Its splits in the first level are splits of an $s = 0$ recursion, which satisfy the hypotheses of Lemma 2.5 with $c = 1$ (the instance $c = 1$ there), hence with $c = 1.6$. Its splits in the second level are splits of the τ -run of a piece Q with $|Q| \geq P_r$; since $\tau_r \geq 128s_r \log^2 |Q|$ (Proposition 2.11(ii)), Lemma 2.6(b) shows that they satisfy the hypotheses with $c = 1.6$. Every non-leaf node has at least two vertices. So Lemma 2.5 applies to the whole two-level recursion with $c = 1.6$ and $n_0 = n$: by its part (c), $S_r \leq n/(1 - 5.472\epsilon) \leq 1.21n$; by the first bound of its part (a) and $\log M \geq 1$, for $M \geq 2$, $\Delta_{\geq M} \leq 1.6\epsilon(1 + 1/c_{\text{OV}})S_r / \log M$, and $1.6(1 + 1/c_{\text{OV}}) = 5.455 \dots \leq 5.46$ (the rounded constant 3.42 would only give 5.472):

$$\Delta_{\geq M} \leq \frac{5.46 \epsilon S_r}{\log M} \leq \frac{6.61 \epsilon n}{\log M}. \quad (2.2)$$

Applied to the first level alone (an $s = 0$ recursion on n vertices, $c = 1$), the lemma gives $S_r^Q \leq 1.12n$.

(K1) Pre-parts are leaves, so $\sum |Z^0| \leq S_r \leq 1.21n \leq 1.37n$. Every pre-part has at least P_r vertices, so there are at most $1.37n/P_r$ pre-parts. Std_r is a set of pre-parts, and the map from ancestors of round r to pre-parts (a light part to its pre-part, a standalone pre-part to itself) is injective. Summing over $r \leq l - 2$ gives the bound on ν_l .

(K2) The round- r pre-parts are exactly the leaves of the two-level recursion with at least P_r vertices (by (R3) its leaves are the pieces with fewer than P_r vertices and the leaves of the τ -runs; a piece with at least P_r vertices whose τ -run is a single node is such a leaf, hence a pre-part). By Lemma 2.5(b) with $M = P_r \geq 2$, $\mu_r(w) \leq 1 + \text{dup}_{\geq P_r}(w)$ for every w . Hence $\text{dup}_r = \sum_w (\mu_r(w) - 1)^+ \leq \Delta_{\geq P_r} \leq 6.61 \varepsilon n / \log P_r \leq 7.6 \varepsilon n / \log P_r$ by (2.2). Every $w \in D_r$ contributes at least 1 to dup_r , so $|D_r| \leq \text{dup}_r$. Next, $\sum_{Z \in \text{Std}_r} |A_Z| = \sum_{Z \in \text{Std}_r} |Z^0 \cap D_r| \leq \sum_Z |Z^0 \cap D_r| = \sum_{w \in D_r} \mu_r(w) \leq \sum_{w \in D_r} 2(\mu_r(w) - 1) = 2\text{dup}_r$, using $\mu \leq 2(\mu - 1)$ for $\mu \geq 2$.

(K3) If Z fails (L1) then $|Z^0| < 2|S_Z| \leq 2|Z^0 \cap D_r|$; summing and using (K2) gives at most $4\text{dup}_r \leq 30.4 \varepsilon n / \log P_r$. For the second bound, let $u \in \text{Dup}_r^*$ lie in a pre-part Y . Then u lies in another leaf $\mathfrak{L} \neq Y$ of the two-level recursion. Let ν be the lowest common ancestor of the leaves Y and $\mathfrak{L}$; they lie below different children of ν , and u lies in both (vertex sets shrink downwards), so $u \in N_\nu''$ by Lemma 2.3(0); and $|\nu| \geq |Y^0| \geq P_r$. Hence $\text{dup}_{\geq P_r}(u) \geq 1$ and $\mu_r(u) \leq 1 + \text{dup}_{\geq P_r}(u) \leq 2\text{dup}_{\geq P_r}(u)$. Summing over $u \in \text{Dup}_r^*$,

$$\sum_Y |Y^0 \cap \text{Dup}_r^*| = \sum_{u \in \text{Dup}_r^*} \mu_r(u) \leq 2\Delta_{\geq P_r} \leq \frac{13.3 \varepsilon n}{\log P_r} \leq \frac{16 \varepsilon n}{\log P_r}.$$

(F11) Counting pairs (w, Y) with $w \in V(Y)$ gives $\sum_w \text{mult}_r(w) = \sum_Y |V(Y)|$. The injective map $Y \mapsto Y^0$ of (K1) satisfies $V(Y) \subseteq Y^0$, so $\sum_Y |V(Y)| \leq \sum_Z |Z^0| \leq 1.37 n$ and, restricted to the ancestors containing w , it shows $\text{mult}_r(w) \leq \mu_r(w)$. $\square$

2.5 Degree recursion, existence, and tower facts

Proposition 2.14 (degree recursion). *Let l be a round of a valid $\text{HB}^{*\tau+}$ run with $d_l \geq D_*$ (that is, $l \leq R$), and put $x := \lambda_l$. Then $M_l \leq d_l^2$, $\Lambda_l \leq 2x$, $s_l \leq P_l$, and*

$$d_{l+1} \leq 1.21 P_l + 9 s_l \log M_l \leq 6P_l + 9 s_l \log M_l + 1.2 s_l \leq 7 x^{103} \leq x^A < d_l$$

($A = 104$ would suffice; $A = 105$ is used). More precisely, every edge of G_{l+1} is of one of the following kinds:

- (a) an edge deleted by the τ -run of a piece $\mathcal{Q}$ with $|\mathcal{Q}| \geq P_l$; there are at most $4s_l |\mathcal{Q}| \log |\mathcal{Q}|$ of them for each piece and at most $4.5 n s_l \log M_l$ in total;
- (b) an edge of a leaf of the two-level recursion with fewer than P_l vertices (a small leaf of a τ -run, or a piece with $|\mathcal{Q}| < P_l$); such a leaf, with $q < P_l$ vertices, has fewer than $qP_l/2$ edges;
- (c) a guest edge: an edge of X_Z^0 with an end in S_Z , for a light pre-part Z ; there are at most $|Z^0|s_l/2$ of them for each light Z .

Standalone pre-parts, GC-parts included, pass none of the edges of their leaf graphs down; in particular, declaring a pre-part a GC-part (instead of light) can only decrease the set of passed-down edges.

Dependencies: 2.6, 2.9, 2.13, 2.7, 2.3, (F1), 2.5

Proof. Put $\mu := \log x$, where $x = \log d_l \geq \log D_*$; so $\mu \geq \log \log D_*$, and (F1)(a),(b) (Condition (F1)) hold at μ : $\mu \geq 2^8$ and $x = 2^\mu \geq 2^{14} A \mu^3$. Hence $x \geq 2^{256}$, so $x \geq 7$, $x^2 \geq 2^{107}$ and $x^3 \geq 2^{101}$; and $x \geq 2^{14} \cdot 105 \log^3 x$, so $x \geq 21 + 6 \log x$ and $x > 105 \log x$.

Parameters. Let $T := t_l^{\text{HB}} d_l = 2^x x^2$. Then $\log T = x + 2 \log x \leq 2x$, so $2^{16} T \log^4 T \leq 2^{16} \cdot 2^x x^2 \cdot 16x^4 = 2^{20} x^6 2^x \leq 2^{2x-1}$ because $x \geq 21 + 6 \log x$; also $2^{40} \leq 2^{2x-1}$. By (R2), M_l

is at most this maximum plus 1, so $M_l \leq 2^{2x-1} + 1 \leq 2^{2x} = d_l^2$ and $\Lambda_l = \log M_l \leq 2x$. Then $s_l \leq \Lambda_l^{100} + 1 \leq 2^{100} x^{100} + 1 \leq 2^{101} x^{100} \leq x^{103} \leq P_l$.

Classification. Let $e \in E(G_{l+1})$; by (R5), e is an edge of G'_l not assigned at round l . By Lemma 2.3(i) applied to the two-level recursion, e is deleted at some split, or lies in exactly one leaf $\mathfrak{L}$. Splits of the first level delete nothing (Lemma 2.5, instance $c = 1$), so a deleted e is of kind (a). If $|\mathfrak{L}| < P_l$, e is of kind (b). Otherwise $\mathfrak{L}$ is a pre-part Z and $e \in E(X_Z^0)$. If Z is standalone, e is assigned to Z by (R5)(1), which is impossible. If Z is light and e has no end in S_Z , then $e \in E(X_Z)$ is assigned by (R5)(1), again impossible. So e is of kind (c). The same argument shows the assertion about standalone pre-parts and GC-parts.

Counting. (a) By Lemma 2.9(ii), Lemma 2.6(a) applies to the τ -run of $\mathcal{Q}$ and gives at most $4s_l|\mathcal{Q}|\log|\mathcal{Q}| \leq 4s_l|\mathcal{Q}|\log M_l$ deleted edges. By Proposition 2.13, the pieces have total size $S_l^{\mathcal{Q}} \leq 1.12n$, so kind (a) has at most $4.48n s_l \log M_l$ edges. (b) Every leaf of the two-level recursion is non-empty (both children of every split, of the $s = 0$ recursion and of a τ -run, are non-empty). So a leaf with $1 \leq q < P_l$ vertices has at most $q(q-1)/2 < qP_l/2$ edges. (c) Every guest edge of X_Z^0 joins some $u \in Z^0$ to a vertex of S_Z , so it is counted in $\sum_{u \in Z^0} |N_{X_Z^0}(u) \cap S_Z|$; each term is at most $s_l/2$ by (L2), since $X_Z^0 \subseteq G'_l[Z^0]$. The leaves of kind (b) and the light pre-parts are distinct leaves of the two-level recursion, so the sum of q over the leaves of kind (b) plus the sum of $|Z^0|$ over the light pre-parts is at most $S_l \leq 1.21n$ (Proposition 2.13). As $s_l \leq P_l$, kinds (b) and (c) together have at most $\frac{1}{2} \cdot 1.21n \cdot P_l$ edges. Therefore $|E(G_{l+1})| \leq 4.5n s_l \log M_l + 0.605n P_l$, that is, $d_{l+1} \leq 9s_l \log M_l + 1.21P_l$, which is at most $6P_l + 9s_l \log M_l + 1.2s_l$.

Final bounds. Using $P_l \leq x^{103} + 1$, $s_l \leq 2^{101} x^{100}$ and $\log M_l \leq 2x$:

$$6P_l + 9s_l \log M_l + 1.2s_l \leq 6x^{103} + 6 + 9 \cdot 2^{102} x^{101} + 1.2 \cdot 2^{101} x^{100} \leq 6x^{103} + 2^{107} x^{101} \leq 7x^{103},$$

since $x^2 \geq 2^{107}$. Next $7x^{103} \leq x^{104} \leq x^{105}$ as $x \geq 7$, and $x^{105} < 2^x = d_l$ as $105 \log x < x$. $\square$

Proposition 2.15 (existence and termination). *For every graph G a valid $\text{HB}^{*\tau+}$ run exists, and every sequence of admissible choices terminates. More precisely: within a round, (R1) and the first level of (R3) terminate; the τ -runs terminate with any witnesses ($U' \neq \emptyset$ and $n_1, n_2 < m$ at every split); (GC) is a deterministic relabelling that feeds back into neither D_l nor home nor the guest sets; and $d_{l+1} < d_l$ while $d_l \geq D_*$, so R is finite.*

Dependencies: 2.6, 2.14, 2.7, 2.9, 2.5, 2.1, 2.13

Proof. The proofs of Lemma 2.9(ii), Proposition 2.13 and Proposition 2.14 use only the data of the round l in question and $d_l \geq D_*$; so they apply to every round l with $d_l \geq D_*$ of any (possibly unfinished) execution of the procedure.

Fix a round l with $d_l \geq D_*$ (if $d_1 < D_*$ the run stops at once with $R = 0$). (R1) deletes at least one edge per step, so it terminates. In the first level of (R3), a node that is not an $(\varepsilon, 0)$ -expander has a witness (Definition 2.1), and its children have $|U| + |N| < \frac{3}{4}m$ and $m - |U| < m$ vertices (Lemma 2.5, instance $c = 1$); so every root-to-node path has at most n nodes and this level terminates. By Lemma 2.9(ii), $\tau_l \geq 128s_l \log^2 |\mathcal{Q}|$ for every piece, so Lemma 2.6(a) applies to every τ -run with every choice of witnesses: $U' \neq \emptyset$, $n_1, n_2 < m$, and the τ -run terminates. In (R4), D_l , home and the sets S_Z are determined by the pre-parts and the home order alone; conditions (L1), (L2) and (GC) are evaluated afterwards, and their outcome is not used in defining D_l , home or S_Z . (R5) is deterministic. All required choices (witnesses, vertex orders, home orders) exist. So each round is a finite procedure. By Proposition 2.14, $d_{l+1} < d_l$ whenever $d_l \geq D_*$; as $|E(G_l)|$ is a non-negative integer, (R0) stops after finitely many rounds. $\square$

Lemma 2.16 (tower-lacunary sums). *Let $x_0 := \log D_*$.*

- (i) If $X_1, \dots, X_R \geq 0$ and $X_r \leq X_{r+1}/2$ for all $r < R$, then $\sum_{r \leq R} X_r \leq 2X_R$ and $\sum_{r \leq R} (R - r + 1)X_r \leq 4X_R$.
- (ii) For a valid run with $R \geq 1$, the degree recursion gives $\lambda_r \geq 2^{\lambda_{r+1}/A}$ for all $r < R$. Hence, if $F : [x_0, \infty) \rightarrow [0, \infty)$ is non-increasing and $F(2^{x/A}) \leq F(x)/2$ for all $x \geq x_0$, then $\sum_{r \leq R} F(\lambda_r) \leq 2F(\lambda_R) \leq 2F(x_0)$ and $\sum_{r \leq R} (R - r + 1)F(\lambda_r) \leq 4F(x_0)$; for the shifted sums (when $R \geq 3$), $\sum_{l=3}^R F(\lambda_{l-2}) \leq 2F(\lambda_{R-2}) \leq 2F(x_0)$ and $\sum_{l=3}^R (R - l + 1)F(\lambda_{l-2}) \leq 4F(\lambda_{R-2}) \leq 4F(x_0)$.
- (iii) More generally, if $F : [x_0, \infty) \rightarrow [0, \infty)$ satisfies

$$(H) \quad F(y) \leq F(x)/2 \quad \text{whenever } x \geq x_0 \text{ and } y \geq 2^{x/A},$$

then $\sum_{r \leq R} F(\lambda_r) \leq 2F(\lambda_R)$ and $\sum_{r \leq R} (R - r + 1)F(\lambda_r) \leq 4F(\lambda_R)$ (and likewise for $F(\lambda_{l-2})$).

- (iv) Under $(\Gamma 1)$, condition (H) holds for $F(x) = x^{-a}$ (every $a \geq 1/100$), for $F(x) = 1/\log x$ and for $F(x) = (95 \log x + 8)x^{-b}$ (every $b \geq 1$; not used in this paper), all three non-increasing on $[x_0, \infty)$; and for the three functions $F(x) = (2 \log^*(2^x) + 2)/\eta(x)$ with $\eta(x) \in \{x, x^{1/2}, \log x\}$ (note $2 \log^*(2^x) + 2 = 2 \log^* x + 4$ for $x > 1$; not used in this paper). Each of these three satisfies $F(x) \leq 2F(x_0)$ for all $x \geq x_0$ (not used in this paper).

Round- l quantities such as $1/M_l$ or $M_l^3(95 \log \lambda_{l-2} + 8)/\lambda_{l-2}^{95}$ are summed either by (i), once the halving of consecutive terms is known, or by bounding them by $F(\lambda_{l-2})$ for an F as in (ii); for instance the bound $M_l \leq \lambda_{l-2}^{1.6}$, proved in part (b) of the next lemma (this example is not used in the proof of the present lemma), gives $M_l^3(95 \log \lambda_{l-2} + 8)/\lambda_{l-2}^{95} \leq (95 \log \lambda_{l-2} + 8)\lambda_{l-2}^{-90.2}$. (This example is not used in this paper: Lemma 7.16(iv) bounds its round- l terms by a function of its own and sums them by (i).)

Dependencies: 2.14, $(\Gamma 1)$

Proof. (i) By induction $X_r \leq 2^{-(R-r)}X_R$, so $\sum_r X_r \leq X_R \sum_{k \geq 0} 2^{-k} = 2X_R$ and $\sum_r (R - r + 1)X_r \leq X_R \sum_{k \geq 0} (k + 1)2^{-k} = 4X_R$.

(iii) For $r < R$, Proposition 2.14 at round r gives $d_{r+1} \leq \lambda_r^A$, i.e. $\lambda_{r+1} \leq A \log \lambda_r$, i.e. $\lambda_r \geq 2^{\lambda_{r+1}/A}$; and $\lambda_{r+1} \geq x_0$. By (H) with $x = \lambda_{r+1}$ and $y = \lambda_r$, $F(\lambda_r) \leq F(\lambda_{r+1})/2$, and (i) applies to $X_r := F(\lambda_r)$. For sums of $F(\lambda_{l-2})$ over $3 \leq l \leq R$ apply the same to $r = l - 2 \leq R - 2$.

(ii) Let $x \geq x_0$. Then $\mu := \log x \geq \log x_0 = \log \log D_*$, so $(\Gamma 1)(a), (b)$ at μ give $\mu \geq 2^8$ and $x = 2^\mu \geq 2^{14} A \mu^3 \geq A \mu = A \log x$, that is, $2^{x/A} \geq x$. Hence $2^{x/A} \geq x \geq x_0$ for $x \geq x_0$. If F is non-increasing and $F(2^{x/A}) \leq F(x)/2$, then for $y \geq 2^{x/A}$ we get $F(y) \leq F(2^{x/A}) \leq F(x)/2$, which is (H). So (iii) applies, and $F(\lambda_R) \leq F(x_0)$ as $\lambda_R \geq x_0$.

(iv) Let $x \geq x_0$ and $y \geq 2^{x/A}$, and put $\mu := \log x \geq \log \log D_*$. By $(\Gamma 1)(a), (b)$ at μ , $\mu \geq 2^8$ and $x = 2^\mu \geq 2^{14} A \mu^3$; hence $x \geq 2^{256}$, $x/A \geq 101 \log x$, $x \geq 2A \log x$, $x \geq A \log^2 x/4$ and $x/A \geq 2^{12}$, and all of this holds in particular for $x = x_0$. Then $\log y \geq x/A$, and $2^{x/A} \geq x^{101}$ because $x/A \geq 101 \log x$; so $y \geq x^{101} \geq 2^{32}$. We use two facts about $\log^*$: $\log^* z \leq z$ for $z \geq 1$ (if $z \leq 2$ then $\log^* z \leq 1$; if $z > 2$ then $\log^* z = 1 + \log^*(\log z) \leq 1 + \log z \leq z$ by induction), and hence, for $y \geq 4$, $\log^* y = 2 + \log^*(\log \log y) \leq 2 + \log \log y \leq 1 + \log y$.

$$F(x) = x^{-a}, a \geq 1/100: F(y)/F(x) = (x/y)^a \leq x^{-100a} \leq x^{-1} \leq 1/2.$$

$$F(x) = 1/\log x: F(y) = 1/\log y \leq A/x \leq 1/(2 \log x) = F(x)/2, \text{ as } x \geq 2A \log x.$$

$F(x) = (95 \log x + 8)x^{-b}$, $b \geq 1$: $x^{b+1}F'(x) = 95/\ln 2 - b(95 \log x + 8) < 0$, so F is decreasing; and $F(y) \leq y^{-b/2}$, because $95 \log y + 8 \leq y^{1/2}$ for $y \geq 2^{30}$ (true at 2^{30} , and the right side grows faster). Hence $F(y) \leq x^{-50b} \leq x^{-b}/2 \leq F(x)/2$.

$F(x) = (2 \log^* x + 4)/\eta(x)$. Here $F(x) \geq 4/\eta(x)$, so it suffices to show $F(y) \leq 2/\eta(x)$. For $\eta(x) = x$: $F(y) \leq (2 \log y + 6)/y \leq y^{-1/2}$ (the inequality $2 \log y + 6 \leq y^{1/2}$ holds at $y = 2^{12}$, and the

right side grows faster beyond), and $y^{1/2} \geq x^{50} \geq x/2$. For $\eta(x) = x^{1/2}$: $F(y) \leq (2 \log y + 6)/y^{1/2} \leq y^{-1/4}$ (valid for $y \geq 2^{32}$ in the same way), and $y^{1/4} \geq x^{25} \geq x^{1/2}/2$. For $\eta(x) = \log x$: with $u := \log y \geq x/A$, $F(y) \leq (2 \log u + 8)/u \leq u^{-1/2}$ (valid for $u \geq 2^{12}$), and $u^{-1/2} \leq (A/x)^{1/2} \leq 2/\log x$ as $x \geq A \log^2 x/4$.

The bound $F(x) \leq 2F(x_0)$ for the three $\log^$ -functions.* Let $\text{tw}(0) := 1$ and $\text{tw}(j+1) := 2^{\text{tw}(j)}$. Since $\log^{[i]}$ is increasing and $\log^{[j]} \text{tw}(j) = 1$, a real $z > 1$ has $\log^* z = j$ iff $\text{tw}(j-1) < z \leq \text{tw}(j)$. Let $k_0 := \log^* x_0 \geq 1$, let $x \geq x_0$, and write $\log^* x = k_0 + k$ with $k \geq 0$. Then $x_0 \leq \text{tw}(k_0)$ and $x > \text{tw}(k_0 + k - 1)$. If $k = 0$, then $F(x) \leq F(x_0)$ since η is increasing. If $k = 1$, then $2 \log^* x + 4 = 2k_0 + 6 \leq 2(2k_0 + 4)$ and $\eta(x) \geq \eta(x_0)$, so $F(x) \leq 2F(x_0)$. Let $k \geq 2$. We claim $\eta(x) \geq 2^k \eta(x_0)$. Note $\text{tw}(k_0 + 1) = 2^{\text{tw}(k_0)} \geq 2^{x_0}$, and $\text{tw}(j+1) = 2^{\text{tw}(j)} \geq 4 \text{tw}(j)$ whenever $\text{tw}(j) \geq 4$. Hence $x > \text{tw}(k_0 + k - 1) \geq 4^{k-2} 2^{x_0}$, which gives the claim for $\eta(x) = x$ (as $4^{k-2} 2^{x_0} \geq 2^k x_0$ for $k \geq 2$) and for $\eta(x) = x^{1/2}$ (as $2^{k-2} 2^{x_0/2} \geq 2^k x_0^{1/2}$). For $\eta = \log$: $\log x > \text{tw}(k_0 + k - 2)$, which is at least $\text{tw}(k_0) \geq x_0 \geq 4 \log x_0$ if $k = 2$, and at least $4^{k-3} 2^{x_0} \geq 2^k \log x_0$ if $k \geq 3$. Using the claim and $2k_0 + 2k + 4 \leq 2^k(2k_0 + 4)$ (as $2^k \geq 1 + k$), $F(x) \leq (2k_0 + 2k + 4)/(2^k \eta(x_0)) \leq (2k_0 + 4)/\eta(x_0) = F(x_0)$. $\square$

Lemma 2.17 (tower facts). *For every valid $\text{HB}^{*\tau+}$ run with $d_1 \geq D_*$ (so $R \geq 1$):*

(a) $\lambda_r \geq d_{r+1}^{1/A}$ for $r \leq R$; $d_{r+2} \leq \lambda_r$ for $r \leq R-1$; $R-r \leq 2 \log^* d_r - 1$, in particular

$$R-r \leq 2 \log^* d_r + 2 \quad (r \leq R);$$

and $\lambda_r \geq \lambda_{l-2}$ whenever $r \leq l-2 \leq R$.

(b) For $r \leq R$: $M_r \leq d_r^2$; $\lambda_r \leq \Lambda_r \leq 2\lambda_r$; $\lambda_r^{100} \leq s_r \leq 2\Lambda_r^\sigma$; $s_r \leq P_r$; and $L_Y \leq \log M_r \leq 2\lambda_r$ for every ancestor Y of round r . For $3 \leq l \leq R$: $M_l \leq (A \log \lambda_{l-2})^{2A} \leq \lambda_{l-2}^{1.6}$; $P_{l-2} \geq M_l^{13}$; and $P_{l-2}/2 \geq M_l \log^4 M_l$. For $2 \leq l \leq R$: $M_{l-1} \geq 2M_l$. For $r < R$: $P_r \geq 2P_{r+1}$. Moreover $\sum_{l \leq R} 1/M_l \leq 2/D_*$ and $\sum_{l \leq R} \psi(M_l) \leq 2.1 \psi(D_*)$ with $\psi(x) := (6 \log x + 12)/x$. Finally, completing (K1) of Proposition 2.13, the number ν_l of ancestors of rounds at most $l-2$ satisfies $\nu_l \leq 2.74 n/P_{l-2}$ for $3 \leq l \leq R$.

(c) For $r \leq R$: $\tau_r \leq 257 \Lambda_r^{\sigma+2} \leq 2^{\sigma+11} \lambda_r^{\sigma+2}$ and $\tau_r/P_r \leq 2^{\sigma+11}/\lambda_r$; and $\theta_r^{\text{GC}}(Z^0) \geq P_r \lambda_r^{-1/2} > \tau_r$ for every round- r pre-part Z .

(d) For $r \leq l \leq R$: $\lambda_r \geq 2^{2 \log^* d_r + 2} \geq 2^{R-r} \geq 2^{l-r}$.

(e) Put

$$\varepsilon_A := \frac{31 \varepsilon}{C' \log \log D_*},$$

a function of D_* alone (it depends neither on the run nor on n). Then

$$\sum_{l \leq R} \frac{15.2 \varepsilon}{\log P_l} \leq \varepsilon_A.$$

Consequently $\sum_{l \leq R} |D_l| \leq \varepsilon_A n$ and $\sum_{l \leq R} \sum_{Z \in \text{Std}_l} |A_Z| \leq \varepsilon_A n$.

Dependencies: 2.14, 2.16, 2.7, (G1), (G2), 2.11, 2.9, 2.13, 2.8

Proof. Every round $r \leq R$ has $d_r \geq D_*$, so $\log \lambda_r \geq \log \log D_*$, and the items of $(\Gamma 1)$ (Condition $(\Gamma 1)$) hold at $\mu = \log \lambda_r$ for every $r \leq R$; they also hold at $\mu = \log \log d$ for every $d \geq D_*$. Proposition 2.14 applies at every round $r \leq R$.

(a) Proposition 2.14 at round r gives $d_{r+1} \leq \lambda_r^A$. For $r \leq R-1$ it also applies at round $r+1$, so $d_{r+2} \leq \lambda_{r+1}^A$ with $\lambda_{r+1} = \log d_{r+1} \leq A \log \lambda_r$; hence $d_{r+2} \leq (A \log \lambda_r)^A \leq \lambda_r$, because with $u := \log \lambda_r$ we have $A \log(Au) \leq 0.8u \leq u$ by $(\Gamma 1)(c)$ at u ; equivalently, $(A \log \lambda_r)^A \leq \lambda_r$.

Let $k := \log^* d_r$; $k \geq 1$ as $d_r > 1$. Suppose $R \geq r + 2k$. We show $d_{r+2i} \leq \log^{[i]} d_r$ for $0 \leq i \leq k$ by induction: for $i < k$ we have $r + 2i \leq R - 2$, so $d_{r+2i+2} \leq \log d_{r+2i} \leq \log(\log^{[i]} d_r)$, where $\log^{[i]} d_r > 1$ as $i < \log^* d_r$. Hence $d_{r+2k} \leq \log^{[k]} d_r \leq 1 < D_*$, contradicting $r + 2k \leq R$. Therefore $R - r \leq 2k - 1 \leq 2 \log^* d_r + 2$. (The weaker form with “+2” is the one used in later sections.) Finally $d_{j+1} \leq d_j$ for all $j \leq R$ (Proposition 2.11(iii)), so λ is non-increasing in the round.

(b) $M_r \leq d_r^2$, $\Lambda_r \leq 2\lambda_r$ and $s_r \leq P_r$ are in Proposition 2.14; $\Lambda_r \geq \lambda_r$ and $s_r \geq \lambda_r^{100}$ are in (the proof of) Proposition 2.11(i); and $s_r \leq \Lambda_r^\sigma + 1 \leq 2\Lambda_r^\sigma$. For an ancestor Y of round r , $|V(Y)| \leq |Y^0| \leq M_r$ by Lemma 2.9(ii), so $L_Y \leq \log M_r = \Lambda_r \leq 2\lambda_r$.

Let $3 \leq l \leq R$. By Proposition 2.14 at rounds $l-1$ and $l-2$, $d_l \leq \lambda_{l-1}^A$ and $\lambda_{l-1} \leq A \log \lambda_{l-2}$; with $M_l \leq d_l^2$ this gives $M_l \leq (A \log \lambda_{l-2})^{2A}$. With $u := \log \lambda_{l-2}$, $(\Gamma 1)(c)$ at u gives $2A \log(Au) = 210 \log(105u) \leq 1.6u$, so $M_l \leq \lambda_{l-2}^{1.6}$. Hence $M_l^{13} \leq \lambda_{l-2}^{20.8} \leq \lambda_{l-2}^{103} \leq P_{l-2}$. As $\Lambda_l \geq 40$, $\log M_l = \Lambda_l \leq 2\Lambda_l^{1/4} = M_l^{1/4}$, so $M_l \log^4 M_l \leq M_l^2 \leq M_l^{13}/2 \leq P_{l-2}/2$.

For $2 \leq l \leq R$: $M_{l-1} \geq d_{l-1}$ (as in Proposition 2.11(i)), and $M_l \leq d_l^2 \leq \lambda_{l-1}^{2A} = (\log d_{l-1})^{2A}$, with $y := \log d_{l-1}$ and $v := \log y \geq \log \log D_*$, $(\Gamma 1)(a),(b)$ at v give $v \geq 2^8$ and $y = 2^v \geq 2^{14} A v^3 \geq 1 + 2Av$, that is, $2^y \geq 2y^{2A}$; so $M_{l-1} \geq 2M_l$. For $r < R$: $P_{r+1} \leq \lambda_{r+1}^{103} + 1 \leq (A \log \lambda_r)^{103} + 1 \leq \lambda_r + 1 \leq \lambda_r^{103}/2 \leq P_r/2$, because $(A \log \lambda_r)^{103} \leq (A \log \lambda_r)^A \leq \lambda_r$ (the first inequality as $A \log \lambda_r \geq A \geq 1$, since $\lambda_r \geq 2$; the second from the proof of (a)) and $\lambda_r \geq 2$.

By $M_{l-1} \geq 2M_l$ and Lemma 2.16(i) applied to $X_l := 1/M_l$, $\sum_{l \leq R} 1/M_l \leq 2/M_R \leq 2/d_R \leq 2/D_*$. The function ψ is decreasing on $[1, \infty)$, because $\psi'(x) = (6/\ln 2 - 6 \log x - 12)/x^2$ and $6/\ln 2 < 12$; and for $2 \leq l \leq R$,

$$\psi(M_{l-1}) \leq \psi(2M_l) = \frac{6 \log M_l + 18}{2M_l} = \psi(M_l) \left(\frac{1}{2} + \frac{3}{6 \log M_l + 12} \right) \leq 0.512 \psi(M_l),$$

as $\log M_l \geq 40$. Hence $\sum_{l \leq R} \psi(M_l) \leq \psi(M_R)/(1 - 0.512) \leq 2.05 \psi(M_R) \leq 2.1 \psi(D_*)$, since $M_R \geq d_R \geq D_*$.

Finally, by (K1) of Proposition 2.13, $\nu_l \leq 1.37n \sum_{r \leq l-2} 1/P_r$, and $P_r \geq 2P_{r+1}$ for $r < l-2 \leq R$; Lemma 2.16(i) applied to $X_r := 1/P_r$ ($r \leq l-2$) gives $\sum_{r \leq l-2} 1/P_r \leq 2/P_{l-2}$, so $\nu_l \leq 2.74n/P_{l-2}$.

(c) $\tau_r = \lceil 128 s_r \Lambda_r^2 \rceil \leq 128 \cdot 2\Lambda_r^\sigma \Lambda_r^2 + 1 \leq 257 \Lambda_r^{\sigma+2} \leq 257 \cdot 2^{\sigma+2} \lambda_r^{\sigma+2} < 2^{\sigma+11} \lambda_r^{\sigma+2}$, as $257 \cdot 4 < 2^{11}$. Since $C' = \sigma + 3$, $P_r \geq \lambda_r^{\sigma+3}$ and $\tau_r/P_r \leq 2^{\sigma+11}/\lambda_r$. For a round- r pre-part Z , $\theta_r^{\text{GC}}(Z^0) \geq |Z^0| \lambda_r^{-1/2} \geq P_r \lambda_r^{-1/2} \geq \lambda_r^{102.5}$, while $\tau_r < 2^{111} \lambda_r^{102}$; and $\lambda_r^{1/2} > 2^{111}$ because $\log \lambda_r \geq 2^8 > 222$ by $(\Gamma 1)(a)$.

(d) Put $\mu := \log \lambda_r$; so $\mu \geq 2^8$ by $(\Gamma 1)(a)$. As $d_r = 2^{\lambda_r}$ and $\lambda_r = 2^\mu$ with $\lambda_r, \mu > 1$, $\log^* d_r = 2 + \log^* \mu$. By the fact $\log^* z \leq 1 + \log z$ ($z \geq 4$) shown in the proof of Lemma 2.16(iv), $2 \log^* d_r + 2 = 6 + 2 \log^* \mu \leq 8 + 2 \log \mu \leq \mu$, the last step by $(\Gamma 1)(d)$ at μ . So $\lambda_r = 2^\mu \geq 2^{2 \log^* d_r + 2}$, which is at least 2^{R-r} by (a) and at least 2^{l-r} for $l \leq R$.

(e) We have $\log P_l \geq C' \log \lambda_l$. The function $F(x) := 1/(C' \log x)$ is non-increasing and satisfies $F(2^{x/A}) \leq F(x)/2$ for $x \geq \log D_*$ (Lemma 2.16(iv)). By Lemma 2.16(ii),

$$\sum_{l \leq R} \frac{1}{\log P_l} \leq \sum_{l \leq R} F(\lambda_l) \leq 2F(\log D_*) = \frac{2}{C' \log \log D_*},$$

so $\sum_{l \leq R} 15.2 \varepsilon / \log P_l \leq 30.4 \varepsilon / (C' \log \log D_*) \leq \varepsilon_A$. By (K2) of Proposition 2.13, $\sum_l |D_l| \leq \sum_l 7.6 \varepsilon n / \log P_l \leq \varepsilon_A n$ and $\sum_l \sum_{Z \in \text{Std}_l} |A_Z| \leq \sum_l 15.2 \varepsilon n / \log P_l \leq \varepsilon_A n$. $\square$

2.6 Rule (GC), ORIGIN $^\tau$, and parentless mass

Lemma 2.18 (rule GC). *For every valid $\text{HB}^{*\tau+}$ run and every round $r \leq R$:*

- (i) *the pre-parts, their leaf graphs, Dup_r^* , D_r , home_r and the guest sets S_Z are defined in (R1)–(R4) before and independently of condition (GC); (GC) only decides whether a pre-part satisfying (L1) and (L2) is light or standalone;*
- (ii) *for every light part Y of round r and every guest $x \in S_Y$, $e_{X_Y^0}(x, Y) < \theta_r^{\text{GC}}(Y^0)$, where Y denotes the vertex set $Y^0 \setminus S_Y$;*
- (iii) *GC-parts are standalone; for a GC-part Y , every edge of G_r' with both ends in Y^0 is assigned at round r (the edges of X_Y^0 , those between a guest and $Y^0 \setminus S_Y$ included, by (R5)(1)), so no edge of G_{r+1} has both ends in $V(Y) = Y^0$; and every guest $x \in S_Y$ lies in $D_r \subseteq \text{Dup}_r^*$;*
- (iv) *$\theta_r^{\text{GC}}(Z^0) > \tau_r$ for every round- r pre-part Z .*

Dependencies: 2.7, 2.17, 2.12, 2.8

Proof. (i) is the order of definitions in (R3)–(R4): (GC) is evaluated only after the pre-parts, D_r , home_r and the sets S_Z are fixed, and none of these uses the outcome of (GC).

(ii) A light pre-part satisfies (GC): no $x \in S_Y$ has $\theta_r^{\text{GC}}(Y^0)$ or more neighbours in $Y^0 \setminus S_Y$ in X_Y^0 .

(iii) A GC-part fails (GC) and is therefore standalone by (R4). By (R5)(1) all edges of X_Y^0 are assigned to Y . By Lemma 2.12, applied to the ancestor Y (with $V(Y) = Y^0$), no edge of G_{r+1} has both ends in Y^0 ; equivalently, every edge of G_r' inside Y^0 is assigned at round r . A guest $x \in S_Y$ lies in D_r by definition, hence in at least two pre-parts, which are two leaves of the two-level recursion; so $x \in \text{Dup}_r^*$.

(iv) is Lemma 2.17(c). $\square$

Proposition 2.19 (ORIGIN $^\tau$). *Fix a valid $\text{HB}^{*\tau+}$ run, a round $r \leq R$, a round- r pre-part Y and a vertex $u \in Y^0 \setminus \text{Dup}_r^*$.*

- (a) *u lies in a unique $s = 0$ piece $\mathcal{Q}$ and in a unique leaf of the two-level recursion, namely Y (so $Y^0 \subseteq V(\mathcal{Q})$). Every edge ux of G_{r+1} is of exactly one of the following two types:*
 - (β) *$x \notin Y^0$, and ux was deleted by the τ -run of $\mathcal{Q}$ (so $x \in V(\mathcal{Q})$);*
 - (α) *Y is light, $x \in S_Y$, and $ux \in E(X_Y^0)$ (a guest-core edge: it is not deleted and not assigned at round r , and passes down).*

A standalone Y (GC-parts included) has no edges of type (α). Since $E(G_l) \subseteq E(G_{r+1})$ for $l > r$, the same classification applies to the edges ux of G_l .

- (b) *For every vertex h and every round $l > r$,*

$$\#\{u \in Y^0 \setminus \text{Dup}_r^* : hu \in E(G_l) \text{ is of type } (\beta)\} \leq \tau_r - 1.$$

- (c) *$\sum_Y |Y^0 \cap \text{Dup}_r^*| \leq 16 \varepsilon n / \log P_r$, the sum over all round- r pre-parts Y .*

An edge of type (α) has its core end u outside Dup_r^* (it is not true that (α) -edges occur only at vertices of Dup_r^*); a guest x of a light Y is the guest end of fewer than $\theta_r^{\text{GC}}(Y^0)$ edges of type (α) (Lemma 2.18(ii)); and there is no third type.

Dependencies: 2.4, 2.3, 2.9, 2.13, 2.12, 2.7, 2.18, 2.5, 2.11

Proof. (a) As $u \notin \text{Dup}_r^*$, u lies in exactly one leaf of the two-level recursion, and $u \in Y^0$, so that leaf is Y . If u lay in two $s = 0$ pieces, it would lie in a leaf below each of them (Lemma 2.3(0)), i.e. in two leaves; so u lies in a unique piece $\mathcal{Q}$, and Y is a leaf of the τ -run of $\mathcal{Q}$ (by Lemma 2.9(ii) this is a Lemma 14 $^\tau$ run, although only its split structure is used below). By Lemma 2.3(i),(ii) applied to the two-level recursion, every edge of G'_r at u is either an edge of the leaf Y , i.e. of X_Y^0 , or deleted at some split on the path from the root to Y . The splits of the first level delete nothing (Lemma 2.5, instance $c = 1$), so deleted edges at u are deleted by the τ -run of $\mathcal{Q}$ and are edges of $\mathcal{Q}$. A deleted edge ux has $x \notin Y^0$: at the split where it is deleted, u and x go to different children, each to only one, so they do not lie in a common leaf. Hence every edge ux of G'_r with $x \in Y^0$ is an edge of X_Y^0 .

Now let $ux \in E(G_{r+1})$, an edge of G'_r not assigned at round r . Since $u \notin D_r$ (as $D_r \subseteq \text{Dup}_r^*$), Y is the only pre-part containing u , and $u \notin S_Y$.

Case ux deleted. Then $x \notin Y^0$ and $x \in V(\mathcal{Q})$; this is type (β) . (Such an edge is indeed never assigned: a light part or standalone pre-part containing both ends would contain u , hence would be the light part of Y or the pre-part Y itself, neither of which contains $x \notin Y^0$.)

Case $ux \in E(X_Y^0)$. If Y is standalone, ux is assigned to Y by (R5)(1), a contradiction. So Y is light. If $x \notin S_Y$, then $ux \in E(X_Y)$ is assigned by (R5)(1), a contradiction. So $x \in S_Y$: type (α) . (Such an edge is indeed passed down: the only light part that could contain u is Y , which does not contain x , and the only pre-part containing u is the light pre-part Y , so steps (2) and (3) of (R5) do not apply.)

The types are exclusive ($x \notin Y^0$ in (β) , $x \in S_Y \subseteq Y^0$ in (α)). For standalone Y , Lemma 2.12 also shows directly that no edge of G_{r+1} has both ends in $V(Y) = Y^0$. By Proposition 2.11(iii) (or directly from (R1) and (R5)), $E(G_l) \subseteq E(G_{r+1})$ for $l > r$.

(b) Let h be a vertex and $l > r$. By (a), each u counted has hu deleted by the τ -run of the unique piece $\mathcal{Q}$ containing u , and this piece is the piece whose τ -run has Y as a leaf; so all counted edges are deleted edges of one τ -run, and each is of the form hu with $u \in Y^0 \setminus \text{Dup}_r^*$. A vertex lying in two leaves of the τ -run of $\mathcal{Q}$ lies in two leaves of the two-level recursion, so $Y^0 \setminus \text{Dup}_r^* \subseteq Y^0 \setminus \text{Dup}$, where Dup is taken in the τ -run of $\mathcal{Q}$. Lemma 2.4, applied to this τ -run (all of whose splits are τ -rule splits at threshold τ_r) and its leaf Y , bounds the number of such edges by $\lceil \tau_r \rceil - 1 = \tau_r - 1$ (τ_r is an integer by (R2)), and by 0 if $h \in Y^0$. Distinct u give distinct edges hu .

(c) is (K3) of Proposition 2.13.

The final paragraph of the statement: an (α) -edge is an edge ux with $u \in Y^0 \setminus \text{Dup}_r^*$ by the setting of (a); the (α) -edges at a guest x with core ends in $Y \setminus \text{Dup}_r^*$ are edges of X_Y^0 between x and $Y = Y^0 \setminus S_Y$, and there are fewer than $\theta_r^{\text{GC}}(Y^0)$ of them by Lemma 2.18(ii). $\square$

Proposition 2.20 (fresh and parentless mass). *For every valid $\text{HB}^{*\tau+}$ run with $d_1 \geq D_*$:*

- (i) *If x lies in a round- r pre-part, then $H := \text{home}_r(x)$ is an ancestor of round r containing x : $x \in H^0 \setminus S_H = V(H)$ if H is light, and $x \in H^0 = V(H)$ otherwise. Hence, for $Z \in \text{Std}_l$ and $x \in U_Z$, $x \in F_Z$ iff $j_0(x) \geq l - 1$. A vertex is a port of at most one part per round, so it is a fresh port only at rounds $j_0(x)$ and $j_0(x) + 1$, and*

$$\sum_{l \leq R} \sum_{Z \in \text{Std}_l} |F_Z| \leq 2n. \quad (\text{K4})$$

(ii) (admissible parents) If Y is a light part of round $r \leq l - 2$ and Z is a light part of round $l \leq R$, then

$$|Y| \geq P_r/2 \geq P_{l-2}/2 \geq M_l \log^4 M_l \geq |Z| L_Z^4.$$

(iii) Let Bad be any set of light parts. Call a pair (v, Z) , with Z a light part of some round l and $v \in Z$, parentless with respect to Bad if no light part outside Bad of a round at most $l - 2$ contains v . Then the number of such pairs is at most

$$2n + \sum_{Y \in \text{Bad}} |Y| (R - r(Y)).$$

More precisely, a vertex v is in parentless pairs only at its light parts of rounds $j_1(v)$ and $j_1(v) + 1$, and at the later rounds at which the light part of round $j_1(v)$ containing v lies in Bad .

Dependencies: 2.11, 2.17, 2.9, 2.8, 2.7

Proof. (i) By definition $H = \text{home}_r(x)$ is a pre-part containing x . If H is light, then $x \notin S_H$, because S_H consists of vertices whose home is not H ; so $x \in H^0 \setminus S_H = V(H)$, and the light part H is an ancestor of round r . If H is stand-alone, it is an ancestor with $V(H) = H^0 \ni x$.

Let $Z \in \text{Std}_l$ and $x \in U_Z \subseteq Z^0$, so $j_0(x) \leq l$. If $l \leq 2$, then $x \in F_Z$ by definition and $j_0(x) \geq 1 \geq l - 1$. Let $l \geq 3$. If $j_0(x) \leq l - 2$, then $\text{home}_{j_0(x)}(x)$ is an ancestor of round $j_0(x) \leq l - 2$ containing x , so $\text{anc}_l(x) \neq \emptyset$ and $x \notin F_Z$. Conversely, if $\text{anc}_l(x) \neq \emptyset$, then $x \in V(Y) \subseteq Y^0$ for a pre-part Y of a round at most $l - 2$, so $j_0(x) \leq l - 2$. Hence $x \in F_Z$ iff $j_0(x) \geq l - 1$.

The port sets U_Z of distinct $Z \in \text{Std}_l$ are disjoint (Proposition 2.11(iv)), so x is a port of at most one part per round. If x is a fresh port at round l then $j_0(x) \leq l$ (it lies in Z^0) and $j_0(x) \geq l - 1$, so $l \in \{j_0(x), j_0(x) + 1\}$. Summing over x gives $\sum_l \sum_Z |F_Z| \leq 2n$.

(ii) $|Y| \geq P_r/2$ by Proposition 2.11(iv). As λ is non-increasing in the round (Lemma 2.17(a)) and $r \leq l - 2$, $P_r \geq P_{l-2}$. As $l \geq r + 2 \geq 3$, $P_{l-2}/2 \geq M_l \log^4 M_l$ by Lemma 2.17(b). Finally $|Z| \leq |Z^0| \leq M_l$ by Lemma 2.9(ii), and $L_Z = \log |Z| \leq \log M_l$.

(iii) Fix a vertex v that lies in some light part, and let $j_1 := j_1(v)$ and $Y_1(v)$ be the light part of round j_1 containing v . By Proposition 2.11(iv), v lies in at most one light part per round, so the pairs (v, Z) correspond to distinct rounds $l \geq j_1$. If $l \geq j_1 + 2$ and $Y_1(v) \notin \text{Bad}$, then $Y_1(v)$ is a light part outside Bad of round $j_1 \leq l - 2$ containing v , so (v, Z) is not parentless. Hence the parentless pairs of v are at the rounds j_1 and $j_1 + 1$ (at most two pairs), and, if $Y_1(v) \in \text{Bad}$, at rounds l with $j_1 + 2 \leq l \leq R$ (at most $R - j_1$ pairs). Summing over v , the number of parentless pairs is at most

$$2n + \sum_{v: Y_1(v) \in \text{Bad}} (R - r(Y_1(v))) \leq 2n + \sum_{Y \in \text{Bad}} \sum_{v \in Y} (R - r(Y)) = 2n + \sum_{Y \in \text{Bad}} |Y| (R - r(Y)),$$

since $v \in Y_1(v)$.

Part (iii) holds for every set Bad . Its argument uses only that the light part of round $j_1(v)$ containing v is a light part outside Bad ; so, when it is applied, every reason for which a light part is not used as a parent must place that part in Bad . In the later application this means that Bad consists of the light parts that are demoted *or* parent-bad, not only the demoted ones. $\square$

Remark 2.21 (constants of $\text{HB}^{*\tau+}$ versus plain HB^*). Plain HB^* is the procedure of Definition 2.7 in which the second level of (R3) is the recursion of [2, Lemma 14] at $s = s_l$ (no τ -rules) and (GC) is

quantity	plain HB*	HB ^{*τ+} (used here)	where proved
$\sum Z^0 $ over round- r pre-parts	$1.28 n$	$1.37 n$ ($1.21 n$ on the composite tree)	Prop. 2.13, (K1)
duplication at nodes of size $\geq P$	$3.5 \varepsilon S / \log P$	$5.5 \varepsilon S / \log P$	Lemma 2.6(b)
ancestors of rounds $\leq l - 2$ (ν_l)	$2.6 n / P_{l-2}$	$2.74 n / P_{l-2}$	Lemma 2.17(b)
dup_r (and $ D_r $)	$4.5 \varepsilon n / \log P_r$	$7.6 \varepsilon n / \log P_r$	Prop. 2.13, (K2)
$\sum_{Z \in \text{Std}_r} A_Z $	$9 \varepsilon n / \log P_r$	$15.2 \varepsilon n / \log P_r$	Prop. 2.13, (K2)
mass of pre-parts failing (L1)	$18 \varepsilon n / \log P_r$	$30.4 \varepsilon n / \log P_r$	Prop. 2.13, (K3)

Table 4: Overlap constants of plain HB* and of HB^{*τ+}; only the right-hand column is used in this paper.

omitted; HB^{*τ} has the τ -rules but no (GC). The overlap constants change because the separator of a τ -split may exceed the separator N of [2] by $|H_{\text{out}}| + |H_{\text{in}}|$, which raises the charge in Lemma 2.5 from $c = 1$ to $c = 1.6$ at the second level. Table 4 lists the values; only the right-hand column is used in this paper. The left-hand column is recorded for comparison only; it is neither used nor proved here.

Dependencies: 2.13, 2.6, 2.17, 2.7, 2.5

3 Random splitting, linking, and the lending colouring

This section provides the probabilistic tools that every later routing step uses, together with the stage-1 lending data.

- The first subsection proves monotonicity facts and Lemma 15⁺. A uniformly random k -colouring of the edges of an (ε', s) -expander leaves every colour class an $(\varepsilon', s/(2k))$ -expander: the first parameter is untouched and the second loses only the factor $2k$. Lemma 15⁺ replaces [2, Lemma 15] (quoted as Cited result 1.16), which is not used.
- The second subsection proves Theorem 16*. It says that a robust expander is path connected through a random vertex set, in the multiset, for-all form of [2, Definition 7]. The requirement on the robustness is linear in the multiplicity and polynomial in the inverse density. Its proof re-runs the proof of [2, Theorem 16] (Cited result 1.17) with explicit parameters.
- The third subsection proves Lemma HB by one application of Hall's theorem.
- The fourth subsection defines the two-stage lending colouring and the JS labels, verifies the numerical conditions they need (the COL-JV table), and proves Lemma COL.

Every statement of this section is proved in full here, from the results stated in Section 1 and, in the fourth subsection, from the hierarchy of Section 2.

Conventions for this section. Logarithms are to base 2 (Convention 1.3), and $\ln$ is the natural logarithm. For $\rho \in [0, 1]$, a ρ -random subset of a finite set S is a random subset of S that contains each element of S independently with probability ρ . The letter ε' always denotes a generic expansion

parameter in $(0, 1]$. The fixed constant $\varepsilon = 2^{-5}$ of Definition 1.31 enters this section only through the values $\varepsilon_Y \in \{2^{-5}, 2^{-6}\}$ of Definition 2.8. For a graph G , $U \subseteq V(G)$ and $F \subseteq E(G)$, the set $N_{G-F}(U)$ is the outside neighbourhood of U in the graph $G - F$ (Convention 1.3). The ball $B_{G-F}^i(U, V)$ is the set of vertices of V that can be reached from U by a path of length at most i in $G - F$ whose interior vertices all lie in V (Cited result 1.7). Path connectivity (Cited result 1.9) always refers to *multisets* of pairs of distinct vertices; see the remark on multisets at the end of the first subsection.

3.1 Monotonicity and random splitting

Lemma 3.1 (monotonicity and the for-all property). *(i) Let H be an (ε', s) -expander and let $H' \supseteq H$ be a graph with $V(H') = V(H)$. Then H' is an (ε', s) -expander. Moreover, every (ε', s) -expander is an (ε'', s') -expander whenever $\varepsilon'' \leq \varepsilon'$ and $s' \leq s$.*

(ii) Path connectivity is monotone. Let $G \subseteq G'$ be graphs with $V(G) = V(G')$, let $V \subseteq V' \subseteq V(G)$, and suppose $\ell \leq \ell'$ and $t' \leq t$. If G is (ℓ, t) -path connected through V , then G' is (ℓ', t') -path connected through V' .

(iii) Whether a graph X is (ℓ, t) -path connected through a set V depends only on the pair (X, V) . On this event, required paths exist for every multiset $\mathcal{P}$ of pairs of distinct vertices of X in which each vertex lies in at most t pairs. In particular, $\mathcal{P}$ may be chosen after X and V are revealed, as an arbitrary function of them and of any further randomness (adaptively). If several multisets $\mathcal{P}_1, \dots, \mathcal{P}_q$ are given and every vertex lies in at most t pairs of the union multiset $\mathcal{P}_1 + \dots + \mathcal{P}_q$, then one application of the property to this union joins all their pairs by pairwise edge-disjoint paths through V of length at most ℓ (joint routing).

Dependencies: 1.9, 1.12

Proof. (i) Put $n := |V(H)|$. Let $U \subseteq V(H')$ with $1 \leq |U| \leq 2n/3$, and let $F \subseteq E(H')$ with $|F| \leq s|U|$. Then $F \cap E(H)$ is a set of at most $s|U|$ edges of H , and $H - (F \cap E(H)) \subseteq H' - F$. Hence

$$|N_{H'-F}(U)| \geq |N_{H-(F \cap E(H))}(U)| \geq \varepsilon'|U|/\log^2 n$$

by Definition 11 of [2] (Cited result 1.12) applied to H . Both graphs have n vertices, so the logarithm in the bound is the same. The second claim holds because the condition of Cited result 1.12 becomes weaker when ε' or s decreases.

(ii) Let $\mathcal{P}$ be a multiset of pairs of distinct vertices of G' in which every vertex lies in at most $t' \leq t$ pairs. The hypothesis on G applies to $\mathcal{P}$, since $V(G) = V(G')$. It gives pairwise edge-disjoint paths of G , one for each pair, each of length at most $\ell \leq \ell'$ and with all interior vertices in $V \subseteq V'$. These are paths of G' with the required properties.

(iii) By Cited result 1.9, the property says that for every admissible multiset the paths exist. It mentions no other data, so it is a property of (X, V) alone, and on this event it applies to every admissible multiset, however that multiset was chosen. The union $\mathcal{P}_1 + \dots + \mathcal{P}_q$ is itself an admissible multiset, which gives the last sentence. $\square$

Lemma 3.2 (Lemma 15⁺: random splitting without loss). *Let X be an N -vertex (ε', s) -expander with $0 < \varepsilon' \leq 1$, put $L := \log N$, and let $k \geq 1$ be an integer with $s \geq 40kL$. Give every edge of X a colour from $[k]$, independently and uniformly at random, and for $i \in [k]$ let X_i be the graph with vertex set $V(X)$ whose edges are the edges of colour i . Then for each $i \in [k]$, X_i is an $(\varepsilon', s/(2k))$ -expander with probability at least $1 - 2N^{-5}$. Hence all k classes are $(\varepsilon', s/(2k))$ -expanders with probability at least $1 - 2kN^{-5}$.*

Dependencies: 1.12, 1.8

Proof. If $N = 1$, then no set U satisfies $1 \leq |U| \leq 2N/3$, so every graph on one vertex is an expander for all parameters, and there is nothing to prove. Assume $N \geq 2$, so $L \geq 1$. Fix $i \in [k]$ and write $H := X_i$.

Fix $U \subseteq V(X)$ with $1 \leq u := |U| \leq 2N/3$. Put $m := \lceil \varepsilon' u / L^2 \rceil$ and $\mathcal{N} := N_X(U)$. Since $\varepsilon' \leq 1 \leq L^2$, we have $1 \leq m \leq u$.

Step 1 (every small deletion leaves many edges). We claim that

$$e_X(U, \mathcal{N} \setminus \mathcal{T}) > su \quad \text{for every } \mathcal{T} \subseteq \mathcal{N} \text{ with } |\mathcal{T}| \leq m - 1. \quad (3.1)$$

Suppose instead that $e_X(U, \mathcal{N} \setminus \mathcal{T}) \leq su$ for some such $\mathcal{T}$, and let F be the set of edges of X between U and $\mathcal{N} \setminus \mathcal{T}$. Then $|F| \leq su$, and every neighbour of U in $X - F$ outside U lies in $\mathcal{T}$. Hence $|N_{X-F}(U)| \leq m - 1 < \varepsilon' u / L^2$, which contradicts the expansion of X (Cited result 1.12).

Step 2 (what a failure looks like). Suppose that U violates the $(\varepsilon', s/(2k))$ -condition in H . Then there is $F \subseteq E(H)$ with $|F| \leq su/(2k)$ and $|N_{H-F}(U)| < \varepsilon' u / L^2$. Since $|N_{H-F}(U)|$ is an integer, it is at most $m - 1$. Put $\mathcal{T} := N_{H-F}(U)$. Since $H \subseteq X$, we have $\mathcal{T} \subseteq \mathcal{N}$ and $|\mathcal{T}| \leq m - 1$, and every edge of H between U and $\mathcal{N} \setminus \mathcal{T}$ lies in F . Hence $e_H(U, \mathcal{N} \setminus \mathcal{T}) \leq su/(2k)$.

Step 3 (one pair $(U, \mathcal{T})$). Fix $\mathcal{T} \subseteq \mathcal{N}$ with $|\mathcal{T}| \leq m - 1$. Since U and $\mathcal{N} \setminus \mathcal{T}$ are disjoint, the edges of X between them are $e := e_X(U, \mathcal{N} \setminus \mathcal{T})$ distinct edges, and $e > su$ by (3.1). Each of them lies in H independently with probability $1/k$. So $Z := e_H(U, \mathcal{N} \setminus \mathcal{T}) \sim \text{Bin}(e, 1/k)$, with mean $\mu_Z := e/k > su/k$. The event $\{Z \leq su/(2k)\}$ is contained in $\{Z < \mu_Z/2\}$. By the lower Chernoff bound (Cited result 1.8) with $\delta = 1/2$,

$$\mathbb{P}(Z \leq su/(2k)) \leq e^{-\mu_Z/8} \leq e^{-su/(8k)} \leq e^{-5uL},$$

where the last step uses $s \geq 40kL$.

Step 4 (union bound). There are at most N^u sets U of size u . The number of sets $\mathcal{T}$ with $|\mathcal{T}| \leq m - 1$ is at most $\sum_{j \leq m} N^j \leq N^m \leq N^u$. Since $e^{-5uL} = N^{-5u \log e} \leq N^{-7.2u}$, the probability that H is not an $(\varepsilon', s/(2k))$ -expander is at most

$$\sum_{u \geq 1} N^{2u} \cdot N^{-7.2u} \leq \frac{N^{-5.2}}{1 - N^{-5.2}} \leq 2N^{-5}.$$

The final statement follows by a union bound over the k colours. Neither ε' nor the logarithm L changes, because X_i has the same vertex set as X . $\square$

Remark 3.3 (multisets). In [2, Definition 7] (Cited result 1.9), the collection $\mathcal{P}$ is a *multiset* of pairs of distinct vertices.

- The same pair may occur several times, and each occurrence needs a path of its own.
- The endpoints are arbitrary vertices of the graph; they need not lie in V .
- The proof of [2, Lemma 9] (Cited result 1.11) indexes the pairs by $i \in [r]$, so it applies verbatim to multisets.
- Its counting step also holds when pairs are counted with multiplicity. The step says that a maximal subfamily $I' \subseteq I$ of pairwise vertex-disjoint pairs satisfies $2t|I'| \geq |I|$. It holds because every pair of I meets one of the $2|I'|$ vertices covered by I' , and each vertex lies in at most t pairs, counted with multiplicity.

All path-connectivity statements of this paper are meant in this multiset sense, and the proofs of this subsection and the next treat the pairs as an indexed family. Repeated pairs genuinely occur later: in the routing inside the vortex devices of Section 4, and in the joint routing of the JS-LC systems in Section 6.

Dependencies: 1.9, 1.11

3.2 Linking through random sets: Theorem 16*

This subsection proves Theorem 16*, stated at its end. The route is that of [2, Section 4]:

1. well-expanding sets grow into a random set by sprinkling (Lemma 17* below, modelled on [2, Lemma 17]);
2. every set contains a well-expanding subset (Proposition 18* and Lemma 19* below, modelled on [2, Proposition 18 and Lemma 19]);
3. ball expansion implies multiset linking (Lemma 9 _{ρ} below, modelled on [2, Lemma 9]);
4. a random edge-splitting (Lemma 3.2) assembles these.

Every parameter is explicit, and the density ρ and the multiplicity t are free.

Local parameters. Let $n \geq 2$ be the number of vertices of the graph at hand, $L := \log n$, $\varepsilon' \in [2^{-7}, 1]$ its expansion parameter, $\rho \in (0, 1]$ a density and $t \geq 1$ a multiplicity. The following quantities are local to this subsection. All carry a star subscript; the letters $\ell, g, p, q, d, \lambda, \Delta, \sigma, \theta, \mu, \bar{s}, K, M, t_0$ without a star keep the meaning they have elsewhere. Throughout, $\ell_*, d_*, \lambda_*, \Delta_*, M_*$ and K_* are integers.

$$\begin{aligned}
\ell_* &:= \lfloor 2^{10} L^3 \rfloor, & g_* &:= \frac{\varepsilon'}{8L^2}, & q_* &:= 0.9\rho, & p_* &\in (0, 1] \text{ given by } (1 - p_*)^{\ell_* - 1} = \frac{1 - \rho}{1 - 0.9\rho}, \\
d_* &:= \lceil 1/p_* \rceil, & \lambda_* &:= \lceil 6/p_* \rceil, & \Delta_* &:= \lambda_* + \lceil d_* \lambda_*/3 \rceil, & \sigma_* &:= \frac{24L^2}{\varepsilon'}, \\
\theta_* &:= \frac{2^{19} \ell_*^2 L^3}{\varepsilon' \rho^2}, & \mu_* &:= \frac{\varepsilon'}{6\theta_* L^2}, & \bar{s}_* &:= \frac{2^{28} t L^8}{\rho}, & M_* &:= \lceil 2.1/\rho \rceil, & K_* &:= \left\lceil \frac{6 \bar{s}_* \theta_* L^2}{\varepsilon'} \right\rceil.
\end{aligned} \tag{3.2}$$

The integer t_{0*} is defined inside the proof of Lemma 9 _{ρ} below. We record six elementary facts, each with its proof. Write $p := p_*$.

- (S1) $2^{10} L^3 - 1 < \ell_* \leq 2^{10} L^3$ and $\ell_* \geq 2^{10}$. *Proof:* $L \geq 1$ because $n \geq 2$.
- (S2) The number p_* exists, is unique, and satisfies $p_* \geq 0.1\rho/\ell_*$, so $p_*^{-2} \leq 100\ell_*^2 \rho^{-2}$. If $V_1, \dots, V_{\ell_*}$ are independent random subsets of a set S , where V_i is p_* -random for $i < \ell_*$ and V_{ℓ_*} is q_* -random, then $V_1 \cup \dots \cup V_{\ell_*}$ is a ρ -random subset of S . *Proof:* since $\ell_* \geq 2$, the map $x \mapsto (1 - x)^{\ell_* - 1}$ is a continuous decreasing bijection of $[0, 1]$ onto $[0, 1]$, and the value $(1 - \rho)/(1 - 0.9\rho)$ lies in $[0, 1]$. Hence $p_* \in (0, 1]$ exists and is unique. By Bernoulli's inequality, $1 - (\ell_* - 1)p \leq (1 - p)^{\ell_* - 1} = 1 - 0.1\rho/(1 - 0.9\rho)$. Hence $(\ell_* - 1)p \geq 0.1\rho$, which gives $p \geq 0.1\rho/(\ell_* - 1) \geq 0.1\rho/\ell_*$. For the last claim, an element avoids the union with probability $(1 - p)^{\ell_* - 1}(1 - 0.9\rho) = 1 - \rho$, independently over elements.
- (S3) $\Delta_* \leq 2p^{-2} + 8.34p^{-1} + 2.34$. Hence $\Delta_* \leq 3p^{-2}$ if $p \leq 0.11$, and $\Delta_* \leq 13p^{-2}$ for every $p \in (0, 1]$. *Proof:* $d_* \leq p^{-1} + 1$ and $\lambda_* \leq 6p^{-1} + 1$, so $d_* \lambda_* \leq 6p^{-2} + 7p^{-1} + 1$. Then $\Delta_* \leq \lambda_* + d_* \lambda_*/3 + 1 \leq (6p^{-1} + 1) + (2p^{-2} + \frac{7}{3}p^{-1} + \frac{1}{3}) + 1 = 2p^{-2} + \frac{25}{3}p^{-1} + \frac{7}{3}$. If $p \leq 0.11$, then $8.34p^{-1} \leq 0.92p^{-2}$ and $2.34 \leq 0.03p^{-2}$. If $p \leq 1$, then $8.34p^{-1} + 2.34 \leq 10.68p^{-2}$.

(S4) $8d_*\lambda_* \leq 112p^{-2} \leq \theta_*$ and $\theta_* \leq 2^{39}L^9/(\varepsilon'\rho^2) \leq 2^{46}L^9\rho^{-2}$. *Proof:* $8d_*\lambda_* \leq 8(6+7+1)p^{-2}$ since $p \leq 1$. By (S2), $112p^{-2} \leq 11200\ell_*^2\rho^{-2} \leq 2^{19}\ell_*^2L^3/(\varepsilon'\rho^2)$. The upper bound uses $\ell_*^2 \leq 2^{20}L^6$ and $\varepsilon' \geq 2^{-7}$.

(S5) $K_* \geq \bar{s}_*/\mu_*$; $K_* \leq (6 \cdot 2^{81} + 1)tL^{19}\rho^{-3} \leq 2^{83.6}tL^{19}\rho^{-3}$; $2K_*(\theta_* + 1) \leq 2^{130.6}tL^{28}\rho^{-5}$; and $40K_*L \leq 2K_*(\theta_* + 1)$. *Proof:* $\bar{s}_*/\mu_* = 6\bar{s}_*\theta_*L^2/\varepsilon' \leq K_*$. By (S4), $6\bar{s}_*\theta_*L^2/\varepsilon' \leq 6 \cdot 2^{28}tL^8\rho^{-1} \cdot 2^{39}L^9\rho^{-2} \cdot L^2 \cdot 2^{14} = 6 \cdot 2^{81}tL^{19}\rho^{-3}$, since $\varepsilon'^{-2} \leq 2^{14}$. The rounding adds at most $1 \leq tL^{19}\rho^{-3}$. Likewise $\theta_* + 1 \leq (2^{46} + 1)L^9\rho^{-2}$, and $\log(2(6 \cdot 2^{81} + 1)(2^{46} + 1)) < 130.6$. Finally $\theta_* \geq 2^{19}L^3 \geq 20L$.

(S6) $\sigma_* \geq 24$, $\lambda_*p \geq 6$ and $1 \leq d_*p \leq 1 + p \leq 2$. *Proof:* immediate from (3.2), since $\varepsilon' \leq 1 \leq L$.

The values σ_* , λ_* , Δ_* , d_* of (3.2) satisfy the parameter hypotheses of Proposition 13* below: $\sigma_* = 24L^2/\varepsilon'$ holds with equality, and $\Delta_* - \lambda_* = \lceil d_*\lambda_*/3 \rceil \geq d_*\lambda_*/3 = 8d_*\lambda_*L^2/(\varepsilon'\sigma_*)$. The proposition itself is stated for arbitrary numbers with these properties.

Proposition 3.4 (Proposition 13*: stars or bipartite). *Let G be an n -vertex (ε', s_1) -expander with $n \geq 2$ and $2^{-7} \leq \varepsilon' \leq 1$, and put $L := \log n$. Let $W \subseteq V(G)$ with $1 \leq |W| \leq 2n/3$, and let $F \subseteq E(G)$ with $|F| \leq s_1|W|/4$. Let $\sigma_* > 0$ be real and let $\lambda_*, d_* \geq 1$ and $\Delta_* \geq \lambda_*$ be integers such that*

$$\sigma_* \geq \frac{24L^2}{\varepsilon'}, \quad \Delta_* - \lambda_* \geq \frac{8d_*\lambda_*L^2}{\varepsilon'\sigma_*}, \quad s_1 \geq 8d_*\lambda_*.$$

Then $G - F$ contains one of the following:

- (a) *at least $|W|/\sigma_*$ vertex-disjoint stars, each with exactly λ_* leaves, its centre in W and all its leaves in $V(G) \setminus W$; or*
- (b) *a bipartite subgraph $H \subseteq G - F$ with sides W and $X \subseteq V(G) \setminus W$, such that $|X| \geq \varepsilon'|W|/(2L^2)$, every vertex of X has degree exactly d_* in H , and every vertex of W has degree at most Δ_* in H .*

Dependencies: 1.13, 1.14, 1.12

Proof. We follow the proof of [2, Proposition 13] (Cited result 1.14) with general parameters. For $U \subseteq V(G)$ and an integer $d \geq 1$, write $N_{G-F,d}(U)$ for the set of vertices outside U that have at least d neighbours in U in the graph $G - F$. We use [2, Proposition 12] (Cited result 1.13): if $|U| \leq 2n/3$, $|F| \leq s_1|U|/2$ and $0 < d \leq s_1$, then $|N_{G-F}(U)| \geq s_1|U|/(2d)$ or $|N_{G-F,d}(U)| \geq \varepsilon'|U|/L^2$.

Assume that (a) fails. Take a maximal collection of vertex-disjoint stars in $G - F$, each with exactly λ_* leaves, its centre in W and its leaves in $V(G) \setminus W$. Let $\text{Cen} \subseteq W$ be the set of centres and $\text{Lvs} \subseteq V(G) \setminus W$ the set of leaves. Since (a) fails, $|\text{Cen}| < |W|/\sigma_*$, and $|\text{Lvs}| = \lambda_*|\text{Cen}| < \lambda_*|W|/\sigma_*$. By maximality, every $w \in W \setminus \text{Cen}$ has at most $\lambda_* - 1$ neighbours in $G - F$ in $V(G) \setminus (W \cup \text{Lvs})$. (Otherwise a new star with centre w and λ_* such leaves could be added. It is disjoint from the others, because $w \notin \text{Cen}$, $w \notin \text{Lvs}$ and its leaves avoid $W \cup \text{Lvs}$.) Every vertex of $N_{G-F}(W \setminus \text{Cen})$ lies in Cen , in Lvs , or in $V(G) \setminus (W \cup \text{Lvs})$. Hence

$$|N_{G-F}(W \setminus \text{Cen})| < \frac{(1 + \lambda_*)|W|}{\sigma_*} + (\lambda_* - 1)|W| \leq \left(\frac{1 + \lambda_*}{24} + \lambda_* - 1 \right)|W|, \quad (3.3)$$

where we used $\sigma_* \geq 24L^2/\varepsilon' \geq 24$.

The greedy bipartite graph. Label the vertices of $V(G) \setminus W$ as $v_1, \dots, v_{r'}$ and let H_0 be the graph on W with no edges. For $i = 1, \dots, r'$, check whether v_i has d_* neighbours in $G - F$ inside W , each of degree less than Δ_* in H_{i-1} . If so, let H_i be H_{i-1} together with v_i and these d_* edges, and put v_i into X . Otherwise put $H_i := H_{i-1}$. Finally let $H := H_{r'}$.

- $H \subseteq G - F$ is bipartite with sides W and X .
- Every vertex of X has degree exactly d_* in H .
- Every vertex of W has degree at most Δ_* in H : a degree increases, by one, only when it is below the integer Δ_* .

It remains to show $|X| \geq \varepsilon'|W|/(2L^2)$.

Saturated vertices. Let $\text{Sat} := \{w \in W \setminus \text{Cen} : \deg_H(w) = \Delta_*\}$. A vertex $w \in \text{Sat}$ has at most $\lambda_* - 1$ neighbours of $G - F$ in $V(G) \setminus (W \cup \text{Lvs})$. Hence it has at most $\lambda_* - 1$ H -neighbours in $X \setminus \text{Lvs}$, and so more than $\Delta_* - \lambda_*$ H -neighbours in $X \cap \text{Lvs}$. Every vertex of X has exactly d_* H -neighbours, so

$$(\Delta_* - \lambda_*)|\text{Sat}| \leq e_H(\text{Sat}, X \cap \text{Lvs}) \leq d_*|\text{Lvs}| < \frac{d_*\lambda_*|W|}{\sigma_*}.$$

The hypothesis on $\Delta_* - \lambda_*$ forces $\Delta_* > \lambda_*$, and it gives $|\text{Sat}| < \varepsilon'|W|/(8L^2)$. Put $\text{Bl} := \text{Cen} \cup \text{Sat}$. Then

$$|\text{Bl}| < \frac{|W|}{\sigma_*} + \frac{\varepsilon'|W|}{8L^2} \leq \frac{\varepsilon'|W|}{24L^2} + \frac{\varepsilon'|W|}{8L^2} = \frac{\varepsilon'|W|}{6L^2} \leq \frac{|W|}{6}, \quad (3.4)$$

so $|W \setminus \text{Bl}| \geq 5|W|/6$.

Applying Proposition 12. The set $U := W \setminus \text{Bl}$ has $1 \leq |U| \leq 2n/3$, and $|F| \leq s_1|W|/4 \leq s_1|U|/2$. Also $1 \leq d_* \leq s_1$, since $s_1 \geq 8d_*\lambda_*$. By Cited result 1.13 one of two cases holds.

Case 1: $|N_{G-F}(U)| \geq s_1|U|/(2d_*)$. Then $|N_{G-F}(U)| \geq s_1|W|/(3d_*) \geq 8\lambda_*|W|/3$. Let $x \in N_{G-F}(U)$. Either $x \notin W \setminus \text{Cen}$, and then $x \in N_{G-F}(W \setminus \text{Cen})$; or $x \in W \setminus \text{Cen}$ with $x \notin U$, and then $x \in \text{Sat} \subseteq W$. Hence $|N_{G-F}(W \setminus \text{Cen})| \geq (8\lambda_*/3 - 1)|W|$. Comparing with (3.3) gives $8\lambda_*/3 < (1 + \lambda_*)/24 + \lambda_*$, that is, $39\lambda_* < 1$. This contradicts $\lambda_* \geq 1$.

Case 2: $|N_{G-F,d_*}(U)| \geq \varepsilon'|U|/L^2$. Let $v \in N_{G-F,d_*}(U) \setminus \text{Bl}$. Then $v \notin W$, since $v \notin U$ and $v \notin \text{Bl}$, and v has at least d_* neighbours of $G - F$ in U . Vertices of U lie in $W \setminus \text{Cen}$ and not in Sat , so they have H -degree below Δ_* , and their degree in H_{i-1} is at most that. So when $v = v_i$ was processed, a star was available, and $v \in X$. Therefore, using (3.4),

$$|X| \geq |N_{G-F,d_*}(U)| - |\text{Bl}| \geq \frac{\varepsilon'}{L^2} \cdot \frac{5|W|}{6} - \frac{\varepsilon'|W|}{6L^2} = \frac{2\varepsilon'|W|}{3L^2} \geq \frac{\varepsilon'|W|}{2L^2}.$$

So (b) holds. $\square$

Lemma 3.5 (Lemma 17*: sprinkling at density ρ). *Let G be an n -vertex (ε', s_1) -expander with $n \geq 2$ and $2^{-7} \leq \varepsilon' \leq 1$, let $\rho \in (0, 1]$, take the parameters of (3.2) with $L := \log n$, and assume $s_1 \geq 8d_*\lambda_*$. Let $V_1, \dots, V_{\ell_*}$ be independent random subsets of $V(G)$, where V_i is p_* -random for $i < \ell_*$ and V_{ℓ_*} is q_* -random, and put $V := V_1 \cup \dots \cup V_{\ell_*}$. By (S2), V is a ρ -random subset of $V(G)$. Call $U \subseteq V(G)$ well-expanding if $|N_G(U)| \geq \theta_*|U|$. Then for every well-expanding U and every $F \subseteq E(G)$ with $|F| \leq |U|$,*

$$\mathbb{P}\left(|B_{G-F}^{\ell_*}(U, V)| \leq |V|/2\right) \leq \exp(-7|U|L).$$

Since this event depends only on V , the same bound holds for every ρ -random subset V of $V(G)$.

Dependencies: 3.4, 1.18, 1.27, 1.8, 1.25

Proof. The structure is that of the proof of [2, Lemma 17] (Cited result 1.18). The batch probabilities are adapted to the density ρ , and the concentration step in case (b) uses the Bernstein inequality for bounded differences (Lemma 1.27).

If $U = \emptyset$ the bound is 1, so let $u := |U| \geq 1$.

Step 0: n and ρn are large. Since U is well-expanding, $\theta_* u \leq |N_G(U)| < n$. Hence $n > \theta_* \geq 2^{19} \ell_*^2 L^3 / \rho^2$ (as $\varepsilon' \leq 1$), and so $\rho n \geq \rho^2 n > 2^{19} \ell_*^2 L^3 \geq 2^{39}$ by (S1). Also $7uL < 7nL/\theta_* \leq 7\rho^2 n / (2^{19} \ell_*^2 L^2) \leq 2^{-36} \rho n$ (using $\varepsilon' \leq 1$, $\rho \leq 1$ and $\ell_* \geq 2^{10}$).

Step 1: the reached sets. For $0 \leq i \leq \ell_*$, let $\mathcal{B}_i$ be the set of vertices $v \in V(G)$ that can be reached from U by a path in $G - F$ of length at most i whose interior vertices (if any) lie in $V_1 \cup \dots \cup V_{i-1}$. The vertices of $\mathcal{B}_i$ themselves need not lie in any V_j . Then $\mathcal{B}_0 = U$, $\mathcal{B}_1 = U \cup N_{G-F}(U)$ and $\mathcal{B}_0 \subseteq \mathcal{B}_1 \subseteq \dots \subseteq \mathcal{B}_{\ell_*}$. The set $\mathcal{B}_i$ is determined by U , F and $V_1, \dots, V_{i-1}$, so it is independent of $V_i, \dots, V_{\ell_*}$.

(O1) Let $v \in N_{G-F}(\mathcal{B}_i)$ have a neighbour w in $G - F$ with $w \in \mathcal{B}_i \cap V_i$. Then $v \in \mathcal{B}_{i+1}$. Indeed, take a path from U to w as in the definition of $\mathcal{B}_i$. The vertex v does not lie on it, since otherwise an initial segment would put v into $\mathcal{B}_i$. Appending the edge wv gives a path of length at most $i + 1$ whose interior lies in $V_1 \cup \dots \cup V_i$.

(O2) $\mathcal{B}_{\ell_*} \cap V_{\ell_*} \subseteq \mathcal{B}_{G-F}^{\ell_*}(U, V)$, since the interiors lie in V and the end lies in $V_{\ell_*} \subseteq V$.

(O3) $|\mathcal{B}_1| \geq \theta_* u$. Deleting one edge removes at most one vertex from an outside neighbourhood, so $|N_{G-F}(U)| \geq |N_G(U)| - |F| \geq (\theta_* - 1)u$.

Step 2: growth into a fixed set. Let $1 \leq i \leq \ell_* - 1$, and let $W \subseteq V(G)$ be a fixed set with $\mathcal{B}_1 \subseteq W$ and $|W| \leq 2n/3$. Put

$$A_i(W) := \{v \in N_{G-F}(W) : v \text{ has a neighbour in } G - F \text{ that lies in } W \cap V_i\}.$$

We claim that

$$\mathbb{P}(|A_i(W)| < g_* |W|) \leq e^{-18uL}. \quad (3.5)$$

By (O3), $|W| \geq \theta_* u$. Also $|F| \leq u \leq |W| \leq s_1 |W|/4$, because $s_1 \geq 8d_* \lambda_* \geq 48$. So Proposition 3.4 applies to W and F with the values (3.2), and gives one of two cases.

Case (a): stars. Let Cen be the set of centres of at least $|W|/\sigma_*$ vertex-disjoint stars as in (a). Every leaf of a star whose centre lies in V_i belongs to $A_i(W)$, and distinct stars have distinct leaves. Now $|\text{Cen} \cap V_i| \sim \text{Bin}(|\text{Cen}|, p_*)$, so the Chernoff bound (Cited result 1.8, $\delta = 1/2$) gives $\mathbb{P}(|\text{Cen} \cap V_i| < p_* |\text{Cen}|/2) \leq \exp(-p_* |\text{Cen}|/8)$. Outside this event, by (S6),

$$|A_i(W)| \geq \lambda_* p_* |\text{Cen}|/2 \geq 3|\text{Cen}| \geq 3|W|/\sigma_* = g_* |W|.$$

For the exponent, (S2) and $|W| \geq \theta_* u$ give

$$\begin{aligned} \frac{p_* |\text{Cen}|}{8} &\geq \frac{p_* \varepsilon' |W|}{192 L^2} \geq \frac{0.1 \rho}{\ell_*} \cdot \frac{2^{19} \ell_*^2 L^3 u}{192 \rho^2 L^2} = \frac{0.1 \cdot 2^{19} \ell_* L u}{192 \rho} \\ &\geq 273 \ell_* u L \quad (\text{using } \rho \leq 1) \geq 18uL. \end{aligned}$$

Case (b): a bipartite graph. Let $H \subseteq G - F$ be bipartite with sides W and X as in (b). Every $x \in X$ with an H -neighbour in V_i lies in $A_i(W)$, because $x \notin W$ and $H \subseteq G - F$. Let $Z := |\{x \in X : N_H(x) \cap V_i \neq \emptyset\}|$. For $x \in X$, by (S6), $\mathbb{P}(N_H(x) \cap V_i = \emptyset) = (1 - p_*)^{d_*} \leq e^{-p_* d_*} \leq e^{-1}$, so $\mathbb{E}Z \geq (1 - e^{-1})|X| \geq 0.63|X|$.

The count Z is a function of the independent indicators $\xi_w := \mathbf{1}[w \in V_i]$, $w \in W$, each Bernoulli with parameter p_* . Order W as $w_1, \dots, w_{|W|}$. For $y = (y_1, \dots, y_{|W|}) \in \{0, 1\}^{|W|}$ let $\Psi(y)$ be the number of $x \in X$ such that $y_k = 1$ for some k with $w_k \in N_H(x)$. Since H is bipartite with sides W and X , we have $N_H(x) \subseteq W$ for $x \in X$, so $Z = \Psi(\xi_{w_1}, \dots, \xi_{w_{|W|}})$. If $y, y' \in \{0, 1\}^{|W|}$ differ only in the k -th coordinate, then only the vertices $x \in N_H(w_k)$ can change their contribution to Ψ , each by at most 1, so $|\Psi(y) - \Psi(y')| \leq \deg_H(w_k) \leq \Delta_*$ by Proposition 3.4(b). Using $1 - p_* \leq 1$, $\deg_H(w_k) \leq \Delta_*$, $\sum_{w \in W} \deg_H(w) = e(H) = d_*|X|$ (every $x \in X$ has exactly d_* neighbours in H , all in W) and (S6),

$$\sum_k p_*(1 - p_*) \deg_H(w_k)^2 \leq p_* \Delta_* \sum_{w \in W} \deg_H(w) = p_* d_* \Delta_* |X| \leq 2 \Delta_* |X| =: \beta,$$

and $\beta > 0$, since $|X| \geq \varepsilon' |W| / (2L^2) > 0$ (as $|W| \geq \theta_* u > 0$) and $\Delta_* \geq \lambda_* \geq 1$. Apply the second bound of Lemma 1.27 with $m := |W|$, $I_k := \xi_{w_k}$, $p_k := p_*$, $c_k := \deg_H(w_k)$, with Δ_* in the role of the bound b of that lemma, this β and $a := 0.3|X|$. Since $2(\beta + \Delta_* a/3) = 4\Delta_* |X| + 0.2\Delta_* |X|$ and $0.09/4.2 > 1/47$, it gives

$$\mathbb{P}(Z \leq \mathbb{E}Z - 0.3|X|) \leq \exp\left(-\frac{0.09|X|^2}{4\Delta_* |X| + 0.2\Delta_* |X|}\right) \leq \exp\left(-\frac{|X|}{47\Delta_*}\right).$$

Since $\mathbb{E}Z - 0.3|X| \geq 0.33|X|$, we get $\mathbb{P}(Z < 0.33|X|) \leq \exp(-|X|/(47\Delta_*))$. Outside this event, $|A_i(W)| \geq Z \geq 0.33|X| \geq 0.165 \varepsilon' |W| / L^2 \geq g_* |W|$.

For the exponent, $|X| \geq \varepsilon' |W| / (2L^2)$ and $|W| \geq \theta_* u$ give

$$\frac{|X|}{47\Delta_*} \geq \frac{\varepsilon' \theta_* u}{94\Delta_* L^2} = \frac{2^{19} \ell_*^2 L u}{94 \Delta_* \rho^2}.$$

- If $p_* \leq 0.11$, then (S3) and (S2) give $\Delta_* \leq 3p_*^{-2} \leq 300 \ell_*^2 \rho^{-2}$, and the exponent is at least $2^{19} u L / 28200 \geq 18uL$ (indeed $2^{19}/28200 > 18.59$; this is the tightest inequality of the lemma).
- If $p_* > 0.11$, then (S3) gives $\Delta_* \leq 13p_*^{-2} < 13/0.0121 < 1075$, and the exponent is at least $2^{19} \ell_*^2 u L / (94 \cdot 1075)$ (using $\rho \leq 1$) $\geq 5 \ell_*^2 u L \geq 18uL$.

The large- p_* regime occurs only for ρ extremely close to 1. It is handled directly by the bound $\Delta_* \leq 13p_*^{-2}$, so no coupling with a smaller density is needed. This proves (3.5).

Step 3: all rounds. For $1 \leq i \leq \ell_* - 1$ let E_i be the event that $|\mathcal{B}_i| \geq 2n/3$ or $|\mathcal{B}_{i+1} \setminus \mathcal{B}_i| \geq g_* |\mathcal{B}_i|$. Condition on $V_1, \dots, V_{i-1}$. This fixes $W := \mathcal{B}_i \supseteq \mathcal{B}_1$, while V_i is still p_* -random (by the independence of V_i from $(V_1, \dots, V_{i-1})$). If $|W| \geq 2n/3$ then E_i holds. Otherwise $A_i(W) \subseteq \mathcal{B}_{i+1} \setminus \mathcal{B}_i$ by (O1), since $A_i(W) \cap W = \emptyset$, and (3.5) gives $\mathbb{P}(E_i^c \mid V_1, \dots, V_{i-1}) \leq e^{-18uL}$. Hence, using $u \geq 1$, (S1), and the fact that $L^3 e^{-11L} \leq e^{-11}$ for $L \geq 1$,

$$\mathbb{P}\left(\bigcup_{i < \ell_*} E_i^c\right) \leq (\ell_* - 1) e^{-18uL} \leq 2^{10} L^3 e^{-11L} e^{-7uL} \leq 0.02 e^{-7uL}.$$

Step 4: growth to two thirds. Suppose all E_i hold, and suppose $|\mathcal{B}_{\ell_*}| < 2n/3$. Then $|\mathcal{B}_i| < 2n/3$ for all $i \leq \ell_*$, so $|\mathcal{B}_{i+1}| \geq (1 + g_*) |\mathcal{B}_i|$ for $1 \leq i \leq \ell_* - 1$, and $|\mathcal{B}_{\ell_*}| \geq (1 + g_*)^{\ell_* - 1}$. Since $g_* \leq 1/8$, we have $\ln(1 + g_*) \geq g_* - g_*^2/2 \geq 15g_*/16$. By (S1) and $\varepsilon' \geq 2^{-7}$, $(\ell_* - 1)g_* > (2^{10} L^3 - 2) 2^{-10} L^{-2} \geq L - 2^{-9}$. Hence $\ln((1 + g_*)^{\ell_* - 1}) \geq \frac{15}{16}(L - 2^{-9}) \geq L \ln 2 = \ln n$, so $|\mathcal{B}_{\ell_*}| \geq n$, a contradiction. Therefore $|\mathcal{B}_{\ell_*}| \geq 2n/3$.

Step 5: the last batch. Condition on $V_1, \dots, V_{\ell_* - 1}$ with $|\mathcal{B}_{\ell_*}| \geq 2n/3$. The set V_{ℓ_*} is still 0.9ρ -random, so $|\mathcal{B}_{\ell_*} \cap V_{\ell_*}|$ is binomial with mean at least $0.6\rho n$. The lower-tail Chernoff bound for a

mean bound $\mu := 0.6\rho n \leq \mathbb{E}|\mathcal{B}_{\ell_*} \cap V_{\ell_*}|$ (Cited result 1.25(a)) with $\delta = 0.09$ gives $\mathbb{P}(|\mathcal{B}_{\ell_*} \cap V_{\ell_*}| < 0.546\rho n) \leq \exp(-0.0081 \cdot 0.6\rho n/2) \leq \exp(-0.00243\rho n)$. Since V is ρ -random, $|V| \sim \text{Bin}(n, \rho)$, and $\mathbb{P}(|V| > 1.09\rho n) \leq \exp(-0.0081\rho n/3) = \exp(-0.0027\rho n)$. Outside both events, $|\mathcal{B}_{\ell_*} \cap V_{\ell_*}| \geq 0.546\rho n > 0.545\rho n \geq |V|/2$, and then $|B_{G-F}^{\ell_*}(U, V)| > |V|/2$ by (O2). The probability of the two events is at most $2e^{-\rho n/412} \leq 2e^{-\rho n/413} e^{-7uL} \leq 0.01 e^{-7uL}$. Here we used Step 0: $7uL \leq 2^{-36}\rho n \leq \rho n(1/412 - 1/413)$ and $\rho n > 2^{39}$.

Conclusion. If $|B_{G-F}^{\ell_*}(U, V)| \leq |V|/2$, then either some E_i fails, or (by Step 4) $|\mathcal{B}_{\ell_*}| \geq 2n/3$ and one of the two events of Step 5 occurs. The total probability is at most $0.03 e^{-7uL} \leq e^{-7uL}$. The final sentence of the lemma holds because the event is a function of the random set V alone, and V is ρ -random. $\square$

Lemma 3.6 (Proposition 18* and Lemma 19*). *Let G be an n -vertex (ε', s_1) -expander with $n \geq 2$ and $2^{-7} \leq \varepsilon' \leq 1$, let $\rho \in (0, 1]$, take the parameters of (3.2) with $L := \log n$, and assume $s_1 \geq \theta_* + 1$.*

- (i) (Proposition 18*.) *Every $U \subseteq V(G)$ with $1 \leq |U| \leq 2n/3$ contains a set U' with $|N_G(U')| \geq \theta_*|U'|$ and $|U'| \geq \varepsilon'|U|/(3\theta_*L^2)$.*
- (ii) (Lemma 19*.) *Let V be a ρ -random subset of $V(G)$. With probability at least $1 - n^{-6}$ the following holds: for every nonempty $U \subseteq V(G)$ and every $F \subseteq E(G)$ with $|F| \leq \mu_*|U|$,*

$$|B_{G-F}^{\ell_*}(U, V)| > |V|/2.$$

The hypothesis is $s_1 \geq \theta_ + 1$ and not $s_1 \geq \theta_*$. The maximality step of [2, Proposition 18] (Cited result 1.19) produces vertices with fewer than $\theta_* + 1$ neighbours outside, so the edge set F built there satisfies $|F| < (\theta_* + 1)|U|$.*

Dependencies: 1.19, 1.20, 3.5, 1.12

Proof. By (S4), $\theta_* \geq 8d_*\lambda_*$, so $s_1 \geq 8d_*\lambda_*$ and Lemma 3.5 applies to G . Also $\theta_* \geq 1$.

(i) We follow the proof of [2, Proposition 18]. Let $U' \subseteq U$ be maximal subject to $|N_G(U')| \geq \theta_*|U'|$; the empty set qualifies. If $U' = U$ we are done, since $\varepsilon'/(3\theta_*L^2) \leq 1$. Otherwise let $v \in U \setminus U'$, and let a_v be the number of neighbours of v outside $U' \cup N_G(U')$. Then

$$|N_G(U' \cup \{v\})| = |N_G(U')| - \mathbf{1}[v \in N_G(U')] + a_v.$$

By maximality this is less than $\theta_*(|U'| + 1)$. Two consequences follow:

- $a_v < \theta_*(|U'| + 1) - |N_G(U')| + 1 \leq \theta_* + 1$;
- $|N_G(U')| < \theta_*(|U'| + 1) + 1$.

Let F be the set of edges vx with $v \in U \setminus U'$ and $x \notin U' \cup N_G(U')$. Then $|F| \leq \sum_{v \in U \setminus U'} a_v < (\theta_* + 1)|U| \leq s_1|U|$. Every vertex $x \notin U$ with a neighbour in U in $G - F$ lies in $N_G(U')$: if the neighbour is in U' this is clear, and if it is in $U \setminus U'$ then, since the edge is not in F , $x \in U' \cup N_G(U')$ with $x \notin U'$. By the expansion of G (Cited result 1.12),

$$\varepsilon'|U|/L^2 \leq |N_{G-F}(U)| \leq |N_G(U')| < \theta_*(|U'| + 1) + 1.$$

If $U' = \emptyset$, then the middle term is 0, which is impossible because $|U| \geq 1$. So $|U'| \geq 1$, and then $\theta_*(|U'| + 1) + 1 \leq 3\theta_*|U'|$ since $\theta_* \geq 1$. Hence $|U'| \geq \varepsilon'|U|/(3\theta_*L^2)$.

- (ii) We follow the proof of [2, Lemma 19] (Cited result 1.20).

The union bound. Consider all pairs (U', F') with U' well-expanding and nonempty, $F' \subseteq E(G)$ and $|F'| \leq |U'|$. Put $e_G := |E(G)| \leq n^2/2$; since G has minimum degree $> s_1 \geq 1$, $e_G \geq 2$. The number of pairs with $|U'| = u$ is at most $n^u \sum_{f=0}^u e_G^f \leq n^u \cdot 2e_G^u \leq n^{3u}$. By Lemma 3.5, each pair fails $|B_{G-F'}^{\ell_*}(U', V)| > |V|/2$ with probability at most $e^{-7uL} = n^{-7u \log e} \leq n^{-10.09u}$. Hence, with probability at least

$$1 - \sum_{u \geq 1} n^{3u} n^{-10.09u} \geq 1 - \frac{n^{-7.09}}{1 - n^{-7.09}} \geq 1 - n^{-6},$$

every such pair satisfies the ball bound. Fix an outcome in this event.

Deterministic consequence. Let U be nonempty, and let $F \subseteq E(G)$.

- *Case $|U| \leq 2n/3$ and $|F| \leq 2\mu_*|U|$.* By (i) there is a well-expanding $U' \subseteq U$ with $|U'| \geq \varepsilon'|U|/(3\theta_*L^2) = 2\mu_*|U| \geq |F|$. The ball $B_{G-F}^{\ell_*}(\cdot, V)$ is monotone in its first argument, so $|B_{G-F}^{\ell_*}(U, V)| \geq |B_{G-F}^{\ell_*}(U', V)| > |V|/2$.
- *Case $|U| > 2n/3$ and $|F| \leq \mu_*|U|$.* Choose $\bar{U} \subseteq U$ with $n/2 \leq |\bar{U}| \leq 2n/3$. This is possible for every $n \geq 2$: the interval $[n/2, 2n/3]$ contains an integer, as one checks directly for $n \leq 5$, and for $n \geq 6$ it has length at least 1. Then $|F| \leq \mu_*n \leq 2\mu_*|\bar{U}|$, and the previous case gives $|B_{G-F}^{\ell_*}(U, V)| \geq |B_{G-F}^{\ell_*}(\bar{U}, V)| > |V|/2$.

In both cases the conclusion of (ii) holds. $\square$

Lemma 3.7 (Lemma 9 $_{\rho}$: multiset linking from ball expansion). *Let G be an n -vertex graph with $n \geq 2^{30}$, put $L := \log n$, and let $t \geq 1$ and $\rho \in (0, 1]$. Take ℓ_* , $\bar{s}_*$ and $M_* = \lceil 2.1/\rho \rceil$ from (3.2). Let $V \subseteq V(G)$ with $|V| \geq \rho n/2$, and suppose $\rho n \geq 84$. Then $|V| \geq n/M_* + 2$. Suppose that*

$$|B_{G-F}^{\ell_*}(U, V)| > |V|/2$$

for every nonempty $U \subseteq V(G)$ and every $F \subseteq E(G)$ with $|F| \leq \bar{s}_|U|$. Then G is $(2^{12}L^4, t)$ -path connected through V in the multiset sense of [2, Definition 7] (Cited result 1.9). That is, every multiset of pairs of distinct vertices of G in which each vertex lies in at most t pairs can be joined by pairwise edge-disjoint paths of length at most $2^{12}L^4$ whose interior vertices lie in V .*

Dependencies: 1.11, 1.10, 1.29, 1.9, 3.3

Proof. First, $n/M_* \leq \rho n/2.1$ and $\rho n/2 - \rho n/2.1 = \rho n/42 \geq 2$, so $|V| \geq n/M_* + 2$. Next, $1 \leq \ell_* \leq 2^{10}L^3 \leq n$ for $n \geq 2^{30}$. We re-run the proof of [2, Lemma 9] (Cited result 1.11) with its constants changed, and with Haxell's theorem (Cited result 1.29) in place of the Aharoni–Haxell theorem [2, Theorem 6]. It uses [2, Proposition 8] (Cited result 1.10): if $1 \leq \ell, t_0 \leq n$, $|V| \geq 4t_0 - 2$, and every set of t_0 vertices has a ball of radius ℓ in V of size greater than $|V|/2$, then among any $2t_0 - 1$ pairs of vertices, all $4t_0 - 2$ of them distinct, some pair is joined by a path through V of length at most $4\ell \log n$.

Let $\mathcal{P} = (\{x_i, y_i\})_{i \in [r]}$ be a multiset of pairs of distinct vertices, indexed by $[r]$, in which every vertex lies in at most t pairs (counted with multiplicity; see Remark 3.3). Put $h := \lfloor 4\ell_*L \rfloor$; then $h \geq 1$, since $\ell_* \geq 1$ and $L \geq 1$. For $i \in [r]$, let $\mathcal{Y}_i$ be the set of $x_i y_i$ -paths in G that have length at most h and all interior vertices in V . Every $P \in \mathcal{Y}_i$ has at most h edges.

Claim: *for every nonempty $I \subseteq [r]$ and every $F \subseteq E(G)$ with $|F| < h^2|I|$, there are $j \in I$ and $P \in \mathcal{Y}_j$ with $E(P) \cap F = \emptyset$. Let $I' \subseteq I$ be maximal such that the pairs $\{x_i, y_i\}$, $i \in I'$, are pairwise vertex-disjoint; $I' \neq \emptyset$ because $I \neq \emptyset$. By maximality every pair indexed by I contains one of the $2|I'|$ vertices covered by I' , and each vertex lies in at most t pairs. Hence $|I| \leq 2t|I'|$. Also $|I'| \leq n/2$. Put $t_{0*} := \lceil |I'|/(2M_*) \rceil$. Then:*

- $1 \leq t_{0*}$ and $2t_{0*} - 1 \leq |I'|$ (if $|I'| \leq 2M_*$ then $t_{0*} = 1$; otherwise $2t_{0*} - 1 \leq |I'|/M_* + 1 \leq |I'|$, as $M_* \geq 3$);
- $4t_{0*} - 2 \leq n/M_* + 2 \leq |V|$;
- $t_{0*} \leq n$;
- $|F| < h^2|I| \leq 16(\ell_*L)^2 \cdot 2t|I'| \leq 64M_*t t_{0*}(\ell_*L)^2$.

Since $M_* \leq 2.1/\rho + 1 \leq 3.1/\rho$ and $(\ell_*L)^2 \leq 2^{20}L^8$, we get $64M_*t(\ell_*L)^2 \leq 2^{28}tL^8/\rho = \bar{s}_*$. Hence $|F| < \bar{s}_*t_{0*}$. So every set U of t_{0*} vertices satisfies $|F| \leq \bar{s}_*|U|$, and therefore $|B_{G-F}^{\ell_*}(U, V)| > |V|/2$ by hypothesis.

Choose $2t_{0*} - 1$ indices of I' ; their $4t_{0*} - 2$ endpoints are distinct. Apply Cited result 1.10 to $G - F$, with $\ell := \ell_*$ and $t_0 := t_{0*}$. It gives, for one of these indices j , an x_jy_j -path P in $G - F$ through V of length at most $4\ell_*L$, hence at most h , as lengths are integers. So $j \in I' \subseteq I$, $P \in \mathcal{Y}_j$ and $E(P) \cap F = \emptyset$. This proves the claim.

Now let $\mathcal{H}$ be the hypergraph with vertex set $[r] \sqcup E(G)$ (a disjoint union) whose edges are the sets $\{i\} \cup E(P)$ with $i \in [r]$ and $P \in \mathcal{Y}_i$. Every edge of $\mathcal{H}$ meets $[r]$ in exactly one vertex and $E(G)$ in at most h vertices. We apply Cited result 1.29 to $\mathcal{H}$, with $[r]$ in the role of its class $\mathcal{X}$, $E(G)$ in the role of its other class, and with $h \geq 1$ in place of its parameter q . We check its hypothesis. Let $I \subseteq [r]$ and $Z \subseteq E(G)$ with $|Z| \leq (2h - 1)(|I| - 1)$. Then $I \neq \emptyset$, since otherwise $|Z| \leq 1 - 2h < 0$. As $h^2 - (2h - 1) = (h - 1)^2 \geq 0$ and $|I| - 1 \geq 0$,

$$|Z| \leq (2h - 1)(|I| - 1) \leq h^2(|I| - 1) < h^2|I|.$$

So the claim, with $F := Z$, gives $j \in I$ and $P \in \mathcal{Y}_j$ with $E(P) \cap Z = \emptyset$. The edge $e := \{j\} \cup E(P)$ of $\mathcal{H}$ satisfies $e \cap [r] = \{j\} \subseteq I$ and, as $j \notin E(G) \supseteq Z$, $e \cap Z = E(P) \cap Z = \emptyset$. (The claim covers every Z with $|Z| < h^2|I|$, which is more than the hypothesis of Cited result 1.29 requires.) By Cited result 1.29, $\mathcal{H}$ has a set of pairwise disjoint edges such that every $i \in [r]$ lies in one of them. For $i \in [r]$ let e_i be an edge of this set containing i . By construction $e_i = \{k\} \cup E(Q)$ for some $k \in [r]$ and $Q \in \mathcal{Y}_k$, so $e_i \cap [r] = \{k\}$; as $i \in e_i \cap [r]$, we get $k = i$, and we put $P_i := Q$, so that $E(P_i) \subseteq e_i$. For $i \neq i'$ we have $e_i \cap [r] = \{i\} \neq \{i'\} = e_{i'} \cap [r]$, so $e_i \neq e_{i'}$, and therefore $e_i \cap e_{i'} = \emptyset$, as both lie in the set of pairwise disjoint edges. Hence the paths P_i , $i \in [r]$, are pairwise edge-disjoint x_iy_i -paths through V of length at most $h \leq 4\ell_*L \leq 2^{12}L^4$. Repeated pairs receive distinct paths, because the paths are indexed by i . $\square$

Theorem 3.8 (Theorem 16*: linking through random sets; for-all multiset form). *Let X be an N -vertex (ε', s) -expander with $\varepsilon' \in [2^{-7}, 1]$, and put $L := \log N$. Let $\rho \in (0, 1]$ and $t \geq 1$, and let V be a ρ -random subset of $V(X)$. Here X is fixed; if X was itself produced at random, V is independent of that randomness. Suppose that*

$$\rho N \geq L^2 \quad \text{and} \quad s \geq 2^{135} t L^{28} \rho^{-5}.$$

Then, with probability at least $1 - 2^{86} t L^{19} \rho^{-3} N^{-3}$, the graph X is $(2^{12}L^4, t)$ -path connected through V . That is, for every multiset $\mathcal{P}$ of pairs of distinct vertices of X (endpoints anywhere, in V or not) in which each vertex lies in at most t pairs, there are pairwise edge-disjoint paths of X , one for each pair of $\mathcal{P}$, each joining its pair, of length at most $2^{12}L^4$, and with all interior vertices in V . By Lemma 3.1(iii), $\mathcal{P}$ may be chosen after V is revealed. Moreover:

- (a) *The constant 2^{86} in the failure probability is absolute. No lower bound on N is needed.*

- (b) If $N \geq 2$, the hypotheses force $\rho N > 2^{27} L^{5.6} N^{4/5}$, because $s < N$. In particular the hypothesis $\rho N \geq L^2$ is implied by the others; it is kept only for comparison with [2, Theorem 16]. The last batch of the sprinkling argument needs ρN large compared with L times the size of the well-expanding sets involved. This is automatic (Step 0 of the proof of Lemma 3.5), so the final-stage union bound needs no further hypothesis.
- (c) The requirement on s is exactly linear in t and uniform in $\rho \in (0, 1]$. Apart from Step 0 of the proof below, which uses the full hypothesis on s to obtain (b), $N \geq 2^{30}$ and $\rho N \geq 84$ (from (b), Step 5 then draws $e^{-\rho N/8} \leq N^{-3}$), the proof uses only $s \geq 2K_*(\theta_* + 1)$, and $2K_*(\theta_* + 1) \leq 2^{130.6} t L^{28} \rho^{-5}$ by (S5). No coupling with a smaller density is used, also when ρ is close to 1.

Dependencies: 3.2, 3.4, 3.5, 3.6, 3.7, 3.1, 1.12, 1.8

Proof. The proof is an assembly, following the proof of [2, Theorem 16] (Cited result 1.17). The loss-free splitting Lemma 3.2 takes the place of [2, Lemma 15].

Step 0: N and ρN are large. If $N = 1$ there are no pairs of distinct vertices, and there is nothing to prove. Let $N \geq 2$, so $L \geq 1$. Taking $U = \{v\}$ in Cited result 1.12 shows that X has minimum degree greater than s . Hence $N > s \geq 2^{135} t L^{28} \rho^{-5} \geq 2^{135}$. Then $\rho^5 N > \rho^5 s \geq 2^{135} L^{28}$, so $\rho > 2^{27} L^{5.6} N^{-1/5}$, which is (b). In particular $\rho N \geq 84$ and $N \geq 2^{30}$. Take the parameters (3.2) with $n := N$ and the given ε' , ρ and t .

Step 1: splitting. Colour $E(X)$ with K_* colours, uniformly and independently of V , and let $X_1, \dots, X_{K_*}$ be the colour classes, each with vertex set $V(X)$. By (S5), $s \geq 2K_*(\theta_* + 1) \geq 40K_*L$. So by Lemma 3.2, with probability at least $1 - 2K_*N^{-5}$, every X_i is an $(\varepsilon', s/(2K_*))$ -expander, and $s/(2K_*) \geq \theta_* + 1$.

Step 2: balls in each class. Condition on a colouring in which all classes are such expanders; V is still ρ -random. Each X_i has N vertices and parameter ε' , so the parameters (3.2) for X_i are those for X . By Lemma 3.6(ii) with $s_1 := s/(2K_*)$ and a union bound over i , with probability at least $1 - K_*N^{-6}$ the following holds for every i : for every nonempty U and every $F_i \subseteq E(X_i)$ with $|F_i| \leq \mu_*|U|$, we have $|B_{X_i - F_i}^{\ell_*}(U, V)| > |V|/2$.

Step 3: the size of V . Since $|V| \sim \text{Bin}(N, \rho)$, the Chernoff bound (Cited result 1.8) gives $\mathbb{P}(|V| < \rho N/2) \leq e^{-\rho N/8}$.

Step 4: assembly. Assume the events of Steps 1–3. Let U be nonempty and let $F \subseteq E(X)$ with $|F| \leq \bar{s}_*|U|$. The classes partition $E(X)$, so some i has $|F \cap E(X_i)| \leq |F|/K_* \leq \bar{s}_*|U|/K_* \leq \mu_*|U|$ by (S5). Since $X_i - (F \cap E(X_i)) \subseteq X - F$, balls can only grow when passing to $X - F$, and

$$|B_{X-F}^{\ell_*}(U, V)| \geq |B_{X_i - (F \cap E(X_i))}^{\ell_*}(U, V)| > |V|/2.$$

So the hypotheses of Lemma 3.7 hold for $G := X$: $N \geq 2^{30}$, $|V| \geq \rho N/2$ and $\rho N \geq 84$. Hence X is $(2^{12}L^4, t)$ -path connected through V , in the multiset sense.

Step 5: probability. The failure probability is at most $2K_*N^{-5} + K_*N^{-6} + e^{-\rho N/8}$. By (b), $\rho N/8 > 2^{24}N^{4/5} \geq 3 \ln N$, so $e^{-\rho N/8} \leq N^{-3}$. By Lemma 3.1(iii), the random colouring of Step 1, which is auxiliary, does not affect the event, which depends only on (X, V) . So the event fails with probability at most

$$3K_*N^{-5} + N^{-3} \leq (3 \cdot 2^{83.6} + 1)tL^{19}\rho^{-3}N^{-3} \leq 2^{86}tL^{19}\rho^{-3}N^{-3},$$

using (S5) and $tL^{19}\rho^{-3} \geq 1$. This proves the main statement and (a).

For (c): apart from Step 0, whose consequences $N \geq 2^{30}$ and $\rho N \geq 84$ (used in Step 4) and $e^{-\rho N/8} \leq N^{-3}$ (used in Step 5, through (b)) rest on the full hypothesis $s \geq 2^{135} t L^{28} \rho^{-5}$, the requirements on s used above are $s \geq 40K_*L$ (Step 1) and $s/(2K_*) \geq \theta_* + 1$ (Step 2). Both follow from $s \geq 2K_*(\theta_* + 1)$, and $2K_*(\theta_* + 1) \leq 2^{130.6} t L^{28} \rho^{-5}$ by (S5). Only $\bar{s}_*$, and hence K_* , depends on t , and linearly up to rounding. The parameters (3.2) are defined for every $\rho \in (0, 1]$. The regime $p_* > 0.11$, reached only for ρ very close to 1, is treated directly in Lemma 3.5 through $\Delta_* \leq 13p_*^{-2}$ (fact (S3)). $\square$

3.3 Candidate sets of bounded multiplicity

Lemma 3.9 (Lemma HB: candidate sets of bounded multiplicity; Hall argument). *Let X be an N -vertex (ε', s) -expander with $N \geq 2$ and $2^{-7} \leq \varepsilon' \leq 1$, and put $L := \log N$. Let m be an integer with $1 \leq m$ and $2m \leq s$, and put $b := \lceil 2mL^2/\varepsilon' \rceil$. Thus $b = \lceil 64L^2m \rceil$, $\lceil 128L^2m \rceil$ or $\lceil 256L^2m \rceil$ at $\varepsilon' = 2^{-5}$, 2^{-6} or 2^{-7} . Then every vertex w has a set $\mathcal{A}(w) \subseteq N_X(w)$ with $|\mathcal{A}(w)| = m$, such that every vertex of X lies in at most b of the sets $\mathcal{A}(w)$, $w \in V(X)$.*

Dependencies: 1.28, 1.12

Proof. For $w \in V(X)$, its degree is $d_X(w) = |N_X(w)|$. As $N \geq 2$, we have $L \geq 1$ and $d_X(w) \geq \delta(X) > s \geq 2m$ for every w (Cited result 1.12); in particular $d_X(w) - m > m \geq 1$ (not needed below). Let $\vec{E}(X) := \{(w, u) \in V(X) \times V(X) : wu \in E(X)\}$ be the set of ordered pairs along edges (every edge gives two), and for $w \in V(X)$ let $\vec{E}_X(w) := \{(w, u) : u \in N_X(w)\}$. Then $|\vec{E}_X(w)| = d_X(w)$, and distinct elements of $\vec{E}_X(w)$ have distinct second coordinates.

The auxiliary bipartite graph. Its left side is $\vec{E}(X)$. Its right side is the set of slots

$$\mathcal{B} := \{(u, j, 1) : u \in V(X), j \in [b]\} \cup \{(w, i, 2) : w \in V(X), i \in [d_X(w) - m]\};$$

$(u, j, 1)$ is a *load slot* of u , and $(w, i, 2)$ is a *spare slot* of w . A pair $a = (w, u) \in \vec{E}(X)$ is joined to the b load slots of its second coordinate u and to the $d_X(w) - m$ spare slots of its first coordinate w :

$$\mathcal{T}(a) := \{(u, j, 1) : j \in [b]\} \cup \{(w, i, 2) : i \in [d_X(w) - m]\}.$$

Reduction to Hall's condition. Suppose that $\chi : \vec{E}(X) \rightarrow \mathcal{B}$ is injective and $\chi(a) \in \mathcal{T}(a)$ for every $a \in \vec{E}(X)$. Fix $w \in V(X)$. If $a \in \vec{E}_X(w)$ and $\chi(a)$ is a spare slot, then $\chi(a)$ is one of the $d_X(w) - m$ spare slots of w ; as χ is injective, at most $d_X(w) - m$ elements of $\vec{E}_X(w)$ are of this kind. So at least $d_X(w) - (d_X(w) - m) = m$ elements $(w, u) \in \vec{E}_X(w)$ have $\chi(w, u)$ a load slot, and then it is a load slot of u . Let $\mathcal{A}(w)$ be the set of second coordinates of m such elements (the first m in a fixed ordering of $V(X)$). Then $\mathcal{A}(w) \subseteq N_X(w)$, and $|\mathcal{A}(w)| = m$ because distinct elements of $\vec{E}_X(w)$ have distinct second coordinates. Now fix $u \in V(X)$. If $u \in \mathcal{A}(w)$, then $\chi(w, u)$ is one of the b load slots of u . The pairs (w, u) with distinct w are distinct and χ is injective, so $u \in \mathcal{A}(w)$ holds for at most b vertices w . This is the statement. By Hall's theorem (Cited result 1.28, applied to the finite index set $\vec{E}(X)$ and the finite sets $\mathcal{T}(a)$), such a χ exists provided

$$|\mathcal{E}| \leq \left| \bigcup_{a \in \mathcal{E}} \mathcal{T}(a) \right| \quad \text{for every } \mathcal{E} \subseteq \vec{E}(X). \quad (3.6)$$

Proof of (3.6). Fix $\mathcal{E} \subseteq \vec{E}(X)$. Let $\mathcal{S}$ be the set of first coordinates and $\mathcal{R}$ the set of second coordinates of the elements of $\mathcal{E}$. For $w \in \mathcal{S}$ put $\mathcal{E}_w := \mathcal{E} \cap \vec{E}_X(w)$, so that $1 \leq |\mathcal{E}_w| \leq d_X(w)$ and $|\mathcal{E}| = \sum_{w \in \mathcal{S}} |\mathcal{E}_w|$. The union in (3.6) consists of the $b|\mathcal{R}|$ load slots of the vertices of $\mathcal{R}$ and

the $\sum_{w \in \mathcal{S}} (d_X(w) - m)$ spare slots of the vertices of $\mathcal{S}$, and all these slots are distinct. So (3.6) is equivalent to

$$\sum_{w \in \mathcal{S}} (|\mathcal{E}_w| - d_X(w) + m) \leq b|\mathcal{R}|. \quad (3.7)$$

Let $\mathcal{S}' := \{w \in \mathcal{S} : d_X(w) - |\mathcal{E}_w| \leq m - 1\}$. A summand of (3.7) with $w \in \mathcal{S} \setminus \mathcal{S}'$ is at most 0, and one with $w \in \mathcal{S}'$ is at most m , as $|\mathcal{E}_w| \leq d_X(w)$. So the left side of (3.7) is at most $m|\mathcal{S}'|$, and (3.7) holds if $\mathcal{S}' = \emptyset$ (this includes $\mathcal{E} = \emptyset$). Assume $\mathcal{S}' \neq \emptyset$.

Let $w \in \mathcal{S}'$ and $u \in N_X(w) \setminus \mathcal{R}$. Then $(w, u) \in \vec{E}_X(w) \setminus \mathcal{E}_w$, since otherwise $u \in \mathcal{R}$; distinct such u give distinct pairs (w, u) . As $|\vec{E}_X(w) \setminus \mathcal{E}_w| = d_X(w) - |\mathcal{E}_w| \leq m - 1$, we get

$$|N_X(w) \setminus \mathcal{R}| \leq m - 1 \quad \text{for every } w \in \mathcal{S}'. \quad (3.8)$$

If $|\mathcal{S}'| \leq 2N/3$, put $\mathcal{U} := \mathcal{S}'$. Otherwise let $\mathcal{U} \subseteq \mathcal{S}'$ be any set with $|\mathcal{U}| = \lceil |\mathcal{S}'|/2 \rceil$; then $1 \leq |\mathcal{U}| \leq \lceil N/2 \rceil \leq 2N/3$, because $N \geq 2$ (for $N = 2$, $\lceil N/2 \rceil = 1 \leq 4/3$; for $N \geq 3$, $\lceil N/2 \rceil \leq (N+1)/2 \leq 2N/3$). In both cases

$$1 \leq |\mathcal{U}| \leq 2N/3 \quad \text{and} \quad |\mathcal{U}| \geq |\mathcal{S}'|/2.$$

Let F be the set of edges $wu \in E(X)$ with $w \in \mathcal{U}$ and $u \notin \mathcal{R}$. Every edge of F is of the form wu with $w \in \mathcal{U}$ and $u \in N_X(w) \setminus \mathcal{R}$, so, by (3.8), $|F| \leq \sum_{w \in \mathcal{U}} |N_X(w) \setminus \mathcal{R}| \leq (m-1)|\mathcal{U}| \leq s|\mathcal{U}|$, since $m-1 < 2m \leq s$. Let $v \in N_{X-F}(\mathcal{U})$: then $v \notin \mathcal{U}$ and $vw \in E(X) \setminus F$ for some $w \in \mathcal{U}$. As $w \in \mathcal{U}$ and $vw \notin F$, we have $v \in \mathcal{R}$. Hence $N_{X-F}(\mathcal{U}) \subseteq \mathcal{R}$, and the expansion of X (Cited result 1.12 with $n = N$, applied to $\mathcal{U}$ and F) gives

$$|\mathcal{R}| \geq |N_{X-F}(\mathcal{U})| \geq \frac{\varepsilon' |\mathcal{U}|}{L^2} \geq \frac{\varepsilon' |\mathcal{S}'|}{2L^2}.$$

Since $b \geq 2mL^2/\varepsilon'$, it follows that

$$b|\mathcal{R}| \geq \frac{2mL^2}{\varepsilon'} \cdot \frac{\varepsilon' |\mathcal{S}'|}{2L^2} = m|\mathcal{S}'|,$$

which is at least the left side of (3.7). This proves (3.6), and hence the lemma. $\square$

3.4 The lending colouring

Fix a valid $\text{HB}^{*\tau+}$ run on G (Definition 2.7). Every ancestor Y (Definition 2.8) splits the edges of its expander H_Y into two halves:

- an *own* half, which only Y 's own vortex device uses;
- a *lend* half, which is coloured into exclusive classes, each lent to exactly one later consumer.

The random data of this subsection form part of stage 1 of the randomness. The other stage-1 data, introduced in later sections, are drawn independently of them.

Definition 3.10 (stage-1 lending data: the two-stage colouring COL-JV and the JS labels). For every ancestor Y of round r , with $V(Y)$, H_Y , ε_Y , s_Y and $L_Y = \log |V(Y)|$ as in Definition 2.8, the random data (i)–(iv) below are drawn. The data of distinct ancestors are independent.

- (i) *Own/Lend split*. Every edge of H_Y independently lies in Own_Y or in Lend_Y , with probability $1/2$ each.

(ii) *Lent classes.* Consider the three index families

$$\begin{aligned}\mathcal{I}_U(Y) &:= \{(l, c, \varsigma) : r+2 \leq l \leq R, c \in [4], \varsigma \in [T_Y^{\text{sl}}]\} \text{ if } Y \text{ is light, where } T_Y^{\text{sl}} := \lceil L_Y^2 \rceil, \\ \mathcal{I}_U(Y) &:= \emptyset \text{ if } Y \text{ is standalone,} \\ \mathcal{I}_{\text{JS}}(Y) &:= \{(l, j) : r+2 \leq l \leq R, 0 \leq j < K_l^{\text{JS}}\}, \text{ where } K_l^{\text{JS}} := M_l^2, \\ \mathcal{I}_{\text{JV}}(Y) &:= \{l : r+2 \leq l \leq R\}.\end{aligned}$$

These families are regarded as pairwise disjoint (their indices carry the tags U, JS, JV). Put $k_{\text{lend}}(Y) := |\mathcal{I}_U(Y)| + |\mathcal{I}_{\text{JS}}(Y)| + |\mathcal{I}_{\text{JV}}(Y)|$.

- If $r \leq R-2$, every edge of Lend_Y independently receives an index chosen uniformly from $\mathcal{I}_U(Y) \sqcup \mathcal{I}_{\text{JS}}(Y) \sqcup \mathcal{I}_{\text{JV}}(Y)$. The edges with index $(l, c, \varsigma) \in \mathcal{I}_U(Y)$ form the *U-lent class* $\mathcal{L}_{Y,l,c,\varsigma}^U$. The edges with index $(l, j) \in \mathcal{I}_{\text{JS}}(Y)$ form the *JS-lent class* $\mathcal{L}_{Y,l,j}^{\text{JS}}$. The edges with index $l \in \mathcal{I}_{\text{JV}}(Y)$ form the *JV-lent class* $\mathcal{L}_{Y,l}^{\text{JV}}$. Every class is regarded as a graph with vertex set $V(Y)$.
 - If $r \geq R-1$, all three families are empty, $k_{\text{lend}}(Y) = 0$, and Lend_Y has no classes.
- (iii) *Own classes (light Y only).* Put $J_Y := \lfloor \log(L_Y/8) \rfloor$ and $k_{\text{own}} := 4J_Y + 1$. Every edge of Own_Y independently receives one of k_{own} labels, chosen uniformly. This yields the own classes $R_{j,c}$ ($0 \leq j < J_Y, c \in [4]$) and M of Y , each regarded as a graph on $V(Y)$. They are the own classes of the four-phase P-vortex run on Y in later sections. For standalone Y , Own_Y is not subdivided at stage 1.
- (iv) *JS labels.* For every $r+2 \leq l \leq R$ and every $y \in V(Y)$, independently, draw a label $\text{lab}_{Y,l}(y) \in \{*\} \cup \{0, 1, \dots, K_l^{\text{JS}} - 1\}$ with $\mathbb{P}(\text{lab}_{Y,l}(y) = j) = \rho_l := M_l^{-4}$ for each j , so that $\mathbb{P}(\text{lab}_{Y,l}(y) = *) = 1 - M_l^{-2}$. Put $T_j(Y, l) := \{y \in V(Y) : \text{lab}_{Y,l}(y) = j\}$.

Formally, every edge e of H_Y carries three independent uniform random variables (a fair bit, a lent index and an own label), and the labels of (iv) are independent of all edge variables. Consequently:

- colours of distinct edges are independent;
- the colourings of distinct ancestors are independent;
- the labels are independent of the colours;
- given Lend_Y , stage (ii) is a uniform $k_{\text{lend}}(Y)$ -colouring of Lend_Y , and given Own_Y , stage (iii) is a uniform k_{own} -colouring of Own_Y ;
- for fixed (l, j) , the set $T_j(Y, l)$ is a ρ_l -random subset of $V(Y)$, independent of the colouring, and the sets $T_j(Y, l)$, $0 \leq j < K_l^{\text{JS}}$, are pairwise disjoint.

Derived quantities. For $k_{\text{lend}}(Y) \geq 1$ put $p_Y := 1/(2k_{\text{lend}}(Y))$. For each fixed l with $r+2 \leq l \leq R$, this is the probability that a fixed edge of H_Y lies in $\mathcal{L}_{Y,l}^{\text{JV}}$. Also put $t_l^{\text{JS}} := 2M_l + 2$. For a light ancestor Y of round r put $t_Y := \lceil \lambda_r^{1.6} \rceil$, the *U-multiplicity parameter* of Y : it is the multiplicity bound t at which the U-lent classes of Y are path connected (Lemma 3.13(c)). It depends only on the round of Y , and $t_Y \geq M_l$ for every $r+2 \leq l \leq R$, because $M_l \leq \lambda_{l-2}^{1.6} \leq \lambda_r^{1.6}$ (Lemma 2.17(a),(b)).

Consumers (property COL(g)). If $r \leq R-2$, each part of the data has exactly one consumer:

- Own_Y , together with its own classes if Y is light, is used only by the vortex device of Y itself. This is the transparent P-vortex if Y is standalone. If Y is light, it is the four-phase P-vortex, or VX^+ if Y is demoted.
- $\mathcal{L}_{Y,l,c,\varsigma}^{\text{U}}$ is used only by the Theorem-U chaining of the U-bundle $\text{Bdl}_{l,c}$, in slot ς .
- $\mathcal{L}_{Y,l,j}^{\text{JS}}$ is used only by the JS-LC systems of the pair (Y, l) , at junction j .
- $\mathcal{L}_{Y,l}^{\text{JV}}$ is used only by the round- l J-consumer.

Lent edges that their consumer does not use are junk for Y 's own device: they are added to the edge set that this device decomposes. If $r \geq R - 1$, then Own_Y has the same consumer, and Lend_Y has no classes and no other consumer: all of Lend_Y is junk for Y 's own device. Indeed $\text{Lent}(Y) = \emptyset$, since it collects edges used at rounds $l \geq r + 2 > R$, so $E^{\text{rem}}(Y) = E_r(Y)$ for standalone Y and $H_0(Y) = E_r(Y)$ for light non-demoted Y (Construction 6.17); a demoted Y runs on $E_r(Y)$ itself.

Dependencies: 2.8, 2.11, 2.17

The graphs H_Y of all ancestors Y of all rounds are pairwise edge-disjoint (Proposition 2.11(iii)). So the colourings of distinct ancestors act on disjoint edge sets. The label distribution in (iv) is well defined because $K_l^{\text{JS}} \rho_l = M_l^{-2} \leq 1$.

Lemma 3.11 (the COL-JV table). *Assume condition (Γ1). Let Y be an ancestor of round $r \leq R$, and write $\lambda := \lambda_r$ and $\mu := \log \lambda$; if $r \leq R - 2$, write also $M := M_{r+2}$. Then:*

- (i) *If $r \geq R - 1$, then $k_{\text{lend}}(Y) = 0$. If $r \leq R - 2$, then $|\mathcal{I}_{\text{U}}(Y)| \leq 12L_Y^3$, $|\mathcal{I}_{\text{JS}}(Y)| \leq \frac{4}{3}M^2$ and $|\mathcal{I}_{\text{JV}}(Y)| = R - r - 1 \leq 2\log^* d_r + 1$. Hence*

$$k_{\text{lend}}(Y) \leq 12L_Y^3 + \frac{4}{3}M^2 + 2\log^* d_r + 2 \leq 24L_Y^3 + \frac{4}{3}M^2 \leq \bar{k} \leq \lambda^{3.3},$$

and consequently $p_Y \geq \lambda^{-4}$. The JV family has $R - r - 1 \leq 2\log^ d_r + 1$ members. Its size is not bounded by an absolute constant, since d_1 may be of order n ; the count “+16” must not be used.*

- (ii) *Every requirement in column 2 and every inequality in column 3 of Table 5 holds. (Column 4 is not part of this assertion: its triples serve only Lemma 3.12, and the inequalities $a_i\mu - b_i - c_i \log(A\mu) \geq 0$ that they define need not hold at μ .) More precisely, for each row every inequality in column 3 holds at every $\mu \geq \log \log D_*$ by (Γ1)(f); in particular it holds at $\mu = \log \lambda_r$ and, for rows 2, 11 and 12, also at $\mu = \log \lambda_{l-2}$ for every round $3 \leq l \leq R$, both of which are at least $\log \log D_*$. At the value of λ fixed in the caption of Table 5, these inequalities imply the requirement in column 2 (for row 4, together with row 9). So every requirement holds, in every round. The scope of the rows is as follows.*

- Rows 1, 3 and 8 (rule GC, the three levels of Lemma 15⁺, and the own-device thresholds) hold for every round $r \leq R$, that is, also for $r \in \{R - 1, R\}$: row 1 for every round- r pre-part, and rows 3 and 8 for every ancestor Y of round r . They involve neither M_{r+2} nor any lent class; for $r \geq R - 1$ the lent level of row 3 is void, because Y has no lent classes.
- Rows 4–7 and 10 concern lent classes or p_Y . They are asserted for $r \leq R - 2$; for $r \geq R - 1$ there are no lent classes and p_Y is not defined. Row 9 holds for every $r \leq R$, trivially when $k_{\text{lend}}(Y) = 0$.

No.	Requirement (use)	Sufficient inequality in $\lambda = 2^\mu$	(a_i, b_i, c_i)
1	$\theta_r^{\text{GC}}(Z^0) > \tau_r$ (rule GC)	$\lambda^{102.5} > 257(\lambda + 6\mu + 20)^{102}$; crude form $\lambda^{1/2} > 2^{111}$	(0.5, 112, 0)
2	$P_{l-2} \geq M_l^{13}$, with $\lambda := \lambda_{l-2}$	$\lambda^{103} \geq M^{13}$	(103, 0, 26A)
3	Lemma 15 ⁺ , lent level: $s_Y/4 \geq 40 k_{\text{lend}} L_Y$	$\lambda^{99} \geq 640 k$	(96, 19, 4A)
4	T16* for U-lent classes: $s_Y/(8k_{\text{lend}}) \geq 2^{135} t_Y L_Y^{28} (12L_Y^5)^5$, $t_Y = \lceil \lambda^{1.6} \rceil$	$\lambda^{42.1} \geq 2^{193} \cdot 12^5$ (with row 9) [†]	(42.1, 211, 0)
5	T16* for JS-lent classes: $s_Y/(8k_{\text{lend}}) \geq 2^{135} (3M) L_Y^{28} M^{20}$	$\lambda^{72} \geq 3 \cdot 2^{167} \cdot \bar{k} \bar{M}^{21}$	(69, 178, 46A)
6	T16* failures over $\mathcal{I}_{\text{JS}}(Y)$ sum to at most $ V(Y) ^{-2}/4$	$\lambda^{84} \geq 2^{110} \bar{M}^{15}$	(84, 110, 30A)
7	T16* failures over $\mathcal{I}_{\text{U}}(Y)$ sum to at most $ V(Y) ^{-2}/4$	$\lambda^{64.4} \geq 2^{127} \cdot 12^4$	(64.4, 142, 0)
8	own devices: (a) $s_r/4 \geq 2^{150} L_Y^{42}$; (b) $s_r/4 \geq 2^{146} L_Y^{38} \log L_Y$ (the threshold of VX^+ for standalone parts; not used, as VX^+ is applied only to demoted light parts, Lemma 5.8; cf. Remark 4.5); (c) $s_r/2 \geq 2^{151} L_Y^{38} \log L_Y$; (d) $s' := s_r/(16k_{\text{own}}) \geq 2^{145} L_Y^{41}$ and $s' \geq 2 \lceil L_Y^6 \rceil$	(a) $\lambda^{58} \geq 2^{194}$; (b) $\lambda^{62} \geq 2^{186}(\mu + 1)$; (c) $\lambda^{62} \geq 2^{190}(\mu + 1)$; (d) $\lambda^{58} \geq 2^{191}$ and $\lambda^{99} \geq 64((2\lambda)^6 + 1)$	(58, 194, 1)
9	$k_{\text{lend}}(Y) \leq \lambda^{3.3}$	$\bar{k} \leq \lambda^{3.3}$	(0.3, 9, 4A)
10	$p_Y \geq \lambda^{-4}$	$2\bar{k} \leq \lambda^4$	(1, 10, 4A)
11	$\lambda_{l-2}^{95}/(8M_l^2) \geq 2^{10} M_l^{10}$, with $\lambda := \lambda_{l-2}$	$\lambda^{95} \geq 2^{13} \bar{M}^{12}$	(95, 13, 24A)
12	$M_l \leq \lambda_{l-2}^{1.6}$, with $\lambda := \lambda_{l-2}$	$\bar{M} \leq \lambda^{1.6}$	(1.6, 0, 2A)
13	(G*)	$\lambda^{36} \geq 2^{240} (A\mu)^{46A}$	(36, 240, 46A)

Table 5: The COL-JV table (Lemma COL-JV below). Here $\lambda := \lambda_r$ and $\mu := \log \lambda$, except in rows 2, 11 and 12, where $\lambda := \lambda_{l-2}$ for a round $3 \leq l \leq R$. We write $\bar{M} := (A\mu)^{2A}$, with $A = 105$, and $\bar{k} := 192\lambda^3 + \frac{4}{3}\bar{M}^2$. Column 3 is a sufficient form of the requirement, obtained from the standing bounds (B1)–(B7) of the proof. Column 4 gives, for row i , reals $a_i > 0$ and $b_i, c_i \geq 0$ such that, for $\mu \geq 6$, the inequality $a_i\mu - b_i - c_i \log(A\mu) \geq 0$ implies every inequality in column 3 of row i (Lemma 3.12, which has no hypothesis); hence every column-3 inequality holds for all sufficiently large μ , and (Γ1)(f) requires it for every $\mu \geq \log \log D_*$. The inequalities $a_i\mu - b_i - c_i \log(A\mu) \geq 0$ defined by column 4 are cruder than column 3; they are not part of (Γ1), are not asserted in Lemma COL-JV, and may fail at some $\mu \geq \log \log D_*$ although (Γ1) holds. No galactic threshold is evaluated: the thresholds of Lemma 3.12 are explicit, but their numerical values are never computed. Rows for off-route devices are omitted. [†]Row 4: column 3 implies the column-2 requirement together with row 9.

- Rows 2, 11 and 12 concern λ_{l-2} for rounds $3 \leq l \leq R$, and row 13 is (G^*) ; they do not depend on Y .

Dependencies: (F1), (G^*) , 2.17, 3.10, 2.8, 2.11, 3.2, 3.8

Proof. Every round $r \leq R$ has $d_r \geq D_*$, so $\mu = \log \lambda_r \geq \log \log D_*$, and the items of (F1) hold at μ . The same holds for λ_{l-2} with $3 \leq l \leq R$. We use the following standing bounds.

- (B1) $L_Y \leq \log M_r \leq 2\lambda$ (Lemma 2.17(b)).
- (B2) If $r \leq R - 2$: $M_l \leq M \leq \bar{M}$ for $r + 2 \leq l \leq R$, and $M_{l+1} \leq M_l/2$ for $r + 2 \leq l < R$ (Lemma 2.17(b)). Likewise $M_l \leq (A \log \lambda_{l-2})^{2A}$ for every $3 \leq l \leq R$.
- (B3) $s_r \geq \lambda^{100}$ and $s_r \leq 2\lambda_r^{100}$ (Lemma 2.17(b)), and $s_Y \geq s_r/2$ (Definition 2.8).
- (B4) $|V(Y)| \geq P_r/2 \geq \lambda^{103}/2$. A standalone Y has $|V(Y)| = |Y^0| \geq P_r$, and a light Y has $|Y| \geq |Y^0|/2 \geq P_r/2$ (Proposition 2.11(iv)); and $P_r = \lceil \lambda^{C'} \rceil$ with $C' = 103$.
- (B5) $R - r \leq 2 \log^* d_r + 2$ and $2^{2 \log^* d_r + 2} \leq \lambda$ (Lemma 2.17(a),(d)), so $2 \log^* d_r + 2 \leq \mu$.
- (B6) $\Lambda_r \leq \lambda + 6\mu + 20$. Indeed, with $T := t_r^{\text{HB}} d_r = \lambda^2 2^\lambda$ we have $\log T = \lambda + 2\mu$. Put $B := 2^{16} T \log^4 T$. As $\lambda = 2^\mu \geq 2^{256}$ ($\mu \geq 2^8$ by (F1)(a)) and $\log T = \lambda + 2\mu \geq 1$, we have $B \geq 2^{16} T \geq 2^{16+\lambda} > 2^{161} \geq 2^{40}$, so $M_r \leq B + 1$ by (R2) (Definition 2.7), and $\Lambda_r = \log M_r \leq \log B + \log(1 + 1/B) \leq \log B + 1/(B \ln 2) \leq \log B + 2^{-160}$, since $1/(B \ln 2) < 2^{-161}/0.69 < 2^{-160}$. Next $\log B = 16 + \lambda + 2\mu + 4 \log(\lambda + 2\mu)$, and $\log(\lambda + 2\mu) = \mu + \log(1 + 2\mu/\lambda) \leq \mu + 2\mu/(\lambda \ln 2) \leq \mu + 0.0025$, because $\mu \geq 2^8$ and $2\mu/\lambda = 2\mu 2^{-\mu} \leq 2^{-240}$. Hence $\Lambda_r \leq \lambda + 6\mu + 16.01 + 2^{-160} \leq \lambda + 6\mu + 20$.
- (B7) If $r \leq R - 2$ and $r + 2 \leq l \leq R$: $t_l^{\text{JS}} = 2M_l + 2 \leq 3M_l \leq 3M$ (as $M_l \geq 2^{40}$), and $\rho_l^{-1} = M_l^4 \leq M^4$.

Only (B2) and (B7) involve M , and they are used only for $r \leq R - 2$. (B1) and (B3)–(B6) hold for every round $r \leq R$.

(i) *Counting.* If $r \geq R - 1$, the three index families are empty (Definition 3.10(ii)), so $k_{\text{lend}}(Y) = 0$. Let now $r \leq R - 2$.

- $|\mathcal{I}_U(Y)| = 4T_Y^{\text{sl}}(R - r - 1) \leq 4(L_Y^2 + 1)(2 \log^* d_r + 1)$. By (B5) and (B4), $2 \log^* d_r + 1 < \mu \leq L_Y$, since $L_Y \geq 103\mu - 1$. Hence $|\mathcal{I}_U(Y)| \leq 4(L_Y^2 + 1)L_Y \leq 12L_Y^3$.
- $|\mathcal{I}_{\text{JS}}(Y)| = \sum_{l=r+2}^R M_l^2 \leq M^2 \sum_{i \geq 0} 4^{-i} = \frac{4}{3} M^2$ by (B2).
- $|\mathcal{I}_{\text{JV}}(Y)| = R - r - 1 \leq 2 \log^* d_r + 1$ by (B5).

The second inequality of the display holds because $2 \log^* d_r + 2 \leq \mu \leq 12L_Y^3$. The third follows from (B1) and (B2). The fourth is the sufficient inequality of row 9, verified in (ii) below. Then $p_Y = 1/(2k_{\text{lend}}(Y)) \geq 1/(2\lambda^{3.3}) \geq \lambda^{-4}$, and row 10 gives the same conclusion directly.

(ii) *The rows.* In each row, the passage from column 2 to column 3 uses only (B1)–(B7) and the fact that the requirement is monotone in L_Y , M_l , k_{lend} , s_r and $|V(Y)|$ in the appropriate direction. Rows 1, 3 and 8 use only (B1), (B3)–(B6), Lemma 2.17(c) and the bound $k_{\text{lend}}(Y) \leq \bar{k}$ of (i), which is trivial when $k_{\text{lend}}(Y) = 0$; so they hold for every round $r \leq R$. Rows 4–7, 9 and 10 are derived for $r \leq R - 2$, where (B2) and (B7) are available; for $r \geq R - 1$, row 9 reads $0 \leq \lambda^{3.3}$. We now give the derivations.

1. For a round- r pre-part Z^0 we have $|Z^0| \geq P_r \geq \lambda^{103}$, so $\theta_r^{\text{GC}}(Z^0) = \lceil |Z^0| \lambda^{-1/2} \rceil \geq \lambda^{102.5}$, and $\tau_r \leq 128s_r \Lambda_r^2 + 1 \leq 257\Lambda_r^{102}$ by (B3). Now use (B6). The crude form uses $\tau_r \leq 2^{111} \lambda^{102}$ (Lemma 2.17(c)).
2. $P_{l-2} \geq \lambda_{l-2}^{103}$, and $M_l \leq \bar{M}$ by (B2).
3. $s_Y/4 \geq \lambda^{100}/8$ and $40k_{\text{lend}}L_Y \leq 80\lambda\bar{k}$. The other two levels of Lemma 3.2 are $s_Y \geq 80L_Y$ (the Own/Lend split, $k = 2$) and $s_r/8 \geq 40k_{\text{own}}L_Y$ (own classes). They follow from $\lambda^{100}/2 \geq 160\lambda$ and $\lambda^{100}/8 \geq 160\lambda^2$, using $k_{\text{own}} \leq L_Y \leq 2\lambda$, and both hold for every $\mu \geq 1$.
4. U-lent classes exist only for light Y . Theorem 3.8, with $t = t_Y = \lceil \lambda^{1.6} \rceil \leq 2\lambda^{1.6}$ and $\rho \geq 1/(12L_Y^5)$, requires $s \geq 2^{135} t L_Y^{28} \rho^{-5}$, and $2^{135} t L_Y^{28} \rho^{-5} \leq 2^{136} 12^5 \lambda^{1.6} L_Y^{53}$. With $s = s_Y/(8k_{\text{lend}}) \geq \lambda^{100}/(16\bar{k})$ and $L_Y \leq 2\lambda$, it suffices that $\lambda^{100} \geq 2^{4+136+53} 12^5 \bar{k} \lambda^{54.6}$, i.e. $\lambda^{45.4} \geq 2^{193} \cdot 12^5 \cdot \bar{k}$. By row 9, $\bar{k} \leq \lambda^{3.3}$, so the sufficient inequality of column 3, $\lambda^{42.1} \geq 2^{193} \cdot 12^5$, implies it. (Row 9 holds at μ under (Γ1); so column 3 implies column 2 under (Γ1).)
5. Theorem 3.8 with $t = t_l^{\text{JS}} \leq 3M$ and $\rho = \rho_l \geq M^{-4}$ requires $s \geq 2^{135} \cdot 3M \cdot L_Y^{28} M^{20}$. With $s \geq \lambda^{100}/(16\bar{k})$, $L_Y \leq 2\lambda$ and $M \leq \bar{M}$, this becomes $\lambda^{100} \geq 3 \cdot 2^{4+135+28} \bar{k} \bar{M}^{21} \lambda^{28}$.
6. The failure bound of Theorem 3.8 is $2^{86} t L_Y^{19} \rho^{-3} N^{-3}$ per class, with $N := |V(Y)|$. Summed over the at most $\frac{4}{3}M^2$ JS classes, with $t \leq 3M$, $\rho^{-3} \leq M^{12}$ and $L_Y \leq 2\lambda$, it is at most $2^{107} \lambda^{19} M^{15} N^{-3}$. This is at most $N^{-2}/4$ if $N \geq 2^{109} \lambda^{19} M^{15}$, which by (B4) follows from $\lambda^{84} \geq 2^{110} \bar{M}^{15}$. This line needs (Γ1): the class count and the multiplicity grow with M , which is bounded only by $\bar{M} = (A\mu)^{2A}$, not by a fixed power of λ for moderate μ . It does not follow from $C' > 34$ alone.
7. There are at most $12L_Y^3$ U-classes, each with $t = t_Y \leq 2\lambda^{1.6}$ and $\rho^{-3} \leq 1728L_Y^{15}$. The sum is at most $12L_Y^3 \cdot 2^{86} \cdot 2\lambda^{1.6} L_Y^{19} \cdot 12^3 L_Y^{15} N^{-3} = 12^4 \cdot 2^{87} \lambda^{1.6} L_Y^{37} N^{-3} \leq 12^4 \cdot 2^{124} \lambda^{38.6} N^{-3}$. This is at most $N^{-2}/4$ if $N \geq 12^4 \cdot 2^{126} \lambda^{38.6}$, which by (B4) follows from $\lambda^{103}/2 \geq 12^4 \cdot 2^{126} \lambda^{38.6}$, i.e. $\lambda^{64.4} \geq 2^{127} \cdot 12^4$.
8. By (B3), $s_r/4 \geq \lambda^{100}/4$ and $s_r/2 \geq \lambda^{100}/2$. Also $k_{\text{own}} \leq L_Y \leq 2\lambda$, so $s' \geq \lambda^{99}/32$, and $\log L_Y \leq \mu + 1$. The right sides are increasing in L_Y . Substituting $L_Y \leq 2\lambda$ gives the four inequalities.
- 9.–10. By the first three inequalities of (i), $k_{\text{lend}}(Y) \leq \bar{k}$. So $\bar{k} \leq \lambda^{3.3}$ suffices for row 9, and, since $p_Y = 1/(2k_{\text{lend}}) \geq 1/(2\bar{k})$, $2\bar{k} \leq \lambda^4$ suffices for row 10.
11. $\lambda^{95}/(8M_l^2) \geq 2^{10} M_l^{10}$ is equivalent to $\lambda^{95} \geq 2^{13} M_l^{12}$; use (B2).
- 12.–13. Row 12 is (B2) together with the inequality shown. Row 13 is the condition (G*) of Condition 1.33 itself.

□

The next lemma shows that every column-3 inequality of Table 5 holds for all sufficiently large μ . It has no hypothesis. It is used only in the proof of Lemma 7.21, to show that item (f) of (Γ1) can be satisfied.

Lemma 3.12 (eventual form of the COL-JV table). *Call an inequality in a real variable μ of type (E) if it reads $a\mu - b - c \log(A\mu) \geq 0$ with reals $a > 0$ and $b, c \geq 0$ (here $A = 105$ and $\log = \log_2$).*

- (i) An inequality of type (E) holds for every $\mu \geq \max\{1, v_0^2\}$, where $v_0 := (c + (c^2 + a(b + c \log A))^{1/2})/a$. In particular, it holds for all sufficiently large μ .
- (ii) Let $\mu \geq 6$ be real, and put $\lambda := 2^\mu$, $\bar{M} := (A\mu)^{2A}$ and $\bar{k} := 192\lambda^3 + \frac{4}{3}\bar{M}^2$. Let $1 \leq i \leq 13$, and let (a_i, b_i, c_i) be the triple in column 4 of row i of Table 5. If $a_i\mu - b_i - c_i \log(A\mu) \geq 0$, then every inequality in column 3 of row i holds at λ .
- (iii) Consequently, for each row of Table 5, every inequality in column 3 holds at $\lambda = 2^\mu$ for all sufficiently large real μ .

The lemma concerns explicit functions of the real variable μ only. It involves no graph, no run of the hierarchy and no condition on D_* .

Dependencies: none (the inequalities are those printed in Table 5, which is not a statement).

Proof. (i) Let $\mu \geq \max\{1, v_0^2\}$ and $v := \mu^{1/2}$, so that $v \geq 1$ and $v \geq v_0$. For $v \geq 1$ we have $2^v \geq 1 + v > v$: equality $2^v = 1 + v$ holds at $v = 1$, and for $v \geq 1$ the derivative $2^v \ln 2 \geq 2 \ln 2 > 1$ of the left side exceeds the derivative 1 of the right side. Hence $\log v < v$, and $\log \mu = 2 \log v \leq 2v$. As $c \geq 0$,

$$a\mu - b - c \log(A\mu) = av^2 - (b + c \log A) - c \log \mu \geq av^2 - 2cv - (b + c \log A).$$

The right side is a quadratic polynomial in v with leading coefficient $a > 0$. Its discriminant $4c^2 + 4a(b + c \log A)$ is non-negative, as $b, c \geq 0$ and $\log A > 0$, and its larger root is v_0 . So it is non-negative for every $v \geq v_0$, and the inequality of type (E) holds at μ .

(ii) We use three elementary bounds, valid for $\mu \geq 6$. First, $\log \mu \leq \log(A\mu)$ and $\log(\mu + 1) \leq 1 + \log \mu \leq 1 + \log(A\mu)$ (as $\mu + 1 \leq 2\mu$), and $\log(A\mu) \geq 0$. Second, $6\mu + 20 \leq 2^\mu$ (at $\mu = 6$ this reads $56 \leq 64$, and for $\mu \geq 6$ the derivative $2^\mu \ln 2 \geq 44$ of the right side exceeds the derivative 6 of the left side), so $\log(2^\mu + 6\mu + 20) \leq \log(2^{\mu+1}) = \mu + 1$; and $(2\lambda)^6 + 1 \leq 2 \cdot (2\lambda)^6 = 2^7 \lambda^6$. Third, both summands of $\bar{k}$ are at least 1 (as $\lambda \geq 1$ and $A\mu \geq 1$), and $x + y \leq 2 \max\{x, y\} \leq 2xy$ for $x, y \geq 1$; so $\bar{k} \leq 2 \cdot 192\lambda^3 \cdot \frac{4}{3}\bar{M}^2 = 2^9 \lambda^3 \bar{M}^2$ and

$$\log \bar{k} \leq 3\mu + 4A \log(A\mu) + 9.$$

Also $\log \bar{M} = 2A \log(A\mu)$. For an inequality of column 3 put $\varphi := \log(\text{left side}) - \log(\text{right side})$; the inequality holds if and only if $\varphi \geq 0$ (in row 1, where it is strict, if and only if $\varphi > 0$). The bounds above give, row by row:

1. $\varphi = 102.5\mu - \log 257 - 102 \log(2^\mu + 6\mu + 20) \geq 102.5\mu - \log 257 - 102(\mu + 1) \geq 0.5\mu - 111$, as $\log 257 < 9$; so $\varphi \geq (0.5\mu - 112) + 1 > 0$ if $0.5\mu - 112 \geq 0$. The crude form $\lambda^{1/2} > 2^{111}$ reads $\mu > 222$, which also holds if $0.5\mu - 112 \geq 0$, since then $\mu \geq 224$;
2. $\varphi = 103\mu - 13 \log \bar{M} = 103\mu - 26A \log(A\mu)$;
3. $\varphi = 99\mu - \log 640 - \log \bar{k} \geq 96\mu - 19 - 4A \log(A\mu)$, as $\log 640 < 10$;
4. $\varphi = 42.1\mu - 193 - 5 \log 12 \geq 42.1\mu - 211$, as $5 \log 12 < 18$;
5. $\varphi = 72\mu - \log 3 - 167 - \log \bar{k} - 21 \log \bar{M} \geq 69\mu - 178 - 46A \log(A\mu)$, as $\log 3 < 2$;
6. $\varphi = 84\mu - 110 - 15 \log \bar{M} = 84\mu - 110 - 30A \log(A\mu)$;
7. $\varphi = 64.4\mu - 127 - 4 \log 12 \geq 64.4\mu - 142$, as $4 \log 12 < 15$;

8. put $E := 58\mu - 194 - \log(A\mu)$. The five inequalities of row 8 hold if the following five differences are non-negative: $58\mu - 194 \geq E$; $62\mu - 186 - \log(\mu + 1) \geq 62\mu - 187 - \log(A\mu) = E + 4\mu + 7$; $62\mu - 190 - \log(\mu + 1) \geq 62\mu - 191 - \log(A\mu) = E + 4\mu + 3$; $58\mu - 191 \geq 58\mu - 194 \geq E$; and $99\mu - 6 - \log((2\lambda)^6 + 1) \geq 99\mu - 6 - (7 + 6\mu) = 93\mu - 13 > 0$. So all five hold if $E \geq 0$;
9. $\varphi = 3.3\mu - \log \bar{k} \geq 0.3\mu - 9 - 4A \log(A\mu)$;
10. $\varphi = 4\mu - 1 - \log \bar{k} \geq \mu - 10 - 4A \log(A\mu)$;
11. $\varphi = 95\mu - 13 - 12 \log \bar{M} = 95\mu - 13 - 24A \log(A\mu)$;
12. $\varphi = 1.6\mu - \log \bar{M} = 1.6\mu - 2A \log(A\mu)$;
13. $\varphi = 36\mu - 240 - 46A \log(A\mu)$.

In each row the lower bound obtained is $a_i\mu - b_i - c_i \log(A\mu)$ for the triple (a_i, b_i, c_i) of column 4 (in row 1 it is this quantity plus 1, and in row 8 it is at least $E = a_8\mu - b_8 - c_8 \log(A\mu)$ for the four inequalities that depend on it). So if $a_i\mu - b_i - c_i \log(A\mu) \geq 0$, every inequality in column 3 of row i holds at λ . This proves (ii).

(iii) Fix a row i , and let v_0 be as in (i) for $(a, b, c) = (a_i, b_i, c_i)$. By (i), the inequality $a_i\mu - b_i - c_i \log(A\mu) \geq 0$ holds for every $\mu \geq \max\{1, v_0^2\}$; by (ii), every inequality in column 3 of row i holds at every $\mu \geq \max\{6, v_0^2\}$. This threshold is explicit, but its numerical value is never computed: only its existence is used. $\square$

Lemma 3.13 (Lemma COL). *Assume condition (F1). Let Y be an ancestor of round r , with the stage-1 lending data of Definition 3.10. If $r \geq R - 1$, then $k_{\text{lend}}(Y) = 0$ and every statement below about lent classes is vacuous.*

- (a) Own_Y and Lend_Y are $(\varepsilon_Y, s_Y/4)$ -expanders on $V(Y)$. Every lent class, as a graph on $V(Y)$, is an $(\varepsilon_Y, s_Y/(8k_{\text{lend}}(Y)))$ -expander. If Y is light, every own class $R_{j,c}$, M , as a graph on $V(Y)$, is a $(2^{-6}, s_r/(16k_{\text{own}}))$ -expander.
- (b) For all $r + 2 \leq l \leq R$ and $0 \leq j < K_l^{\text{JS}}$, the class $\mathcal{L}_{Y,l,j}^{\text{JS}}$ is $(2^{12}L_Y^4, t_l^{\text{JS}})$ -path connected through $T_j(Y, l)$. By Lemma 3.1(iii), one application routes any multiset of pairs of distinct vertices in which every vertex lies in at most t_l^{JS} pairs. For instance, the union of the junction- j pair multisets of several systems, with multiplicities counted over the union, is joined by pairwise edge-disjoint paths of $\mathcal{L}_{Y,l,j}^{\text{JS}}$ of length at most $2^{12}L_Y^4$ with interiors in $T_j(Y, l)$.
- (c) Let Y be light. Let $(V_{l,c,\varsigma})_{(l,c,\varsigma) \in \mathcal{I}_U(Y)}$ be random subsets of $V(Y)$, independent of the stage-1 lending data of Y , such that each $V_{l,c,\varsigma}$ is a $\rho_{l,c,\varsigma}$ -random subset of $V(Y)$ with $\rho_{l,c,\varsigma} \geq 1/(12L_Y^5)$. The family may be dependent across indices. Then, with probability at least $1 - |V(Y)|^{-2}/2$ over the lending data and the sets, every U -lent class $\mathcal{L}_{Y,l,c,\varsigma}^U$ is $(2^{12}L_Y^4, t_Y)$ -path connected through $V_{l,c,\varsigma}$, where $t_Y = \lceil \lambda_r^{1.6} \rceil$ (Definition 3.10).
- (e) The own-device thresholds hold. If Y is standalone, then $s_Y/4 \geq 2^{150}L_Y^{42}$ and $s_Y/4 \geq 2^{146}L_Y^{38} \log L_Y$. On (a), Own_Y is therefore a spanning $(2^{-5}, s_r/4)$ -expander of $V(Y)$ whose robustness meets the thresholds of the transparent P -vortex and of VX^+ (the second of these is not used: VX^+ is applied only to demoted light parts, Lemma 5.8; cf. Remark 4.5). If Y is light, then $s_r/2 \geq 2^{151}L_Y^{38} \log L_Y$; this is the threshold for VX^+ at expansion 2^{-6} on $H_Y = X_Y$, which is a $(2^{-6}, s_r/2)$ -expander. Moreover, on (a) every own class is a $(2^{-6}, s')$ -expander with $s' := s_r/(16k_{\text{own}}) \geq 2^{145}L_Y^{41}$ and $2\lceil L_Y^6 \rceil \leq s'$.

(g) (*COL(g).*) The consumers listed in Definition 3.10 are exclusive. The own classes partition Own_Y , and if $r \leq R - 2$, the lent classes are pairwise edge-disjoint and partition Lend_Y . So no edge of H_Y is available to two consumers, and every lent edge that its consumer does not use is junk for Y 's own device. If $r \geq R - 1$, then Lend_Y has no classes, and all of Lend_Y is junk for Y 's own device (Definition 3.10, Consumers).

Items (a), (b) and (e) hold simultaneously with probability at least $1 - |V(Y)|^{-2}/2$ over the stage-1 lending data of Y . For every family as in (c), items (a), (b), (c) and (e) hold simultaneously with probability at least $1 - |V(Y)|^{-2}$. Item (g) always holds. The failure unions are bounded through rows 6 and 7 of Table 5, which need $(\Gamma 1)$. There are no items (d) and (f); the letters follow the references used in later sections.

Dependencies: 3.10, 3.2, 3.8, 3.1, 3.11, 2.17, 2.11, 2.8, $(\Gamma 1)$

Proof. Write $N := |V(Y)|$, $L := L_Y$, $k := k_{\text{lend}}(Y)$, $\lambda := \lambda_r$ and $\mu := \log \lambda$. By Proposition 2.11(i), H_Y is an (ε_Y, s_Y) -expander on $V(Y)$. Here $(\varepsilon_Y, s_Y) = (2^{-5}, s_r)$ if Y is standalone and $(2^{-6}, s_r/2)$ if Y is light (Definition 2.8), and $s_r \geq \lambda^{100}$ (Lemma 2.17(b)). The standing bounds (B1) and (B3)–(B6) of the proof of Lemma 3.11 are available in every round, and (B2), (B7) when $r \leq R - 2$. Rows 1, 3, 8 and 9 of Table 5 hold for every $r \leq R$, and the other rows involving Y hold for $r \leq R - 2$ (Lemma 3.11(ii)); for $r \geq R - 1$ only rows 3, 8 and 9 are used below. In particular $N \geq \lambda^{103}/2$ and $L \leq 2\lambda$.

(a) Stage (i) is a uniform 2-colouring of $E(H_Y)$. By Lemma 3.2 with $k = 2$, which needs $s_Y \geq 80L$ (row 3 of Table 5), Own_Y and Lend_Y are $(\varepsilon_Y, s_Y/4)$ -expanders on $V(Y)$, except with probability at most $4N^{-5}$.

Condition on stage (i) with Lend_Y such an expander. Stage (ii) is then a uniform k -colouring of Lend_Y . By Lemma 3.2, which needs $s_Y/4 \geq 40kL$ (row 3), every lent class is an $(\varepsilon_Y, s_Y/(8k))$ -expander on $V(Y)$, except with probability at most $2kN^{-5}$.

If Y is light, condition on Own_Y being a $(2^{-6}, s_r/8)$ -expander. Stage (iii) is a uniform k_{own} -colouring. Lemma 3.2, which needs $s_r/8 \geq 40k_{\text{own}}L$ (row 3), makes every own class an expander with parameters 2^{-6} and $s_r/(16k_{\text{own}})$, except with probability at most $2k_{\text{own}}N^{-5}$.

Using row 9 ($k \leq \lambda^{3.3}$), $k_{\text{own}} \leq L \leq 2\lambda$ and $N^3 \geq \lambda^{309}/8$, the total failure probability of (a) is at most

$$2(2 + k + k_{\text{own}})N^{-5} \leq 2(2 + \lambda^{3.3} + 2\lambda)N^{-5} \leq N^{-2}/4.$$

(b) Condition on a stage-(i)/(ii) outcome in which (a) holds. The labels are independent of the colouring, so for fixed (l, j) the set $T_j(Y, l)$ is still a ρ_l -random subset of $V(Y)$. The class $\mathcal{L}_{Y,l,j}^{\text{JS}}$ is now a fixed N -vertex (ε_Y, s) -expander with $s := s_Y/(8k)$ and $\varepsilon_Y \in [2^{-7}, 1]$. Apply Theorem 3.8 with $t := t_l^{\text{JS}}$ and $\rho := \rho_l$. Its hypotheses hold:

- $s \geq 2^{135}tL^{28}\rho^{-5}$ by row 5;
- $\rho N \geq L^2$, since by row 2 at $\lambda = \lambda_r$ (Lemma 3.11(ii)), $\bar{M}^4 \leq \lambda^{31.7}$, so $\rho_l N \geq \bar{M}^{-4}\lambda^{103}/2 \geq \lambda^{71}/2 \geq 4\lambda^2 \geq L^2$.

Each class fails with probability at most $2^{86}t_l^{\text{JS}}L^{19}\rho_l^{-3}N^{-3}$. By row 6, the sum over $\mathcal{I}_{\text{JS}}(Y)$ is at most $N^{-2}/4$. The final sentence of (b) is Lemma 3.1(iii).

(c) Condition as in (b). The sets $V_{l,c,\varsigma}$ are independent of the colouring. For a fixed index, apply Theorem 3.8 to the fixed class $\mathcal{L}_{Y,l,c,\varsigma}^{\text{U}}$, an N -vertex $(2^{-6}, s_r/(16k))$ -expander, with $t := t_Y = \lceil \lambda^{1.6} \rceil$ (so $1 \leq t \leq 2\lambda^{1.6}$) and $\rho := \rho_{l,c,\varsigma} \geq 1/(12L^5)$. Its hypotheses hold:

- $s_r/(16k) \geq 2^{135}tL^{28}\rho^{-5}$ by row 4 (with row 9), since $\rho^{-5} \leq (12L^5)^5$;

- $\rho N \geq N/(12L^5) \geq \lambda^{103}/(24(2\lambda)^5) \geq L^2$.

Each index fails with probability at most $2^{86}tL^{19}(12L^5)^3N^{-3}$. The theorem is applied to one index at a time, so dependence across indices is irrelevant. By row 7, the union over $\mathcal{I}_U(Y)$ is at most $N^{-2}/4$. Together with the failure probability of (a), (c) fails with probability at most $N^{-2}/2$.

(e) The inequalities are row 8 of Table 5, with $s_Y = s_r$ for standalone Y . On (a), Own_Y is a $(2^{-5}, s_r/4)$ -expander if Y is standalone, and the own classes are $(2^{-6}, s_r/(16k_{\text{own}}))$ -expanders if Y is light. Also $H_Y = X_Y$ is a $(2^{-6}, s_r/2)$ -expander for light Y , by Proposition 2.11(i).

(g) This holds by construction. If $r \leq R - 2$, each edge of Lend_Y receives exactly one index; if $r \geq R - 1$, Lend_Y has no classes and is junk for Y 's own device. Each edge of Own_Y receives exactly one own label.

Probabilities. By the bounds above, (a) or (b) fails with probability at most $N^{-2}/4 + N^{-2}/4 = N^{-2}/2$, and (e) holds on (a). Adding the U-union of (c) gives at most $3N^{-2}/4 \leq N^{-2}$. $\square$

4 Vortex tools

This section proves three absorption statements for a single vertex set carrying a spanning robust expander: Lemma TPV (the transparent P-vortex, used on standalone pre-parts), Lemma PV (the four-phase P-vortex, used on light parts) and Theorem VX^+ (vortex absorption with arbitrary extra edges, used on demoted light parts). All three are proved from Theorem 16* (Theorem 3.8), Lemma HB (Lemma 3.9), Bucić–Montgomery's Corollary 22 (Cited result 1.22) and the elementary long-cycle bound (Fact 1.6); Lemma TPV and Theorem VX^+ also use Lemma 15⁺ (Lemma 3.2), while Lemma PV takes its own classes as given expanders and does not. None of them refers to the hierarchy: each is a statement about one vertex set Z , one expander on Z and edge sets inside Z . In each of them the random labels and the good event are fixed *before* the edge set to be decomposed is revealed, and the conclusion then holds for *every* admissible edge set; this is the form in which the later sections use them.

Throughout the section, graphs are simple (Convention 1.3); this is used, for instance, in Corollary 22, in the distinctness of the far ends $u_1 \neq u_2$ of a cherry (two distinct edges at w have distinct far ends), in the fact that a closed trail in a simple graph has length at least 3, in the per-vertex bound $\deg_{H_0}(x) \leq |Z| - 1$ of Lemma PV(c), and in the finishes of the three runs, through Fact 1.6, whose proof uses that a vertex of degree at least d has at least d neighbours and that a graph on h vertices has at most $\binom{h}{2}$ edges.

4.1 Conventions and elementary observations

Convention 4.1 (vortex-local notation). Inside one run of Lemma TPV, Lemma PV or Theorem VX^+ below (a *vortex run*), Z denotes the vertex set of the run, $N := |Z|$ and $L := \log_2 |Z|$. All run-local objects are written in sans-serif: the number of steps J ; the reservoir colour classes $R_{j,c}$ and the reserve class M ; the level $\text{lv}(v) \in \{0, 1, \dots, J\}$ and the label $\text{k}(v)$ of a vertex; the sets U_j (vertices still active before step j), W_j (vertices retiring at step j , for $0 \leq j < J$; the vertices of level J stay in U_J and are treated by the finish), Z_j (reserve far ends of step j) and $V_{j,c}$ (connector interiors of step j and phase c); the graphs H_j (edges not yet covered before step j) and edge sets F_j (edges of H_j meeting W_j); the partition $Z = P \sqcup Q$ of Lemma TPV; the reserve sets C_w^j (or C_w); the parameters $\mathbf{m}$, $\mathbf{b}$ (of Lemma HB) and $\mathbf{t}$ (the multiplicity used with Theorem 16*); the path families $\mathcal{P}_{j,c}^\vee$ (produced by Corollary 22) and $\tilde{\mathcal{P}}_{j,c}^\vee$ (after moving ends and splitting). The letter j is the step index and c the phase. The number of colours of a run is written k , and the good events of a run are called (G1)–(G5) (numbered alike in the three runs; a run need not use all five); these letters,

the invariants (Inv1)–(Inv3) and the path properties (P1)–(P4) (and (P5) in Lemma PV) are local to the run (the good events (Gk) are unrelated to the galactic conditions (Γk) of Condition 1.33). None of these symbols has a meaning outside the run in which it is defined. Auxiliary symbols that are local to a single estimate inside one of the three proofs (such as χ , $\mathcal{K}_{j,c}$, V' and I_u below) are written in *italic* or calligraphic type, not in sans-serif.

Dependencies: none.

The three proofs below share five elementary observations. They are recorded here once, with tags, and proved in full; they are parts of the proofs of Lemma TPV, Lemma PV and Theorem VX⁺ below, not separate statements.

Trail splitting (S). *Let $T = x_0x_1 \cdots x_q$ ($q \geq 0$) be a trail (a walk with pairwise distinct edges) in a simple graph and put $\text{rep}(T) := (q+1) - |\{x_0, \dots, x_q\}|$. Then $E(T)$ is the disjoint union of the edge sets of at most $\text{rep}(T)$ cycles, each of length at least 3, and of at most one path; the path is present if and only if $x_0 \neq x_q$, and then its ends are x_0 and x_q . Every vertex of every piece is a vertex of T . In particular, if the inner vertices $x_1, \dots, x_{q-1}$ are pairwise distinct, then $\text{rep}(T) \leq 2$ and T splits into at most two cycles and at most one path, so into at most three pieces.*

Proof. Induction on q . If $\text{rep}(T) = 0$, all x_i are distinct, so T is empty ($q = 0$) or a path with ends $x_0 \neq x_q$. Otherwise choose $i < i'$ with $x_i = x_{i'}$ and $i' - i$ minimal. Then $x_i, x_{i+1}, \dots, x_{i'-1}$ are pairwise distinct, and $i' - i \geq 3$: $i' - i = 1$ would be a loop, and $i' - i = 2$ would give $x_i x_{i+1} = x_{i+1} x_{i+2}$, the same edge twice, which a trail does not contain. So $C := x_i x_{i+1} \cdots x_{i'}$ is a cycle of length $i' - i \geq 3$. Delete it: $T' := x_0 \cdots x_i x_{i'+1} \cdots x_q$ is again a trail (its step from x_i to $x_{i'+1}$ is the edge $x_{i'} x_{i'+1}$ of T), $E(T) = E(C) \sqcup E(T')$, and T' begins at x_0 and ends at x_q (if $i' = q$ it ends at $x_i = x_q$). The vertices $x_{i+1}, \dots, x_{i'-1}$ are pairwise distinct and differ from x_i , so at most $i' - i - 1$ distinct vertices disappear while $i' - i$ entries are removed; hence $\text{rep}(T') \leq \text{rep}(T) - 1$, and the induction hypothesis applied to T' finishes the proof. If $x_1, \dots, x_{q-1}$ are pairwise distinct, then $|\{x_0, \dots, x_q\}| \geq q - 1$, so $\text{rep}(T) \leq 2$. $\diamond$

Path ends (E). *Let $\mathcal{P}$ be a decomposition of an edge set F into (non-trivial) paths. For every vertex v the number of paths of $\mathcal{P}$ having v as an end is congruent to $\deg_F(v)$ modulo 2. If every vertex is an end of at most two paths of $\mathcal{P}$ and all edges of F have both ends in a set W , then $|\mathcal{P}| \leq |W|$.*

Proof. A path contributes 2 to the degree of each of its inner vertices and 1 to the degree of each of its two (distinct) ends. For the second part count path ends: $2|\mathcal{P}| \leq 2|W|$. $\diamond$

By Cited result 1.22, every (simple) graph, in particular every edge set F , has a decomposition $\mathcal{P}$ as in (E) in which every vertex is an end of at most two paths; we call it a *Corollary-22 decomposition* of F .

Monotone comparison (MC). *Let S be a finite set and let $\mathcal{K}$ be a family of subsets of S that is closed upwards: $T \in \mathcal{K}$ and $T \subseteq T' \subseteq S$ imply $T' \in \mathcal{K}$. For $r = (r_x)_{x \in S} \in [0, 1]^S$ put*

$$\Psi_{\mathcal{K}}(r) := \sum_{T \in \mathcal{K}} \prod_{x \in T} r_x \prod_{x \in S \setminus T} (1 - r_x).$$

If $p, q \in [0, 1]^S$ satisfy $p_x \leq q_x$ for every $x \in S$, then $\Psi_{\mathcal{K}}(p) \leq \Psi_{\mathcal{K}}(q)$. If V is a random subset of S such that the events $\{x \in V\}$ ($x \in S$) are mutually independent, then $\mathbb{P}(V \in \mathcal{K}) = \Psi_{\mathcal{K}}(q^V)$, where $q^V \in [0, 1]^S$ is given by $q_x^V := \mathbb{P}(x \in V)$. In particular, let $\rho \in [0, 1]$, let V be a random subset of S such that the events $\{x \in V\}$ ($x \in S$) are mutually independent and $\mathbb{P}(x \in V) \geq \rho$ for every $x \in S$, and let V' be a ρ -random subset of S , that is, a random subset of S that contains each element of S independently with probability ρ . Then $\mathbb{P}(V \notin \mathcal{K}) \leq \mathbb{P}(V' \notin \mathcal{K})$ (V and V' need not be defined on the same probability space; only the two probabilities are compared).

Proof. For the comparison write $S = \{x_1, \dots, x_{|S|}\}$, and for $0 \leq i \leq |S|$ let $q^{(i)} \in [0, 1]^S$ agree with q at $x_1, \dots, x_i$ and with p at $x_{i+1}, \dots, x_{|S|}$. Then $q^{(0)} = p$ and $q^{(|S|)} = q$, and for $1 \leq i \leq |S|$ the vectors $q^{(i-1)}$ and $q^{(i)}$ agree except possibly at x_i , where $q_{x_i}^{(i-1)} = p_{x_i} \leq q_{x_i} = q_{x_i}^{(i)}$. So it suffices to show that $\Psi_{\mathcal{K}}$ is non-decreasing in each coordinate when the other coordinates are fixed. Fix $y \in S$ and values $r_x \in [0, 1]$ ($x \in S \setminus \{y\}$), and for $z \in [0, 1]$ let $r(z) \in [0, 1]^S$ have entry r_x at $x \neq y$ and entry z at y . Every subset of S is, for exactly one $T \subseteq S \setminus \{y\}$, equal to T or to $T \cup \{y\}$. With $\omega(T) := \prod_{x \in T} r_x \prod_{x \in S \setminus (T \cup \{y\})} (1 - r_x) \geq 0$ this gives

$$\begin{aligned} \Psi_{\mathcal{K}}(r(z)) &= \sum_{T \subseteq S \setminus \{y\}} \omega(T) \left((1 - z) \mathbf{1}[T \in \mathcal{K}] + z \mathbf{1}[T \cup \{y\} \in \mathcal{K}] \right) \\ &= \sum_{T \subseteq S \setminus \{y\}} \omega(T) \left(\mathbf{1}[T \in \mathcal{K}] + z (\mathbf{1}[T \cup \{y\} \in \mathcal{K}] - \mathbf{1}[T \in \mathcal{K}]) \right). \end{aligned}$$

As $\mathcal{K}$ is closed upwards, $\mathbf{1}[T \cup \{y\} \in \mathcal{K}] \geq \mathbf{1}[T \in \mathcal{K}]$ for every T , so every summand, and hence $\Psi_{\mathcal{K}}(r(z))$, is a non-decreasing function of z . For the formula, let V be as stated. For $T \subseteq S$ the event $\{V = T\}$ is the intersection of the events $\{x \in V\}$ ($x \in T$) and $\{x \notin V\}$ ($x \in S \setminus T$); these events are mutually independent, because replacing some of finitely many mutually independent events by their complements preserves mutual independence. So $\mathbb{P}(V = T) = \prod_{x \in T} q_x^V \prod_{x \in S \setminus T} (1 - q_x^V)$, and summing over the pairwise disjoint events $\{V = T\}$, $T \in \mathcal{K}$, gives $\mathbb{P}(V \in \mathcal{K}) = \Psi_{\mathcal{K}}(q^V)$. For the last claim, the events $\{x \in V'\}$ ($x \in S$) are mutually independent and $q_x^{V'} = \rho \leq q_x^V$ for every $x \in S$; so, by the formula and the comparison, $\mathbb{P}(V' \in \mathcal{K}) = \Psi_{\mathcal{K}}(q^{V'}) \leq \Psi_{\mathcal{K}}(q^V) = \mathbb{P}(V \in \mathcal{K})$. Taking complements gives the claim. $\diamond$

Binomial domination (B). Let $X = \sum_{i=1}^{m'} I_i$ be a sum of independent indicator variables with $\mathbb{P}(I_i = 1) \geq p$ for every i , and let $a > 0$ with $m'p \geq 2a$. Then $\mathbb{P}(X < a) \leq \exp(-m'p/8)$.

Proof. As $m'p \geq 2a > 0$, we have $m' \geq 1$ and $0 < p \leq \mathbb{P}(I_1 = 1) \leq 1$. The set $V := \{i \in [m'] : I_i = 1\}$ is a random subset of $[m']$ whose events $\{i \in V\} = \{I_i = 1\}$ ($i \in [m']$) are mutually independent, each of probability at least p , and $X = |V|$. The family $\mathcal{K} := \{T \subseteq [m'] : |T| \geq a\}$ is closed upwards. Let V' be a p -random subset of $[m']$; then $|V'| \sim \text{Bin}(m', p)$, with mean $\mu = m'p \geq 2a$. By observation (MC) (with $S := [m']$ and $\rho := p$), $\mathbb{P}(X < a) = \mathbb{P}(V \notin \mathcal{K}) \leq \mathbb{P}(V' \notin \mathcal{K}) = \mathbb{P}(|V'| < a)$, and by the lower tail of Cited result 1.8 with $\delta = 1/2$, $\mathbb{P}(|V'| < a) \leq \mathbb{P}(|V'| < \mu/2) \leq \exp(-\mu/8)$. $\diamond$

Closing (C). Let Q be a path of length at least 2 with ends $x \neq y$, and let Q' be an x - y path whose inner vertices avoid $V(Q)$ and which shares no edge with Q . Then $Q \cup Q'$ is a cycle of length at least 3.

Proof. The only common vertices of Q and Q' are x and y , so $Q \cup Q'$ is a cycle; its length is $|E(Q)| + |E(Q')| \geq 2 + 1$. $\diamond$

Finite probability spaces. Every random object of this section takes finitely many values. The labels of a run are finitely many independent random variables, each with finitely many values: in Lemma TPV and Theorem VX⁺ a uniform colouring of $E(O)$ with k colours, and in all three statements the levels, with values in $\{0, 1, \dots, J\}$, and the labels k . The ρ -random subsets of a finite set S used with (MC) and (B) may be taken to be the identity map on the finite probability space of all subsets $T \subseteq S$, where T has probability $\prod_{x \in T} \rho \prod_{x \in S \setminus T} (1 - \rho)$ (an empty product is 1); identifying each T with its indicator vector, this space is the product $\{0, 1\}^S$ of $|S|$ copies of the two-point space $\{0, 1\}$ in which 1 has probability ρ . So every probability and expectation in this section is a finite sum over a finite product of finite probability spaces. Where a proof below fixes the colouring of a run, the probability over the remaining labels, which are independent of the

colouring, is a finite sum as well, and the unconditional probability is the average of these sums over the finitely many (equally likely) colourings.

The absolute constant N_0 (Definition 1.31) is chosen so large that the finitely many explicit inequalities listed as *size conditions* in the three proofs of this section hold for every $N \geq N_0$; each of them is an inequality between explicit functions of N that is true for all sufficiently large N . In every application in this paper the vertex set of a vortex run is a pre-part or a light part, which has at least N_0 vertices by condition (F3) of Condition 1.33.

4.2 Lemma TPV: the transparent P-vortex

Lemma 4.2 (Lemma TPV (transparent P-vortex)). *Let Z be a set of $N \geq N_0$ vertices and $L := \log_2 N$. Let O be a spanning (ε_O, s) -expander on Z with $\varepsilon_O \in [2^{-7}, 2^{-5}]$ (only $\varepsilon_O \geq 2^{-7}$ is used in the proof) and $s \geq 2^{150} L^{42}$, and let $Z = P \sqcup Q$ be a partition. Put*

$$J := \lfloor \log_2(L/6) \rfloor, \quad k := 3J + 1, \quad m := \lceil L^6 \rceil, \quad b := \lceil 2^8 L^2 m \rceil, \quad t := 2^{10} L^8,$$

(b is the bound of Lemma HB at the worst case $\varepsilon_O = 2^{-7}$), and fix, as a function of O only, sets $\mathcal{A}(w) \subseteq N_O(w)$ ($w \in Z$) with $|\mathcal{A}(w)| = m$ such that every vertex lies in at most b of them (they exist by Lemma 3.9; see the proof). The labels of the run are the following independent random variables:

- (a) a uniform colouring of $E(O)$ with the k colours $R_{j,c}$ ($0 \leq j < J$, $c \in [3]$) and M ; the colour classes are regarded as spanning subgraphs of O on Z and denoted by the same letters;
- (b) for every $v \in P$ a level $lv(v) \in \{0, 1, \dots, J\}$ with $\mathbb{P}(lv(v) \geq j) = 2^{-j}$ for $0 \leq j \leq J$ (that is, $\mathbb{P}(lv(v) = j) = 2^{-(j+1)}$ for $0 \leq j < J$ and $\mathbb{P}(lv(v) = J) = 2^{-J}$);
- (c) for every $v \in Z$ a label $k(v) \in \{0, 1, 2, 3\}$ with $\mathbb{P}(k(v) = 0) = 1/2$ and $\mathbb{P}(k(v) = c) = 1/6$ for $c \in [3]$.

Put $U_j := Q \cup \{v \in P : lv(v) \geq j\}$, $W_j := \{v \in P : lv(v) = j\}$, $Z_j := \{u \in U_{j+1} : k(u) = 0\}$ and $V_{j,c} := \{u \in U_{j+1} : k(u) = c\}$. There is an event $\mathcal{G}_{\text{TPV}}$, determined by (Z, O, P) and the labels only, with

$$\mathbb{P}(\mathcal{G}_{\text{TPV}}) \geq \frac{1}{2} - \eta_{\text{TPV}}(N), \quad \eta_{\text{TPV}}(N) := 2LN^{-5} + 2^{96}L^{31}N^{-3} + NL e^{-3L^4/8} \leq \frac{1}{100}, \quad (4.1)$$

such that on $\mathcal{G}_{\text{TPV}}$ the following holds for every edge set E with $E(O) \subseteq E$ all of whose edges have both ends in Z : there is a partition $E = E^V \sqcup E^Q$ with the properties

- (T1) every edge of E with both ends in P lies in E^V ;
- (T2) every edge of E^V with an end in Q is an edge of O ;
- (T3) E^V decomposes into at most $169|P|$ objects, and $E^V = \emptyset$ if $P = \emptyset$;
- (T4) (no arcs) all objects of (T3) consist of edges of E ; every path formed during the run is closed into a cycle within the step in which it is formed, so the run leaves no path (“arc”) to be closed by any other device.

The partition and the decomposition are obtained from E and the labels by a deterministic procedure.

Dependencies: 3.2, 3.8, 3.9, 3.1, 1.22, 1.6, 1.8, 1.26, (F3), 4.1, 1.3, 1.31.

Proof. If $\mathbf{P} = \emptyset$, let $\mathcal{G}_{\text{TPV}}$ be the sure event and put $E^V := \emptyset$, $E^Q := E$; then (T1) is vacuous and (T2)–(T4) hold trivially. From now on $\mathbf{P} \neq \emptyset$.

Size conditions. We use: (i) $L \geq 2^{10}$; (ii) $N \geq L^3$; (iii) $4 \cdot 48 \leq L$ (a consequence of (i)); (iv) $\eta_{\text{TPV}}(N) \leq 1/100$. They hold for $N \geq N_0$ by the choice of N_0 .

Parameters. By the definition of J , $6 \cdot 2^J \leq L < 12 \cdot 2^J$, so

$$2^{-J} \geq 6/L, \quad 2^{-J} < 12/L, \quad 1 \leq J \leq \log_2 L, \quad k = 3J + 1 \leq L \quad (4.2)$$

(the last two by (i)). Since $s \geq 2^{150} L^{42} \geq 2 \lceil L^6 \rceil = 2\mathbf{m}$, Lemma 3.9 applied to $X := O$, $\varepsilon' := \varepsilon_O \geq 2^{-7}$ and $m := \mathbf{m}$ gives sets $\mathcal{A}(w) \subseteq N_O(w)$, $|\mathcal{A}(w)| = \mathbf{m}$, such that every vertex lies in at most $\lceil 2\mathbf{m}L^2/\varepsilon_O \rceil \leq \lceil 2^8 L^2 \mathbf{m} \rceil = \mathbf{b}$ of them; we fix the lexicographically first such family (for a fixed ordering of Z), so that it is a function of O . Moreover

$$2 + \mathbf{b} \leq 2 + 2^8 L^2 (L^6 + 1) + 1 \leq 2^{10} L^8 = \mathbf{t}. \quad (4.3)$$

Directly from the definitions: $U_0 = Z$; $U_{j+1} \subseteq U_j$ and $U_j = U_{j+1} \sqcup W_j$; $W_j \subseteq \mathbf{P}$ and $W_j \cap U_{j+1} = \emptyset$; and $Z_j, V_{j,1}, V_{j,2}, V_{j,3}$ partition U_{j+1} .

Good events. Let $\mathcal{G}_{\text{TPV}}$ be the intersection of the following four events.

- (G1) Every colour class ($R_{j,c}$ and $\mathbf{M}$) is a spanning $(\varepsilon_O, s/(2k))$ -expander on Z . Since $s \geq 2^{150} L^{42} \geq 40kL$, Lemma 3.2 (with $X := O$ and k colours) shows that each class fails with probability at most $2N^{-5}$; so (G1) fails with probability at most $2kN^{-5} \leq 2LN^{-5}$. Note that $s/(2k) \geq s/(2L) \geq 2^{149} L^{41}$.
- (G2) For all $0 \leq j < J$ and $c \in [3]$, the class $R_{j,c}$ is $(2^{12} L^4, \mathbf{t})$ -path connected through $V_{j,c}$ (for every multiset of pairs of distinct vertices of Z in which each vertex lies in at most $\mathbf{t}$ pairs there are edge-disjoint paths of $R_{j,c}$ of length at most $2^{12} L^4$ joining the pairs, with all inner vertices in $V_{j,c}$). Fix a colouring χ in (G1) and a pair (j, c) , and let $\mathcal{K}_{j,c}$ be the family of sets $T \subseteq Z$ such that $R_{j,c}$ is $(2^{12} L^4, \mathbf{t})$ -path connected through T . It is determined by χ . It is closed upwards by Lemma 3.1(ii), applied with $G = G' := R_{j,c}$, $\ell = \ell' := 2^{12} L^4$, $t = t' := \mathbf{t}$ and with $T \subseteq T' \subseteq Z = V(R_{j,c})$ as the two sets through which path connectivity is taken (the colour classes are spanning). For $v \in Z$ the event $v \in V_{j,c}$ depends only on $\mathbf{k}(v)$ and, if $v \in \mathbf{P}$, on $\mathbf{lv}(v)$, so these events are mutually independent over v and independent of the colouring, and

$$\mathbb{P}(v \in V_{j,c}) = \begin{cases} 1/6 \geq 1/L, & v \in \mathbf{Q}, \\ 2^{-(j+1)}/6 \geq 2^{-J}/6 \geq 1/L, & v \in \mathbf{P}, \end{cases}$$

by (4.2) and size condition (i). Let V' be a $(1/L)$ -random subset of Z (each vertex independently with probability exactly $1/L$), and apply Theorem 3.8 to $X := R_{j,c}$ (an N -vertex $(\varepsilon_O, s/(2k))$ -expander with $\varepsilon_O \in [2^{-7}, 1]$), $V := V'$, $\rho := 1/L$ and $t := \mathbf{t}$: its hypotheses $\rho N = N/L \geq L^2$ (by size condition (ii)) and $s/(2k) \geq 2^{149} L^{41} \geq 2^{135} \cdot 2^{10} L^8 \cdot L^{28} \cdot L^5 = 2^{135} \mathbf{t} L^{28} \rho^{-5}$ hold. So $\mathbb{P}(V' \notin \mathcal{K}_{j,c}) \leq 2^{86} \mathbf{t} L^{19} \rho^{-3} N^{-3} = 2^{96} L^{30} N^{-3}$. By observation (MC) (with $S := Z$, $\mathcal{K} := \mathcal{K}_{j,c}$, the random set $V_{j,c}$ in place of V , and $\rho := 1/L$), also $\mathbb{P}(V_{j,c} \notin \mathcal{K}_{j,c}) \leq 2^{96} L^{30} N^{-3}$, the probability being over the labels (b) and (c): for the fixed colouring χ , $R_{j,c}$ fails to be $(2^{12} L^4, \mathbf{t})$ -path connected through $V_{j,c}$ with probability at most $2^{96} L^{30} N^{-3}$. As the labels (b), (c) are independent of the colouring, a union bound over the $3J$ pairs (j, c) shows that the probability that (G1) holds and (G2) fails is at most $\sum_{\chi} \mathbb{P}(\text{the colouring is } \chi) \cdot 3J \cdot 2^{96} L^{30} N^{-3} \leq 2^{96} L^{31} N^{-3}$, the sum running over the colourings χ in (G1), and using $3J \leq L$. By Lemma 3.1(iii) the event (G2) depends only on the classes $R_{j,c}$ and the sets $V_{j,c}$, and on it the pair multisets may be chosen after all labels are revealed, as they will be below.

- (G4) For every $0 \leq j < J$ and every $w \in P$ the set $C_w^j := \{wu : u \in \mathcal{A}(w), wu \in M, u \in Z_j\}$ has at least 6 elements. For $u \in \mathcal{A}(w)$ let $I_u := \mathbf{1}[wu \in M \text{ and } u \in Z_j]$. Since $u \neq w$, the variables I_u ($u \in \mathcal{A}(w)$) depend on pairwise distinct edges wu and distinct vertices u , so they are independent; and $\mathbb{P}(I_u = 1) = \frac{1}{k} \cdot \mathbb{P}(u \in U_{j+1}) \cdot \frac{1}{2} \geq \frac{1}{L} \cdot 2^{-J} \cdot \frac{1}{2} \geq \frac{3}{L^2}$ by (4.2) (for $u \in Q$, $\mathbb{P}(u \in U_{j+1}) = 1$). As $m \cdot 3/L^2 \geq 3L^4 \geq 12$, observation (B) gives $\mathbb{P}(|C_w^j| < 6) \leq e^{-3L^4/8}$. A union bound over the at most $NJ \leq NL$ pairs (w, j) bounds the failure probability of (G4) by $NLe^{-3L^4/8}$.
- (G5) $\sum_{j < J} |P \cap U_j| \leq 8|P|$ and $|P \cap U_J| \leq 48|P|/L$. Here $\mathbb{E}|P \cap U_j| = 2^{-j}|P|$, so $\mathbb{E} \sum_{j < J} |P \cap U_j| < 2|P|$ and, by (4.2), $\mathbb{E}|P \cap U_J| < 12|P|/L$. By Markov's inequality (Cited result 1.26) each of the two inequalities fails with probability less than $1/4$.

All four events are determined by (Z, O, P) (through O , the sets $\mathcal{A}(w)$ and the partition) and the labels; none involves E . Summing the failure probabilities gives (4.1). From now on we fix labels in $\mathcal{G}_{TPV}$ and an arbitrary edge set $E \supseteq E(O)$ inside Z .

The process. All choices below (reserved edges, classes, Corollary-22 decompositions, connectors, the final decomposition) are made by fixed rules, so the procedure is deterministic given E and the labels. An edge is *used* once it has been placed into an output object. Put $H_0 := \{e \in E : e \text{ has both ends in } P\}$ and, for $j \geq 0$, $H_{j+1} := H_j \setminus \{\text{edges used at step } j\}$; thus H_j is the set of unused edges of H_0 . Before step j ($0 \leq j \leq J$) we maintain:

- (Inv1) every edge of H_j has both ends in $P \cap U_j$;
- (Inv2) no edge of $R_{j',c}$ with $j' \geq j$, $c \in [3]$ and both ends in $U_{j'+1}$ has been used;
- (Inv3) no M -edge wu with $w \in W_{j'}$ for some $j' \geq j$ and $u \in U_{j'+1}$ has been used.

For $j = 0$ nothing has been used and $U_0 = Z$, so (Inv1)–(Inv3) hold.

Step j ($0 \leq j < J$). Let F_j be the set of edges of H_j meeting W_j . By (Inv1), every edge of F_j has both ends in $P \cap U_j$.

- (i) *Reserve.* For each $w \in W_j$ choose six distinct edges $e_w^{c,i} \in C_w^j$ ($c \in [3]$, $i \in [2]$); this is possible by (G4), since $w \in P$. Each $e_w^{c,i} = wu$ is an M -edge with $w \in W_j$ and $u \in Z_j \subseteq U_{j+1}$, so it is unused by (Inv3), and w is its only end in W_j ; hence reserved edges of distinct w are distinct. If $u \in P$ then $e_w^{c,i} \in H_0$ (as $E(O) \subseteq E$), so $e_w^{c,i} \in H_j$ and $e_w^{c,i} \in F_j$; if $u \in Q$ it is an edge of O outside H_0 . The Z_j -end u of a reserved edge is called its *far end*.
- (ii) *Classes.* For every non-reserved edge of F_j choose an end $w \in W_j$ (either one if both ends lie in W_j), call the other end x , and give the edge a class $\gamma \in [3] \setminus \{k(x)\}$ (possible as $k(x)$ is a single value). Let $F_{j,c}$ be the set of non-reserved edges of class c . Then

$$V(F_{j,c}) \cap V_{j,c} = \emptyset \quad (c \in [3]) : \quad (4.4)$$

an end in W_j is not in $U_{j+1} \supseteq V_{j,c}$, and the other end x of a class- c edge has $k(x) \neq c$.

- (iii) *Paths.* Let $\mathcal{P}_{j,c}^\vee$ be a Corollary-22 decomposition of $F_{j,c}$. As $V(F_{j,c}) \subseteq P \cap U_j$, observation (E) gives $|\mathcal{P}_{j,c}^\vee| \leq |P \cap U_j|$.
- (iv) *Ends in W_j .* For $w \in W_j$ and $c \in [3]$ let $d \in \{0, 1, 2\}$ be the number of paths of $\mathcal{P}_{j,c}^\vee$ having w as an end.

- If $d = 2$, append $e_w^{c,1}$ at w to one of these paths and $e_w^{c,2}$ at w to the other.

- If $d = 1$, append $e_w^{c,1}$ at w to that path and output $e_w^{c,2}$ as a single edge.
- If $d = 0$, form the *cherry* $u_1 w u_2$ from $e_w^{c,1} = w u_1$ and $e_w^{c,2} = w u_2$ ($u_1 \neq u_2$, since the two edges are distinct).

This produces, for each c , a family of *trails*: every path of $\mathcal{P}_{j,c}^\vee$ with exactly one reserved edge appended at each of its ends lying in W_j (and nothing appended at its other ends), and the cherries. If a path has both ends in W_j , the two appended edges belong to distinct vertices of W_j and are distinct. In every trail $x_0 x_1 \cdots x_q$ the inner vertices $x_1, \dots, x_{q-1}$ are vertices of one path of $\mathcal{P}_{j,c}^\vee$ (or the centre w of a cherry), hence pairwise distinct, and the edges are pairwise distinct.

(v) *Splitting*. By observation (S) every trail splits into at most two cycles (which are output) and at most one path, with the same two ends as the trail; let $\tilde{\mathcal{P}}_{j,c}^\vee$ be the set of these paths. They satisfy:

- (P1) no vertex of a path in $\tilde{\mathcal{P}}_{j,c}^\vee$ lies in $V_{j,c}$: its vertices are vertices of $F_{j,c}$ (by (4.4)), cherry centres (in W_j) or far ends (in Z_j , where $k = 0$);
- (P2) the two ends of a path in $\tilde{\mathcal{P}}_{j,c}^\vee$ are distinct and lie in U_{j+1} : an end of a trail is either an end, not in W_j , of a path of $\mathcal{P}_{j,c}^\vee$, hence in $(P \cap U_j) \setminus W_j \subseteq U_{j+1}$, or a far end, in $Z_j \subseteq U_{j+1}$; distinctness is part of (S);
- (P3) every path in $\tilde{\mathcal{P}}_{j,c}^\vee$ has length at least 2: each of its edges meets W_j (edges of F_j by definition, reserved edges at w), while by (P2) neither of its ends lies in W_j , so no single edge can join its two ends;
- (P4) every vertex x is an end of at most $2 + b \leq t$ paths of $\tilde{\mathcal{P}}_{j,c}^\vee$: each trail yields at most one path, whose ends are ends of the trail; x is an end of at most two paths of $\mathcal{P}_{j,c}^\vee$; and x is the far end of a reserved edge $e_w^{c,i}$ only if $x \in \mathcal{A}(w)$, which holds for at most b vertices w , while for each such w at most one of the two distinct edges $e_w^{c,1}, e_w^{c,2}$ at w has far end x . Then use (4.3).

(vi) *Closing*. For each $c \in [3]$ the ends of the paths of $\tilde{\mathcal{P}}_{j,c}^\vee$ form a multiset of pairs of distinct vertices in which every vertex lies in at most t pairs (P2, P4). By (G2) there are pairwise edge-disjoint paths $Q' \subseteq R_{j,c}$, one for each $Q \in \tilde{\mathcal{P}}_{j,c}^\vee$, joining the two ends of Q , of length at most $2^{12} L^4$, with all inner vertices in $V_{j,c}$. All vertices of Q' lie in U_{j+1} (ends by (P2), inner vertices in $V_{j,c} \subseteq U_{j+1}$), so every edge of Q' is an $R_{j,c}$ -edge with both ends in U_{j+1} ; it is unused by (Inv2), it is not in F_j (which meets W_j), and it is not reserved (reserved edges are M-edges). The inner vertices of Q' avoid $V(Q)$ by (P1), and Q has length at least 2 by (P3); so by observation (C), $Q \cup Q'$ is a cycle, which is output.

The edges used at step j are: all of F_j (every non-reserved edge lies on a path of some $\mathcal{P}_{j,c}^\vee$, every reserved edge is appended, in a cherry or a single edge, and every trail edge ends in a split cycle or in a closed cycle), the reserved edges with far end in Q , and the connector edges. Within the step the output objects are pairwise edge-disjoint: trails of one class are edge-disjoint and classes are disjoint; connectors of one class are edge-disjoint by (G2), connectors of distinct classes lie in distinct colour classes, and connectors meet neither F_j nor the reserved edges. No edge used at step j was used before: $F_j \subseteq H_j$, reserved edges by (Inv3), connectors by (Inv2).

The invariants persist. (Inv1) for $j + 1$: an edge of H_{j+1} is an edge of H_j not in F_j , so it has both ends in $(P \cap U_j) \setminus W_j = P \cap U_{j+1}$. (Inv2), (Inv3) for $j + 1$: step j used only edges meeting W_j and $R_{j,-}$ -edges. An edge protected by (Inv2) or (Inv3) for an index $j' \geq j + 1$ has both ends in

$U_{j'} \subseteq U_{j+1}$ (in (Inv3), $w \in W_{j'} \subseteq U_{j'}$), so it does not meet W_j , and it is an $R_{j',c}$ -edge with $j' \neq j$ or an M-edge; so it was not used at step j .

Cost of step j . There are at most $3|W_j|$ single edges (at most one per pair (w, c)). For each c there are at most $|\mathcal{P}_{j,c}^v| + |W_j| \leq |P \cap U_j| + |W_j|$ trails, each giving at most three objects (at most two split cycles and at most one closed cycle). So step j outputs at most

$$3|W_j| + 3 \cdot 3(|P \cap U_j| + |W_j|) = 12|W_j| + 9|P \cap U_j| \quad (4.5)$$

objects. The sets W_j ($j < J$) are pairwise disjoint subsets of P , so $\sum_{j < J} |W_j| \leq |P|$, and $\sum_{j < J} |P \cap U_j| \leq 8|P|$ by (G5). Hence, by (4.5), the steps $0 \leq j < J$ together output at most $12|P| + 9 \cdot 8|P| = 84|P|$ objects.

Finish. By (Inv1) for $j = J$, every edge of H_J has both ends in $P \cap U_J$, and $|P \cap U_J| \leq 48|P|/L$ by (G5). Apply Fact 1.6(b) with $F := H_J$, $W := P \cap U_J$, $\beta := 48$ and $\alpha := |P| \leq N$; its hypothesis $N > 1$ holds as $L \geq 2^{10}$ by size condition (i), and its hypothesis $4\beta \leq L$ is size condition (iii). So H_J decomposes into at most $48|P|$ objects, which are output. The total number of objects is at most $84|P| + 48|P| = 132|P| \leq 169|P|$.

Conclusion. Let E^V be the set of used edges and $E^Q := E \setminus E^V$. Every used edge lies in $H_0 \subseteq E$ or is a reserved or connector edge, which lies in $E(O) \subseteq E$; so $E^V \subseteq E$ and the output is a decomposition of E^V into at most $169|P|$ objects, which is (T3). (T1): every edge of E with both ends in P lies in H_0 , and every edge of H_0 is used (edges leave H_j only by being used, and H_J is used in the finish). (T2): an edge of E^V with an end in Q is not in H_0 , so it is a reserved edge or a connector edge, hence an edge of O . (T4): all vertices of the Corollary-22 paths lie in P (their edges lie in H_0), every path of $\tilde{\mathcal{P}}_{j,c}^v$ is closed in step j , and the finish outputs only cycles and single edges; no path is left open.

Remark. Property (P4) bounds the pair multiplicity by $2 + b$ (the cruder bound $2 + 2b$ is also below t , so nothing changes quantitatively). Property (P3) is what makes $Q \cup Q'$ a cycle of length at least 3; the case of a single edge closed by a single connector edge cannot occur. $\square$

4.3 Lemma PV: the four-phase P-vortex

In Lemma TPV every path is closed inside the run. In the P-vortex a set R_t of *rooted* vertices never retires, and paths whose two ends are rooted are not closed inside the run: they are returned as *arcs*, each carrying one of four phases, to be closed later by a different device through sets of vertices that are marked by an external phase map φ . The only property of φ that the lemma uses is the hypothesis (E1') below.

Lemma 4.3 (Lemma PV (the four-phase P-vortex)). *Let Z be a set of $N \geq N_0$ vertices and $L := \log_2 N$. Put*

$$J := \lfloor \log_2(L/8) \rfloor, \quad m := \lceil L^6 \rceil, \quad b := \lceil 2^7 L^2 m \rceil, \quad t := 2^9 L^8.$$

The following data are fixed (not random):

- a spanning $(2^{-6}, s_O)$ -expander O on Z with $2m \leq s_O$, and sets $\mathcal{A}(w) \subseteq N_O(w)$ ($w \in Z$) with $|\mathcal{A}(w)| = m$ such that every vertex lies in at most b of them (such sets exist by Lemma 3.9 with $\varepsilon' = 2^{-6}$ and $m = m$);
- pairwise edge-disjoint graphs $R_{j,c}$ ($0 \leq j < J$, $c \in [4]$) and M on Z , the own classes, each a spanning $(2^{-6}, s')$ -expander on Z with $s' \geq 2^{145} L^{41}$;
- a partition $Z = R_t \sqcup P_l$ into rooted and parentless vertices;

- an external phase map $\varphi : Z \rightarrow [4] \cup \{*\}$.

For $w \in Z$ let $C_w := \{wu : u \in \mathcal{A}(w), wu \in E(M), \varphi(u) = *\}$, and assume

(E1') $|C_w| \geq L^5/8$ for every $w \in Pl$.

The labels of the run are the following independent random variables: for every $v \in Pl$ a level $lv(v) \in \{0, 1, \dots, J\}$ with $\mathbb{P}(lv(v) \geq j) = 2^{-j}$ for $0 \leq j \leq J$ (that is, $\mathbb{P}(lv(v) = j) = 2^{-(j+1)}$ for $0 \leq j < J$ and $\mathbb{P}(lv(v) = J) = 2^{-J}$); for every $v \in Z$ a label $k(v) \in \{0, 1, 2, 3, 4\}$ with $\mathbb{P}(k(v) = 0) = 1/2$ and $\mathbb{P}(k(v) = c) = 1/8$ for $c \in [4]$. Put

$$\begin{aligned} U_j &:= Rt \cup \{v \in Pl : lv(v) \geq j\}, & W_j &:= \{v \in Pl : lv(v) = j\}, \\ Z_j &:= \{u \in U_{j+1} : k(u) = 0\}, & V_{j,c} &:= \{u \in U_{j+1} : k(u) = c\}. \end{aligned}$$

Let $\mathcal{G}_{PV}$ be the event that the following three conditions hold:

(G2) for all $0 \leq j < J$ and $c \in [4]$, $R_{j,c}$ is $(2^{12}L^4, t)$ -path connected through $V_{j,c}$;

(G4) for all $0 \leq j < J$ and $w \in Pl$, the set $C_w^j := \{wu \in C_w : u \in Z_j\}$ has at least 8 elements;

(G5) $\sum_{j < J} |Pl \cap U_j| \leq 8|Pl|$ and $|Pl \cap U_J| \leq 64|Pl|/L$.

The event $\mathcal{G}_{PV}$ is determined by the own classes, the sets $\mathcal{A}(w)$, the partition, φ and the labels, and

$$\mathbb{P}(\mathcal{G}_{PV}) \geq \frac{1}{2} - \eta_{PV}(N), \quad \eta_{PV}(N) := 2^{95}L^{31}N^{-3} + NL e^{-L^4/16} \leq \frac{1}{100}. \quad (4.6)$$

On $\mathcal{G}_{PV}$, for every edge set H_0 all of whose edges have both ends in Z and which contains $E(M)$ and every $E(R_{j,c})$, there is a partition $H_0 = H^{obj} \sqcup H^{arc}$ and a family $\mathfrak{A}$ of paths (arcs) such that:

- (a) H^{obj} decomposes into at most $369|Pl|$ objects;
- (b) H^{arc} is the disjoint union of the edge sets of the arcs; the arcs are pairwise edge-disjoint paths of length at least 1, each has two distinct ends, both in Rt , and each carries a phase $c \in [4]$ such that every vertex x of a phase- c arc, ends included, satisfies $\varphi(x) \neq c$;
- (c) $|\mathfrak{A}| \leq 4(J+1)N$, and every vertex x is an end of at most $\deg_{H_0}(x) \leq |Z| - 1$ arcs;
- (d) if $Rt = \emptyset$ then $\mathfrak{A} = \emptyset$; if $Pl = \emptyset$ then $H^{obj} = \emptyset$.

The partition, the decomposition and the arcs are obtained from H_0 and the labels by a deterministic procedure.

Dependencies: 3.8, 3.9, 3.1, 1.22, 1.6, 1.8, 1.26, (Γ3), 4.1, 4.2, 1.3, 1.31.

Proof. The proof follows that of Lemma 4.2, with three changes: rooted vertices never retire; the classing uses four phases and also avoids the external phases of both ends; and paths with two rooted ends that do not come from cherries are returned as arcs instead of being closed.

Size conditions. We use: (i) $L \geq 2^{10}$; (ii) $N \geq L^3$; (iii) $4 \cdot 64 \leq L$ (a consequence of (i)); (iv) $\eta_{PV}(N) \leq 1/100$. They hold for $N \geq N_0$ by the choice of N_0 .

Parameters. By the definition of J , $8 \cdot 2^J \leq L < 16 \cdot 2^J$, so

$$2^{-J} \geq 8/L, \quad 2^{-J} < 16/L, \quad 1 \leq J \leq \log_2 L, \quad 4J \leq L \quad (4.7)$$

(the last two by (i)). Lemma 3.9 with $\varepsilon' = 2^{-6}$ and $m = \mathbf{m} \leq s_O/2$ gives sets as in the statement with multiplicity at most $\lceil 2\mathbf{m}L^2/2^{-6} \rceil = \mathbf{b}$. Since $\mathbf{b} \leq 2^7 L^2(L^6 + 1) + 1$,

$$2 + \mathbf{b} \leq 2^7 L^8 + 2^7 L^2 + 3 \leq 2^9 L^8 = \mathbf{t}. \quad (4.8)$$

As in Lemma 4.2: $U_0 = Z$, $U_{j+1} \subseteq U_j$, $U_j = U_{j+1} \sqcup W_j$, $W_j \subseteq \text{Pl}$, $W_j \cap U_{j+1} = \emptyset$, $\text{Rt} \subseteq U_j$ for every j , and $Z_j, V_{j,1}, \dots, V_{j,4}$ partition U_{j+1} .

Probability of $\mathcal{G}_{\text{PV}}$. If $\text{Pl} = \emptyset$, (G4) and (G5) hold surely, and the bound for (G2) below covers this case as well; in the items (G4) and (G5) below we assume $\text{Pl} \neq \emptyset$.

(G2) Fix (j, c) , and let $\mathcal{K}_{j,c}$ be the family of sets $T \subseteq Z$ such that $R_{j,c}$ is $(2^{12}L^4, \mathbf{t})$ -path connected through T . It is fixed, since the own classes are fixed data. It is closed upwards by Lemma 3.1(ii), applied with $G = G' := R_{j,c}$, $\ell = \ell' := 2^{12}L^4$, $t = t' := \mathbf{t}$ and with $T \subseteq T' \subseteq Z = V(R_{j,c})$ as the two sets through which path connectivity is taken (the own classes are spanning). For $v \in Z$ the event $v \in V_{j,c}$ depends only on $k(v)$ and, if $v \in \text{Pl}$, on $\text{lv}(v)$, so these events are mutually independent over v ; and $\mathbb{P}(v \in V_{j,c}) = 1/8 \geq 1/L$ if $v \in \text{Rt}$ and $\mathbb{P}(v \in V_{j,c}) = 2^{-(j+1)}/8 \geq 2^{-J}/8 \geq 1/L$ if $v \in \text{Pl}$, by (4.7) and size condition (i). Let V' be a $(1/L)$ -random subset of Z . Theorem 3.8 applies to $X := R_{j,c}$ (a $(2^{-6}, s')$ -expander), $V := V'$, $\rho := 1/L$, $t := \mathbf{t}$, because $\rho N \geq L^2$ by size condition (ii) and $s' \geq 2^{145}L^{41} \geq 2^{135} \cdot 2^9 L^8 \cdot L^{28} \cdot L^5 = 2^{135} \mathbf{t} L^{28} \rho^{-5}$; so $\mathbb{P}(V' \notin \mathcal{K}_{j,c}) \leq 2^{86} \cdot 2^9 L^8 \cdot L^{19} \cdot L^3 \cdot N^{-3} = 2^{95} L^{30} N^{-3}$. By observation (MC) (with $S := Z$, $\mathcal{K} := \mathcal{K}_{j,c}$, the random set $V_{j,c}$ in place of V , and $\rho := 1/L$), also $\mathbb{P}(V_{j,c} \notin \mathcal{K}_{j,c}) \leq 2^{95} L^{30} N^{-3}$. $R_{j,c}$ fails to be $(2^{12}L^4, \mathbf{t})$ -path connected through $V_{j,c}$ with probability at most $2^{95} L^{30} N^{-3}$. There are $4J \leq L$ pairs (j, c) . By Lemma 3.1(iii) the pair multisets may be chosen after the labels are revealed.

(G4) Fix $j < J$ and $w \in \text{Pl}$. The set C_w is fixed and has at least $L^5/8$ elements by (E1'). For $wu \in C_w$ the events $u \in Z_j$ depend on the labels of distinct vertices $u \neq w$, so they are independent, and $\mathbb{P}(u \in Z_j) = \frac{1}{2}\mathbb{P}(u \in U_{j+1}) \geq \frac{1}{2}\min(1, 2^{-J}) \geq 4/L$ by (4.7). As $(L^5/8)(4/L) = L^4/2 \geq 16$, observation (B) gives $\mathbb{P}(|C_w^j| < 8) \leq e^{-L^4/16}$. There are at most $NJ \leq NL$ pairs (w, j) .

(G5) $\mathbb{E} \sum_{j < J} |\text{Pl} \cap U_j| < 2|\text{Pl}|$ and $\mathbb{E}|\text{Pl} \cap U_J| = 2^{-J}|\text{Pl}| < 16|\text{Pl}|/L$ by (4.7); by Markov's inequality (Cited result 1.26) each inequality of (G5) fails with probability less than $1/4$.

Summing gives (4.6). The three events involve only the own classes, C_w (hence $\mathcal{A}(w)$, $\mathbf{M}$ and φ), the partition and the labels; none involves H_0 . Fix labels in $\mathcal{G}_{\text{PV}}$ and an edge set H_0 as in the statement.

The process. All choices are made by fixed rules. An edge is *used* once it has been placed into an output object or into an arc. Put $H_{j+1} := H_j \setminus \{\text{edges used at step } j\}$, where $H_0 := H_0$. Before step j ($0 \leq j \leq J$) we maintain:

- (Inv1) every edge of H_j has both ends in U_j ;
- (Inv2) every edge of $R_{j',c}$ with $j' \geq j$, $c \in [4]$ and both ends in $U_{j'+1}$ lies in H_j ;
- (Inv3) every $\mathbf{M}$ -edge wu with $w \in W_{j'}$ for some $j' \geq j$ and $u \in U_{j'+1}$ lies in H_j .

They hold for $j = 0$ because $U_0 = Z$ and H_0 contains all own classes.

Step j ($0 \leq j < J$). Let F_j be the set of edges of H_j meeting W_j ; by (Inv1) all their ends lie in U_j .

- (i) *Reserve.* For each $w \in W_j$ choose eight distinct edges $e_w^{c,i} \in C_w^j$ ($c \in [4]$, $i \in [2]$), possible by (G4). Each is an M-edge wu with $w \in W_j$, $u \in Z_j \subseteq U_{j+1}$, so it lies in H_j by (Inv3), hence in F_j ; w is its only end in W_j , so reserved edges of distinct w are distinct. Its far end u satisfies $k(u) = 0$ and $\varphi(u) = *$.
- (ii) *Phases.* For every non-reserved edge of F_j choose an end $w \in W_j$, call the other end x , and give the edge a phase $\gamma \in [4] \setminus \{k(x), \varphi(x), \varphi(w)\}$; at most three of the four values are excluded, so a phase exists. Let $F_{j,c}$ be the set of non-reserved edges of phase c . Then, for every $c \in [4]$,

$$V(F_{j,c}) \cap V_{j,c} = \emptyset \quad \text{and} \quad \varphi(y) \neq c \text{ for every } y \in V(F_{j,c}), \quad (4.9)$$

the first as in (4.4), the second because both ends of a phase- c edge were constrained.

- (iii) *Paths.* Let $\mathcal{P}_{j,c}^v$ be a Corollary-22 decomposition of $F_{j,c}$; by (E), $|\mathcal{P}_{j,c}^v| \leq |U_j| \leq N$.
- (iv) *Ends in W_j .* Exactly as in step (iv) of the proof of Lemma 4.2, for every $w \in W_j$ and $c \in [4]$: if w is an end of two paths of $\mathcal{P}_{j,c}^v$, append $e_w^{c,1}$ to one and $e_w^{c,2}$ to the other at w ; if of one, append $e_w^{c,1}$ and output $e_w^{c,2}$ as a single edge; if of none, form the cherry $u_1 w u_2$ from $e_w^{c,1} = w u_1$ and $e_w^{c,2} = w u_2$. The resulting trails have pairwise distinct inner vertices and pairwise distinct edges.
- (v) *Splitting.* By (S) each trail splits into at most two cycles (output) and at most one path with the same ends as the trail; let $\tilde{\mathcal{P}}_{j,c}^v$ be the set of these paths. Exactly as in the proof of Lemma 4.2 (with (4.9) and (4.8) in place of (4.4) and (4.3)): (P1) no vertex of a path in $\tilde{\mathcal{P}}_{j,c}^v$ lies in $V_{j,c}$; (P2) its two ends are distinct and lie in U_{j+1} (an end is an end of a path of $\mathcal{P}_{j,c}^v$ not in W_j , hence in $U_j \setminus W_j = U_{j+1}$, or a far end); (P3) it has length at least 2; (P4) every vertex is an end of at most $2 + b \leq t$ paths of $\tilde{\mathcal{P}}_{j,c}^v$. Moreover:
 - (P5) if $Q \in \tilde{\mathcal{P}}_{j,c}^v$ does not come from a cherry, then $\varphi(y) \neq c$ for every vertex y of Q : the vertices of its trail are vertices of $F_{j,c}$ (use (4.9)) and far ends (where $\varphi = *$).
- (vi) *Arcs and closing.* A path $Q \in \tilde{\mathcal{P}}_{j,c}^v$ that does not come from a cherry and has both ends in R_t is declared an *arc of phase c* ; its edges are put into H^{arc} . Every other path of $\tilde{\mathcal{P}}_{j,c}^v$ (one with an end in Pl , or one coming from a cherry) is closed: the multiset of their end pairs has every vertex in at most t pairs by (P4), so by (G2) there are pairwise edge-disjoint connectors $Q' \subseteq R_{j,c}$ joining the ends of these paths, of length at most $2^{12}L^4$, with inner vertices in $V_{j,c}$. As in step (vi) of the proof of Lemma 4.2, every connector edge is an $R_{j,c}$ -edge with both ends in U_{j+1} , so it lies in H_j by (Inv2), is not in F_j and is not reserved; by (P1), (P3) and (C), $Q \cup Q'$ is a cycle, which is output.

Closing cherries inside the run (rather than declaring a cherry with two rooted ends an arc) guarantees (b) for the centre w of the cherry, for which $\varphi(w) = c$ is not excluded.

The edges used at step j are exactly the edges of F_j and the connector edges, all of them edges of H_j ; as in the proof of Lemma 4.2, the objects and arcs of the step are pairwise edge-disjoint. The invariants persist by the argument given there: an edge of H_{j+1} lies in $H_j \setminus F_j$, so has both ends in $U_j \setminus W_j = U_{j+1}$; step j used only edges meeting W_j and $R_{j,-}$ -edges, while edges protected by (Inv2), (Inv3) for indices $j' \geq j+1$ have both ends in U_{j+1} and are not $R_{j,-}$ -edges.

Cost of step j . There are at most $4|W_j|$ single edges. For a phase c , a trail produces objects only if (1) it is a cherry or has an appended edge, or (2) it is a path of $\mathcal{P}_{j,c}^v$ without appended

edges having an end in Pl . A trail of type (1) contains at least one of the at most $2|W_j|$ reserved edges of phase c used in (iv), and each of these edges lies in exactly one trail (a trail may contain two of them: a cherry, or a path of $\mathcal{P}_{j,c}^\vee$ with both ends in W_j); so there are at most $2|W_j|$ such trails. An end in Pl of a trail of type (2) is not in W_j (ends in W_j receive appended edges), so it lies in $\text{Pl} \cap U_{j+1}$, and each vertex is an end of at most two paths of $\mathcal{P}_{j,c}^\vee$; so there are at most $2|\text{Pl} \cap U_{j+1}|$ such trails. Every remaining trail is a path of $\mathcal{P}_{j,c}^\vee$ with both ends in Rt and without appended edges; it is not split ($\text{rep} = 0$), and it becomes an arc. Each trail of type (1) or (2) gives at most three objects (at most two split cycles and at most one closed cycle). Hence, using $W_j \sqcup (\text{Pl} \cap U_{j+1}) = \text{Pl} \cap U_j$, step j outputs at most

$$4|W_j| + 4 \cdot 3(2|\text{Pl} \cap U_{j+1}| + 2|W_j|) \leq 28|\text{Pl} \cap U_j| \quad (4.10)$$

objects. (No separate allowance is needed for split cycles of trails with two rooted ends: by the case analysis just given such trails are either of type (1), and already counted, or not split at all. The bound $28|\text{Pl} \cap U_j|$ is the one used in (a) below.)

Finish. By (Inv1) for $j = J$, every edge of H_J has both ends in $U_J = \text{Rt} \sqcup \text{Pl}_J$, where $\text{Pl}_J := \text{Pl} \cap U_J$. Split $H_J = E_1 \sqcup E_2$, where E_1 is the set of edges with both ends in Pl_J ; every edge of E_2 has an end in Rt .

- By (G5), $|\text{Pl}_J| \leq 64|\text{Pl}|/L$. Apply Fact 1.6(b) with $F := E_1$, $W := \text{Pl}_J$, $\beta := 64$ and $\alpha := |\text{Pl}| \leq N$; its hypothesis $N > 1$ holds as $L \geq 2^{10}$ by size condition (i), and its hypothesis $4\beta \leq L$ is size condition (iii). So E_1 decomposes into at most $64|\text{Pl}|$ objects, which are output.
- Give every edge $xy \in E_2$ a phase $c \in [4] \setminus \{\varphi(x), \varphi(y)\}$ (at most two values are excluded) and let $E_{2,c}$ be the set of edges of phase c . Take a Corollary-22 decomposition of each $E_{2,c}$. Let T be one of its paths. If an end p of T lies in Pl_J , the edge of T at p lies in E_2 , so its other end lies in Rt ; output this edge as a single edge (*stripping*). After stripping at the ends of T lying in Pl_J , what remains of T has no edges or is a path of length at least 1 with two distinct ends in Rt (every end of T lies in $\text{Rt} \sqcup \text{Pl}_J$, and a stripped end is replaced by a vertex of Rt); in the latter case it is declared an arc of phase c . Every vertex y of such an arc is an end of an edge of phase c , so $\varphi(y) \neq c$. Since every vertex of Pl_J is an end of at most two paths of each decomposition, at most $2|\text{Pl}_J|$ edges are stripped per phase, i.e. at most $8|\text{Pl}_J| \leq 16|\text{Pl}|$ single edges in total (also $8|\text{Pl}_J| \leq 8|\text{Pl}|$; the weaker factor 16 suffices, and nothing depends on it).

Conclusion. Every edge of H_0 is used exactly once: edges leave H_j only by being used, H_J is used entirely in the finish, and every used edge lies in H_0 (reserved edges lie in $F_j \subseteq H_j$ and connector edges in H_j). Let H^{arc} be the union of the edge sets of the arcs and $H^{\text{obj}} := H_0 \setminus H^{\text{arc}}$, the union of the output objects.

(a) By (4.10) and (G5), the steps $0 \leq j < J$ together output at most $28 \sum_{j < J} |\text{Pl} \cap U_j| \leq 28 \cdot 8|\text{Pl}| = 224|\text{Pl}|$ objects, and the finish outputs at most $64|\text{Pl}| + 16|\text{Pl}|$ objects. So the number of objects is at most $224|\text{Pl}| + 64|\text{Pl}| + 16|\text{Pl}| = 304|\text{Pl}| \leq 369|\text{Pl}|$.

(b) Arcs are pairwise edge-disjoint (each edge is used once), they are paths of length at least 1 with two distinct ends in Rt (by (P2) and the rule in (vi), and by the finish), and every vertex of a phase- c arc has $\varphi \neq c$ (by (P5), since arcs never come from cherries, and by the finish).

(c) At step j , each arc of phase c comes from a distinct path of $\mathcal{P}_{j,c}^\vee$, so there are at most N of them; in the finish there are at most $|U_J| \leq N$ arcs per phase by (E). Hence $|\mathcal{A}| \leq 4JN + 4N = 4(J+1)N$. For the per-vertex bound, let $x \in Z$. By (b) the arcs are pairwise edge-disjoint paths with two distinct ends, and their edges lie in $H^{\text{arc}} \subseteq H_0$. An arc with end x contains exactly one

edge at x (a path meets each of its two ends in exactly one edge), and distinct arcs are edge-disjoint; so x is an end of at most $\deg_{H_0}(x)$ arcs. Every edge of H_0 at x joins x to a vertex of $Z \setminus \{x\}$, and distinct edges at x have distinct other ends (edge sets are sets of edges of a simple graph, Convention 1.3(a)); hence $\deg_{H_0}(x) \leq |Z| - 1$. (Counting only ends of Corollary-22 paths and far ends of reserved edges, which would bound the arc ends at x by $8J + 8 + \mathbf{b}$, does not suffice. That count misses the arcs left by stripping in the finish: such an arc can end at a vertex $x \in \text{Rt}$ that is an interior vertex of its Corollary-22 path, and Corollary 22 does not limit the number of such paths at x . The bound through degrees covers all arcs.)

(d) Arcs have their ends in Rt , so there are none if $\text{Rt} = \emptyset$. If $\text{Pl} = \emptyset$ then every W_j is empty, the steps use nothing, $\text{Pl}_j = \emptyset$, and $E_1 = \emptyset$; no edge is stripped, and all of $H_0 = H_j = E_2$ is partitioned into arcs, so $H^{\text{obj}} = \emptyset$. $\square$

4.4 Theorem VX^+ : vortex absorption with arbitrary extra edges

Theorem 4.4 (Theorem VX^+ (sharpened vortex absorption with arbitrary extra edges)). *Let Z be a set of $N \geq N_0$ vertices, $L := \log_2 N$, and let O be a spanning (ε_O, s) -expander on Z , where either*

$$\varepsilon_O = 2^{-5} \quad \text{and} \quad s \geq 2^{146} L^{38} \log_2 L, \quad \text{or} \quad \varepsilon_O = 2^{-6} \quad \text{and} \quad s \geq 2^{151} L^{38} \log_2 L.$$

Put $J := \lfloor \log_2(L/6) \rfloor$, $k := 3J + 1$, $\mathbf{m} := \lceil L^3 \rceil$, $\mathbf{b} := \lceil 2\mathbf{m}L^2/\varepsilon_O \rceil$ (that is, $\lceil 64L^2\mathbf{m} \rceil$, resp. $\lceil 128L^2\mathbf{m} \rceil$) and $\mathbf{t} := 2^8 L^5$, resp. $\mathbf{t} := 2^9 L^5$. The labels of the run are independent: a uniform colouring of $E(O)$ with the k colours $\mathbf{R}_{j,c}$ ($0 \leq j < J$, $c \in [3]$) and $\mathbf{M}$; for every $v \in Z$ a level $\text{lv}(v) \in \{0, 1, \dots, J\}$ with $\mathbb{P}(\text{lv}(v) \geq j) = 2^{-j}$ for $0 \leq j \leq J$ (that is, $\mathbb{P}(\text{lv}(v) = j) = 2^{-(j+1)}$ for $0 \leq j < J$ and $\mathbb{P}(\text{lv}(v) = J) = 2^{-J}$); for every $v \in Z$ a label $\mathbf{k}(v) \in \{0, 1, 2, 3\}$ with $\mathbb{P}(\mathbf{k}(v) = 0) = 1/2$ and $\mathbb{P}(\mathbf{k}(v) = c) = 1/6$ ($c \in [3]$). There is an event $\mathcal{G}_{\text{VX}}$, determined by O and the labels only, with

$$\mathbb{P}(\mathcal{G}_{\text{VX}}) \geq 1 - \eta_{\text{VX}}(N) \geq \frac{1}{2}, \quad \text{where} \quad (4.11)$$

$$\eta_{\text{VX}}(N) := 2LN^{-5} + 2^{95}L^{28}N^{-3} + NL e^{-3L^2/(32 \log_2 L)} + L e^{-N/(5000L)},$$

such that on $\mathcal{G}_{\text{VX}}$ the following holds simultaneously for every graph G with $V(G) = Z$ and $E(O) \subseteq E(G)$ (the edges of $E(G) \setminus E(O)$ are arbitrary): $E(G)$ decomposes into at most

$$38.4N + 13N \leq 52N \leq 80N$$

objects, all of which are cycles except at most $N + 13N/L = N + o(N)$ single edges. In particular $f(G) \leq 80N$.

Dependencies: 3.2, 3.8, 3.9, 3.1, 1.22, 1.6, 1.8, (F3), 4.1, 1.3, 1.31.

Proof. Size conditions. We use: (i) $L \geq 2^{10}$; (ii) $N \geq L^3$; (iii) $4 \cdot 13 \leq L$ (a consequence of (i)); (iv) $\eta_{\text{VX}}(N) \leq 1/100$. They hold for $N \geq N_0$ by the choice of N_0 .

Parameters. By the definition of J , $6 \cdot 2^J \leq L < 12 \cdot 2^J$, so $2^{-J} \geq 6/L$, $2^{-J} < 12/L$, $1 \leq J$ and $J + 1 \leq L$; moreover $k = 3J + 1 \leq 3 \log_2 L + 1 \leq 4 \log_2 L$, so $2k \leq 8 \log_2 L$ and $k \leq L$. Since $s \geq 2 \lceil L^3 \rceil = 2\mathbf{m}$, Lemma 3.9 (with $\varepsilon' = \varepsilon_O \geq 2^{-7}$, $m = \mathbf{m}$) gives sets $\mathcal{A}(w) \subseteq N_O(w)$, $|\mathcal{A}(w)| = \mathbf{m}$, with every vertex in at most $\mathbf{b}$ of them; we fix the lexicographically first such family. As $\mathbf{m} \leq L^3 + 1$, we have $\mathbf{b} \leq 2^6 L^5 + 2^6 L^2 + 1$ (resp. $\mathbf{b} \leq 2^7 L^5 + 2^7 L^2 + 1$), hence

$$2 + \mathbf{b} \leq 2L^2\mathbf{m}/\varepsilon_O + 3 \leq \mathbf{t}. \quad (4.12)$$

Put $\mathbf{U}_j := \{v : \text{lv}(v) \geq j\}$, $\mathbf{W}_j := \mathbf{U}_j \setminus \mathbf{U}_{j+1}$, $\mathbf{Z}_j := \{u \in \mathbf{U}_{j+1} : \mathbf{k}(u) = 0\}$ and $\mathbf{V}_{j,c} := \{u \in \mathbf{U}_{j+1} : \mathbf{k}(u) = c\}$; thus $\mathbf{U}_0 = Z$, $\mathbf{W}_j \cap \mathbf{U}_{j+1} = \emptyset$ and $\mathbf{Z}_j, \mathbf{V}_{j,1}, \mathbf{V}_{j,2}, \mathbf{V}_{j,3}$ partition $\mathbf{U}_{j+1}$.

Good events. Let $\mathcal{G}_{\text{VX}}$ be the intersection of:

- (G1) every colour class is a spanning $(\varepsilon_O, s/(2k))$ -expander on Z . As $s \geq 40kL$, Lemma 3.2 bounds the failure probability by $2kN^{-5} \leq 2LN^{-5}$. Note $s/(2k) \geq s/(8 \log_2 L) \geq 2^{143}L^{38}$ (resp. $\geq 2^{148}L^{38}$).
- (G2) every $R_{j,c}$ is $(2^{12}L^4, t)$ -path connected through $V_{j,c}$. Here the event $v \in V_{j,c}$ depends only on $(lv(v), k(v))$, so these events are mutually independent over v and independent of the colouring, and $\mathbb{P}(v \in V_{j,c}) = 2^{-(j+1)}/6 \geq 2^{-J}/6 \geq 1/L$ for every v . Fix a colouring χ in (G1) and a pair (j, c) , and let $\mathcal{K}_{j,c}$ be the family of sets $T \subseteq Z$ such that $R_{j,c}$ is $(2^{12}L^4, t)$ -path connected through T . It is determined by χ . It is closed upwards by Lemma 3.1(ii), applied with $G = G' := R_{j,c}$, $\ell = \ell' := 2^{12}L^4$, $t = t' := t$ and with $T \subseteq T' \subseteq Z = V(R_{j,c})$ as the two sets through which path connectivity is taken (the colour classes are spanning). Let V' be a $(1/L)$ -random subset of Z and apply Theorem 3.8 with $X := R_{j,c}$, $V := V'$, $\rho := 1/L$, $t := t$: its hypotheses are $N/L \geq L^2$ (by size condition (ii)) and $s/(2k) \geq 2^{135}tL^{28}L^5$, i.e. $2^{143}L^{38}$ for $t = 2^8L^5$ and $2^{144}L^{38}$ for $t = 2^9L^5$, both covered by (G1). So $\mathbb{P}(V' \notin \mathcal{K}_{j,c}) \leq 2^{86}tL^{19}L^3N^{-3} \leq 2^{95}L^{27}N^{-3}$. By observation (MC) (with $S := Z$, $\mathcal{K} := \mathcal{K}_{j,c}$, the random set $V_{j,c}$ in place of V , and $\rho := 1/L$), also $\mathbb{P}(V_{j,c} \notin \mathcal{K}_{j,c}) \leq 2^{95}L^{27}N^{-3}$, the probability being over the levels and the labels k . As these are independent of the colouring, a union bound over the $3J \leq L$ pairs (j, c) shows that the probability that (G1) holds and (G2) fails is at most $\sum_{\chi} \mathbb{P}(\text{the colouring is } \chi) \cdot 3J \cdot 2^{95}L^{27}N^{-3} \leq 2^{95}L^{28}N^{-3}$, the sum running over the colourings χ in (G1). By Lemma 3.1(iii), pair multisets may be chosen after the labels are revealed.
- (G3) $|U_j| \leq 1.01 \cdot 2^{-j}N$ for $0 \leq j \leq J$. $|U_j|$ is binomial with mean $2^{-j}N \geq 2^{-J}N \geq 6N/L$; by Cited result 1.8 with $\delta = 1/100$ it exceeds 1.01 times its mean with probability at most $e^{-10^{-4} \cdot 6N/(3L)} \leq e^{-N/(5000L)}$; there are $J+1 \leq L$ values of j .
- (G4) for every $0 \leq j < J$ and every $w \in Z$ the set $C_w^j := \{wu : u \in \mathcal{A}(w), wu \in M, u \in Z_j\}$ has at least 9 elements. For $u \in \mathcal{A}(w)$ the events $\{wu \in M, u \in Z_j\}$ are independent (distinct edges wu and distinct vertices $u \neq w$), each of probability $\frac{1}{k} \cdot 2^{-(j+1)} \cdot \frac{1}{2} \geq \frac{1}{4 \log_2 L} \cdot \frac{3}{L}$. As $m \cdot \frac{3}{4L \log_2 L} \geq \frac{3L^2}{4 \log_2 L} \geq 18$, observation (B) bounds the failure probability for one pair (w, j) by $e^{-3L^2/(32 \log_2 L)}$; there are at most NL pairs.

These events are determined by O (through the colouring and $\mathcal{A}(\cdot)$) and the labels; summing the failure probabilities gives (4.11), and $\eta_{VX}(N) \leq 1/100 \leq 1/2$ by (iv). Fix labels in $\mathcal{G}_{VX}$ and a graph G with $V(G) = Z$ and $E(O) \subseteq E(G)$.

The process. All choices are made by fixed rules. Put $H_0 := E(G)$ and let H_{j+1} be H_j minus the edges used at step j . Before step j ($0 \leq j \leq J$) we maintain: (Inv1) every edge of H_j has both ends in U_j ; (Inv2) every $R_{j',c}$ -edge with $j' \geq j$ and both ends in $U_{j'+1}$ lies in H_j ; (Inv3) every M -edge wu with $w \in W_{j'}$ for some $j' \geq j$ and $u \in U_{j'+1}$ lies in H_j . They hold for $j = 0$ since $H_0 \supseteq E(O)$ and $U_0 = Z$.

Step j ($0 \leq j < J$). Let F_j be the set of edges of H_j meeting W_j . By (Inv3), $C_w^j \subseteq F_j$ for every $w \in W_j$; every edge of C_w^j has exactly one end in W_j (its other end lies in $Z_j \subseteq U_{j+1}$), so the sets C_w^j ($w \in W_j$) are pairwise disjoint.

- (a) *Parity.* For every $w \in W_j$ with $\deg_{F_j}(w)$ odd, output one edge of C_w^j as a single edge (the *parity edge* of w). Let F'_j be F_j minus the parity edges. As each parity edge has exactly one end in W_j , $\deg_{F'_j}(w)$ is even for every $w \in W_j$.
- (b) *Reserved and flexible edges.* For every $w \in W_j$ choose eight further distinct edges of C_w^j (possible by (G4), since $|C_w^j| \geq 9$): *reserved* edges $e_w^{c,1}, e_w^{c,2}$ ($c \in [3]$) and *flexible* edges f_w, f'_w .

These *chosen* edges of distinct w are distinct; the end of a chosen edge in Z_j is its *far end*, and $k = 0$ there.

- (c) *Classes.* Every edge of F'_j that is not chosen receives, once and for all, a class: choose an end $w \in W_j$, call the other end x , and pick $\gamma \in [3] \setminus \{k(x)\}$. For $w \in W_j$ let $d_c(w)$ be the number of such classed edges of class c at w (an edge with both ends in W_j counts at both ends). Then $d_1(w) + d_2(w) + d_3(w) = \deg_{F'_j}(w) - 8$ is even, so the number of odd $d_c(w)$ is 0 or 2. If $d_c(w)$ and $d_{c'}(w)$ are odd ($c \neq c'$), give f_w class c and f'_w class c' ; otherwise give both class 1. A flexible edge has only one end in W_j , so these choices at different w do not interact. Let $F_{j,c}$ be the set of class- c edges (classed and flexible; reserved edges excluded). Then for all $w \in W_j$ and $c \in [3]$, $\deg_{F_{j,c}}(w)$ is even, and $V(F_{j,c}) \cap V_{j,c} = \emptyset$ (ends in W_j are outside U_{j+1} , other ends x of classed edges have $k(x) \neq c$, far ends have $k = 0$).
- (d) *Paths.* Let $\mathcal{P}_{j,c}^v$ be a Corollary-22 decomposition of $F_{j,c}$. By (E) and (Inv1), $|\mathcal{P}_{j,c}^v| \leq |U_j|$, and every $w \in W_j$ is an end of an even number, hence of 0 or 2, paths of $\mathcal{P}_{j,c}^v$.
- (e) *Moving ends out of W_j .* For $w \in W_j$ and $c \in [3]$: if w is an end of two paths of $\mathcal{P}_{j,c}^v$, append $e_w^{c,1}$ at w to one and $e_w^{c,2}$ at w to the other; if of none, form the cherry $u_1 w u_2$ from $e_w^{c,1} = w u_1$, $e_w^{c,2} = w u_2$ ($u_1 \neq u_2$). Every resulting trail $x_0 \cdots x_q$ has pairwise distinct edges and pairwise distinct inner vertices (those of one path of $\mathcal{P}_{j,c}^v$, or the centre of a cherry). By (S) it splits into at most two cycles, which are output, and at most one path with the ends $x_0 \neq x_q$; let $\tilde{\mathcal{P}}_{j,c}^v$ be the set of these paths. Every $Q \in \tilde{\mathcal{P}}_{j,c}^v$ satisfies: (P1) $V(Q) \cap V_{j,c} = \emptyset$ (its vertices are vertices of $F_{j,c}$, cherry centres in W_j , or far ends); (P2) its ends are distinct and lie in U_{j+1} (ends of paths of $\mathcal{P}_{j,c}^v$ outside W_j , which lie in $U_j \setminus W_j = U_{j+1}$, or far ends in Z_j); (P3) its length is at least 2 (all its edges meet W_j , its ends do not); (P4) every vertex x is an end of at most $2 + b$ paths of $\tilde{\mathcal{P}}_{j,c}^v$ (at most two ends of paths of $\mathcal{P}_{j,c}^v$, plus at most one reserved edge $e_w^{c,i}$ with far end x for each of the at most b vertices w with $x \in \mathcal{A}(w)$; each trail yields at most one path, with the trail's ends).
- (f) *Closing.* For each c , the end pairs of $\tilde{\mathcal{P}}_{j,c}^v$ form a multiset of pairs of distinct vertices with every vertex in at most $2 + b \leq t$ pairs, by (P2), (P4) and (4.12). By (G2) there are pairwise edge-disjoint connectors $Q' \subseteq R_{j,c}$ joining them, of length at most $2^{12}L^4$, with inner vertices in $V_{j,c} \subseteq U_{j+1}$. All vertices of a connector lie in U_{j+1} , so its edges are $R_{j,c}$ -edges with both ends in U_{j+1} : they lie in H_j by (Inv2), are not in F_j and are not chosen edges (which are M-edges). By (P1), (P3) and (C), $Q \cup Q'$ is a cycle, which is output.

Every edge of F_j is used exactly once at step j (parity edges as single edges; reserved edges appended or in cherries; classed and flexible edges on paths of $\mathcal{P}_{j,c}^v$; every trail edge ends in a split cycle or a closed cycle), connectors of one class are edge-disjoint and connectors of distinct classes lie in distinct colour classes, and connectors avoid F_j . The invariants persist: an edge of H_{j+1} lies in $H_j \setminus F_j$, so both its ends lie in U_{j+1} ; step j used only edges meeting W_j and $R_{j,-}$ -edges, whereas the edges protected by (Inv2), (Inv3) for indices $j' \geq j + 1$ have both ends in U_{j+1} and are $R_{j',-}$ or M-edges.

Cost. Step j outputs at most $|W_j|$ parity edges and, for each c , at most $|\mathcal{P}_{j,c}^v| + |W_j| \leq |U_j| + |W_j|$ trails, each giving at most three objects. As $W_j \subseteq U_j$, this is at most $|W_j| + 9(|U_j| + |W_j|) \leq 19|U_j|$ objects. By (Inv1), after the last step every edge of H_j has both ends in U_j , and $|U_j| \leq 1.01 \cdot 2^{-j}N < 1.01 \cdot 12N/L \leq 13N/L$ by (G3) and $2^{-j} < 12/L$. By Fact 1.6(b) with $F := H_j$, $W := U_j$, $\beta := 13$ and $\alpha := N$ (its hypothesis $N > 1$ holds as $L \geq 2^{10}$ by size condition (i), and its hypothesis $4\beta \leq L$ is size condition (iii)), H_j decomposes into at most $13N$ objects, at most $|U_j| \leq 13N/L$ of which are

single edges. Every edge of $E(G) = H_0$ is used exactly once. By (G3) the total number of objects is at most

$$\sum_{j < J} 19 \cdot 1.01 \cdot 2^{-j} N + 13N \leq 38.4N + 13N \leq 52N.$$

The only single edges are the parity edges, at most $\sum_j |W_j| \leq N$ in total since the sets W_j are disjoint, and the at most $13N/L$ single edges of the finish; all other objects are cycles. So there are $N + o(N)$ single edges; the finish may add single edges beyond the at most N parity edges. $\square$

Remark 4.5 (Lemma TPV as a vortex-absorption statement; not used by the main argument). Let O be a spanning (ε_O, s) -expander on a set Z of $N \geq N_0$ vertices with $\varepsilon_O \in [2^{-7}, 2^{-5}]$ and $s \geq 2^{150} L^{42}$. Apply Lemma 4.2 with $P := Z$ and $Q := \emptyset$. On $\mathcal{G}_{\text{TPV}}$, for every edge set $E \supseteq E(O)$ inside Z , property (T1) gives $E^V = E$, and (T3) gives a decomposition of E into at most $169N$ objects. So Lemma 4.2 contains an absorption statement of the same kind as Theorem 4.4, with 169 in place of 80 and the threshold $2^{150} L^{42}$ in place of $2^{146} L^{38} \log_2 L$ (resp. $2^{151} L^{38} \log_2 L$); for $\varepsilon_O = 2^{-6}$ both thresholds apply to the same expanders once $s \geq 2^{150} L^{42}$. Consequently, wherever Theorem 4.4 is applied in this paper to an expander that also meets the threshold of Lemma 4.2, either result may be used; the choice changes only the coefficient (80 or 169) of the vertex mass to which the result is applied. In this paper Theorem 4.4 is applied only to parts whose total mass enters the final bound through a term εn with ε depending only on D_* and tending to 0 as $D_* \rightarrow \infty$, so the choice does not affect any constant of the cost line. This remark is not used in the proofs; the paper applies Theorem 4.4.

Dependencies: 4.2, 4.4.

5 Light parts: Theorem U machinery and K-RED

This section treats the light parts of the hierarchy $\text{HB}^{*\tau+}$. Each light part runs the four-phase P-vortex of Lemma 4.3 on its own edges. The paths that the vortex cannot close inside the part (the *arcs*, whose two ends are rooted in light parts two or more rounds back) are chained over one quotient multigraph whose nodes are all admissible parents of all earlier rounds. The connectors run through lent colour classes of the parents, inside random vertex sets (*zones*) that are pairwise disjoint over all parents of all rounds. Lemma K-RED, the last lemma of this section, collects the result: all edges outside the standalone pre-parts and outside the externally lent edges decompose into $(D_*/2 + 745 + \varepsilon_K(D_*))n + 80 \text{ dem} + 369 \text{ lp}$ objects, where *dem* and *lp* are stage-1 quantities whose weighted sum has expectation at most $\varepsilon_U(D_*)n$.

Setting of this section. Throughout, G is a graph on $n \geq N_0$ vertices with $d_1 \geq D_*$; D_* satisfies (T1)–(T4) (Condition 1.33); a valid $\text{HB}^{*\tau+}$ run on G is fixed (Definition 2.7); and the stage-1 lending data of Definition 3.10 are drawn. Every lemma below is stated for every outcome of the stage-1 data that it does not name; probabilities are taken only over the random labels named in each statement. For a light part Y we write $r = r(Y)$ for its round, $L_Y = \log_2 |Y|$, $H_Y = X_Y$, and we use $J_Y = \lfloor \log_2(L_Y/8) \rfloor$ and $k_{\text{own}} = 4J_Y + 1 \leq L_Y$ from Definition 3.10(iii). We shall use repeatedly the following consequences of Section 2: for a light part Y of round r ,

$$|Y| \geq P_r/2 \geq \lambda_r^{103}/2, \quad L_Y \geq 103 \log_2 \lambda_r - 1 \geq 102 \log_2 \lambda_r, \quad R - r \leq 2 \log^* d_r + 2 \leq \log_2 \lambda_r, \quad (5.1)$$

by Proposition 2.11(iv), $P_r = \lceil \lambda_r^{C'} \rceil$ with $C' = 103$, and Lemma 2.17(a),(d). In particular $R - r \leq L_Y/102$. Moreover, for every vertex v ,

$$\sum_{Y \ni v} L_Y^{-2} < \frac{1}{2}, \quad (5.2)$$

the sum over all light parts Y containing v . Indeed, by Proposition 2.11(iv), v lies in at most one light part per round; let Y_r denote the light part of round r containing v , if there is one. By (5.1), $L_{Y_r}^{-2} \leq (102 \log_2 \lambda_r)^{-2}$. For $r < R$, Lemma 2.17(a) gives $\lambda_r \geq d_{r+1}^{1/A} = 2^{\lambda_{r+1}/A}$, hence

$$\log_2 \lambda_r \geq \lambda_{r+1}/A = 2^{\log_2 \lambda_{r+1}/A} \geq 2 \log_2 \lambda_{r+1},$$

because $x := \log_2 \lambda_{r+1} \geq \log_2 \log_2 D_*$, so that $(\Gamma 1)(a),(b)$ (Condition (F1)) at x give $x \geq 2^8$ and $2^x \geq 2^{14} A x^3 \geq 2 A x$. Hence $\log_2 \lambda_r \geq 2^{R-r} \log_2 \lambda_R \geq 2^{R-r} \cdot 2^8$, by $(\Gamma 1)(a)$ at $\log_2 \lambda_R$, and

$$\sum_{Y \ni v} L_Y^{-2} \leq \sum_{r \leq R} (102 \log_2 \lambda_r)^{-2} \leq (102 \cdot 2^8)^{-2} \sum_{k \geq 0} 4^{-k} \leq \frac{4}{3} (102 \cdot 2^8)^{-2} < \frac{1}{2}.$$

5.1 Zones

Definition 5.1 (vertex choice, sublabels and zones; stage 1b). For a light part Y of round r (Definition 2.8) put $\varpi_Y := L_Y^{-2}$ and $T_Y^{\text{sl}} := \lceil L_Y^2 \rceil$ (as in Definition 3.10). Independently for every vertex v of G , and independently of all other stage-1 data (the colourings and the JS labels of Definition 3.10, and any further stage-1 labels), draw

$$\text{choice}(v) \in \{\text{none}\} \cup \{Y : Y \text{ a light part with } v \in Y \text{ and } r(Y) \leq R - 2\},$$

with $\mathbb{P}(\text{choice}(v) = Y) = \varpi_Y$ for each such Y and $\text{choice}(v) = \text{none}$ with the remaining probability (which is at least $1/2$, since $\sum_Y \varpi_Y \leq \sum_{Y \ni v} L_Y^{-2} < 1/2$ by (5.2), the first sum over the light parts Y available to $\text{choice}(v)$). If $\text{choice}(v) = Y$, the vertex v also draws, independently and uniformly, a *sublabel*

$$(l, c, \varsigma) \quad \text{with} \quad r(Y) + 2 \leq l \leq R, \quad c \in [4], \quad \varsigma \in [T_Y^{\text{sl}}].$$

The sublabels available to Y are exactly the indices of the family $\mathcal{I}_U$ of Y in Definition 3.10(ii). For such an index put

$$\text{Zone}_{Y,l,c,\varsigma} := \{v \in Y : \text{choice}(v) = Y \text{ and the sublabel of } v \text{ is } (l, c, \varsigma)\}.$$

For a round l and a vertex v we say that v carries a *round- l zone* if $\text{choice}(v) \neq \text{none}$ and the sublabel of v has first coordinate l ; the *phase* of that zone is then the second coordinate c . Equivalently, v carries a round- l zone of phase c iff $v \in \text{Zone}_{Y,l,c,\varsigma}$ for some light part Y and some ς . A *round- l phase- c zone* is any set $\text{Zone}_{Y,l,c,\varsigma}$.

Dependencies: 2.8, 3.10, 2.11, 2.17, (F1)

Lemma 5.2 (zones). (i) For every vertex v , $\sum_Y \varpi_Y \leq 1/2$, the sum over all light parts $Y \ni v$.

(ii) For every light part Y of round $r \leq R - 2$ and every sublabel (l, c, ς) of Y , the set $\text{Zone}_{Y,l,c,\varsigma}$ is exactly a ρ_Y -random subset of Y (each vertex of Y independently, product measure), where

$$\rho_Y := \frac{\varpi_Y}{(R - r - 1) \cdot 4 \cdot T_Y^{\text{sl}}} \geq \frac{1}{12 L_Y^5}.$$

- (iii) All zones $\text{Zone}_{Y,l,c,\varsigma}$, over all light parts Y and all sublabels (l, c, ς) of Y , are pairwise disjoint.
- (iv) The family of all zones is independent of the colourings (i)–(iii) and of the JS labels (iv) of Definition 3.10.

Dependencies: 5.1, 2.11, 2.17, (T1), 3.10

Proof. (i) Since $\varpi_Y = L_Y^{-2}$, this is (5.2), which is proved in the setting paragraph of this section. In particular the distribution of $\text{choice}(v)$ in Definition 5.1 is well defined, and $\mathbb{P}(\text{choice}(v) = \text{none}) \geq 1/2$.

(ii) Let Y have round $r \leq R - 2$ and fix a sublabel (l, c, ς) of Y . The number of sublabels of Y is $(R - r - 1) \cdot 4 \cdot T_Y^{\text{sl}}$, since l ranges over the $R - r - 1 \geq 1$ values $r + 2, \dots, R$. For $v \in Y$,

$$\begin{aligned} \mathbb{P}(v \in \text{Zone}_{Y,l,c,\varsigma}) &= \mathbb{P}(\text{choice}(v) = Y) \cdot \mathbb{P}(\text{sublabel } (l, c, \varsigma) \mid \text{choice}(v) = Y) \\ &= \frac{\varpi_Y}{(R - r - 1) \cdot 4 \cdot T_Y^{\text{sl}}} = \rho_Y. \end{aligned}$$

The event $\{v \in \text{Zone}_{Y,l,c,\varsigma}\}$ is a function of the pair $(\text{choice}(v), \text{sublabel of } v)$, and these pairs are independent over v . So $\text{Zone}_{Y,l,c,\varsigma}$ contains each vertex of Y independently with probability exactly ρ_Y . For the lower bound, (5.1) gives $R - r - 1 \leq L_Y/102$, and $T_Y^{\text{sl}} = \lceil L_Y^2 \rceil \leq 2L_Y^2$. Hence

$$\rho_Y \geq \frac{L_Y^{-2}}{(L_Y/102) \cdot 4 \cdot 2L_Y^2} = \frac{102}{8L_Y^5} \geq \frac{1}{12L_Y^5}.$$

(iii) Each vertex v has a single value $\text{choice}(v)$ and, if it is not none, a single sublabel. So v lies in at most one set $\text{Zone}_{Y,l,c,\varsigma}$, namely the one with $Y = \text{choice}(v)$ and (l, c, ς) its sublabel. Distinct zones are therefore disjoint.

(iv) The zones are functions of the variables $\text{choice}(v)$ and of the sublabels, which by Definition 5.1 are drawn independently of the colourings and of the JS labels. $\square$

5.2 Stage-2 statuses of light parts and their probabilities

Definition 5.3 (stage-2 statuses of light parts). Fix an outcome of the stage-1 data (the colourings and labels of Definition 3.10 and the choices and sublabels of Definition 5.1). Let Y be a light part of round r . Then $H_Y = X_Y$ is a $(2^{-6}, s_r/2)$ -expander on Y (Definition 2.8 and Proposition 2.11(i)). Put $\mathbf{m}_Y := \lceil L_Y^6 \rceil$ and $\mathbf{b}_Y := \lceil 2^7 L_Y^2 \mathbf{m}_Y \rceil$. Since $s_r \geq \lambda_r^{100}$ and $L_Y \leq 2\lambda_r$ (Lemma 2.17(b)), we have $2\mathbf{m}_Y \leq 2^8 \lambda_r^6 + 2 \leq s_r/2$. So Lemma 3.9, applied to X_Y with $\varepsilon' = 2^{-6}$ and $m = \mathbf{m}_Y$ (then $b = \lceil 2^7 L_Y^2 \mathbf{m}_Y \rceil = \mathbf{b}_Y$), provides sets $\mathcal{A}_Y(w) \subseteq N_{X_Y}(w)$, $w \in Y$, with $|\mathcal{A}_Y(w)| = \mathbf{m}_Y$ such that every vertex lies in at most $\mathbf{b}_Y$ of them. We fix one such family by a fixed rule; it depends only on the run. Let $\mathbf{M}_Y$ denote the own colour class $\mathbf{M}$ of Y (Definition 3.10(iii)).

- (E1) holds for Y if for every round l with $r \leq l \leq R$ and every $w \in Y$:

$$\#\{u \in \mathcal{A}_Y(w) : wu \in \mathbf{M}_Y \text{ and } u \text{ carries no round-}l \text{ zone}\} \geq L_Y^5/8.$$

Only the instance $l = r$ is used later in this section; the instances $l > r$ (the child rounds of Y) are included so that one event serves every round, and they are paid for in the failure bound below.

- Y is *demoted* if (E1) fails for Y or a COL(a) event fails for Y , that is, if one of Own_Y , Lend_Y is not a $(2^{-6}, s_r/8)$ -expander on Y , or some lent class of Y is not a $(2^{-6}, s_{\text{lent}})$ -expander on Y with $s_{\text{lent}} := s_r/(16k_{\text{lend}}(Y))$, or some own class of Y is not a $(2^{-6}, s_{\text{own}})$ -expander on Y with $s_{\text{own}} := s_r/(16k_{\text{own}})$ (Lemma 3.13(a) with $\varepsilon_Y = 2^{-6}$, $s_Y = s_r/2$).

- Y is *parent-bad* if for some (l, c, ς) with $r + 2 \leq l \leq R$, $c \in [4]$, $\varsigma \in [T_Y^{\text{sl}}]$, the class $\mathcal{L}_{Y,l,c,\varsigma}^U$ is not $(2^{12}L_Y^4, t_Y)$ -path connected through $\text{Zone}_{Y,l,c,\varsigma}$ (in the multiset sense of Cited result 1.9), where $t_Y = \lceil \lambda_r^{1.6} \rceil$ is the U-multiplicity parameter of Definition 3.10. If $r \geq R - 1$, the family $\mathcal{I}_U$ of Y is empty (Definition 3.10(ii)), so Y has no sublabels and no zones and is never parent-bad; Lemma 5.2(ii) is then vacuous, and Lemma 5.4(c) for Y reduces to (b).
- A *good parent* is a light part that is neither demoted nor parent-bad. Put

$$\text{Bad} := \{\text{demoted light parts}\} \cup \{\text{parent-bad light parts}\}.$$

- For a non-demoted light part Z of round l put

$$\text{Rt}(Z) := Z \cap \bigcup \{Y : Y \text{ a good parent of round } \leq l - 2\}, \quad \text{Pl}(Z) := Z \setminus \text{Rt}(Z),$$

and give every $v \in \text{Rt}(Z)$ the fixed parent $\text{par}(v)$, the good parent of the smallest round $\leq l - 2$ that contains v . (It is unique: v lies in at most one light part per round, Proposition 2.11(iv).) Since v lies in at most one light part of round l , $\text{par}(v)$ depends only on v and on the round l of the part in which it is used.

- $\text{dem} := \sum_{Y \text{ demoted}} |Y|$ and $\text{lp} := \sum_{Y \in \text{Bad}} |Y| (R - r(Y))$.

A pair (v, Z) with Z a light part of round l and $v \in \text{Pl}(Z)$ is exactly a pair that is parentless with respect to the set Bad in the sense of Proposition 2.20(iii): no light part outside Bad of a round $\leq l - 2$ contains v . Hence, by Proposition 2.20(iii),

$$\sum_Z |\text{Pl}(Z)| \leq 2n + \text{lp}, \tag{5.3}$$

the sum over all non-demoted light parts Z of all rounds. The statuses (E1), demoted, parent-bad, good parent, the set Bad , the sets $\text{Rt}(Z)$, $\text{Pl}(Z)$, the map par , and the numbers dem , lp are deterministic functions of the stage-1 outcome: (E1) depends only on the colouring of Y and on the choices and sublabels, $\text{COL}(a)$ only on the colouring of Y , and parent-bad only on the colouring of Y and the zones of Y . None of them depends on any edge set formed later (remainders, junk, lent edges used by consumers) or on any later random choice.

Dependencies: 5.1, 3.10, 3.9, 2.20, 2.8, 2.11, 2.17, 1.9

In (5.3), note that a vertex v can be parentless only in its light parts at rounds $j_1(v)$ and $j_1(v) + 1$, and at those later rounds at which its light part of round $j_1(v)$ lies in Bad , i.e. is demoted or parent-bad. Both kinds of bad parts are counted in lp .

Lemma 5.4 (bad probabilities). *Let Y be a light part of round r . Over the stage-1 randomness:*

- (a) $\mathbb{P}((E1) \text{ fails for } Y) \leq 3L_Y |Y| \exp(-L_Y^5/32);$
- (b) $\mathbb{P}(Y \text{ is demoted}) \leq |Y|^{-2}/2;$
- (c) $\mathbb{P}(Y \in \text{Bad}) \leq |Y|^{-2}.$

Dependencies: 3.13, 3.9, 5.2, 5.3, 1.25, 3.10, 3.2, 3.11, 2.11, 2.17, (F1)

Proof. (a) Fix a round l with $r \leq l \leq R$ and a vertex $w \in Y$. For $u \in \mathcal{A}_Y(w)$ let χ_u be the indicator of the event that $wu \in \mathbf{M}_Y$ and u carries no round- l zone. The sets $\mathcal{A}_Y(w)$ are fixed (Definition 5.3), so the only randomness in χ_u is the colour of the edge wu and the pair (choice, sublabel) of the vertex u . The edges wu , $u \in \mathcal{A}_Y(w)$, are distinct, colours of distinct edges are independent (Definition 3.10), the pairs (choice, sublabel) of distinct vertices are independent, and the two families are independent of each other (Lemma 5.2(iv)). Hence the χ_u , $u \in \mathcal{A}_Y(w)$, are independent. By the uniform split and the uniform own colouring of Definition 3.10(i),(iii), $\mathbb{P}(wu \in \mathbf{M}_Y) = \frac{1}{2} \cdot \frac{1}{k_{\text{own}}} \geq \frac{1}{2L_Y}$, because $k_{\text{own}} = 4J_Y + 1 \leq L_Y$. By Lemma 5.2(i), $\mathbb{P}(u \text{ carries a round-}l \text{ zone}) \leq \mathbb{P}(\text{choice}(u) \neq \text{none}) \leq 1/2$. Hence $\mathbb{P}(\chi_u = 1) \geq 1/(4L_Y)$, and the count $\text{cnt}_{w,l} := \sum_{u \in \mathcal{A}_Y(w)} \chi_u$ is a sum of m_Y independent indicators with mean $\mu \geq m_Y/(4L_Y) \geq L_Y^5/4$. By the Chernoff lower tail for sums of independent indicators (Cited result 1.25) with $\delta = 1/2$,

$$\mathbb{P}(\text{cnt}_{w,l} < L_Y^5/8) \leq \mathbb{P}(\text{cnt}_{w,l} < \mu/2) \leq \exp(-\mu/8) \leq \exp(-L_Y^5/32).$$

The number of rounds l with $r \leq l \leq R$ is $R - r + 1 \leq 2 \log^* d_r + 3 \leq L_Y/102 + 1 \leq 3L_Y$ by (5.1). A union bound over these rounds and over the $|Y|$ vertices w gives (a).

(b) COL(a) is the conclusion of three applications of Lemma 15⁺ (Lemma 3.2) in the proof of Lemma 3.13: with $k = 2$ to X_Y (giving Own_Y , Lend_Y); with $k = k_{\text{lend}}(Y)$ to Lend_Y (the lent classes); with $k = k_{\text{own}}$ to Own_Y (the own classes). The hypotheses $s \geq 40kL$ of these applications are row 3 of Table 5 (Lemma 3.11). If $r \geq R - 1$, then $k_{\text{lend}}(Y) = 0$ and Lend_Y has no classes (Definition 3.10(ii)), so only the applications with $k = 2$ and $k = k_{\text{own}}$ occur. The second and third colourings are uniform given the split (Definition 3.10(ii),(iii)), so Lemma 15⁺ applies to them conditionally on the split, on the event that Lend_Y (respectively Own_Y) is a $(2^{-6}, s_r/8)$ -expander; if that event fails, COL(a) has already failed in the first application. In each application each colour class fails with probability at most $2|Y|^{-5}$. Summing over the $2 + k_{\text{lend}}(Y) + k_{\text{own}}$ classes,

$$\mathbb{P}(\text{COL(a) fails for } Y) \leq 2(2 + k_{\text{lend}}(Y) + k_{\text{own}})|Y|^{-5} \leq 6|Y|^{-4},$$

since $k_{\text{lend}}(Y) \leq \lambda_r^{3,3} \leq |Y|$ if $r \leq R - 2$ (Lemma 3.11(i) and (5.1)), $k_{\text{lend}}(Y) = 0$ if $r \geq R - 1$, and $k_{\text{own}} \leq L_Y \leq |Y|$. By (a) and $|Y| = 2^{L_Y}$, $\mathbb{P}((\text{E1}) \text{ fails}) \leq 3L_Y 2^{L_Y} e^{-L_Y^5/32} \leq 2^{-4L_Y} = |Y|^{-4}$, because $\ln(3L_Y) + 5L_Y \ln 2 \leq L_Y^5/32$ for $L_Y \geq 10$ (and $L_Y \geq 102 \cdot 2^8$ by (5.1) and (F1)(a)). Hence $\mathbb{P}(Y \text{ demoted}) \leq 7|Y|^{-4} \leq |Y|^{-2}/2$.

(c) By Lemma 5.2(ii),(iv), for every sublabel (l, c, ς) of Y the set $V_{l,c,\varsigma} := \text{Zone}_{Y,l,c,\varsigma} \subseteq V(Y)$ contains each vertex of Y independently with probability $\rho_Y \geq 1/(12L_Y^5)$, independently of the colouring of Y . (The zones of Y are dependent across indices, since they are disjoint; Lemma 3.13(c) allows this.) So Lemma 3.13(c), applied with this family, gives $\mathbb{P}(Y \text{ parent-bad}) \leq |Y|^{-2}/2$. With (b), $\mathbb{P}(Y \in \text{Bad}) \leq \mathbb{P}(Y \text{ demoted}) + \mathbb{P}(Y \text{ parent-bad}) \leq |Y|^{-2}$. $\square$

Lemma 5.5 (expected demotion and lost-parent mass). *Over the stage-1 randomness,*

$$\mathbb{E}[80 \text{ dem} + 369 \text{ lp}] \leq \varepsilon_U(D_*) n, \quad \text{where} \quad \varepsilon_U(D_*) := 2462 (\log_2 D_*)^{-205}.$$

Moreover $\varepsilon_U(D_*) \rightarrow 0$ as $D_* \rightarrow \infty$.

Dependencies: 5.4, 5.3, 2.13, 2.17, 2.16, (F1), 2.11

Proof. By linearity and Definition 5.3,

$$\mathbb{E}[80 \text{ dem} + 369 \text{ lp}] = \sum_Y |Y| \left(80 \mathbb{P}(Y \text{ demoted}) + 369 (R - r(Y)) \mathbb{P}(Y \in \text{Bad}) \right),$$

the sum over all light parts Y . By Lemma 5.4(b),(c), both probabilities are at most $|Y|^{-2}$; by (5.1), $R - r \leq 2 \log^* d_r + 2 \leq \log_2 \lambda_r$. Hence the term of Y is at most

$$|Y|^{-1} (80 + 369 (2 \log^* d_r + 2)) \leq |Y|^{-1} 449 \log_2 \lambda_r \leq 898 \lambda_r^{-103} \log_2 \lambda_r,$$

using $|Y| \geq \lambda_r^{103}/2$. Light parts of round r are ancestors of round r , and there are at most $1.37n/P_r \leq 1.37n\lambda_r^{-103}$ of them (Proposition 2.13, (K1), with the $\text{HB}^{*\tau+}$ constant). Therefore round r contributes at most $1.37 \cdot 898 n \lambda_r^{-206} \log_2 \lambda_r \leq 1231 n \lambda_r^{-205}$. Let $F(x) := x^{-205}$ on $[\log_2 D_*, \infty)$. It is non-increasing, and $F(2^{x/A}) = 2^{-205x/A} \leq x^{-205}/2 = F(x)/2$ for $x \geq \log_2 D_*$. Indeed, put $\mu := \log_2 x \geq \log_2 \log_2 D_*$; then $(\Gamma 1)(a),(b)$ (Condition $(\Gamma 1)$) at μ give $\mu \geq 2^8$ and $x = 2^\mu \geq 2^{14} A \mu^3$, so $x/A \geq 2^{14} \mu^3 \geq \mu + 1 = \log_2 x + 1$ and $2^{-205x/A} \leq 2^{-205} x^{-205} \leq x^{-205}/2$. (This is also the case $a = 205$ of Lemma 2.16(iv).) By Lemma 2.16(ii), $\sum_{r \leq R} F(\lambda_r) \leq 2F(\log_2 D_*)$. So the total is at most $2 \cdot 1231 n (\log_2 D_*)^{-205} = \varepsilon_U(D_*)n$. Finally, $\varepsilon_U(D_*) = 2462 (\log_2 D_*)^{-205} \rightarrow 0$ as $D_* \rightarrow \infty$. $\square$

How Lemma 5.5 is used. It is an expectation bound over stage 1 only. It is used through a single Markov bound, with multiplier 3, on one combined weighted stage-1 functional that contains $80 \text{ dem} + 369 \text{ lp}$ as a summand (not through separate Markov bounds for dem and lp : with multiplier 2 each, the two failure probabilities could add up to 1, and no outcome would be guaranteed). All statements below hold for every stage-1 outcome; the bounds on dem and lp enter only through that functional.

5.3 The child side

Lemma 5.6 (child side). *Fix a stage-1 outcome and a non-demoted light part Z of round l . Apply Lemma 4.3 to the vertex set Z with the following data:*

- $O := H_Z = X_Z$, and the sets $\mathcal{A}(w) := \mathcal{A}_Z(w)$ of Definition 5.3 (so $\mathbf{m} = \lceil L_Z^6 \rceil$, $\mathbf{b} = \lceil 2^7 L_Z^2 \mathbf{m} \rceil$);
- as own classes, the classes $R_{j,c}$ ($0 \leq j < J_Z$, $c \in [4]$) and $\mathbf{M} = \mathbf{M}_Z$ of the colouring of Own_Z in Definition 3.10(iii);
- $\text{Rt} := \text{Rt}(Z)$ and $\text{Pl} := \text{Pl}(Z)$ (Definition 5.3);
- $\varphi(v) := c$ if v carries a round- l zone and c is its phase, and $\varphi(v) := *$ if v carries no round- l zone.

Then all hypotheses of Lemma 4.3 hold, including $(E1')$. Let $\mathcal{G}_Z$ be its good event G_{PV} for these data. It is an event over the stage-3 labels of Z (the levels $\mathbf{lv}$ on $\text{Pl}(Z)$ and the labels $\mathbf{k}$ on Z); it is determined by the stage-1 outcome and these labels; and its conditional probability given the stage-1 outcome is at least $1/2 - o(1) \geq 1/3$. On $\mathcal{G}_Z$ the following holds for every edge set $H_0(Z)$ with $\text{Own}_Z \subseteq H_0(Z) \subseteq E_l(Z)$: there is a partition $H_0(Z) = H^{\text{obj}}(Z) \sqcup H^{\text{arc}}(Z)$ such that

- $H^{\text{obj}}(Z)$ decomposes into at most $369|\text{Pl}(Z)|$ objects;
- $H^{\text{arc}}(Z)$ is the edge set of a family of pairwise edge-disjoint paths (the arcs of Z), each with at least one edge and both ends in $\text{Rt}(Z)$, and each carrying a phase $c \in [4]$, such that every vertex of a phase- c arc, ends included, lies in no round- l phase- c zone;
- Z has at most $14(J_Z + 1)|Z|$ arcs, and every vertex x is an end of at most $\deg_{H_0(Z)}(x) \leq |Z| - 1$ arcs of Z ;

(d) if $\text{Rt}(Z) = \emptyset$, in particular if $l \leq 2$, then Z has no arcs.

Dependencies: 4.3, 5.3, 5.2, 3.13, 2.11, 3.10, (Γ3)

Proof. Hypotheses. Put $N := |Z|$ and $L := L_Z = \log_2 N$.

- $N \geq P_l/2 \geq N_0$, by Proposition 2.11(iv) and (Γ3) (Condition (Γ3)).
- $O = X_Z$ is a spanning $(2^{-6}, s_l/2)$ -expander on Z (Proposition 2.11(i)), and $2\mathbf{m} \leq s_l/2$ (Definition 5.3). The sets $\mathcal{A}_Z(w)$ are Lemma-HB sets of O with $\mathbf{m} = \lceil L^6 \rceil$ and $\mathbf{b} = \lceil 2^7 L^2 \mathbf{m} \rceil$, as Lemma 4.3 requires.
- Z is not demoted, so COL(a) holds for Z : every own class is a spanning $(2^{-6}, s')$ -expander on Z with $s' := s_l/(16k_{\text{own}})$ (Lemma 3.13(a)). By Lemma 3.13(e), $s' \geq 2^{145} L^{41}$ and $2\lceil L^6 \rceil \leq s'$. The index range of the own classes is $0 \leq j < J_Z = \lfloor \log_2(L/8) \rfloor$, $c \in [4]$, which is the range $0 \leq j < J$ of Lemma 4.3. The own classes are fixed in stage 1, before the stage-3 labels of Z are drawn.
- $\text{Rt}(Z) \sqcup \text{Pl}(Z) = Z$ by definition.
- φ is a well-defined map $Z \rightarrow [4] \cup \{*\}$, because every vertex carries at most one zone (Lemma 5.2(iii)).
- (E1'). For $u \in Z$ we have $\varphi(u) = *$ iff u carries no round- l zone. Since Z is not demoted, (E1) holds for Z ; its instance at the round $l = r(Z)$ of Z states that for every $w \in Z$, in particular for every $w \in \text{Pl}(Z)$,

$$\#\{u \in \mathcal{A}_Z(w) : wu \in M_Z, \varphi(u) = *\} \geq L^5/8,$$

which is (E1').

The good event. By Lemma 4.3, G_{PV} depends only on the own classes, on Rt , on φ and on the stage-3 labels of Z . The own classes are stage-1 data; $\text{Rt}(Z)$ is a deterministic function of stage 1 (Definition 5.3); φ is a function of the choices and sublabeled (stage 1). So $\mathcal{G}_Z$ is determined by the stage-1 outcome and the stage-3 labels of Z . Its probability is at least $1/2 - o(1)$ by Lemma 4.3, and the $o(1)$ term is at most $1/100$ by (Γ3), so the probability is at least $1/3$. In particular $\mathcal{G}_Z$ does not depend on $H_0(Z)$.

Conclusion. Let $H_0(Z)$ satisfy $\text{Own}_Z \subseteq H_0(Z) \subseteq E_l(Z)$. Every edge of $E_l(Z)$ has both ends in Z (Proposition 2.11(iii)), and Own_Z is the union of the own classes. So $H_0 := H_0(Z)$ is an edge set inside Z containing all own classes, and on G_{PV} Lemma 4.3 gives $H_0(Z) = H^{\text{obj}} \sqcup H^{\text{arc}}$ with its properties (a)–(d). By Lemma 4.3(b), every arc is a path with at least one edge and has two distinct ends. Now:

- (a) is Lemma 4.3(a), since $369 = 369$.
- (b): by Lemma 4.3(b) the arcs are pairwise edge-disjoint paths with both ends in Rt , each with a phase $c \in [4]$, and every vertex x of a phase- c arc has $\varphi(x) \neq c$. If x lay in a round- l phase- c zone $\text{Zone}_{Y,l,c,\zeta}$, then x would carry a round- l zone of phase c , i.e. $\varphi(x) = c$. So x lies in no round- l phase- c zone.
- (c): Lemma 4.3(c) bounds the number of arcs by $4(J+1)N \leq 14(J+1)N = 14(J_Z+1)|Z|$ (the weaker constant 14 is kept so that the constants of this section are unchanged) and the number of arcs ending at a vertex x by $\deg_{H_0}(x) = \deg_{H_0(Z)}(x) \leq N - 1 = |Z| - 1$.

- (d): Lemma 4.3(d) gives no arcs when $\text{Rt} = \emptyset$. If $l \leq 2$ there is no round $\leq l-2$, so $\text{Rt}(Z) = \emptyset$.

Since $\mathcal{G}_Z$ does not depend on $H_0(Z)$, on $\mathcal{G}_Z$ the conclusion holds simultaneously for all admissible $H_0(Z)$; in particular it holds for whatever set $H_0(Z)$ the later processing order of this section produces. The stage-3 labels of distinct parts are independent, so outcomes in the events $\mathcal{G}_Z$ can be fixed part by part; no union bound over parts is used. $\square$

5.4 The parent side: chaining over U-bundles

For a round $l \geq 3$ put $\bar{J}_l := \max\{J_Z : Z \text{ a light part of round } l\}$ (and $\bar{J}_l := 0$ if there is none), and let ν_l be the number of ancestors of rounds $\leq l-2$, as in Proposition 2.13.

Lemma 5.7 (parent side: multi-parent chaining over U-bundles). *Fix a stage-1 outcome and, for every non-demoted light part Z , a stage-3 outcome in the event $\mathcal{G}_Z$ of Lemma 5.6. Fix a round l with $3 \leq l \leq R$ and, for every non-demoted light part Z of round l , an edge set $H_0(Z)$ with $\text{Own}_Z \subseteq H_0(Z) \subseteq E_l(Z)$; the arcs of Z are those given by Lemma 5.6 for this $H_0(Z)$. For $c \in [4]$ the U-bundle $\text{Bdl}_{l,c}$ is the set of all phase- c arcs of all non-demoted light parts of round l ; $E(\text{Bdl}_{l,c})$ denotes the union of their edge sets. Then for every $c \in [4]$ there is a set $\text{Lent}_{l,c}^U$ of edges, disjoint from $E(\text{Bdl}_{l,c})$, such that $E(\text{Bdl}_{l,c}) \cup \text{Lent}_{l,c}^U$ decomposes into cycles and single edges, where:*

- (i) $\text{Lent}_{l,c}^U \subseteq \bigcup_Y \bigcup_{\varsigma \in [T_Y^{\text{sl}}]} \mathcal{L}_{Y,l,c,\varsigma}^U$, the union over the good parents Y of rounds $\leq l-2$; $\text{Lent}_{l,c}^U$ is the union of the edge sets of pairwise edge-disjoint connectors, one per transition: a connector for a transition at the good parent Y with slot ς is a path in $\mathcal{L}_{Y,l,c,\varsigma}^U$ of length at most $2^{12} L_Y^4$ that joins the two vertices of the transition and has all its interior vertices in $\text{Zoney}_{l,c,\varsigma}$;
- (ii) the number of objects, summed over $c \in [4]$, is at most

$$4 \cdot 14 (\bar{J}_l + 1) M_l (M_l + 1) \nu_l + \text{cap}_l, \quad \text{cap}_l \leq \frac{14 (\bar{J}_l + 1) n}{(102 \log_2 \lambda_{l-2})^2},$$

where cap_l is the number of visit-capping pieces; and

$$\sum_{l=3}^R \left(4 \cdot 14 (\bar{J}_l + 1) M_l (M_l + 1) \nu_l + \text{cap}_l \right) \leq \varepsilon_{\text{ch}}(D_*) n,$$

where

$$\varepsilon_{\text{ch}}(D_*) := \frac{614}{D_*} + \frac{\log_2(2A \log_2(A \log_2 \log_2 D_*))}{371 (\log_2 \log_2 D_*)^2};$$

- (iii) no other edge is used: every object consists of edges of $E(\text{Bdl}_{l,c}) \cup \text{Lent}_{l,c}^U$, and every such edge lies in exactly one object.

The sets $\text{Lent}_{l,c}^U$ for distinct pairs (l, c) are pairwise disjoint, and the objects and $\text{Lent}_{l,c}^U$ depend only on the stage-1 outcome and on the bundle $\text{Bdl}_{l,c}$. Moreover $\varepsilon_{\text{ch}}(D_*) \rightarrow 0$ as $D_* \rightarrow \infty$.

Dependencies: 5.6, 5.3, 5.2, 3.10, 3.11, 3.1, 1.30, 2.13, 2.17, 2.11, (F1), 1.9

Proof. Fix l and c ; the letter Z ranges over the non-demoted light parts of round l and Y over the good parents of rounds $\leq l-2$.

Step 0: facts used.

- (F-a) $|Z| \leq M_l$, since $Z \subseteq Z^0$ and $|Z^0| \leq M_l$ (Proposition 2.11(ii)). Hence $L_Z \leq \log_2 M_l$, and $J_Z + 1 \leq \log_2 L_Z - 2 \leq \log_2 \log_2 M_l$.

- (F-b) (*Not used.*) If Y has round $r \leq l-2$ then $L_Y \geq L_Z$. Indeed $|Y| \geq P_r/2$ (Proposition 2.11(iv)), $P_r \geq P_{l-2}$ (as $P_{r'} \geq 2P_{r'+1}$) and $P_{l-2}/2 \geq M_l \log^4 M_l$ (both by Lemma 2.17(b)), so $|Y| \geq M_l \geq |Z|$. ((F-b) is not used in this proof: the multiplicity bound of Step 5 is $M_l - 1$, which does not involve L_Z . It is recorded for completeness.)
- (F-c) Light parts of round l are pairwise vertex-disjoint (Proposition 2.11(iv)), and the edge sets $E_l(Z)$, $E_r(Y)$ of distinct light parts are pairwise disjoint (Proposition 2.11(iii)).
- (F-d) By Lemma 5.6(b),(c), every arc of Z is a path with at least one edge, hence with two distinct ends; both ends lie in $\text{Rt}(Z)$; the arcs of Z are pairwise edge-disjoint; Z has at most $14(J_Z + 1)|Z| \leq \text{ncl} := 14(\bar{J}_l + 1)M_l$ arcs; every vertex v is an end of at most $\deg_{H_0(Z)}(v) \leq |Z| - 1 \leq M_l - 1$ arcs of Z (the last step by (F-a)); every vertex of an arc of Z lies in Z (its edges lie in $H_0(Z) \subseteq E_l(Z)$, Proposition 2.11(iii)); and every vertex of a phase- c arc lies in no round- l phase- c zone.

For an end v of an arc of Z we have $v \in \text{Rt}(Z)$, so $\text{par}(v)$ is defined; it is a good parent of round $\leq l-2$ containing v (Definition 5.3).

Step 1: classes. For each Z list its phase- c arcs in a fixed order as $a_{Z,1}, \dots, a_{Z,\text{na}(Z)}$, with $\text{na}(Z) \leq \text{ncl}_l$. For $1 \leq i \leq \text{ncl}_l$ the *class* i is $\mathcal{A}_i := \{a_{Z,i} : \text{na}(Z) \geq i\}$. Each class contains at most one arc of each Z , so by (F-c) the arcs of one class are pairwise vertex-disjoint. Every arc of $\text{Bdl}_{l,c}$ lies in exactly one class.

Step 2: the quotient multigraphs. For each class i let $\mathcal{M}_{l,c,i}$ be the multigraph (loops allowed) whose vertices are the good parents of rounds $\leq l-2$, with one edge e_a joining $\text{par}(v)$ and $\text{par}(v')$ for each arc $a \in \mathcal{A}_i$ with ends v, v' (a loop if $\text{par}(v) = \text{par}(v')$). Good parents are ancestors of rounds $\leq l-2$, so $\mathcal{M}_{l,c,i}$ has at most ν_l vertices, and $\nu_l \leq 2.74n/P_{l-2}$ (Lemma 2.17(b), which completes (K1) of Proposition 2.13).

Step 3: T-join and Euler trails. Let Odd_i be the set of vertices of odd degree in $\mathcal{M}_{l,c,i}$ (a loop adds 2 to the degree). Every component contains an even number of them. Fix a spanning forest $\mathcal{F}_i$ of $\mathcal{M}_{l,c,i}$ (it contains no loop). By Cited result 1.30, applied to a spanning tree of each component, there is a set $\mathcal{T}_i \subseteq E(\mathcal{F}_i)$ in which every vertex has odd degree iff it lies in Odd_i . Then all degrees of $\mathcal{M}_{l,c,i} - \mathcal{T}_i$ are even, and $|\mathcal{T}_i| \leq |E(\mathcal{F}_i)| \leq \nu_l - 1$. For every arc a with $e_a \in \mathcal{T}_i$, output each edge of a as a single edge; a is a path in Z and so has at most $|Z| - 1 \leq M_l - 1$ edges. Every component of $\mathcal{M}_{l,c,i} - \mathcal{T}_i$ that has an edge is connected with all degrees even, so it has a closed Euler trail (Cited result 1.30; the statement is valid for multigraphs with loops). This gives at most ν_l closed trails for class i , which together use every edge of $\mathcal{M}_{l,c,i} - \mathcal{T}_i$ exactly once.

We record a closed trail $\mathcal{W}$ as a cyclic sequence $(a_1, \dots, a_k)$, $k \geq 1$ (in this proof j indexes positions along a trail), of distinct arcs of one class, each with an orientation: a_j is traversed from its end v_j^- to its end v_j^+ , and $\text{par}(v_j^+) = \text{par}(v_{j+1}^-)$ for all j (indices modulo k). The *transition* at position j is the pair (v_j^+, v_{j+1}^-) , and its *node* is $\text{par}(v_j^+)$.

Claim 5.7.1 (transitions). *In every closed trail of this kind, every transition consists of two distinct vertices of its node Y .*

Proof. Both vertices have parent Y , so both lie in Y . If $k \geq 2$, then $a_j \neq a_{j+1}$ are distinct arcs of one class, which are vertex-disjoint (Step 1), so $v_j^+ \neq v_{j+1}^-$. If $k = 1$, the transition is (v_1^+, v_1^-) , the two ends of one arc, which are distinct by (F-d). $\square$

Step 4: visit capping. For a closed trail $\mathcal{W}$ and a node Y let $\text{vis}_Y(\mathcal{W})$ be the number of transitions of $\mathcal{W}$ at Y . If $\text{vis}_Y(\mathcal{W}) > T_Y^{\text{sl}}$, *split* $\mathcal{W}$ at Y as follows. Write $\text{vis} := \text{vis}_Y(\mathcal{W})$. Rotate the cyclic sequence so that its last transition (position k) is at Y , and let $j_1 < \dots < j_{\text{vis}} = k$ be the positions

of the transitions at Y . They cut $\mathcal{W}$ into vis consecutive segments $(a_1, \dots, a_{j_1})$, $(a_{j_1+1}, \dots, a_{j_2})$, $\dots$, $(a_{j_{\text{vis}}-1+1}, \dots, a_{j_{\text{vis}}})$; each segment begins with an arc whose first end has parent Y and ends with an arc whose last end has parent Y . Group consecutive segments into $\lceil \text{vis}/T_Y^{\text{sl}} \rceil$ blocks of at most T_Y^{sl} segments. A block $(a_p, \dots, a_q)$ is again a closed trail of the same kind: its transitions are the transitions of $\mathcal{W}$ at positions $p, \dots, q-1$ together with the new transition (v_q^+, v_p^-) at Y . By the argument of the transition claim, $v_q^+ \neq v_p^-$ (two distinct arcs of one class, or the two ends of one arc). The block has as many transitions at Y as it has segments, hence at most T_Y^{sl} ; and each transition of $\mathcal{W}$ at a node $Y' \neq Y$ lies in exactly one block. So a split at Y replaces $\mathcal{W}$ by $\lceil \text{vis}/T_Y^{\text{sl}} \rceil \leq 1 + \text{vis}/T_Y^{\text{sl}}$ closed trails, partitions the arcs of $\mathcal{W}$ among them, leaves every transition at every other node unchanged, and does not increase $\text{vis}_{Y'}$ of any trail for $Y' \neq Y$.

Process the nodes Y one after another in a fixed order and split every current trail at the current node. Since the number of Y -transitions of a trail never increases through splits at other nodes, at the end every trail $\mathcal{W}$ satisfies $\text{vis}_Y(\mathcal{W}) \leq T_Y^{\text{sl}}$ for every Y . When Y is processed, the number of new trails is at most $\sum_{\mathcal{W}} \text{vis}_Y(\mathcal{W})/T_Y^{\text{sl}}$, summed over the trails present at that time; and $\sum_{\mathcal{W}} \text{vis}_Y(\mathcal{W})$, the total number of transitions at Y in class i of phase c , is not changed by any split. Call it $\text{tr}_{c,i,Y}$ (for the fixed l).

A closed trail with k arcs has exactly k transitions, and splits preserve the total number of arcs. Hence, summing over all nodes Y , phases $c \in [4]$ and classes i , the total number of transitions $\sum_{Y,c,i} \text{tr}_{c,i,Y}$ is at most the number of arcs of all non-demoted light parts of round l , which by (F-d), (F-a) and (F-c) is at most $\sum_Z 14(J_Z + 1)|Z| \leq 14(\bar{J}_l + 1)n$ (the parts Z are pairwise disjoint, so $\sum_Z |Z| \leq n$). Every node Y has round $r \leq l-2$, so $T_Y^{\text{sl}} \geq L_Y^2 \geq (102 \log_2 \lambda_r)^2 \geq (102 \log_2 \lambda_{l-2})^2$, by (5.1) and $\lambda_r \geq \lambda_{l-2}$ (Lemma 2.17(a)). Therefore the number of visit-capping pieces over all c and i is

$$\text{cap}_l \leq \sum_{Y,c,i} \frac{\text{tr}_{c,i,Y}}{T_Y^{\text{sl}}} \leq \frac{14(\bar{J}_l + 1)n}{(102 \log_2 \lambda_{l-2})^2}.$$

Step 5: slots and pair multisets. For every final trail $\mathcal{W}$ and every node Y , give the transitions of $\mathcal{W}$ at Y pairwise distinct slots $\varsigma \in [T_Y^{\text{sl}}]$, by a fixed rule; this is possible because $\text{vis}_Y(\mathcal{W}) \leq T_Y^{\text{sl}}$. For each good parent Y of round $\leq l-2$ and each $\varsigma \in [T_Y^{\text{sl}}]$ let $\mathfrak{P}_{Y,\varsigma}$ be the multiset of the transitions at Y with slot ς , over all final trails of all classes i (for the fixed l, c), each regarded as an unordered pair. By the transition claim its elements are pairs of distinct vertices of $Y = V(\mathcal{L}_{Y,l,c,\varsigma}^U)$; the same pair may occur several times, so $\mathfrak{P}_{Y,\varsigma}$ is a multiset as in Cited result 1.9. A vertex v occurs in a pair at Y only if $\text{par}(v) = Y$. We bound the number of its occurrences.

- *Each arc end lies in at most one transition.* In a closed trail $(a_1, \dots, a_k)$ of the kind recorded in Step 3, the transitions are the k pairs (v_j^+, v_{j+1}^-) (indices modulo k); so the end v_j^+ of a_j lies in the transition at position j only, and the end v_j^- in the transition at position $j-1$ only (for $k=1$ both ends lie in the single transition (v_1^+, v_1^-) , once each). Thus every end of every arc of a trail lies in exactly one transition of that trail. This applies to every final trail, including the blocks created in Step 4: a block $(a_p, \dots, a_q)$ is a closed trail of the same kind, and its transitions are exactly the pairs (v_j^+, v_{j+1}^-) , $p \leq j < q$, together with (v_q^+, v_p^-) ; the new transition carries the two ends v_q^+ and v_p^- , whose transitions at positions q and $p-1$ of $\mathcal{W}$ are not transitions of the block. The final trails partition the arcs that are not output in Step 3, and the arcs output in Step 3 lie on no trail. Hence every arc end lies in at most one transition over all final trails of all classes.
- *Counting.* By the transition claim, v is at most one of the two vertices of any pair, and each occurrence of v in a pair is an arc end at v (the end v_j^+ of a_j or the end v_{j+1}^- of a_{j+1}). By the first bullet, distinct occurrences are distinct arc ends, and an arc has at most one end

equal to v , since its two ends are distinct (F-d). So the number of pairs of $\bigcup_{\varsigma} \mathfrak{P}_{Y,\varsigma}$ containing v , counted with multiplicity, is at most the number of arcs of $\text{Bdl}_{l,c}$ having v as an end. By (F-d) and (F-c), all of them are arcs of the unique round- l light part $Z(v)$ containing v , and there are at most $|Z(v)| - 1 \leq M_l - 1$ of them.

- *Comparison with t_Y .* Let $r = r(Y) \leq l - 2$. By Lemma 2.17(b) (row 12 of Table 5) and Lemma 2.17(a), $M_l \leq \lambda_{l-2}^{1.6} \leq \lambda_r^{1.6} \leq \lceil \lambda_r^{1.6} \rceil = t_Y$ (Definition 3.10).

So v lies in at most $M_l - 1 < t_Y$ pairs of $\bigcup_{\varsigma} \mathfrak{P}_{Y,\varsigma}$, and in particular of each $\mathfrak{P}_{Y,\varsigma}$.

Step 6: routing. Let Y be a good parent of round $r \leq l - 2$ and $\varsigma \in [T_Y^{\text{sl}}]$. Then (l, c, ς) is an index of $\mathcal{I}_U$ for Y , since $r + 2 \leq l \leq R$. As Y is not parent-bad, $\mathcal{L}_{Y,l,c,\varsigma}^U$ is $(2^{12}L_Y^4, t_Y)$ -path connected through $\text{Zone}_{Y,l,c,\varsigma}$ (Definition 5.3). This property depends only on the pair $(\mathcal{L}_{Y,l,c,\varsigma}^U, \text{Zone}_{Y,l,c,\varsigma})$, which is fixed by the stage-1 outcome, so it may be applied to the multiset $\mathfrak{P}_{Y,\varsigma}$, which is formed afterwards (Lemma 3.1(iii)). By Step 5 every vertex lies in at most $M_l - 1 \leq t_Y$ of its pairs. Hence there are pairwise edge-disjoint paths $\mathcal{C}_{\pi} \subseteq \mathcal{L}_{Y,l,c,\varsigma}^U$, one for each element π of $\mathfrak{P}_{Y,\varsigma}$ (counted with multiplicity), such that $\mathcal{C}_{\pi}$ joins the two vertices of π , has length at most $2^{12}L_Y^4$, and has all its interior vertices in $\text{Zone}_{Y,l,c,\varsigma}$. These paths are the connectors. Define $\text{Lent}_{l,c}^U$ as the union of the edge sets of all connectors, over all Y and ς . This proves the inclusion in (i) and the property of the connectors in (i).

Step 7: the cycles. Let $\mathcal{W} = (a_1, \dots, a_k)$ be a final trail, and let $\mathcal{C}_j$ be the connector of its transition (v_j^+, v_{j+1}^-) , which lies at the node Y_j and has slot ς_j . Let $C(\mathcal{W})$ be the closed walk that traverses a_1 from v_1^- to v_1^+ , then $\mathcal{C}_1$ from v_1^+ to v_2^- , then a_2 , then $\mathcal{C}_2$, and so on, and returns to v_1^- along $\mathcal{C}_k$.

Claim 5.7.2 (cycles). *$C(\mathcal{W})$ is a cycle of G of length at least 3.*

Proof. (1) The arcs $a_1, \dots, a_k$ are paths with pairwise disjoint vertex sets, since they are distinct arcs of one class (Step 1). (2) The interior vertices of $\mathcal{C}_j$ lie in $\text{Zone}_{Y_j,l,c,\varsigma_j}$, which is a round- l phase- c zone; every vertex of every arc of $\text{Bdl}_{l,c}$ lies in no such zone (F-d). So no interior vertex of a connector lies on an arc of $\mathcal{W}$. (3) For $j \neq j'$ the interiors of $\mathcal{C}_j$ and $\mathcal{C}_{j'}$ are disjoint: if $Y_j \neq Y_{j'}$, zones of distinct light parts are disjoint; if $Y_j = Y_{j'}$, then $\varsigma_j \neq \varsigma_{j'}$ (Step 5), and the two zones are again disjoint (Lemma 5.2(iii) in both cases). (4) Each $\mathcal{C}_j$ is a path from v_j^+ to v_{j+1}^- , and these two vertices are distinct (transition claim), so $\mathcal{C}_j$ has at least one edge. By (1)–(4), $C(\mathcal{W})$ visits each of its vertices once before closing, so it is a cycle with at least two edges. Suppose it had exactly two. Then $k = 1$, a_1 is the single edge $v_1^- v_1^+$ and $\mathcal{C}_1$ is the single edge $v_1^+ v_1^-$. As G is simple, these are the same edge. But $a_1 \subseteq E_l(Z)$ for a round- l light part Z , while $\mathcal{C}_1 \subseteq E(H_{Y_1}) \subseteq E_r(Y_1)$ with $r = r(Y_1) \leq l - 2$, and these edge sets are disjoint (F-c). This is a contradiction, so the length is at least 3. $\square$

Step 8: exact cover. The objects for (l, c) are the single edges of Step 3 and the cycles $C(\mathcal{W})$ of Step 7.

- *Arc edges.* Arcs of one Z are pairwise edge-disjoint (F-d), and arcs of distinct Z lie in the disjoint sets $E_l(Z)$ (F-c). Each arc lies in exactly one class; there it is either an edge $e_a \in \bar{\mathcal{T}}_i$, and then its edges are output as single edges, or it lies on exactly one Euler trail and, after the splits (which partition the arcs of a trail), on exactly one final trail $\mathcal{W}$, and then its edges lie on $C(\mathcal{W})$ only.
- *Connector edges.* For fixed (Y, ς) the connectors are pairwise edge-disjoint (Step 6). Connectors of distinct (Y, ς) lie in distinct classes $\mathcal{L}_{Y,l,c,\varsigma}^U$: for a fixed Y these are distinct colour

classes of one colouring, and for distinct Y they lie in the disjoint sets $E(H_Y) \subseteq E_{r(Y)}(Y)$ (F-c). Each connector belongs to exactly one transition and hence to exactly one cycle $C(\mathcal{W})$.

- *Arc edges versus connector edges.* Arc edges lie in sets $E_l(Z)$ and connector edges in sets $E_r(Y)$ with $r \leq l-2$; these are disjoint (F-c). So $\text{Lent}_{l,c}^U \cap E(\text{Bdl}_{l,c}) = \emptyset$.

Hence the objects partition $E(\text{Bdl}_{l,c}) \cup \text{Lent}_{l,c}^U$, and no other edge is used; this is (iii). For distinct pairs $(l, c) \neq (l', c')$ the sets $\text{Lent}_{l,c}^U$ and $\text{Lent}_{l',c'}^U$ are disjoint: within one good parent Y they lie in distinct colour classes of one colouring (the index (l, c, ς) differs), and for distinct parents $Y \neq Y'$ they lie in the disjoint sets $E(H_Y) \subseteq E_{r(Y)}(Y)$ and $E(H_{Y'}) \subseteq E_{r(Y')}(Y')$ (F-c). Every choice made above is a fixed rule applied to the stage-1 outcome and to $\text{Bdl}_{l,c}$.

Step 9: counting. For a fixed class i , Step 3 outputs at most $(\nu_l - 1)(M_l - 1)$ single edges and at most ν_l closed trails, together at most $\nu_l M_l$ objects before capping. There are at most $\text{ncl}_l = 14(\bar{J}_l + 1)M_l$ classes and 4 phases. With the capping pieces of Step 4, the number of objects summed over c is at most

$$4 \cdot 14(\bar{J}_l + 1)M_l \cdot \nu_l M_l + \text{cap}_l \leq 4 \cdot 14(\bar{J}_l + 1)M_l(M_l + 1)\nu_l + \text{cap}_l,$$

which is (ii) for the round l . For the sum over l , first let $3 \leq l \leq R$. By (F-a), $\bar{J}_l + 1 \leq \log_2 \log_2 M_l \leq M_l$; moreover $M_l + 1 \leq 2M_l$, $\nu_l \leq 2.74n/P_{l-2}$ (Step 2), and $P_{l-2} \geq M_l^{13}$ (Lemma 2.17(b)). So

$$4 \cdot 14(\bar{J}_l + 1)M_l(M_l + 1)\nu_l \leq 56 \cdot 2 \cdot 2.74n M_l^3/M_l^{13} \leq 307n/M_l,$$

and $\sum_l 307n/M_l \leq 614n/D_*$ because $\sum_l 1/M_l \leq 2/D_*$ (Lemma 2.17(b)).

Next, the capping pieces. By Step 4 and (F-a), $\bar{J}_l + 1 \leq \log_2 \log_2 M_l$. Moreover $M_l \leq (A \log \lambda_{l-2})^{2A}$ (Lemma 2.17(b)), so that $\log_2 M_l \leq 2A \log_2 (A \log \lambda_{l-2})$. With

$$\eta(x) := \frac{\log_2(2A \log_2(A \log_2 x))}{(\log_2 x)^2} \quad (x \geq \log_2 D_*),$$

this gives $\text{cap}_l \leq \frac{14}{102^2} n \eta(\lambda_{l-2}) \leq \frac{n}{743} \eta(\lambda_{l-2})$. We claim that η is non-increasing on $[\log_2 D_*, \infty)$ and that $\eta(2^{x/A}) \leq \eta(x)/2$ there. Write $x_1 := \log_2 x \geq \log_2 \log_2 D_*$, so that $x_1 \geq 2^8$ by (Γ1)(a) at x_1 , $x_2 := \log_2(Ax_1) \geq 14$ and $x_3 := \log_2(2Ax_2) \geq 11$, so that $\eta = x_3/x_1^2$. Then

$$\frac{d}{dx_1} \ln \eta = \frac{1}{x_3 x_2 x_1 (\ln 2)^2} - \frac{2}{x_1} < 0.$$

Further, $\log_2(A \log_2 2^{x/A}) = \log_2 x = x_1$, so $\eta(2^{x/A}) = A^2 \log_2(2Ax_1)/x^2$, while $\eta(x)/2 = x_3/(2x_1^2)$ with $x_3 \geq 1$. Since $x = 2^{x_1}$ and $2A^2 x_1^2 \log_2(2Ax_1) \leq 4A^3 x_1^3 \leq (2^{14} A x_1^3)^2 \leq 2^{2x_1}$ (by $\log_2 z \leq z$ and (Γ1)(b) at x_1), we get $\eta(2^{x/A}) \leq \eta(x)/2$. As $\lambda_{r'} \geq 2^{\lambda_{r'+1}/A}$ (Lemma 2.17(a)), $\eta(\lambda_{r'}) \leq \eta(2^{\lambda_{r'+1}/A}) \leq \eta(\lambda_{r'+1})/2$ for $r' < R$. Hence $\sum_{l=3}^R \eta(\lambda_{l-2}) = \sum_{r'=1}^{R-2} \eta(\lambda_{r'}) \leq 2\eta(\lambda_{R-2}) \leq 2\eta(\log_2 D_*)$, because $\lambda_{R-2} \geq \log_2 D_*$. Altogether

$$\sum_{l=3}^R \text{cap}_l \leq \frac{2}{743} \eta(\log_2 D_*) n \leq \frac{\log_2(2A \log_2(A \log_2 \log_2 D_*))}{371 (\log_2 \log_2 D_*)^2} n.$$

Adding the two sums gives $\varepsilon_{\text{ch}}(D_*)n$, which is (ii). Finally $\eta(\log_2 D_*) \rightarrow 0$ and $614/D_* \rightarrow 0$ as $D_* \rightarrow \infty$, so $\varepsilon_{\text{ch}}(D_*) \rightarrow 0$. $\square$

5.5 Demoted parts and Lemma K-RED

Lemma 5.8 (fallback for demoted light parts). *Fix a stage-1 outcome and let Y be a demoted light part of round r . Then Y is not a good parent; in particular no class $\mathcal{L}_Y^U$ of Y is used in Lemma 5.7. Let $\mathcal{G}_Y^{\text{VX}}$ be the good event of Theorem 4.4 for $O := X_Y$, an event over the VX^+ labels of Y (stage 3) that depends only on X_Y and these labels and has probability at least $1 - \eta_{\text{VX}}(|Y|) \geq 1/2$. On $\mathcal{G}_Y^{\text{VX}}$, for every edge set E with $E(X_Y) \subseteq E \subseteq E_r(Y)$, the set E decomposes into at most $80|Y|$ objects.*

Dependencies: 4.4, 2.11, 3.11, 2.17, (Γ1), (Γ3), 5.3, 5.7

Proof. A demoted part lies in Bad, so it is not a good parent (Definition 5.3), and Lemma 5.7(i) uses only classes of good parents. Put $N := |Y|$ and $L := L_Y$. By Proposition 2.11(i),(iv), X_Y is a spanning $(2^{-6}, s_r/2)$ -expander on Y and $N \geq P_r/2 \geq N_0$ ((Γ3)). The threshold of Theorem 4.4 at $\varepsilon = 2^{-6}$ is $s_r/2 \geq 2^{151} L^{38} \log_2 L$; this is row 8(c) of Table 5 (Lemma 3.11), and it holds in every round $r \leq R$, including $r \geq R - 1$. Explicitly, $s_r \geq \lambda_r^{100}$ (Proposition 2.11(i)) and $L \leq 2\lambda_r$ (Lemma 2.17(b), valid for every ancestor of every round $r \leq R$) give $2^{152} L^{38} \log_2 L \leq 2^{190} \lambda_r^{38} \cdot 2\lambda_r = 2^{191} \lambda_r^{39} \leq \lambda_r^{100} \leq s_r$, since $\log_2 \lambda_r \geq 2^8$ ((Γ1)(a)). Theorem 4.4 provides random labels and a good event depending only on $O = X_Y$, of probability at least $1 - \eta_{\text{VX}}(N) \geq 1/2$, since $N \geq N_0$ by (Γ3). On this event its conclusion holds simultaneously for every graph on the vertex set Y that contains O . Every edge of $E_r(Y)$ has both ends in Y (Proposition 2.11(iii)), so for $E(X_Y) \subseteq E \subseteq E_r(Y)$ the graph (Y, E) contains O , and Theorem 4.4 gives $f(E) \leq 80N = 80|Y|$. $\square$

Put $K^{\text{std}} := \bigsqcup_l \bigsqcup_{Z \in \text{Std}_l} E_l(Z)$. Here Std_l is the deterministic family of Definition 2.7 (it contains the GC-parts and the pre-parts failing (L1), and it does *not* contain the random demotions of Definition 5.3); so K^{std} is a function of the run alone, and every demoted light part is still a light part.

Lemma 5.9 (Lemma K-RED). *Fix the valid run, a stage-1 outcome, and stage-3 outcomes in the good events of all light parts: the event $\mathcal{G}_Z$ of Lemma 5.6 for every non-demoted light part Z , and the event $\mathcal{G}_Z^{\text{VX}}$ of Lemma 5.8 for every demoted light part Z . Let $(\text{Lent}^{\text{ext}}(Y))_Y$ be any family, indexed by the light parts Y , such that for every light part Y*

$$\text{Lent}^{\text{ext}}(Y) \subseteq \left(\bigcup_{l,j} \mathcal{L}_{Y,l,j}^{\text{JS}} \right) \cup \left(\bigcup_l \mathcal{L}_{Y,l}^{\text{JV}} \right), \quad \text{and} \quad \text{Lent}^{\text{ext}}(Y) = \emptyset \text{ if } Y \text{ is demoted.}$$

Process the rounds $l = R, R - 1, \dots, 1$ (children before parents). At round l :

- (1) *for every non-demoted light part Z of round l , let $\text{Lent}^U(Z) := E(H_Z) \cap \bigcup_{l' \geq l+2, c \in [4]} \text{Lent}_{l',c}^U$ be the set of U-lent edges of Z used by Lemma 5.7 at the rounds already processed, put*

$$H_0(Z) := E_l(Z) \setminus (\text{Lent}^U(Z) \cup \text{Lent}^{\text{ext}}(Z)),$$

and decompose $H_0(Z)$ by Lemma 5.6 into $H^{\text{obj}}(Z)$ (as objects) and arcs;

- (2) *for every demoted light part Z of round l , decompose $E_l(Z)$ by Lemma 5.8;*
- (3) *if $l \geq 3$, apply Lemma 5.7 to the bundles $\text{Bdl}_{l,c}$, $c \in [4]$, formed from the arcs of step (1).*

Output the long cycles $\bigcup_l \text{Cyc}_l$ and the edges of E_0 (as single edges) directly. Then the objects produced form a decomposition of

$$E(G) \setminus K^{\text{std}} \setminus \bigcup_Y \text{Lent}^{\text{ext}}(Y)$$

into at most

$$(D_*/2 + 745 + \varepsilon_K(D_*))n + 80 \text{ dem} + 369 \text{ lp}$$

objects, where $\varepsilon_K(D_*) := \varepsilon_{\text{ch}}(D_*)$. Itemized: long cycles and E_0 , at most $n + D_*n/2$; child vortices, at most $369 \sum_Z |\text{Pl}(Z)| \leq 369(2n + \text{lp})$; U-chaining, at most $\varepsilon_{\text{ch}}(D_*)n$; demoted parts, at most 80 dem. The exact sum of these items is at most $(D_*/2 + 739 + \varepsilon_K(D_*))n + 80 \text{ dem} + 369 \text{ lp}$; the constant 745 is kept for the cost line. No edge of K^{std} and no edge of any $\text{Lent}^{\text{ext}}(Y)$ is used. Moreover, the objects produced at round l depend only on the stage-1 and stage-3 outcomes, on the sets $\text{Lent}^{\text{ext}}(Z)$ of the light parts Z of round l , and on what was produced at the rounds $> l$.

Dependencies: 5.6, 5.7, 5.8, 2.11, 2.20, 2.12, 3.10, 5.3

Proof. The edge partition. By Proposition 2.11(iii),

$$E(G) = \bigsqcup_l \text{Cyc}_l \sqcup E_0 \sqcup \bigsqcup_{Y \text{ light}} E_{r(Y)}(Y) \sqcup K^{\text{std}},$$

where Cyc_l is read as an edge set. For every light Y , $\text{Lent}^{\text{ext}}(Y) \subseteq \text{Lend}_Y \subseteq E(H_Y) \subseteq E_{r(Y)}(Y)$ (Definition 3.10 and Proposition 2.11(iii)). Hence the set to be decomposed is

$$\mathcal{D} := \bigsqcup_l \text{Cyc}_l \sqcup E_0 \sqcup \bigsqcup_{Y \text{ light}} (E_{r(Y)}(Y) \setminus \text{Lent}^{\text{ext}}(Y)).$$

The edges of a light part Y of round r have both ends in Y (Proposition 2.11(iii)). The two kinds of edges combined by Lemma 5.7 are separated also by Lemma 2.12: an arc edge of a round- l part lies in $E(G_l)$, so it does not have both ends in any parent Y of round $\leq l-2$, whereas every connector edge of Y has both ends in Y . The construction below applies Lemma 5.8 and Lemma 5.6 only to edge sets of light parts, and Lemma 5.7 only to arc edges and U-lent edges of light parts; it never applies anything to an edge set $E_l(Z)$ with $Z \in \text{Std}_l$.

The construction is well defined. At round l , the set $\text{Lent}^{\text{U}}(Z)$ of a round- l light part Z consists of edges used by Lemma 5.7 at rounds $l' \geq l+2$, all of which were processed before round l ; Lemma 5.7 at a round l' uses classes only of good parents of rounds $\leq l'-2$, so no later round adds edges of Z to any $\text{Lent}^{\text{U}}_{l',c}$. Thus $\text{Lent}^{\text{U}}(Z)$ is final when Z is processed. The hypotheses of Lemma 5.6 hold for $H_0(Z)$: $H_0(Z) \subseteq E_l(Z)$, and $\text{Lent}^{\text{U}}(Z) \cup \text{Lent}^{\text{ext}}(Z) \subseteq \text{Lend}_Z$ (the U-lent classes and the JS- and JV-lent classes are colour classes of Lend_Z , Definition 3.10(ii)), which is disjoint from Own_Z ; so $\text{Own}_Z \subseteq H_0(Z)$. The stage-3 outcome of Z lies in $\mathcal{G}_Z$, which does not depend on $H_0(Z)$. The hypotheses of Lemma 5.7 at round l are those of Lemma 5.6 for the round- l parts. For a demoted Z of round l , $\text{Lent}^{\text{ext}}(Z) = \emptyset$ by assumption and no edge of Z lies in any $\text{Lent}^{\text{U}}_{l',c}$ (Lemma 5.8), so all of $E_l(Z)$ is available, and $E(X_Z) \subseteq E_l(Z)$ (Proposition 2.11(iii)); Lemma 5.8 applies with $E := E_l(Z)$. The objects produced at round l are functions of the stage-1 and stage-3 outcomes, of $\text{Lent}^{\text{ext}}(Z)$ and $\text{Lent}^{\text{U}}(Z)$ for the round- l parts Z , and of the bundles $\text{Bdl}_{l,c}$, which are formed from these; and $\text{Lent}^{\text{U}}(Z)$ is determined by the rounds $> l$. This proves the last assertion.

Every edge of $\mathcal{D}$ lies in exactly one object, and no other edge is used.

- Long cycles and E_0 are output directly, once each.
- Let Y be a demoted light part of round r . All of $E_r(Y)$ is covered by the objects of Lemma 5.8, each edge once. No other object uses an edge of $E_r(Y)$: the objects of Lemmas 5.6 and 5.8 for another part Z' use only edges of $E_{r(Z')}(Z')$; the objects of Lemma 5.7 use only arc edges of non-demoted parts and U-lent edges of good parents (Lemma 5.7(i),(iii)).

- Let Y be a non-demoted light part of round r . The U-lent classes are disjoint from the JS- and JV-lent classes, so $E_r(Y) \setminus \text{Lent}^{\text{ext}}(Y) = \text{Lent}^{\text{U}}(Y) \sqcup H_0(Y)$. An edge of $\text{Lent}^{\text{U}}(Y)$ lies in $\text{Lent}_{l,c}^{\text{U}}$ for exactly one pair (l, c) (these sets are pairwise disjoint), with $l \geq r + 2$, and is covered by exactly one object of Lemma 5.7 at (l, c) . An edge $e \in H_0(Y)$ lies either in $H^{\text{obj}}(Y)$, and is covered by exactly one of its objects, or on exactly one arc of Y , of some phase c ; that arc belongs to $\text{Bdl}_{r,c}$ (and $r \geq 3$, by Lemma 5.6(d)), and e is covered by exactly one object of Lemma 5.7 at (r, c) . No other object uses e : it is not in any $\text{Lent}_{l',c'}^{\text{U}}$ (those edges were removed as $\text{Lent}^{\text{U}}(Y)$), it is an arc edge only of Y 's own arcs, and objects of other parts use their own edges.
- Edges of K^{std} and of the sets $\text{Lent}^{\text{ext}}(Y)$ are used by no object: Lemma 5.8 is applied only to demoted light parts, which have $\text{Lent}^{\text{ext}} = \emptyset$; Lemma 5.6 is applied to $H_0(Z)$, which excludes $\text{Lent}^{\text{ext}}(Z)$; Lemma 5.7 uses arc edges (contained in the sets $H_0(Z)$) and U-lent edges (disjoint from JS- and JV-lent classes); and all these edges lie in the edge sets of light parts, which are disjoint from K^{std} . Zones are vertex sets and carry no edges.

Every object is a cycle of length at least 3 or a single edge, by Lemmas 5.6, 5.7 and 5.8 and the definition of Cyc_l .

Cost.

- Long cycles: at most n ; single edges of E_0 : $|E_0| < D_*n/2$ (Proposition 2.11(iii)).
- Child vortices: by Lemma 5.6(a), at most $369 \sum_Z |\text{Pl}(Z)|$ over the non-demoted light parts Z . By (5.3), which is Proposition 2.20(iii) applied with the set Bad of Definition 5.3 (demoted or parent-bad parts), this is at most $369(2n + \text{lp}) = 738n + 369 \text{lp}$.
- U-chaining: by Lemma 5.7(ii), at most $\varepsilon_{\text{ch}}(D_*)n$ over all rounds $l \geq 3$ and phases c .
- Demoted parts: by Lemma 5.8, at most $80 \sum_{Y \text{ demoted}} |Y| = 80 \text{dem}$.

The total is at most $(D_*/2 + 1 + 738 + \varepsilon_{\text{ch}}(D_*))n + 80 \text{dem} + 369 \text{lp} = (D_*/2 + 739 + \varepsilon_K(D_*))n + 80 \text{dem} + 369 \text{lp}$, which is at most the stated bound with 745 in place of 739. $\square$

The for-all form and the lent edges. Lemma 5.9 holds for *every* family $(\text{Lent}^{\text{ext}}(Y))$ of the stated kind, and the objects of round l depend on that family only through the sets $\text{Lent}^{\text{ext}}(Z)$ of the round- l parts. The lemma is applied later with $\text{Lent}^{\text{ext}}(Y)$ equal to the set of JS-lent edges of Y used on output cycles of junction chaining together with the JV-lent junction edges of Y used by lifted cycles, at rounds $\geq r(Y) + 2$; such a family is of the stated kind, and it is fixed before round $r(Y)$ is processed, which is all that the last assertion of the lemma requires. With the used U-lent edges this gives the full set of used lent edges of Y , all of which are excluded from $H_0(Y)$; in particular, used JV junction edges are removed from H_0 and are not covered twice. Unused lent edges of Y stay in $H_0(Y)$ as junk, which Lemma 5.6 accepts because it holds for every $H_0(Y)$ between Own_Y and $E_{r(Y)}(Y)$.

Remark 5.10 (the constants 369 and 745). (a) The constant $369 = 369$ of Lemma 4.3 is an upper bound for $28 \cdot 8 + 64 + 16 = 304$. The proof of Lemma 4.3 shows that a vortex step j costs at most $4|W_j| + 12(2|\text{Pl} \cap U_{j+1}| + 2|W_j|) \leq 28|\text{Pl} \cap U_j|$ objects (equation (4.10), using $\text{Pl} \cap U_j = W_j \sqcup (\text{Pl} \cap U_{j+1})$). The good event (G5) gives $\sum_j |\text{Pl} \cap U_j| \leq 8|\text{Pl}|$, so all steps together cost at most $28 \cdot 8|\text{Pl}| = 224|\text{Pl}|$. The finish costs at most $64|\text{Pl}|$ objects for the edges with both ends in $\text{Pl} \cap U_j$ (Fact 1.6(b), applied to at most $64|\text{Pl}|/L$ vertices) and at most $8|\text{Pl} \cap U_j| \leq 16|\text{Pl}|$ stripped single edges (proof of Lemma 4.3; the bound $8|\text{Pl} \cap U_j| \leq 8|\text{Pl}|$ also holds, and the weaker factor

16 suffices; nothing depends on it). So at most $304|\text{Pl}|$ objects are output; the constant 369 is not optimized, and it is the value used in the cost line.

(b) The constant 745 = 745 exceeds the exact accounting $739 = 739 = 1 + 2 \cdot 369$ of Lemma 5.9 by 6, so $745 \geq 1 + 2 \cdot 369 + \varepsilon_K(D_*)$ whenever $\varepsilon_K(D_*) \leq 6$. This holds under $(\Gamma 4)$ (Condition $(\Gamma 4)$), whose left-hand side contains ε_K as a summand. The bound of Lemma 5.9 carries ε_K separately in any case, so this slack is not needed; the value 745 is the one used in the cost line.

(c) The constants 369 and 745 depend on the hierarchy only through three facts: each vertex lies in at most one light part per round (Proposition 2.11(iv); used in Lemma 5.2(i), in the vertex-disjointness of the arcs of one class, and in (5.3)); the parentless count $\sum_Z |\text{Pl}(Z)| \leq 2n + \text{lp}$ (Proposition 2.20(iii); used in (5.3); the size comparison $|Y| \geq |Z|L_Z^4$ of Proposition 2.20(ii) is not used here, cf. (F-b) in the proof of Lemma 5.7); and the overlap counts, namely at most $1.37n/P_r$ ancestors of round r (Proposition 2.13, (K1)) and $\nu_l \leq 2.74n/P_{l-2}$ (Lemma 2.17(b)), which enter only the terms $\varepsilon_U(D_*)n$ and $\varepsilon_{\text{ch}}(D_*)n$. The value 369 is combinatorics of the four-phase P-vortex and does not depend on colour probabilities; 80 is the constant of Theorem 4.4; and neither constant depends on the Markov multiplier used to fix the stage-1 outcome.

Dependencies: 4.3, 5.9, 2.11, 2.20, 2.13, 2.17, 4.4, $(\Gamma 4)$, 1.6

6 Chaining in standalone parts: HCC-P and JS-LC; hub concentration; the MIX-C assembly

This section contains the part of the proof that treats the standalone pre-parts of a valid $\text{HB}^{*\tau+}$ run. It has four parts.

- A purely combinatorial *routing engine*: Lemma GATE, clusters, Lemma MED, Lemma EQ-LPT, Lemma PAR and Lemma HCC-P. HCC-P closes layered systems of clusters into few cycles through junction paths.
- The *hub-concentration* bounds CONC and CONC-L. For every valid $\text{HB}^{*\tau+}$ run and every designation they give $\sum_{(Y,l)} m_{Y,l} \leq \varepsilon_{\text{CONC}}(D_*)n$.
- The stage-2 *lending statuses*, and Lemma JS-LC, which chains the beads of all standalone parts of one round through lent classes of ancestors two or more rounds back. The single edges left over form the *J-set* J_l , whose interface is Lemma J^+ .
- The *MIX-C assembly*. For every J-consumer (a rule that disposes of the J-set, defined in the last subsection) it turns all of this into a decomposition of $E(G)$ with an explicit cost.

There are no VX-parts ($\mathcal{V} = \emptyset$): every standalone pre-part is chained.

Throughout, G is the input graph (Convention 1.3). Orientations, directed paths and directed cycles are those of digraphs obtained by orienting edges of G ; each edge receives exactly one direction. A digraph is *acyclic* if it has no directed cycle.

6.1 The routing engine: GATE, clusters, MED, EQ-LPT, PAR, HCC-P

Lemma 6.1 (Lemma GATE, forward direction). *Let $F \subseteq E(G)$ and let $\vec{F}$ be an orientation of F with $d_{\vec{F}}^+(v) = d_{\vec{F}}^-(v)$ at every vertex v . Let $\mathcal{W}$ be a set of arcs of $\vec{F}$ such that $\vec{F} - \mathcal{W}$ is acyclic. Then F decomposes into at most $|\mathcal{W}|$ cycles of G . More precisely, $\vec{F}$ is the arc-disjoint union of directed cycles, each of length at least 3, each the orientation of a cycle of G , and each containing an arc of $\mathcal{W}$; there are at most $|\mathcal{W}|$ of them.*

Dependencies: 1.4

Proof. We first show that $\vec{F}$ is an arc-disjoint union of directed cycles of length at least 3, by induction on the number of arcs. If $\vec{F}$ has no arc there is nothing to prove.

Otherwise let $v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_k$ ($k \geq 1$) be a directed path in $\vec{F}$ with pairwise distinct vertices and with k maximal. The vertex v_k has in-degree at least 1, so by balance it has an out-arc $v_k \rightarrow x$. By maximality of the path, $x = v_i$ for some $i \leq k$. Since G has no loops, $i \neq k$. So $C := v_i \rightarrow v_{i+1} \rightarrow \cdots \rightarrow v_k \rightarrow v_i$ is a directed cycle with distinct vertices, of length $k - i + 1 \geq 2$.

Suppose the length were 2. Then $i = k - 1$, and $\vec{F}$ would contain the arcs $v_{k-1} \rightarrow v_k$ and $v_k \rightarrow v_{k-1}$. These arcs come from two distinct edges of F with the same ends, which is impossible since G is simple. So C has length at least 3, and its underlying edge set is a cycle of G .

Deleting the arcs of C lowers in- and out-degree by 1 at every vertex of C and changes nothing elsewhere. So the remaining digraph is still balanced, and the induction hypothesis applies to it.

Let $C_1, \dots, C_p$ be the directed cycles obtained. Their underlying cycles partition F , so they form a decomposition of F into cycles in the sense of Definition 1.4. Each C_i contains an arc of $\mathcal{W}$, since otherwise C_i would be a directed cycle of $\vec{F} - \mathcal{W}$. The C_i are arc-disjoint, so $p \leq |\mathcal{W}|$. $\square$

Definition 6.2 (clusters). A *cluster* is a triple $\mathfrak{K} = (A_{\mathfrak{K}}, U_{\mathfrak{K}}, B_{\mathfrak{K}})$ with the following parts.

- $A_{\mathfrak{K}}$ (the *hubs* of $\mathfrak{K}$) and $U_{\mathfrak{K}}$ (the *ports* of $\mathfrak{K}$) are disjoint vertex sets. Put $V(\mathfrak{K}) := A_{\mathfrak{K}} \cup U_{\mathfrak{K}}$.
- $B_{\mathfrak{K}} \subseteq E(G[A_{\mathfrak{K}} \cup U_{\mathfrak{K}}])$ is the set of *beads*. No bead has both ends in $A_{\mathfrak{K}}$.

The *bead count* is $b(\mathfrak{K}) := \sum_{h \in A_{\mathfrak{K}}} \deg_{B_{\mathfrak{K}}}(h)/2 + e(B_{\mathfrak{K}}[U_{\mathfrak{K}}])$. The cluster $\mathfrak{K}$ is *parity-clean* if every hub has even $B_{\mathfrak{K}}$ -degree.

An *admissible orientation* of $\mathfrak{K}$ is an acyclic orientation of $B_{\mathfrak{K}}$ in which every hub h has $d^+(h) = d^-(h)$. Given an admissible orientation, put for every port u

$$\text{exc}(u) := d^+(u) - d^-(u), \quad \text{exc}(u)^+ := \max(\text{exc}(u), 0), \quad \text{exc}(u)^- := \max(-\text{exc}(u), 0),$$

and define the *load* $\Phi(\mathfrak{K}) := \sum_{u \in U_{\mathfrak{K}}} \text{exc}(u)^+$.

Let $\mathfrak{K}$ be parity-clean and let $\prec$ be a linear order on $U_{\mathfrak{K}}$. The *median orientation* $\text{MED}(\prec)$ orients $B_{\mathfrak{K}}$ as follows.

- At a hub h with $B_{\mathfrak{K}}$ -neighbours $u_1 \prec u_2 \prec \cdots \prec u_{2d}$, it orients $u_i \rightarrow h$ for $i \leq d$ and $h \rightarrow u_i$ for $i > d$. These neighbours are ports, since no bead joins two hubs.
- A port–port bead uv with $u \prec v$ is oriented $u \rightarrow v$ ($\prec$ -upwards).

Every bead is oriented exactly once: a bead at a hub has its other end at a port, so it is oriented by the rule at that hub only.

Dependencies: none

Lemma 6.3 (Lemma MED). *Let $\mathfrak{K}$ be a parity-clean cluster and $\prec$ a linear order on $U_{\mathfrak{K}}$.*

- MED($\prec$) is an admissible orientation. There is a function $\text{pot} : A_{\mathfrak{K}} \cup U_{\mathfrak{K}} \rightarrow (0, 1)$ that strictly increases along every arc of MED($\prec$).*
- For every admissible orientation of $\mathfrak{K}$, in particular for MED($\prec$), and every port u : $|\text{exc}(u)| \leq \deg_{B_{\mathfrak{K}}}(u)$ and $\text{exc}(u) \equiv \deg_{B_{\mathfrak{K}}}(u) \pmod{2}$. Moreover $\sum_{u \in U_{\mathfrak{K}}} \text{exc}(u) = 0$ and $\Phi(\mathfrak{K}) \leq b(\mathfrak{K})$.*

Dependencies: 6.2

Proof. (a) Let $\text{rk} : U_{\mathfrak{K}} \rightarrow \{1, \dots, |U_{\mathfrak{K}}|\}$ be the bijection realising $\prec$, and put $\text{pot}(u) := \text{rk}(u)/(|U_{\mathfrak{K}}| + 1)$ for ports.

For a hub h with $2d \geq 2$ neighbours $u_1 \prec \dots \prec u_{2d}$, put $\text{pot}(h) := (\text{rk}(u_d) + \frac{1}{2})/(|U_{\mathfrak{K}}| + 1)$. Since $\text{rk}(u_d) \leq |U_{\mathfrak{K}}| - 1$, this lies in $(0, 1)$. A hub without beads gets $\text{pot}(h) := \frac{1}{2}$.

Check the arcs.

- An arc $u_i \rightarrow h$ ($i \leq d$) has $\text{rk}(u_i) \leq \text{rk}(u_d) < \text{rk}(u_d) + \frac{1}{2}$.
- An arc $h \rightarrow u_i$ ($i > d$) has $\text{rk}(u_i) \geq \text{rk}(u_{d+1}) \geq \text{rk}(u_d) + 1$, because ranks are distinct integers.
- Port–port arcs go up in rank.

So pot strictly increases along every arc, and $\text{MED}(\prec)$ has no directed cycle. Every hub of degree $2d$ has exactly d in-arcs and d out-arcs. Hence $\text{MED}(\prec)$ is admissible.

(b) Fix an admissible orientation. $\text{exc}(u)$ is a sum of ± 1 over the $\deg_{B_{\mathfrak{K}}}(u)$ beads at u , which gives the bound $|\text{exc}(u)| \leq \deg_{B_{\mathfrak{K}}}(u)$ and the parity claim.

Every arc contributes $+1$ to $d^+ - d^-$ at its tail and -1 at its head, so $\sum_{v \in V(\mathfrak{K})} (d^+(v) - d^-(v)) = 0$. Hubs contribute 0, hence $\sum_{u \in U_{\mathfrak{K}}} \text{exc}(u) = 0$.

Finally,

$$\begin{aligned} \Phi(\mathfrak{K}) &= \sum_u \text{exc}(u)^+ \leq \sum_{u \in U_{\mathfrak{K}}} d^+(u) = \#\{\text{arcs port} \rightarrow \text{hub}\} + e(B_{\mathfrak{K}}[U_{\mathfrak{K}}]) \\ &= \sum_{h \in A_{\mathfrak{K}}} d^-(h) + e(B_{\mathfrak{K}}[U_{\mathfrak{K}}]) = b(\mathfrak{K}). \end{aligned}$$

Here we used that every arc with tail at a port ends at a hub or at a port, and that $d^-(h) = \deg_{B_{\mathfrak{K}}}(h)/2$ at every hub. $\square$

Lemma 6.4 (Lemma EQ-LPT). *Let $\Phi_1 \geq \Phi_2 \geq \dots \geq \Phi_m \geq 0$ be loads of m items, and let $1 \leq k \leq m$. Place the items in the order $1, 2, \dots, m$ into k initially empty layers. Each item goes to a layer of currently minimum load, and to an empty layer whenever one exists (“greedy longest-processing-time placement”). Then all k layers are non-empty, and the final layer loads satisfy $\max_j \Phi_j^{\text{lay}} - \min_j \Phi_j^{\text{lay}} \leq \Phi_1$.*

Dependencies: none

Proof. An empty layer has load 0, which is minimum. So the rule is consistent, and the first k items go to the k distinct empty layers. Since $k \leq m$, every layer is non-empty.

Let j^* be a layer of maximum final load and x the last item placed in it. When x was placed, the load of j^* was at most the load of every layer at that moment. That is at most the final load of every layer, because loads only grow. Hence

$$\max_j \Phi_j^{\text{lay}} = (\text{load of } j^* \text{ before } x) + \Phi_x \leq \min_j \Phi_j^{\text{lay}} + \Phi_1. \quad \square$$

Lemma 6.5 (Lemma PAR, port–port parity). *Let E_{ab} be a bipartite graph with sides V_a, V_b , and let $V(E_{ab})$ be the set of vertices incident with an edge of E_{ab} . Call a connected component of E_{ab} odd if it has an odd number of edges. In each odd component choose one edge that is either a non-bridge or a pendant edge (an edge with an end of degree 1) of that component; such an edge exists. Let J' be the set of chosen edges. Then, for every such choice,*

$$|J'| \leq \#\{\text{odd components}\} \leq |V(E_{ab})|/2,$$

and there is a partition $E_{ab} \setminus J' = S_a \sqcup S_b$ such that every vertex of V_b has even S_a -degree and every vertex of V_a has even S_b -degree.

Dependencies: 1.30

Proof. Existence of the edge. Let C be a component with at least one edge. If C contains a cycle, every edge of that cycle is a non-bridge. Otherwise C is a tree with at least one edge; it has a leaf, and the edge at the leaf is pendant.

The count. Every odd component has at least one edge, hence at least two vertices of $V(E_{ab})$. Components are vertex-disjoint, so there are at most $|V(E_{ab})|/2$ of them.

Even components after deletion. Let C be odd with chosen edge e .

- If e is a non-bridge, $C - e$ is connected and has $|E(C)| - 1$ edges, an even number.
- If e is pendant, $C - e$ consists of an isolated vertex and a connected graph with $|E(C)| - 1$ edges.

Hence every component C' of $E_{ab} \setminus J'$ that has an edge has an even number of edges.

The partition. Fix such a component C' . Prescribe parities $p(v) := 0$ for $v \in V_b \cap V(C')$ and $p(u) := \deg_{C'}(u) \bmod 2$ for $u \in V_a \cap V(C')$. Every edge of C' has exactly one end in V_a , so

$$\sum_{w \in V(C')} p(w) \equiv \sum_{u \in V_a \cap V(C')} \deg_{C'}(u) = |E(C')| \equiv 0 \pmod{2}.$$

So $T := \{w : p(w) = 1\}$ has even size. By Cited result 1.30, a spanning tree of the connected graph C' contains a T -join $S(C')$: an edge set with $\deg_{S(C')}(w)$ odd exactly for $w \in T$.

Put $S_a := \bigcup_{C'} S(C')$ and $S_b := (E_{ab} \setminus J') \setminus S_a$. Then:

- every $v \in V_b$ has even S_a -degree;
- every $u \in V_a$ has $\deg_{S_b}(u) = \deg_{E_{ab} \setminus J'}(u) - \deg_{S_a}(u) \equiv \deg_{C'}(u) - p(u) \equiv 0 \pmod{2}$, where C' is the component of u .

Isolated vertices have degree 0 in both. □

Lemma 6.6 (Lemma HCC-P, layered hub-cluster chaining with path junctions). *Let $k \geq 1$, and suppose the following data are given.*

- Layers $\mathcal{K}_0, \dots, \mathcal{K}_{k-1}$: finite families of clusters, each cluster with a fixed admissible orientation. Put $\mathcal{X}_j := \bigcup_{\mathfrak{K} \in \mathcal{K}_j} U_{\mathfrak{K}}$.
- Junction sets $T_0, \dots, T_{k-1}$: pairwise disjoint vertex sets.
- A padding $\text{pad} : \bigcup_j \mathcal{X}_j \rightarrow \mathbb{Z}_{\geq 0}$, with $\text{pad} \equiv 0$ if $k = 1$.

Define the demands $\text{dm}^-(u) := \text{exc}(u)^- + \text{pad}(u)$ and $\text{dm}^+(u) := \text{exc}(u)^+ + \text{pad}(u)$, and the padded loads $\Phi_j := \sum_{u \in \mathcal{X}_j} \text{dm}^+(u)$. For $k = 1$ put $X^{\text{out}} := \{u \in \mathcal{X}_0 : \text{exc}(u) < 0\}$ and $X^{\text{in}} := \{u \in \mathcal{X}_0 : \text{exc}(u) > 0\}$. Indices of layers and junction sets are taken modulo k . Assume:

(D) the sets $A_{\mathfrak{K}}$ and $U_{\mathfrak{K}}$ (over all clusters of all layers) and $T_0, \dots, T_{k-1}$ are pairwise disjoint;

(P) every cluster is parity-clean;

(JC-P) for each j there is a family $\mathcal{P}_j$ of pairwise edge-disjoint paths of G with the following properties.

- Each path runs from a vertex of $\mathcal{X}_j$ to a vertex of $\mathcal{X}_{j+1}$; for $k = 1$, from X^{out} to X^{in} .
- All interior vertices of each path lie in T_j .
- Every $u \in \mathcal{X}_j$ is the first vertex of exactly $\text{dm}^-(u)$ paths of $\mathcal{P}_j$, and every $v \in \mathcal{X}_{j+1}$ is the last vertex of exactly $\text{dm}^+(v)$ paths of $\mathcal{P}_j$. For $k = 1$ this reads: $u \in X^{\text{out}}$ starts exactly $\text{exc}(u)^-$ paths and $v \in X^{\text{in}}$ ends exactly $\text{exc}(v)^+$ paths.

The edge sets $E(\mathcal{P}_0), \dots, E(\mathcal{P}_{k-1})$ and all bead sets $B_{\mathfrak{R}}$ are pairwise disjoint.

Then:

- (i) all Φ_j are equal to a common value Φ , and $\Phi = |\mathcal{P}_j|$ for every j ;
- (ii) there are sets $F_j \subseteq E(\mathcal{P}_j)$ with $E(\mathcal{P}_j) \setminus F_j \subseteq E(G[T_j])$ such that $\bigcup_{\mathfrak{R}} B_{\mathfrak{R}} \cup \bigcup_j F_j$ decomposes into at most Φ cycles of G , each of length at least $\max(3, k)$.

The edges of $E(\mathcal{P}_j) \setminus F_j$ are called *returned*; they are not used.

Dependencies: 6.1, 6.3, 6.2

Proof. We treat $k \geq 2$ and $k = 1$ together. For $k = 1$, read “ $\mathcal{X}_j$ as first vertices” as X^{out} and “ $\mathcal{X}_{j+1}$ as last vertices” as X^{in} .

Step 0: the loads. Each cluster has an admissible orientation, so $\sum_{u \in U_{\mathfrak{R}}} \text{exc}(u) = 0$ by Lemma 6.3(b). Hence $\sum_{u \in \mathcal{X}_j} \text{exc}(u)^- = \sum_{u \in \mathcal{X}_j} \text{exc}(u)^+$ for every j . Counting first vertices and last vertices of paths gives

$$|\mathcal{P}_j| = \sum_{u \in \mathcal{X}_j} \text{dm}^-(u) = \sum_{u \in \mathcal{X}_j} (\text{exc}(u)^+ + \text{pad}(u)) = \Phi_j, \quad |\mathcal{P}_j| = \sum_{v \in \mathcal{X}_{j+1}} \text{dm}^+(v) = \Phi_{j+1}.$$

Hence $\Phi_j = \Phi_{j+1}$ for all j modulo k , and all padded loads equal $\Phi := |\mathcal{P}_j|$. For $k = 1$ the same count gives $|\mathcal{P}_0| = \sum \text{exc}^- = \sum \text{exc}^+ = \Phi_0$. This proves (i).

Step 1: orient and cancel. Orient every path of $\mathcal{P}_j$ from its first to its last vertex, and let D_j be the resulting set of arcs. The paths are edge-disjoint, so every edge of $E(\mathcal{P}_j)$ receives exactly one direction. We record three facts.

- No vertex of $\mathcal{X}_j$ is an interior vertex of a path of $\mathcal{P}_j$ (interiors lie in T_j , which avoids all ports by (D)). No vertex of $\mathcal{X}_j$ is a last vertex: last vertices lie in $\mathcal{X}_{j+1}$, which is disjoint from $\mathcal{X}_j$ by (D) when $k \geq 2$ (the clusters of different layers are different clusters, and for $k = 2$ the layers $j + 1$ and $j - 1$ coincide but differ from j); for $k = 1$, $X^{\text{out}} \cap X^{\text{in}} = \emptyset$. Symmetrically, no vertex of $\mathcal{X}_{j+1}$ is a first or interior vertex.
- Hence in D_j every $u \in \mathcal{X}_j$ has out-degree $\text{dm}^-(u)$ and in-degree 0, and every $v \in \mathcal{X}_{j+1}$ has in-degree $\text{dm}^+(v)$ and out-degree 0.
- Every $y \in T_j$ is never a first or last vertex (these are ports), and each path through y contributes one in-arc and one out-arc at y . So $d_{D_j}^+(y) = d_{D_j}^-(y)$. All vertices of D_j lie in $\mathcal{X}_j \cup T_j \cup \mathcal{X}_{j+1}$.

A directed cycle of D_j cannot pass through a port, since ports are sources or sinks of D_j . So every directed cycle of D_j has all its vertices in T_j .

Delete directed cycles from D_j one at a time until no directed cycle remains. Call the result D'_j and let F_j be its underlying edge set. Each deletion lowers in- and out-degree by one at the vertices of a cycle inside T_j and touches no arc at a port. So the three facts above hold for D'_j , D'_j is acyclic, and $E(\mathcal{P}_j) \setminus F_j \subseteq E(G[T_j])$.

Step 2: the Eulerian digraph. Let $\vec{F}$ consist of the arcs of the fixed admissible orientations of all clusters together with $D'_0, \dots, D'_{k-1}$. Its underlying edge set is $F := \bigcup_{\mathfrak{R}} B_{\mathfrak{R}} \cup \bigcup_j F_j$.

- These are distinct edges of G . The clusters are vertex-disjoint by (D), so their bead sets are disjoint; the remaining disjointness is part of (JC-P).
- Every hub is balanced. A hub is not a port and lies in no T_j , so it meets only arcs of its own cluster, which are balanced at hubs by admissibility.
- Every $y \in T_j$ is balanced. By (D), y lies in no cluster and in no T_i with $i \neq j$. The arcs of D'_i have both ends in $\mathcal{X}_i \cup T_i \cup \mathcal{X}_{i+1}$, so y meets only arcs of D'_j , which are balanced at y .
- Every port is balanced. Let u be a port of a cluster of layer j , with $k \geq 2$. By (D) and the fact just recalled, u meets arcs of its own cluster, out-arcs of D'_j (as a vertex of $\mathcal{X}_j$) and in-arcs of D'_{j-1} (as a vertex of $\mathcal{X}_{(j-1)+1}$), and no others; $j-1 \neq j$ modulo k . Hence

$$d_{\vec{F}}^+(u) - d_{\vec{F}}^-(u) = \text{exc}(u) + \text{dm}^-(u) - \text{dm}^+(u) = \text{exc}(u) + \text{exc}(u)^- - \text{exc}(u)^+ = 0.$$

For $k = 1$: a port with $\text{exc}(u) < 0$ has $\text{exc}(u)^- = -\text{exc}(u)$ out-arcs in D'_0 ; a port with $\text{exc}(u) > 0$ has $\text{exc}(u)$ in-arcs; a port with $\text{exc}(u) = 0$ has none. In each case the port is balanced.

Step 3: a potential. For each cluster $\mathfrak{K}$ fix a function $\text{pot}_{\mathfrak{K}} : V(\mathfrak{K}) \rightarrow (0, 1)$ that strictly increases along the arcs of its admissible orientation. One exists because the orientation is acyclic: take the position in a topological order divided by $|V(\mathfrak{K})| + 1$ (for MED one may use Lemma 6.3(a)). For each j fix a numbering $\text{tp}_j : T_j \rightarrow \{1, \dots, |T_j|\}$ that increases along the arcs of the acyclic digraph $D'_j[T_j]$. Define

$$\Psi(v) := 3j + \text{pot}_{\mathfrak{K}}(v) \quad (v \in V(\mathfrak{K}), \mathfrak{K} \in \mathcal{K}_j), \quad \Psi(y) := 3j + 1 + \frac{\text{tp}_j(y)}{|T_j| + 1} \quad (y \in T_j).$$

This is well defined by (D). Moreover $\Psi(v) \in (3j, 3j + 1)$ on layer j , and $\Psi(y) \in (3j + 1, 3j + 2)$ on T_j . Now check the arcs.

- Bead arcs increase Ψ .
- An arc of D'_j from $u \in \mathcal{X}_j$ to $y \in T_j$ has $\Psi(u) < 3j + 1 < \Psi(y)$. An arc inside T_j increases tp_j .
- For $j \leq k - 2$, an arc of D'_j with head $v \in \mathcal{X}_{j+1}$ has tail in $T_j \cup \mathcal{X}_j$, and $\Psi(\text{tail}) < 3j + 2 < 3j + 3 < \Psi(v)$.

Let $\mathcal{W}$ be the set of arcs of D'_{k-1} whose head lies in $\mathcal{X}_0$ (for $k = 1$: the arcs of D'_0 whose head is a port). Every arc outside $\mathcal{W}$ increases Ψ , so $\vec{F} - \mathcal{W}$ is acyclic.

An arc of D'_{k-1} whose head is a port is the last arc of its path, because interior vertices lie in T_{k-1} . Each path of $\mathcal{P}_{k-1}$ has exactly one last arc, which ends at a port and is never deleted by the cancellation of Step 1. Hence $|\mathcal{W}| = |\mathcal{P}_{k-1}| = \Phi$.

Step 4: count and length. By Lemma 6.1, F decomposes into at most $|\mathcal{W}| = \Phi$ cycles of G . Each of them is the orientation of a directed cycle C of $\vec{F}$ of length at least 3 containing an arc of $\mathcal{W}$.

For the bound k on the length, call $[3j, 3j + 3)$ the j -th block. Every arc of $\vec{F} - \mathcal{W}$ either has both ends in the same block, or is an arc of some D'_j ($j \leq k - 2$) from block j to block $j + 1$ with head in $\mathcal{X}_{j+1}$. Take a maximal subpath of C that contains no arc of $\mathcal{W}$. It starts at the head of a $\mathcal{W}$ -arc, a vertex of $\mathcal{X}_0$ (block 0), and ends at the tail of a $\mathcal{W}$ -arc, a vertex of $T_{k-1} \cup \mathcal{X}_{k-1}$ (block $k - 1$). Along it Ψ increases, so it contains at least $k - 1$ block-crossing arcs. Together with the $\mathcal{W}$ -arc that follows it, C has at least k arcs. This proves (ii). $\square$

Lemma 6.7 (union of systems). *Let $E_1, \dots, E_q \subseteq E(G)$ be pairwise disjoint, and suppose each E_i decomposes into a_i objects. Then $E_1 \cup \dots \cup E_q$ decomposes into $\sum_i a_i$ objects.*

In particular, suppose several systems satisfy the hypotheses of Lemma 6.6 separately, each with its own layers, junction sets and path families, and the beads and the path edges of distinct systems are pairwise disjoint. Then, with the sets F_j given by Lemma 6.6 for each system, the union over all systems of the beads together with the F_j decomposes into at most the sum of their values Φ of cycles of G . No vertex-disjointness across systems is needed: a vertex may be a hub, a port or a junction vertex in several systems. Hypothesis (D) is used only inside one system, through the potential Ψ of that system.

Dependencies: 1.5, 6.6

Proof. The union of the given decompositions is a family of objects whose edge sets partition $\bigcup_i E_i$, because the E_i are pairwise disjoint. This is the argument of the proof of Fact 1.5(b) (concatenation of decompositions of disjoint edge sets); the statement of Fact 1.5(b) itself is the corresponding bound on f .

For the second statement, apply Lemma 6.6 to each system separately. Its proof uses (D) only for the vertices of that system and of its own junction sets. $\square$

6.2 Designations and hub concentration: CONC and CONC-L

From here on a valid $\text{HB}^{*\tau+}$ run on G is fixed (Definition 2.7), with the notation of Definition 2.8. For an ancestor Y we write $r = r(Y)$ for its round. All logarithms are to base 2.

Definition 6.8 (designation and class data; $\mathcal{V} = \emptyset$). A *designation* δ assigns, for every round $l \geq 3$, to every standalone pre-part $Z \in \text{Std}_l$ and every classed port $u \in Q_Z$ an ancestor $Y(u) \in \text{anc}_l(u)$, the *class* of u . Here δ is any deterministic function of the run. There are no VX-parts: every $Z \in \text{Std}_l$ is treated as a MIX-part. Every classed port u of round l lies in Q_Z for exactly one $Z \in \text{Std}_l$ (port sets of one round are disjoint), and we write Z_u for that part. Since $Y(u) \in \text{anc}_l(u)$, we have $u \in V(Y(u))$ and $r(Y(u)) \leq l - 2$.

For $l \geq 3$, an ancestor Y of round r and any vertex h , define:

- the *class- Y degree* $d_{Y,l}(h) := \#\{hu \in E_l(Z) : Z \in \text{Std}_l, u \in Q_Z, Y(u) = Y\}$, and $m_{Y,l} := \max_h d_{Y,l}(h)$, the maximum over all vertices h of G ;
- $d_{Y,l}^* := \#\{u : u \text{ a classed port of round } l, Y(u) = Y, u \in \text{Dup}_r^*\}$;
- for light Y , the *guest core degree*

$$\alpha_{Y,l} := \max_{x \in S_Y} \#\{u \in Y \setminus \text{Dup}_r^* : u \text{ a classed port of round } l, Y(u) = Y, xu \in E_l(Z_u)\},$$

with $\alpha_{Y,l} := 0$ if $S_Y = \emptyset$;

- the *deterministic bead graph*

$$\mathfrak{B}_{Y,l} := \{hu \in E_l(Z) : Z \in \text{Std}_l, u \in Q_Z, Y(u) = Y, \text{ and } h \notin Q_Z \text{ or } Y(h) \neq Y\};$$

- $\gamma_l := \lfloor P_{l-2}/M_l \rfloor$; the pair (Y, l) is *giant* iff some connected component of $\mathfrak{B}_{Y,l}$ has more than $2\gamma_l$ edges;
- the *class counts* $c_{h,l}^{\text{agg}} := \#\{Y : d_{Y,l}(h) \geq 1\}$; for $Z \in \text{Std}_l$ and a fresh port $x \in F_Z$, $c_x(Z) := \#\{Y(u) : u \in Q_Z, xu \in E_l(Z)\}$; and for $u \in Q_Z$, $c_{\text{pp}}(u) := \#\{Y(v) : v \in Q_Z, uv \in E_l(Z)\}$.

For $l \leq 2$ there are no classed ports ($\text{anc}_l(x) = \emptyset$), and all these quantities vanish. All of them are deterministic functions of the run and δ .

Dependencies: 2.8, 2.7

The following elementary facts about $\log^*$ are used in the proofs of the next two theorems. Put $\mathsf{T}_0 := 1$ and $\mathsf{T}_{i+1} := 2^{\mathsf{T}_i}$. For $x \geq 1$ and an integer $k \geq 0$, $\log^* x \leq k$ iff $x \leq \mathsf{T}_k$. For $x > 1$, $\log^* x = 1 + \log^*(\log x)$. And $\log^* x \leq 1 + \log x$ for all $x \geq 1$: this holds for $x \leq 4$ by inspection, and for $x > 4$ by induction, since $2 + \log \log x \leq 1 + \log x$ when $x \geq 4$.

Put $k_* := \log^* D_*$. By (F1)(a) at $\mu = \log \log D_*$, $\log \log D_* \geq 2^8 > 16$, so $D_* > 2^{65536} = \mathsf{T}_5$ and $k_* \geq 6$.

For $d \geq D_*$ put

$$\mathsf{a}(d) := \frac{2 \log^* d + 2}{\log d}, \quad \mathsf{b}(d) := \frac{2 \log^* d + 1}{\log \log d}, \quad \mathsf{c}(d) := \frac{2 \log^* d + 1}{(\log d)^{1/2}}.$$

We use two facts. First, for every valid run and every $r < R$,

$$\mathsf{a}(d_r) \leq \frac{1}{2} \mathsf{a}(d_{r+1}), \quad \mathsf{b}(d_r) \leq \frac{1}{2} \mathsf{b}(d_{r+1}), \quad \mathsf{c}(d_r) \leq \frac{1}{2} \mathsf{c}(d_{r+1}). \quad (6.1)$$

Second, for every $d \geq D_*$,

$$\mathsf{a}(d) \leq \frac{2k_* + 4}{\log D_*}, \quad \mathsf{b}(d) \leq \frac{2k_* + 3}{\log \log D_*}, \quad \mathsf{c}(d) \leq \frac{2k_* + 3}{(\log D_*)^{1/2}}. \quad (6.2)$$

Proof of (6.1) and (6.2). For (6.1), put $x := \lambda_r = \log d_r$ and $y := \lambda_{r+1}$. Then $y \geq \log D_*$, so $v := \log y \geq \log \log D_*$, and (F1)(a),(b) at v give $v \geq 2^8$ and $y = 2^v \geq 2^{14} A v^3$. Moreover $x \geq d_{r+1}^{1/A} = 2^{y/A}$ by Lemma 2.17(a). Also $\log^* d_r = 1 + \log^* x \leq 2 + \log x$, and $\log^* d_r = 2 + \log^*(\log x) \leq 3 + \log \log x$.

The functions $t \mapsto (2 \log t + 6)/t$, $t \mapsto (7 + 2 \log t)/t$ and $t \mapsto (2 \log t + 5)/t^{1/2}$ are decreasing for $t \geq 4$. Hence:

- $\mathsf{a}(d_r) \leq (2 \log x + 6)/x \leq (2y/A + 6)/2^{y/A} \leq 1/y \leq \mathsf{a}(d_{r+1})/2$;
- with $\mu := \log x \geq y/A$: $\mathsf{b}(d_r) \leq (7 + 2 \log \mu)/\mu \leq A(7 + 2 \log y)/y \leq 1/(2 \log y) \leq \mathsf{b}(d_{r+1})/2$;
- $\mathsf{c}(d_r) \leq (2 \log x + 5)/x^{1/2} \leq (2y/A + 5)/2^{y/(2A)} \leq 1/(2y^{1/2}) \leq \mathsf{c}(d_{r+1})/2$.

The middle inequalities hold because $y/(2A) \geq 2^{13} v^3 \geq 4 + 2v$ and $y \geq 2^{14} A v^3 \geq 2A v(7 + 2v)$. The first gives $2^{y/(2A)} \geq 2^{4+2v} = 16y^2$, hence $y(2y/A + 6) \leq 8y^2 \leq 2^{y/A}$ and $2y^{1/2}(2y/A + 5) \leq 14y^{3/2} \leq 2^{y/(2A)}$; the second is $A(7 + 2 \log y)/y \leq 1/(2 \log y)$.

For (6.2), note $D_* \leq \mathsf{T}_{k_*}$. There are two cases.

- If $\log d \leq \mathsf{T}_{k_*}$, then $d \leq \mathsf{T}_{k_*+1}$, so $\log^* d \leq k_* + 1$, while $\log d \geq \log D_*$ and $\log \log d \geq \log \log D_*$. The three bounds follow.
- If $\log d > \mathsf{T}_{k_*} \geq D_*$, put $x := \log d > D_*$. Then $\log^* d \leq 2 + \log x$ and $\log^* d \leq 3 + \log \log x$. Hence $\mathsf{a}(d) \leq (2 \log x + 6)/x \leq (2 \log D_* + 6)/D_* \leq 1/\log D_*$; similarly $\mathsf{b}(d) \leq (7 + 2 \log \log x)/\log x \leq (7 + 2 \log \log D_*)/\log D_* \leq 1/\log \log D_*$, and $\mathsf{c}(d) \leq (2 \log D_* + 5)/D_*^{1/2} \leq (\log D_*)^{-1/2}$. The last inequality of each of the three chains holds because (F1)(a),(b), applied at $\mu = \log D_*$ and at $\mu = \log \log D_*$ (both at least $\log \log D_*$), give $t \geq 2^8$ and $2^t \geq 2^{14} A t^3$ for $t \in \{\log D_*, \log \log D_*\}$; hence, as $t \geq 1$, $t(2t + 6) \leq t(2t + 7) \leq 9t^2 \leq 2^t$ for both values of t , and $t(2t + 5)^2 \leq 49t^3 \leq 2^t$ for $t = \log D_*$. $\square$

Theorem 6.9 (Theorem CONC, standalone ancestors). *For every valid $\text{HB}^{\tau+}$ run and every designation δ :*

- (i) *for every $Y \in \text{Std}_r$, every $l \geq r+1$ and every vertex h : $\#\{u \in Y^0 \setminus \text{Dup}_r^* : hu \in E(G_l)\} \leq \tau_r - 1$;*
- (ii) *$m_{Y,l} \leq \tau_r - 1 + d_{Y,l}^*$ for every $Y \in \text{Std}_r$ and every l ;*
- (iii) $\sum_{(Y,l): Y \text{ standalone}} m_{Y,l} \leq 2^{\sigma+14} n \frac{\log^* D_*}{\log D_*} + 100 \varepsilon n \frac{\log^* D_*}{C' \log \log D_*}$;
- (iv) *(i)–(iii) hold for every designation, including designations that mix light and standalone classes: (i) bounds, for every vertex h , the number of G_l -edges from h into $Y^0 \setminus \text{Dup}_r^*$, whatever the classes of the ports involved.*

Dependencies: 2.19, 2.17, 2.13, 2.16, 6.8, 2.7, (F1)

Proof. (i) By (R1) and (R5) of Definition 2.7, $G_l \subseteq G_{r+1}$ for $l \geq r+1$. By Proposition 2.19(a), a standalone Y has no edges of type (α) . So every G_{r+1} -edge hu with $u \in Y^0 \setminus \text{Dup}_r^*$ is of type (β) ; in particular $h \notin Y^0$. By Proposition 2.19(b) there are at most $\tau_r - 1$ of them in G_l .

(ii) Fix h . Every edge counted in $d_{Y,l}(h)$ is an edge $hu \in E_l(Z) \subseteq E(G_l)$ with $Y(u) = Y$, hence $u \in V(Y) = Y^0$. Those with $u \notin \text{Dup}_r^*$ number at most $\tau_r - 1$ by (i). Those with $u \in \text{Dup}_r^*$ go to distinct ports u (G is simple) counted by $d_{Y,l}^*$. Taking the maximum over h gives (ii).

(iii) *The τ -term.* $m_{Y,l} = 0$ unless $3 \leq l$ and $r+2 \leq l \leq R$, so at most $R - r - 1$ values of l occur for each Y of round r . We use the looser bound $R - r - 1 < R - r \leq 2 \log^* d_r + 2$ (Lemma 2.17(a)), which matches the numerator of $\mathbf{a}$; the sharper count is not needed. By Proposition 2.13(K1), $|\text{Std}_r| \leq 1.37n/P_r$. By Lemma 2.17(c), $\tau_r/P_r \leq 2^{\sigma+11}/\lambda_r$. Hence

$$\sum_{(Y,l), Y \in \text{Std}} (\tau_r - 1) \leq \sum_{r \leq R} 1.37 \frac{n}{P_r} (2 \log^* d_r + 2) \tau_r \leq 1.37 \cdot 2^{\sigma+11} n \sum_{r \leq R} \mathbf{a}(d_r).$$

By (6.1) and Lemma 2.16(i), $\sum_r \mathbf{a}(d_r) \leq 2\mathbf{a}(d_R)$. By (6.2) and $k_* \geq 6$, $\mathbf{a}(d_R) \leq (2k_* + 4)/\log D_* \leq \frac{8}{3}k_*/\log D_*$. So the τ -term is at most $1.37 \cdot \frac{16}{3} \cdot 2^{\sigma+11} k_* n / \log D_* \leq 2^{\sigma+14} n \log^* D_* / \log D_*$, since $1.37 \cdot 16/3 < 8$.

The d^ -term.* Fix l . A classed port u of round l is counted in $d_{Y,l}^*$ only for $Y = Y(u)$, and then $u \in V(Y) \cap \text{Dup}_{r(Y)}^* \subseteq Y^0 \cap \text{Dup}_{r(Y)}^*$. The map $u \mapsto (Y(u), u)$ is injective. Hence

$$\sum_Y d_{Y,l}^* \leq \sum_{r \leq l-2} \sum_{Y^0 \text{ a pre-part of round } r} |Y^0 \cap \text{Dup}_r^*| \leq \sum_{r \leq l-2} \frac{16\varepsilon n}{\log P_r},$$

by Proposition 2.13(K3). Here distinct ancestors of round r have distinct pre-parts, and a standalone ancestor is its own pre-part.

Summing over l , each r occurs for at most $R - r - 1 \leq 2 \log^* d_r + 1$ values of l . Moreover $\log P_r \geq C' \log \lambda_r = C' \log \log d_r$. Hence

$$\sum_{(Y,l)} d_{Y,l}^* \leq \frac{16\varepsilon n}{C'} \sum_{r \leq R} \mathbf{b}(d_r) \leq \frac{32\varepsilon n}{C'} \mathbf{b}(d_R) \leq \frac{32\varepsilon n(2k_* + 3)}{C' \log \log D_*} \leq \frac{80\varepsilon n \log^* D_*}{C' \log \log D_*},$$

by (6.1), (6.2) and Lemma 2.16(i), using $2k_* + 3 \leq 2.5k_*$ for $k_* \geq 6$. Adding the two terms, via (ii), proves (iii); the constant 100 in (iii) is deliberately looser than the 80 proved here.

(iv) Nothing in (i)–(iii) used how δ chooses among the ancestors in $\text{anc}_l(u)$. The bound (i) is a statement about G_l and Y^0 alone. The bound (ii) holds for every $Y \in \text{Std}_r$ whatever other ports have other classes. The sum in (iii) runs over the pairs (Y, l) with Y standalone, however the light classes are chosen. $\square$

Theorem 6.10 (Theorem CONC-L). *Let Y be a light part of round r of a valid $\text{HB}^{*\tau+}$ run, let $l \geq r+1$, and let h be a vertex.*

- (i) *If $h \notin S_Y$: $\#\{u \in Y \setminus \text{Dup}_r^* : hu \in E(G_l)\} \leq \tau_r - 1$, and there are no such u if $h \in Y$.*
- (ii) *If $h \in S_Y$: every such hu is an edge of X_Y^0 , and their number is at most $e_{X_Y^0}(h, Y)$.*
- (iii) *For every designation, $m_{Y,l} \leq \max(\tau_r - 1, \alpha_{Y,l}) + d_{Y,l}^*$.*
- (iv) *$\alpha_{Y,l} < \theta_r^{\text{GC}}(Y^0)$. For every valid run and every designation, summing over all ancestors (light and standalone, GC-parts included),*

$$\sum_{(Y,l)} m_{Y,l} \leq \varepsilon_{\text{CONC}}(D_*) n,$$

where

$$\varepsilon_{\text{CONC}}(D_*) := 2^{\sigma+15} \frac{\log^* D_*}{\log D_*} + 200\varepsilon \frac{\log^* D_*}{C' \log \log D_*} + \frac{4(2 \log^* D_* + 2)}{(\log D_*)^{1/2}}.$$

$\varepsilon_{\text{CONC}}$ depends only on D_* (and on σ, C', ε), and $\varepsilon_{\text{CONC}}(D_*) \rightarrow 0$ as $D_* \rightarrow \infty$.

Dependencies: 2.19, 2.3, 2.4, 2.18, 2.12, 2.17, 2.16, 6.9, 2.13, 2.7, 6.8, (T1)

Proof. Recall $Y^0 = Y \sqcup S_Y$ and $V(Y) = Y$. Let $u \in Y \setminus \text{Dup}_r^*$ and let $hu \in E(G_l) \subseteq E(G_{r+1})$. By Proposition 2.19(a) (which rests on Lemmas 2.3 and 2.4), hu is of exactly one of two types: (β) , with $h \notin Y^0$; or (α) , with $h \in S_Y$ and $hu \in E(X_Y^0)$. The three cases $h \notin Y^0$, $h \in Y$ and $h \in S_Y$ are exhaustive.

(i) If $h \notin Y^0$, all such edges are of type (β) , and Proposition 2.19(b) bounds their number by $\tau_r - 1$. If $h \in Y$, the edge hu would be an edge of G_l ($l > r$) with both ends in $V(Y)$, which Lemma 2.12 forbids.

(ii) If $h \in S_Y \subseteq Y^0$, the edge cannot be of type (β) , so it is of type (α) and lies in $E(X_Y^0)$. Distinct u give distinct X_Y^0 -edges from h into Y , so their number is at most $e_{X_Y^0}(h, Y)$.

(iii) Fix h . A port u with $Y(u) = Y$ lies in $V(Y) = Y$. The ports in Dup_r^* contribute at most $d_{Y,l}^*$, as in the proof of Theorem 6.9(ii). The others contribute at most $\tau_r - 1$ if $h \notin Y^0$, nothing if $h \in Y$, and at most $\alpha_{Y,l}$ if $h \in S_Y$ (by definition of $\alpha_{Y,l}$, since the counted edges hu lie in $E_l(Z_u)$). Exactly one case applies to h .

(iv) The edges counted in $\alpha_{Y,l}$ at $x \in S_Y$ are distinct X_Y^0 -edges from x into $Y = Y^0 \setminus S_Y$, by (ii). By rule (GC) and Lemma 2.18(ii), $e_{X_Y^0}(x, Y) < \theta_r^{\text{GC}}(Y^0)$. Hence $\alpha_{Y,l} < \theta_r^{\text{GC}}(Y^0)$, and by (iii),

$$m_{Y,l} \leq (\tau_r - 1) + (\theta_r^{\text{GC}}(Y^0) - 1) + d_{Y,l}^* \quad (Y \text{ light}), \quad m_{Y,l} \leq \tau_r - 1 + d_{Y,l}^* \quad (Y \text{ standalone}),$$

the latter by Theorem 6.9(ii); GC-parts are standalone (Lemma 2.18(iii)). We sum the three kinds of terms over all pairs (Y, l) with $r(Y) + 2 \leq l \leq R$ (for other l , $m_{Y,l} = 0$).

τ -terms. The number of ancestors of round r (light and standalone) is at most $1.37n/P_r$ (Proposition 2.13(K1)). The computation in the proof of Theorem 6.9(iii) applies verbatim and gives at most $2^{\sigma+14} n \log^* D_* / \log D_*$.

d-terms.* The injectivity argument in the proof of Theorem 6.9(iii) used only $u \in V(Y(u)) \subseteq Y(u)^0$ and Proposition 2.13(K3), which sums over all pre-parts of round r . So it applies to all ancestors together and gives at most $80\epsilon n \log^* D_*/(C' \log \log D_*)$.

θ^{GC} -terms. For light Y of round r , $\theta_r^{\text{GC}}(Y^0) \leq |Y^0| \lambda_r^{-1/2} + 1$. The pre-parts Y^0 of distinct light parts of round r are distinct, so by Proposition 2.13(K1)

$$\sum_{Y \text{ light, round } r} \theta_r^{\text{GC}}(Y^0) \leq 1.37 n \lambda_r^{-1/2} + 1.37 \frac{n}{P_r} \leq 1.38 n \lambda_r^{-1/2},$$

using $P_r \geq \lambda_r^{C'}$. Each Y occurs for at most $R - r - 1 \leq 2 \log^* d_r + 1$ values of l (Lemma 2.17(a)). By (6.1), (6.2) and Lemma 2.16(i),

$$\sum_{(Y,l), Y \text{ light}} \theta_r^{\text{GC}}(Y^0) \leq 1.38 n \sum_{r \leq R} c(d_r) \leq 2.76 n c(d_R) \leq \frac{2.76 n (2k_* + 3)}{(\log D_*)^{1/2}} \leq \frac{4n(2 \log^* D_* + 2)}{(\log D_*)^{1/2}}.$$

The last step holds because $2.76(2k + 3) \leq 4(2k + 2)$ for all $k \geq 1$.

Adding the three sums gives $\sum_{(Y,l)} m_{Y,l} \leq \varepsilon_{\text{CONC}}(D_*)n$. In fact the first two terms of $\varepsilon_{\text{CONC}}$ carry a spare factor 2.

Each term of $\varepsilon_{\text{CONC}}$ tends to 0 as $D_* \rightarrow \infty$, because $\log^* x$ grows more slowly than every iterate of $\log$. The bound uses nothing about δ beyond $u \in V(Y(u))$, so it holds for every designation. $\square$

6.3 Lending statuses and lost ports

Definition 6.11 (stage-2 lending statuses, retirement sets and the functional X_U). Fix a valid run, a designation δ and a stage-1 outcome: the colourings and JS labels of Definition 3.10, and the zones of Definition 5.1.

- An ancestor Y is *lend-bad* iff Y is a demoted light part (Definition 5.3), or an event of Lemma 3.13(a) fails for Y , or an event of Lemma 3.13(b) fails for Y . Otherwise Y is *lend-good*. Thus for a lend-good Y the classes Own_Y , Lend_Y and all lent classes are the expanders of Lemma 3.13(a), and every $\mathcal{L}_{Y,l,j}^{\text{JS}}$ is $(2^{12}L_Y^4, t_l^{\text{JS}})$ -path connected through $T_j(Y, l)$.
- For $l \geq 3$ and $Z \in \text{Std}_l$, the set of *lost* ports is

$$\text{Lost}_Z := \{u \in Q_Z : Y(u) \text{ is lend-bad, or } \text{lab}_{Y(u),l}(u) \neq *\}.$$

Put $\text{Ret}_Z := A_Z \cup F_Z \cup \text{Lost}_Z$ (the *retirement set* of Z for Lemma 4.2) and $Q_Z^* := Q_Z \setminus \text{Lost}_Z$. For $l \leq 2$ put $\text{Lost}_Z := \emptyset$, $\text{Ret}_Z := A_Z \cup F_Z = Z^0$ and $Q_Z^* := \emptyset$.

Since $A_Z = Z^0 \cap D_l$ and $U_Z = Z^0 \setminus D_l = F_Z \sqcup Q_Z$, we have $Z^0 = \text{Ret}_Z \sqcup Q_Z^*$, and A_Z, F_Z, Lost_Z are pairwise disjoint.

- $O_Z := \text{Own}_Z$ if the ancestor Z is lend-good, and $O_Z := X_Z^0$ otherwise.
- $\text{Lost}_l := \bigcup_{Z \in \text{Std}_l} \text{Lost}_Z$.
- $X_U := 80 \text{ dem} + 369 \text{ lp} + \sum_l (169 + M_l) |\text{Lost}_l|$, with dem and lp as in Definition 5.3.

All of these are deterministic functions of the run, δ and the stage-1 outcome. For every classed port $u \in Q_Z^*$, the class $Y(u)$ is lend-good and $u \notin \bigcup_j T_j(Y(u), l)$.

Dependencies: 3.10, 3.13, 5.3, 6.8, 5.1

Lemma 6.12 (lost ports; the constant (K6)). *Fix a valid run and a designation δ . Probabilities and expectations are over the stage-1 outcome.*

(i) *for every classed port u of round $l \geq 3$,*

$$\mathbb{P}(u \in \text{Lost}) \leq K_l^{\text{JS}} \rho_l + \mathbb{P}(Y(u) \text{ lend-bad}) \leq M_l^{-2} + |V(Y(u))|^{-2} \leq 2M_l^{-2};$$

(K6) $\mathbb{E}|\text{Lost}_l| \leq 2n/M_l^2$ *for every round l ;*

(iii) $\mathbb{E}X_U \leq \varepsilon_U(D_*)n + \sum_l 2n(169 + M_l)/M_l^2 \leq \varepsilon_U(D_*)n + 5.5n/D_*$.

Dependencies: 6.11, 3.13, 5.4, 5.5, 2.11, 2.17, 3.10

Proof. (i) Let $Y := Y(u)$, of round $r \leq l - 2$. The designation is deterministic, so Y is fixed.

The label. By Definition 3.10(iv), $\text{lab}_{Y,l}(u) \neq *$ has probability $K_l^{\text{JS}} \rho_l = M_l^2 \cdot M_l^{-4} = M_l^{-2}$.

Lend-bad. By the union bound, $\mathbb{P}(Y \text{ lend-bad})$ is at most the probability that an event of Lemma 3.13(a) or (b) fails, plus, for light Y , the probability that Y is demoted. The first is at most $|V(Y)|^{-2}/2$ by Lemma 3.13. The second is at most $|Y|^{-2}/2 = |V(Y)|^{-2}/2$ by Lemma 5.4(b), since $V(Y) = Y$ for a light part (Definition 2.8). Hence $\mathbb{P}(Y \text{ lend-bad}) \leq |V(Y)|^{-2}$.

Size of $V(Y)$. By Proposition 2.11(iv), $|V(Y)| \geq P_r/2$. By Lemma 2.17(b), $P_r \geq P_{l-2}$ (as $P_r \geq 2P_{r+1}$) and $P_{l-2} \geq M_l^{13}$. Hence $|V(Y)|^{-2} \leq 4M_l^{-26} \leq M_l^{-2}$.

(K6) The port sets U_Z of distinct $Z \in \text{Std}_l$ are pairwise disjoint (Proposition 2.11(iv)), so round l has at most n classed ports. By (i) and linearity of expectation, $\mathbb{E}|\text{Lost}_l| \leq 2n/M_l^2$.

(iii) By Lemma 5.5, $\mathbb{E}[80 \text{ dem} + 369 \text{ lp}] \leq \varepsilon_U(D_*)n$. By (K6),

$$\mathbb{E} \sum_l (169 + M_l) |\text{Lost}_l| \leq \sum_l \left(\frac{338n}{M_l^2} + \frac{2n}{M_l} \right) \leq \left(2 + \frac{338}{2^{40}} \right) n \sum_l \frac{1}{M_l} \leq \left(2 + \frac{338}{2^{40}} \right) \frac{2n}{D_*} \leq \frac{5.5n}{D_*},$$

using $M_l \geq 2^{40}$ and $\sum_l 1/M_l \leq 2/D_*$ (Lemma 2.17(b)). $\square$

6.4 Junction synchronization: Lemma JS-LC and the J-interface

Lemma 6.13 (Lemma JS-LC; one round $l \geq 3$; cherry split for giant pairs). *Fix a valid run, a designation δ , a stage-1 outcome with its stage-2 data (Definition 6.11), and a round $3 \leq l \leq R$. For every $Z \in \text{Std}_l$ let $B_Z \subseteq E_l(Z)$ be a set of edges, each of which has an end in Q_Z^* . (In the assembly $B_Z := E^Q(Z)$, which has this property by (T1) of Lemma 4.2.)*

Hypothesis: no edge of any class $\mathcal{L}_{Y,l,j}^{\text{JS}}$ ($0 \leq j < K_l^{\text{JS}}$, Y a lend-good ancestor of a round $\leq l-2$) has been used.

Then there are a set $J_l \subseteq \bigcup_Z B_Z$ and a set $\text{Lent}_l^{\text{JS}} \subseteq \bigcup_{Y,j} \mathcal{L}_{Y,l,j}^{\text{JS}}$ such that $\bigcup_Z B_Z \cup \text{Lent}_l^{\text{JS}}$ is partitioned into the single edges of J_l and at most

$$126 \frac{n}{M_l} + 1.5 \sum_{(Y,l) \text{ giant}} m_{Y,l} \text{ cycles.}$$

Here the sum is over the ancestors Y for which the pair (Y,l) is giant (Definition 6.8), and $m_{Y,l}$ is the maximal class- Y degree of Definition 6.8. The proof gives the constant 15 in place of 126; the looser 126 is kept because it is the value used downstream (see the end of Step 8). Each cycle consists of edges of $\bigcup_Z B_Z$ and edges of $\bigcup_j \mathcal{L}_{Y,l,j}^{\text{JS}}$ for a single ancestor Y . Moreover

$$|J_l| \leq \sum_{h \in D_l} c_{h,l}^{\text{agg}} + \sum_{Z \in \text{Std}_l} \left[\sum_{x \in F_Z} c_x(Z) + (M_l - 1) |\text{Lost}_Z| + \frac{1}{2} \sum_{u \in Q_Z} c_{\text{pp}}(u) \right]. \quad (6.3)$$

The right side of (6.3) is computed from the sets $E_l(Z)$, not from the B_Z . The bound (6.3) is recorded for completeness and is not used later: Section 7 bounds J_l through (J1) and (J2) of Lemma 6.15. The edges of $\bigcup_{Y,j} \mathcal{L}_{Y,l,j}^{\text{JS}}$ outside $\text{Lent}_l^{\text{JS}}$, among them the returned edges inside the graphs $G[T_j(Y,l)]$, are not used; they are junk for their owner Y .

Dependencies: 6.5, 6.3, 6.4, 6.6, 6.7, 6.8, 6.11, 3.13, 3.1, 2.12, 2.9, 2.13, 2.17, 2.7, 2.11, 3.10, 1.9

The hypothesis refers only to the classes indexed by the current round l . Classes $\mathcal{L}_{Y,l',j}^{\text{JS}}$ with $l' > l$ of the same ancestors have already been used at earlier steps of the order $R, R-1, \dots$, and that is allowed.

Proof. Write $K_l^{\text{JS}} := K_l^{\text{JS}} = M_l^2$ and $t := t_l^{\text{JS}} = 2M_l + 2$. Recall two facts. For $u \in Q_Z^*$, the class $Y(u)$ is lend-good and $\text{lab}_{Y(u),l}(u) = *$ (Definition 6.11). Every vertex has degree at most $M_l - 1$ in every $E_l(Z)$ (Lemma 2.9(ii)).

Step 1: types. Let $e \in B_Z$. By hypothesis e has an end $u \in Q_Z^*$. Both ends lie in Z^0 , since $E_l(Z) \subseteq E(G[Z^0])$ by (R5). Since $Z^0 = \text{Ret}_Z \sqcup Q_Z^*$, there are two cases for the other end v :

- $v \in \text{Ret}_Z = A_Z \sqcup F_Z \sqcup \text{Lost}_Z$. Then e is a *centre-port edge* with centre v .
- $v \in Q_Z^*$. Then e is a *port-port edge*, and $Y(v) \neq Y(u)$: otherwise both ends of $e \in E(G_l)$ would lie in $V(Y(u))$ with $l > r(Y(u))$, contrary to Lemma 2.12.

By the same lemma, for every edge $hu \in E_l(Z)$ with $u \in Q_Z$ and $Y(u) = Y$, the vertex h lies outside $V(Y)$. In particular $h \notin T_j(Y,l)$, since $T_j(Y,l) \subseteq V(Y)$.

Step 2: parity; the set J_l .

(2a) For each $Z \in \text{Std}_l$ and each unordered pair $\{a,b\}$ of distinct classes, let $E_{ab}(Z)$ be the set of port-port edges of B_Z joining a class- a port of Q_Z^* to a class- b port of Q_Z^* . It is bipartite with sides V_a (class- a ports) and V_b (class- b ports). Apply Lemma 6.5 with any admissible choice of the deleted edges. This gives $J'_{ab}(Z)$ and $E_{ab}(Z) \setminus J'_{ab}(Z) = S_a \sqcup S_b$. The edges of S_a are *assigned to class a* : in them the class- b end is a *pseudo-hub*, of even S_a -degree. Symmetrically, S_b is assigned to class b .

(2b) For each class Y let R_Y^0 consist of all centre-port edges $hu \in B_Z$ ($Z \in \text{Std}_l$) with $Y(u) = Y$, together with all port-port edges assigned to Y in (2a). Every edge of R_Y^0 has exactly one end that is a class- Y port of some Q_Z^* , its *port*; its other end is its *centre*. A centre lies in Ret_Z or is a port of Q_Z^* of a class $\neq Y$. By Step 1, centres of class- Y edges lie outside $V(Y)$, while ports lie in $V(Y)$; so no vertex is both a port and a centre of class Y .

For each Y and each vertex h that is the centre of an odd number of edges of R_Y^0 , move one such edge (any one) into J_l . Let R_Y be the rest: the *realized class- Y beads*. Finally, J_l consists of all edges $J'_{ab}(Z)$ and all moved edges.

Only centres in Ret_Z are ever odd. A pseudo-hub $v \in Q_Z^*$ of class b lies in exactly one part (it is not in D_l). It is a centre of class- Y edges only through the edges of $E_{Yb}(Z)$ assigned to Y , and it has even degree in those by Lemma 6.5. A moved edge hu changes the class- Y centre degree of h only: its other end u is a port, which carries no parity constraint. So after the moves every centre has even degree in every R_Y .

Counting J_l .

- PAR deletions: $\sum_{\{a,b\}} |J'_{ab}(Z)| \leq \frac{1}{2} \sum_{\{a,b\}} |V(E_{ab}(Z))|$. A port $u \in Q_Z^*$ lies in $V(E_{ab}(Z))$ only for $\{a,b\} = \{Y(u), Y(v)\}$ with $uv \in B_Z \subseteq E_l(Z)$ and $v \in Q_Z$. This happens for at most $c_{\text{pp}}(u)$ pairs, so the PAR deletions number at most $\frac{1}{2} \sum_{u \in Q_Z} c_{\text{pp}}(u)$.

- Moves at a hub $h \in D_l$: at most one per class Y for which h is the centre of an edge $hu \in E_l(Z)$ with $u \in Q_Z$ and $Y(u) = Y$, i.e. with $d_{Y,l}(h) \geq 1$. So at most $c_{h,l}^{\text{agg}}$ moves, summed over all parts containing h .
- Moves at a fresh centre $x \in F_Z$: $x \notin D_l$ lies in one part only, so at most $c_x(Z)$ moves.
- Moves at a lost centre $v \in \text{Lost}_Z$: at most the number of edges of $E_l(Z)$ at v , which is at most $M_l - 1$.

This proves (6.3). Every quantity on its right is defined from the sets $E_l(Z)$.

Step 3: realized components. Fix an ancestor Y with $R_Y \neq \emptyset$. Then Y is the class of a port in some Q_Z^* , so Y is lend-good and $r(Y) \leq l - 2$.

Every edge $hu \in R_Y$ lies in $\mathfrak{B}_{Y,l}$: $hu \in E_l(Z)$, $u \in Q_Z^* \subseteq Q_Z$, $Y(u) = Y$, and h is not a class- Y port of Q_Z (a class- Y vertex of Q_Z lies in $V(Y)$, and $h \notin V(Y)$). Hence every connected component of the graph R_Y lies inside a component of $\mathfrak{B}_{Y,l}$ and has at most as many edges. Call a component of R_Y *giant* if it has more than $2\gamma_l$ edges. Giant components exist only if (Y, l) is giant; whether (Y, l) is giant is decided by the deterministic graph $\mathfrak{B}_{Y,l}$.

Every edge of R_Y joins a port (a class- Y port u of some Q_Z^* , with $u \in V(Y) \setminus \bigcup_j T_j(Y, l)$) to a centre (a vertex outside $V(Y)$). The degree of a centre h in R_Y is at most $d_{Y,l}(h) \leq m_{Y,l}$. The degree of a port u in R_Y is at most $\deg_{E_l(Z_u)}(u) \leq M_l - 1$. All edges of R_Y at a port lie in the component of that port.

Step 4: non-giant components; the system $\mathcal{S}_0(Y, l)$. For every non-giant component C of R_Y , let $\mathfrak{K}_C := (\text{centres of } C, \text{ ports of } C, E(C))$.

- This is a cluster: centres and ports are disjoint, and every bead joins a port to a centre. It is parity-clean by Step 2.
- Fix any linear order of its ports and orient $\mathfrak{K}_C$ by MED. By Lemma 6.3, this is admissible and $\Phi(\mathfrak{K}_C) \leq b(\mathfrak{K}_C) = |E(C)|/2 \leq \gamma_l$; here $b(\mathfrak{K}_C) = |E(C)|/2$ because $\mathfrak{K}_C$ has no port–port beads. Also $|\text{exc}(u)| \leq M_l - 1$.
- $\Phi(\mathfrak{K}_C) \geq 1$: an acyclic orientation with an arc has a source, which must be a port of positive excess.

If R_Y has no non-giant component, $\mathcal{S}_0(Y, l)$ is empty and contributes no cycles, with $\Phi(\mathcal{S}_0(Y, l)) := 0$ in this case; the rest of Step 4 assumes that there is at least one cluster. Let $\Sigma\Phi$ and $\Phi_{\max}$ be the sum and the maximum of these loads, and put

$$k_0 := \min(K^{\text{JS}}, \max(1, \lfloor \Sigma\Phi / (2\Phi_{\max}) \rfloor)).$$

Then k_0 is at most the number of clusters. List the clusters in non-increasing order of $\Phi(\mathfrak{K}_C)$, as Lemma 6.4 requires, and layer them by that lemma, with loads $\Phi(\mathfrak{K}_C)$, into k_0 non-empty layers with raw loads Φ_j^{raw} , where $\max_j \Phi_j^{\text{raw}} - \min_j \Phi_j^{\text{raw}} \leq \Phi_{\max}$.

- If $k_0 = 1$: no padding, and the system has depth 1 with load $\Sigma\Phi < 4\Phi_{\max}$.
- If $k_0 \geq 2$: put $\Phi^* := \max_j \Phi_j^{\text{raw}}$ and $\Pi_j := \Phi^* - \Phi_j^{\text{raw}} \leq \Phi_{\max}$. Since $k_0 \leq \Sigma\Phi / (2\Phi_{\max})$, every raw load satisfies $\Phi_j^{\text{raw}} \geq \Sigma\Phi / k_0 - \Phi_{\max} \geq \Phi_{\max} \geq \Pi_j$. By Lemma 6.3(b), $\Phi_j^{\text{raw}} = \frac{1}{2} \sum_{u \in \mathcal{X}_j} |\text{exc}(u)| \leq \frac{1}{2} (M_l - 1) |\mathcal{X}_j|$. So Π_j units of padding can be spread over the ports of layer j with $\text{pad}(u) \leq \lceil (M_l - 1)/2 \rceil \leq \lceil M_l/2 \rceil$. All padded loads then equal $\Phi^* \leq \Sigma\Phi / k_0 + \Phi_{\max}$.

In all cases the common padded load of $\mathcal{S}_0(Y, l)$ satisfies

$$\Phi(\mathcal{S}_0(Y, l)) \leq 1.5 \Sigma\Phi / K^{\text{JS}} + 5\gamma_l. \quad (6.4)$$

Indeed: if $k_0 = K^{\text{JS}}$ then $\Phi_{\max} \leq \Sigma\Phi / (2K^{\text{JS}})$; if $2 \leq k_0 < K^{\text{JS}}$ then $k_0 \geq \Sigma\Phi / (2\Phi_{\max}) - 1 \geq \Sigma\Phi / (4\Phi_{\max})$, so $\Phi^* \leq 5\Phi_{\max}$; if $k_0 = 1$ the load is $< 4\Phi_{\max}$; and $\Phi_{\max} \leq \gamma_l$.

A port u of layer j has demand $\text{dm}^-(u)$ at junction j and $\text{dm}^+(u)$ at junction $j - 1$. Both are at most $|\text{exc}(u)| + \text{pad}(u) \leq (M_l - 1) + \lceil M_l/2 \rceil$.

Step 5: giant components; the cherry split. At every centre h of a giant component of R_Y , pair the (evenly many) edges of R_Y at h arbitrarily into *cherries* $\{hu, hu'\}$, written $u-h-u'$. Since G is simple, $u \neq u'$. Each cherry is a parity-clean cluster ($\{h\}, \{u, u'\}, \{hu, hu'\}$). Its orientation $u \rightarrow h \rightarrow u'$ is admissible, with $\text{exc}(u) = +1$, $\text{exc}(u') = -1$ and load 1. Every edge of a giant component lies in exactly one cherry.

Two cherries *conflict* if they share a vertex. A centre is never an end of a cherry, because centres lie outside $V(Y)$ and ends inside. A cherry at h conflicts with at most $\deg_{R_Y}(h)/2 - 1 \leq m_{Y,l}/2 - 1$ other cherries at h . Each of its two ends u has at most $M_l - 1$ edges in R_Y , each in at most one cherry, so it is an end of at most $M_l - 2$ other cherries. So the conflict degree is at most $m_{Y,l}/2 + 2M_l - 5$. Greedy colouring uses

$$g_{Y,l} \leq m_{Y,l}/2 + 2M_l - 4$$

colours, and each colour class $\mathcal{S}_i(Y, l)$ ($1 \leq i \leq g_{Y,l}$) is a set of pairwise vertex-disjoint cherries. Let η_i be the number of cherries in $\mathcal{S}_i(Y, l)$.

Each class $\mathcal{S}_i(Y, l)$ is made into a layered system of depth

$$k_i := \min(K^{\text{JS}}, \max(1, \lfloor \eta_i/2 \rfloor)).$$

The $\max(1, \cdot)$ is needed: $\eta_i \leq 3$ gives $k_i = 1$. Since $k_i \leq \eta_i$, Lemma 6.4 with unit loads (trivially in non-increasing order) gives k_i non-empty layers whose loads differ by at most 1, i.e. equal $\lfloor \eta_i/k_i \rfloor$ or $\lceil \eta_i/k_i \rceil$. There are three cases.

- $k_i = 1$ ($\eta_i \leq 3$): no padding, X^{out} the ports of excess -1 , X^{in} those of excess $+1$, and load $\Phi(\mathcal{S}_i) = \eta_i \leq 3$.
- $2 \leq k_i = \lfloor \eta_i/2 \rfloor < K^{\text{JS}}$ ($\eta_i \geq 4$): every layer of load $\lfloor \eta_i/k_i \rfloor < \lceil \eta_i/k_i \rceil$ gets padding 1 on one of its ports. The load is $\Phi(\mathcal{S}_i) = \lceil \eta_i / \lfloor \eta_i/2 \rfloor \rceil \leq 3$, since $\eta_i / \lfloor \eta_i/2 \rfloor \leq 2\eta_i / (\eta_i - 1) \leq 8/3$.
- $k_i = K^{\text{JS}}$: padding as in the previous case, and $\Phi(\mathcal{S}_i) = \lceil \eta_i / K^{\text{JS}} \rceil \leq \eta_i / K^{\text{JS}} + 1$.

So $\text{pad} \leq 1$, $\Phi(\mathcal{S}_i) \leq \eta_i / K^{\text{JS}} + 3$, and every port of a cherry system has demand at most 2 at each junction. Put $\text{sc}_{Y,l} := (3g_{Y,l} - 6M_l)^+$ (a symbol local to this proof), and $g_{Y,l} := \text{sc}_{Y,l} := 0$ if R_Y has no giant component. Summing over the classes of (Y, l) ,

$$\sum_i \Phi(\mathcal{S}_i(Y, l)) \leq \frac{\#\{\text{cherries of } (Y, l)\}}{K^{\text{JS}}} + 3g_{Y,l} \leq \frac{\#\{\text{cherries of } (Y, l)\}}{K^{\text{JS}}} + 6M_l + \text{sc}_{Y,l}, \quad (6.5)$$

and $\text{sc}_{Y,l} = (3g_{Y,l} - 6M_l)^+ \leq (1.5m_{Y,l} - 12)^+ \leq 1.5m_{Y,l}$. If R_Y has a giant component, then the component of $\mathfrak{B}_{Y,l}$ containing it has more than $2\gamma_l$ edges (Step 3), so (Y, l) is giant; hence $\text{sc}_{Y,l} = 0$ unless (Y, l) is giant.

Step 6: joint routing. The *systems* of (Y, l) are $\mathcal{S}_0(Y, l)$ (if non-empty) and the cherry classes $\mathcal{S}_i(Y, l)$. For a system $\mathcal{S}$ of depth k_S and a junction $j < k_S$, its *junction- j pairs* are the pairs of a fixed bijection between two multisets:

- if $k_S \geq 2$: the out-units of layer j ($\text{dm}^-(u)$ copies of each $u \in \mathcal{X}_j$) and the in-units of layer $j+1$ modulo k_S ($\text{dm}^+(v)$ copies of each $v \in \mathcal{X}_{j+1}$);
- if $k_S = 1$ and $j = 0$: $\text{exc}(u)^-$ copies of each $u \in X^{\text{out}}$ and $\text{exc}(v)^+$ copies of each $v \in X^{\text{in}}$.

Let $\mathfrak{P}_j(Y, l)$ be the multiset union of the junction- j pairs of all systems of (Y, l) of depth $> j$.

Claim 6.13.1 (the joint-routing facts). *For every lend-good Y with $R_Y \neq \emptyset$ and every $j < K^{\text{JS}}$:*

- (a) *The bijections exist. Every pair consists of two distinct vertices of $V(Y) \setminus \bigcup_{j'} T_{j'}(Y, l)$.*
- (b) *Every edge of R_Y is a bead of exactly one system of (Y, l) . Every port of R_Y is a port of $\mathcal{S}_0(Y, l)$ or of cherry systems, never of both. A port is an end of at most $M_l - 1$ cherries, and so lies in at most $M_l - 1$ cherry systems.*
- (c) *Every vertex lies in at most $\max(2M_l - 2, (M_l - 1) + \lceil M_l/2 \rceil) = 2M_l - 2 \leq t$ pairs of $\mathfrak{P}_j(Y, l)$ (the equality uses $M_l \geq 2$, which holds since $M_l \geq 2^{40}$ by (R2)). If $M_l \geq 4$, the value $2M_l - 2$ can occur only at a port that has demand 2 at junction j in each of $M_l - 1$ cherry systems (for $M_l \in \{2, 3\}$ the bound $(M_l - 1) + \lceil M_l/2 \rceil$ for ports of $\mathcal{S}_0(Y, l)$ also equals $2M_l - 2$).*
- (d) *There are pairwise edge-disjoint paths in $\mathcal{L}_{Y, l, j}^{\text{JS}}$, one for each pair of $\mathfrak{P}_j(Y, l)$ and joining its two vertices. Each has length at most $2^{12}L_Y^4$ and all its interior vertices in $T_j(Y, l)$.*
- (e) *No port or centre of a system of (Y, l) lies in any $T_{j'}(Y, l)$.*

Vertices of systems of *other* pairs (Y', l) may lie in $T_j(Y, l)$ and may be interior vertices of the paths in (d). This is harmless, because hypothesis (D) of Lemma 6.6 is needed only inside one system (Lemma 6.7).

Proof of the claim. (a) In a system of depth $k \geq 2$, the padded loads of all layers are equal (Steps 4 and 5). By Lemma 6.3(b), $\sum_{\mathcal{X}_j} \text{exc}^- = \sum_{\mathcal{X}_j} \text{exc}^+$. So the number of out-units of layer j equals its padded load, which equals the padded load of layer $j+1$, which is the number of its in-units. For depth 1, $\sum \text{exc}^- = \sum \text{exc}^+$.

The two vertices of a pair are distinct: for $k \geq 2$, the layers j and $j+1$ consist of different, pairwise vertex-disjoint clusters (for $k = 2$, layer $j+1$ is layer $j-1 \neq j$); for $k = 1$, $X^{\text{out}} \cap X^{\text{in}} = \emptyset$. The vertices of pairs are class- Y ports $u \in Q_Y^*$. They lie in $V(Y)$ because $Y = Y(u) \in \text{anc}_l(u)$, and outside every $T_{j'}(Y, l)$ because $\text{lab}_{Y, l}(u) = *$.

(b) Every edge of R_Y lies in exactly one component: in one cluster of $\mathcal{S}_0(Y, l)$ if the component is non-giant, and otherwise in exactly one cherry, which lies in exactly one class. All edges of R_Y at a port u lie in the component of u , so u is a port of $\mathcal{S}_0(Y, l)$ or of cherries, not both. Distinct cherries with end u use distinct edges of R_Y at u , so there are at most $M_l - 1$ of them. A class contains at most one of them, since its cherries are vertex-disjoint.

(c) In a system of depth $k \geq 2$, a port of layer j_u occurs in junction- j_u pairs (dm^- times) and in junction- $(j_u - 1)$ pairs (dm^+ times), and $j_u \neq j_u - 1$ modulo k . In a system of depth 1 it occurs only as a member of X^{out} or of X^{in} . So at a fixed junction a port occurs in at most $\max(\text{dm}^-, \text{dm}^+)$ pairs of each system containing it. Centres occur in no pair.

By Steps 4 and 5 and part (b), a port occurs in at most $(M_l - 1) + \lceil M_l/2 \rceil$ pairs (if it belongs to $\mathcal{S}_0$) or at most $2(M_l - 1)$ pairs (if it belongs to cherry systems). For $M_l \geq 2$, $\lceil M_l/2 \rceil \leq M_l - 1$, so both are at most $2M_l - 2 < t = 2M_l + 2$.

(d) Y is lend-good. By Definition 6.11 and Lemma 3.13(b), $\mathcal{L}_{Y, l, j}^{\text{JS}}$ is $(2^{12}L_Y^4, t)$ -path connected through $T_j(Y, l)$, as a graph on $V(Y)$. By Cited result 1.9 (see Remark 3.3) this is a statement about

every multiset of pairs of distinct vertices of $V(Y)$ in which every vertex lies in at most t pairs, with endpoints unrestricted. By Lemma 3.1(iii), the property depends only on $(\mathcal{L}_{Y,l,j}^{\text{JS}}, T_j(Y, l))$, so the multiset may be chosen after the stage-1 outcome and after Steps 1–5. By the hypothesis of the lemma, no edge of $\mathcal{L}_{Y,l,j}^{\text{JS}}$ has been used, so the whole class is available. Apply this to $\mathfrak{P}_j(Y, l)$, which is admissible by (a) and (c).

(e) Ports: see (a). Centres lie outside $V(Y) \supseteq T_{j'}(Y, l)$ (Step 1). $\square$

Step 7: HCC-P per system, and the union. Fix a system $\mathcal{S}$ of (Y, l) of depth $k = k_{\mathcal{S}}$. For $j < k$ let $\mathcal{P}_j(\mathcal{S})$ be the paths of the joint-routing claim, part (d), for the junction- j pairs of $\mathcal{S}$, each oriented from its out-unit to its in-unit. We check the hypotheses of Lemma 6.6 for $\mathcal{S}$, with its layers, junction sets $T_j := T_j(Y, l)$ ($j < k$) and its padding.

(D) The clusters of $\mathcal{S}$ are pairwise vertex-disjoint: distinct components for $\mathcal{S}_0$, vertex-disjoint cherries for a class. Centres lie outside $V(Y)$ and ports inside, so no vertex is both. Neither meets $\bigcup_j T_j(Y, l)$, by part (e) of the joint-routing claim. The sets $T_j(Y, l)$ are pairwise disjoint (Definition 3.10(iv)).

(P) Step 2 for $\mathcal{S}_0$; centres of cherries have degree 2.

(JC-P) By parts (a) and (d) of the joint-routing claim, $\mathcal{P}_j(\mathcal{S})$ is a family of edge-disjoint paths from $\mathcal{X}_j$ to $\mathcal{X}_{j+1}$ (from X^{out} to X^{in} if $k = 1$), with interiors in $T_j(Y, l)$. By the choice of the bijection, every u is the first vertex of exactly $\text{dm}^-(u)$ of them and the last vertex of exactly $\text{dm}^+(u)$. The path families for distinct j lie in distinct classes $\mathcal{L}_{Y,l,j}^{\text{JS}}$, which are disjoint. All of them lie in $\text{Lend}_Y \subseteq E(H_Y) \subseteq E_r(Y)$, while beads lie in $\bigcup_Z E_l(Z)$. These sets are disjoint by the partition of Proposition 2.11(iii).

Lemma 6.6 gives sets $F_j(\mathcal{S}) \subseteq E(\mathcal{P}_j(\mathcal{S}))$ such that the beads of $\mathcal{S}$ together with $\bigcup_j F_j(\mathcal{S})$ decompose into at most $\Phi(\mathcal{S})$ cycles. The returned edges $E(\mathcal{P}_j(\mathcal{S})) \setminus F_j(\mathcal{S})$ lie in $G[T_j(Y, l)]$.

Now consider all systems of all (Y, l) together. Their bead sets are pairwise disjoint (part (b) of the joint-routing claim, and each edge of $\bigcup_Z B_Z \setminus J_l$ lies in exactly one R_Y). Their path edges are pairwise disjoint:

- within one (Y, l) and one j , by the joint routing, part (d) of the claim;
- for different j , because different classes are used;
- for different $Y \neq Y'$, because $E(H_Y) \cap E(H_{Y'}) = \emptyset$ ($E(H_Y) \subseteq E_{r(Y)}(Y)$), and these sets are disjoint by Proposition 2.11(iii)).

Path edges are disjoint from beads, as shown above. By Lemma 6.7, the union decomposes into at most $\sum_{\mathcal{S}} \Phi(\mathcal{S})$ cycles.

Put $\text{Lent}_l^{\text{JS}} := \bigcup_{\mathcal{S}} \bigcup_j F_j(\mathcal{S}) \subseteq \bigcup_{Y,j} \mathcal{L}_{Y,l,j}^{\text{JS}}$. Since $\bigcup_Z B_Z = J_l \sqcup \bigsqcup_Y R_Y$ and $\bigsqcup_Y R_Y$ is the set of all beads of all systems, $\bigcup_Z B_Z \cup \text{Lent}_l^{\text{JS}}$ is partitioned into the single edges of J_l and these cycles. Each cycle lies inside one system, so it uses beads and lent edges of one ancestor Y only.

Step 8: the count. Every bead has exactly one end that is a port of its class (the other end is a centre, possibly a pseudo-hub). Count beads at that end. The classed ports of round l are pairwise distinct vertices, because port sets of one round are disjoint (Proposition 2.11(iv)). Each port has at most $M_l - 1$ beads. So there are at most $n(M_l - 1)$ beads in total. Hence $\sum_Y \Sigma \Phi(\mathcal{S}_0(Y, l)) \leq \frac{1}{2}n(M_l - 1)$ (as $\Phi(\mathfrak{K}_C) \leq |E(C)|/2$), and the total number of cherries is at most $\frac{1}{2}n(M_l - 1)$.

The number of ancestors Y of rounds $\leq l-2$ is $\nu_l \leq 2.74n/P_{l-2}$ (Lemma 2.17(b), which completes (K1) of Proposition 2.13). Moreover $\gamma_l \leq P_{l-2}/M_l$ and $P_{l-2} \geq M_l^{13}$ (Lemma 2.17(b)). By (6.4) and (6.5), the number of cycles is at most

$$\frac{1.5 \cdot \frac{1}{2}n(M_l - 1)}{M_l^2} + 5 \frac{P_{l-2}}{M_l} \cdot \frac{2.74n}{P_{l-2}} + \frac{\frac{1}{2}n(M_l - 1)}{M_l^2} + 6M_l \cdot \frac{2.74n}{P_{l-2}} + \sum_{(Y,l) \text{ giant}} \text{sc}_{Y,l} \leq 15 \frac{n}{M_l} + \sum_{(Y,l) \text{ giant}} \text{sc}_{Y,l}.$$

The four terms are at most $0.75n/M_l$, $13.7n/M_l$, $0.5n/M_l$ and $16.44n/M_l^{12}$, and $14.95 + 16.44 M_l^{-11} \leq 15$ because $M_l \geq 2^{40}$. By Step 5, $\text{sc}_{Y,l} \leq 1.5m_{Y,l}$ for every giant (Y,l) . Since $15 \leq 126$, the number of cycles is at most $126n/M_l + 1.5 \sum_{(Y,l) \text{ giant}} m_{Y,l}$, which proves the lemma as stated.

On the constant. The proof gives the constant 15. The lemma states the looser constant 126 on purpose: 126 is the value carried into Theorem 6.19(c) (the term $252n/D_*$ in ε_M) and from there into Section 7. Any absolute constant would do there, because the term is $O(n/D_*)$. Replacing 126 by 15 would only lower $252/D_*$ to $30/D_*$ in those places.

The sharper bound. Step 8 proves, as a by-product, the sharper bound of at most $15n/M_l + \sum_{(Y,l) \text{ giant}} \text{sc}_{Y,l}$ cycles, where $\text{sc}_{Y,l} = (3g_{Y,l} - 6M_l)^+ \leq (1.5m_{Y,l} - 12)^+$ (Step 5). The stated bound follows from it, and it is the form used by the only consumer, Theorem 6.19(c); nothing downstream uses the sharper bound.

No cost term is paid per layer or per component. The cost of a system is its common padded load $\Phi(\mathcal{S})$. Padding adds paths, not cycles, and returned edges are junk. The only per-(centre, class) costs are the single edges moved into J_l in Step 2, and these are counted in (6.3). $\square$

Remark 6.14 (the star split is not used). For a giant pair (Y,l) one could instead use a “star split” with a concentration threshold τ'_l and a hub-concentration functional HC. That option is not used in this paper, and neither τ'_l nor HC appears. Its cost is bounded only by $\frac{1}{2}\text{HC}_{Y,l}$, which is not controlled here; the cherry split costs at most $\#\{\text{cherries}\}/K_l^{\text{JS}} + 6M_l + 1.5m_{Y,l}$ by (6.5). Neither split emits single edges: all single edges of JS-LC come from Step 2, before any split. So J_l does not depend on this choice, and neither does any property of J_l proved below.

Dependencies: 6.13

Lemma 6.15 (Lemma J⁺: the J-interface). *In Lemma 6.13, the set J_l consists exactly of the deletions of Step 2 of its proof:*

- the PAR deletions, per part and per pair of classes;
- then one bead per (centre, class) with odd realized class degree, where the degree at a hub is aggregated over all standalone parts of round l .

This holds for every choice of which edges are deleted. Steps 3–8 delete nothing.

Moreover $J_l = J_l^{\text{lost}} \sqcup J_l^{\text{hub}} \sqcup J_l^{\text{fr}} \sqcup J_l^{\text{par}}$, where, for $Z \in \text{Std}_l$:

- a J^{par} -edge has both ends in Q_Z^* , with different classes;
- a J^{hub} -edge is an edge hu with $h \in A_Z \subseteq D_l$ and $u \in Q_Z^*$;
- a J^{fr} -edge is an edge xu with $x \in F_Z$ and $u \in Q_Z^*$;
- a J^{lost} -edge is an edge vu with $v \in \text{Lost}_Z$ (a lost centre) and $u \in Q_Z^*$.

The class of a J^{hub} -, J^{fr} - or J^{lost} -edge hu is $Y(u)$. The following properties hold.

- (J1) For each hub $h \in D_l$ and each class Y , J_l contains at most one J^{hub} -edge of class Y at h , aggregated over all parts of round l . For each fresh centre x and each class Y , J_l contains at most one J^{fr} -edge of class Y at x . (The same holds at lost centres.)
- (J2) Every J -edge lying in $E_l(Z)$ has an end in Q_Z^* . For every Z and every vertex, at most $M_l - 1$ J -edges at that vertex lie in $E_l(Z)$. In particular a port or a fresh centre (a vertex outside D_l) carries at most $M_l - 1$ J -edges of round l . Also $|J_l| \leq n(M_l - 1)$.
- (J3) The sets D_l (hubs), $\bigcup_Z F_Z$ (fresh centres) and $\bigcup_Z Q_Z^*$ (classed non-lost ports) are pairwise disjoint. Each vertex outside D_l that lies in a round- l pre-part lies in exactly one such pre-part, so it has one role and, if classed, one class.

In addition, the following interface facts hold.

- (i) The types above are exhaustive and exclusive.
- (ii) (Size cap at fresh centres.) A fresh centre carries at most $M_l - 1$ J^{fr} -edges.
- (iii) (Disjointness from (R5).) $E(H_Y) \subseteq E_r(Y)$ for every ancestor Y of round r , and the sets $E_r(Y)$ (Y light) and $E_l(Z)$ ($Z \in \text{Std}_l$) are pairwise disjoint.
- (iv) $u \in V(Y(u))$ for every classed port u .
- (v) In the colourings of Definition 3.10, colours of distinct edges are independent.

Dependencies: 6.13, 6.5, 6.8, 2.11, 2.12, 2.9, 3.10, 2.7, 6.6, 6.11

Proof. The composition of J_l . In the proof of Lemma 6.13, single edges are created only in Step 2. Steps 3–8 form clusters from the sets R_Y , route paths and apply Lemma 6.6, whose output consists of cycles only. Every bead lies in some system and is covered, and padding adds paths, not edges. Step 2 allowed any admissible choice in Lemma 6.5 and any choice of the moved edge, and nothing below depends on these choices.

Types. A PAR deletion lies in some $E_{ab}(Z)$, so it joins two ports of Q_Z^* of different classes: a J^{par} -edge. A moved edge is a class- Y edge whose centre has odd degree in R_Y^0 . By Step 2 of that proof this centre lies in $\text{Ret}_Z = A_Z \sqcup F_Z \sqcup \text{Lost}_Z$, and the port lies in Q_Z^* . The edge is a J^{hub} -, J^{fr} - or J^{lost} -edge according as the centre lies in $A_Z = Z^0 \cap D_l$, in F_Z or in Lost_Z . These cases are exclusive because A_Z , F_Z , Lost_Z and Q_Z^* are pairwise disjoint (Definition 6.11). This proves (i).

(J1): The moves are made once per pair (centre, class), with the class- Y degree at a hub counted over all parts of round l . PAR deletions are port–port edges and never J^{hub} - or J^{fr} -edges. A fresh centre lies in one part only.

(J2): $J_l \subseteq \bigcup_Z B_Z$, and every edge of B_Z has an end in Q_Z^* . By Lemma 2.9(ii), every vertex has at most $M_l - 1$ edges of $E_l(Z)$ for each Z . A vertex outside D_l lies in at most one round- l pre-part, which gives the “in particular”. For the total, charge every J -edge to an end in Q_Z^* . The classed ports of round l are distinct vertices (Proposition 2.11(iv)), and each receives at most $M_l - 1$ charges. So $|J_l| \leq n(M_l - 1)$.

(J3): $A_Z \subseteq D_l$, while F_Z and Q_Z^* are disjoint subsets of $U_Z = Z^0 \setminus D_l$. The sets U_Z of distinct $Z \in \text{Std}_l$ are pairwise disjoint, and every vertex outside D_l lies in at most one round- l pre-part (Proposition 2.11(iv)).

(ii) is the degree bound of (J2) at a fresh centre. (iii) is Proposition 2.11(iii). (iv) holds because $Y(u) \in \text{anc}_l(u)$ (Definition 6.8), and $\text{anc}_l(u)$ consists of ancestors whose vertex set contains u . (v) holds because every colouring in Definition 3.10 is uniform and independent over edges. By Lemma 2.12, the two ends of a J^{par} -edge indeed have different classes. $\square$

Definition 6.16 (round- l J-consumer). A *J-consumer* is a rule that acts at each round $3 \leq l \leq R$. Given everything constructed at rounds $> l$, the stage-1 outcome and the set J_l (and possibly fresh randomness), it outputs a family $\mathcal{O}_l$ of objects and a set $\text{Lent}_l^{\text{JV}}$ of edges such that:

- (JC1) $\text{Lent}_l^{\text{JV}} \subseteq \bigcup \{\mathcal{L}_{Y,l}^{\text{JV}} : Y \text{ a lend-good ancestor of a round } \leq l-2\}$;
- (JC2) the objects of $\mathcal{O}_l$ are pairwise edge-disjoint cycles and single edges of G , and their union is exactly $J_l \cup \text{Lent}_l^{\text{JV}}$;
- (JC3) $\mathcal{O}_l$ uses no other edge.

By the exclusive-consumer rule $\text{COL}(g)$ of Definition 3.10, the classes $\mathcal{L}_{Y,l}^{\text{JV}}$ are used by nothing except the round- l J-consumer. For example, the rule “ $\mathcal{O}_l :=$ the edges of J_l as single edges, $\text{Lent}_l^{\text{JV}} := \emptyset$ ” is a J-consumer. The J-consumer used for the final count is the junction-vertex round step constructed later in the paper. Here J_l has the properties of Lemma 6.15.

Dependencies: 3.10, 6.15

6.5 The MIX-C assembly

Construction 6.17 (the processing order $R, R-1, \dots, 1$, and the Lent sets). *Input.*

- A valid $\text{HB}^{*\tau+}$ run on G with $n \geq N_0$ and $d_1 \geq D_*$, and a designation δ .
- A stage-1 outcome with its stage-2 data (Definitions 5.3 and 6.11).
- Stage-3 outcomes: labels for Lemma 4.2 for every $Z \in \text{Std}$; child-vortex labels (Lemma 5.6) for every non-demoted light part; labels for Theorem 4.4 (Lemma 5.8) for every demoted light part. Each is fixed in its good event.
- A J-consumer (Definition 6.16).

The *Lent sets* are defined recursively along the construction. For an ancestor Y of round r :

- $\text{Lent}^{\text{U}}(Y)$: the U-lent edges of Y used by Lemma 5.7 at rounds $l \geq r+2$ (light Y only; $\text{Lent}^{\text{U}}(Y) := \emptyset$ for standalone Y);
- $\text{Lent}^{\text{JS}}(Y) := \bigcup_{l \geq r+2} (\text{Lent}_l^{\text{JS}} \cap \bigcup_j \mathcal{L}_{Y,l,j}^{\text{JS}})$, the JS-lent edges of Y on output cycles of JS-LC (returned edges are not included);
- $\text{Lent}^{\text{JV}}(Y) := \bigcup_{l \geq r+2} (\text{Lent}_l^{\text{JV}} \cap E(H_Y))$;
- $\text{Lent}(Y) := \text{Lent}^{\text{U}}(Y) \cup \text{Lent}^{\text{JS}}(Y) \cup \text{Lent}^{\text{JV}}(Y)$.

For standalone Y put $E^{\text{rem}}(Y) := E_r(Y) \setminus \text{Lent}(Y)$. For light non-demoted Y put $H_0(Y) := E_r(Y) \setminus \text{Lent}(Y)$.

The construction. For $l = R, R-1, \dots, 1$, in this order, do:

- (1) For every $Z \in \text{Std}_l$, the set $\text{Lent}(Z)$ is already determined. Its edges lie in lent classes of Z indexed by rounds $\geq l+2$, and by $\text{COL}(g)$ of Definition 3.10 the only consumers of these classes act at steps of rounds $\geq l+2$, which are complete. Apply Lemma 4.2 on its fixed good event, with vertex set Z^0 , expander $O := O_Z$, edge set $E := E^{\text{rem}}(Z)$, $\mathbf{P} := \text{Ret}_Z$ and $\mathbf{Q} := Q_Z^*$. This gives $E^{\text{rem}}(Z) = E^{\text{V}}(Z) \sqcup E^{\text{Q}}(Z)$ and a decomposition of $E^{\text{V}}(Z)$ into at most $169|\text{Ret}_Z|$ objects, which are output.

- (2) Every light part Y of round l that is not demoted runs its child vortex (Lemma 5.6) on $H_0(Y)$. Every demoted light part Y of round l runs Theorem 4.4 on $E_l(Y)$ (Lemma 5.8). If $l \geq 3$, then for every phase $c \in [4]$ the U-bundle $\text{Bdl}_{l,c}$ is chained through the U-classes of good parents of rounds $\leq l-2$ (Lemma 5.7), and the used U-lent edges are recorded. The objects are output.
- (3) If $l \geq 3$: after all round- l applications of Lemma 4.2, apply Lemma 6.13 with $B_Z := E^Q(Z)$ for all $Z \in \text{Std}_l$. Its cycles are output, and $\text{Lent}_l^{\text{JS}}$ is recorded. This yields J_l .
- (4) If $l \geq 3$: apply the J-consumer to J_l . The objects $\mathcal{O}_l$ are output, and $\text{Lent}_l^{\text{JV}}$ is recorded.

Finally, the cycles of every Cyc_l are output, and every edge of E_0 is output as a single edge.

Rounds 1 and 2 have no classed ports ($\text{anc}_l(x) = \emptyset$ for $l \leq 2$), so $J_2 = J_1 = \emptyset$ and steps (3)–(4) are void there.

Dependencies: 4.2, 5.6, 5.7, 5.8, 6.13, 6.16, 6.11, 5.3, 4.4, 3.10

The construction is well defined because each set $\text{Lent}(Y)$ is determined by rounds $\geq r(Y) + 2$, and these precede round $r(Y)$ in the order $R, R-1, \dots, 1$. This is part (i) of the following lemma.

Lemma 6.18 (Lent sets). *In Construction 6.17, for every ancestor Y of round r :*

- (i) $\text{Lent}(Y) \subseteq \text{Lend}_Y$, and $\text{Lent}(Y)$ depends only on the steps at rounds $\geq r+2$. So it is final when Y is processed at round r .
- (ii) If Y is standalone and lend-bad, then $\text{Lent}(Y) = \emptyset$: every port of class Y is lost, so Y has no JS system, and (JC1) excludes its JV classes. If Y is light and demoted, then $\text{Lent}(Y) = \emptyset$.
- (iii) $E^{\text{rem}}(Z) \supseteq E(O_Z)$ for every $Z \in \text{Std}$, and $H_0(Y) \supseteq \text{Own}_Y$ for every non-demoted light Y .
- (iv) Put $\text{Lent}^{\text{ext}}(Y) := \text{Lent}^{\text{JS}}(Y) \cup \text{Lent}^{\text{JV}}(Y)$ for light Y . The family $(\text{Lent}^{\text{ext}}(Y))_Y$ satisfies the hypothesis of Lemma 5.9, and $H_0(Y) = E_r(Y) \setminus (\text{Lent}^{\text{U}}(Y) \cup \text{Lent}^{\text{ext}}(Y))$ is the set $H_0(Y)$ of that lemma. The sets $\text{Lent}^{\text{U}}(Y)$, $\text{Lent}^{\text{JS}}(Y)$ and $\text{Lent}^{\text{JV}}(Y)$ are pairwise disjoint.

Dependencies: 6.17, 6.11, 6.16, 6.13, 5.7, 5.9, 3.10, 6.15, 5.3, 4.2, 5.6, 5.8, 4.4

Proof. (i) By Lemma 5.7(i), the used U-lent edges lie in U-classes $\mathcal{L}_{Y',l,c,\varsigma}^{\text{U}}$ of good parents Y' , so $\text{Lent}^{\text{U}}(Y) \subseteq \bigcup \mathcal{L}_{Y',l,c,\varsigma}^{\text{U}} \subseteq \text{Lend}_Y$. By definition $\text{Lent}^{\text{JS}}(Y) \subseteq \bigcup \mathcal{L}_{Y',l,j}^{\text{JS}} \subseteq \text{Lend}_Y$.

For $\text{Lent}^{\text{JV}}(Y)$: by (JC1), $\text{Lent}_l^{\text{JV}} \subseteq \bigcup_{Y'} \mathcal{L}_{Y',l}^{\text{JV}}$, and $\mathcal{L}_{Y',l}^{\text{JV}} \subseteq E(H_{Y'})$. For $Y' \neq Y$ the sets $E(H_{Y'})$ and $E(H_Y)$ are disjoint (Lemma 6.15(iii)). Hence $\text{Lent}_l^{\text{JV}} \cap E(H_Y) \subseteq \mathcal{L}_{Y,l}^{\text{JV}} \subseteq \text{Lend}_Y$.

The classes $\mathcal{L}_{Y,l,\cdot,\cdot}^{\text{U}}$, $\mathcal{L}_{Y,l,\cdot}^{\text{JS}}$, and $\mathcal{L}_{Y,l}^{\text{JV}}$ exist only for $r+2 \leq l \leq R$ (Definition 3.10(ii)). By COL(g) of Definition 3.10, each is used only by its round- l consumer: the U-bundle (l, c) in step (2), the JS-LC system of (Y, l) in step (3), or the round- l J-consumer in step (4).

No other step uses an edge of $E(H_Y)$ before round r . At a round $l' > r$:

- step (1) uses only edges of $E^{\text{rem}}(Z') \subseteq E_{l'}(Z')$;
- the child vortices and VX^+ use only edges of $E_{l'}(Y')$;
- JS-LC uses beads in $E_{l'}(\cdot)$ and lent edges of classes $\mathcal{L}_{\cdot,l'}^{\text{JS}}$;
- the J-consumer uses only $J_{l'} \cup \text{Lent}_{l'}^{\text{JV}}$ ((JC3)).

The sets $E_{l'}(\cdot)$ are disjoint from $E_r(Y) \supseteq E(H_Y)$ (Lemma 6.15(iii)). So $\text{Lent}(Y)$ is determined by steps of rounds $\geq r + 2$, all of which come before round r .

(ii) Let Y be lend-bad, standalone or light.

- A JS-LC system of (Y, l) exists only if $R_Y \neq \emptyset$ at round l , i.e. only if Y is the class of a port of some Q_Z^* . Since Y is lend-bad, Definition 6.11 puts every port u with $Y(u) = Y$ into Lost_Z , so Y is not the class of any port of any Q_Z^* . Hence $R_Y = \emptyset$ at every round l , no pair is routed in any $\mathcal{L}_{Y,l,j}^{\text{JS}}$, and $\text{Lent}^{\text{JS}}(Y) = \emptyset$.
- By (JC1) and the disjointness used in (i), $\text{Lent}_l^{\text{JV}} \cap E(H_Y) \subseteq \mathcal{L}_{Y,l}^{\text{JV}}$ can be non-empty only if Y is lend-good. So $\text{Lent}^{\text{JV}}(Y) = \emptyset$.
- $\text{Lent}^{\text{U}}(Y) = \emptyset$ for standalone Y by definition. For a demoted light Y it is empty because Lemma 5.7 uses only U-classes of good parents, and a demoted part is not a good parent (Definition 5.3). A demoted light part is lend-bad by Definition 6.11.

(iii) Let $Z \in \text{Std}_l$.

- If Z is lend-good, then $O_Z = \text{Own}_Z$. Moreover $E(\text{Own}_Z) \subseteq E(X_Z^0) = E(H_Z)$, and $E(H_Z) \subseteq E_l(Z)$ by Lemma 6.15(iii). By (i), $\text{Lent}(Z) \subseteq \text{Lend}_Z$, and $\text{Lend}_Z \cap \text{Own}_Z = \emptyset$ (Definition 3.10(i)). So $E(O_Z) \subseteq E_l(Z) \setminus \text{Lent}(Z) = E^{\text{rem}}(Z)$.
- If Z is lend-bad, then $\text{Lent}(Z) = \emptyset$ by (ii), so $E^{\text{rem}}(Z) = E_l(Z) \supseteq E(X_Z^0) = E(O_Z)$.

For a non-demoted light Y : $\text{Own}_Y \subseteq E(X_Y) \subseteq E_r(Y)$ and $\text{Own}_Y \cap \text{Lend}_Y = \emptyset$, so $\text{Own}_Y \subseteq H_0(Y)$ by (i).

(iv) By the argument of (i), $\text{Lent}^{\text{ext}}(Y) \subseteq \bigcup_{l,j} \mathcal{L}_{Y,l,j}^{\text{JS}} \cup \bigcup_l \mathcal{L}_{Y,l}^{\text{JV}}$. By (ii), $\text{Lent}^{\text{ext}}(Y) = \emptyset$ if Y is demoted. $\text{Lent}^{\text{U}}(Y)$ is by definition the set of U-lent edges of Y used by Lemma 5.7, exactly as in Lemma 5.9. So $H_0(Y) = E_r(Y) \setminus \text{Lent}(Y) = E_r(Y) \setminus (\text{Lent}^{\text{U}}(Y) \cup \text{Lent}^{\text{ext}}(Y))$. The three sets lie in distinct colour classes of the colouring of Lend_Y (Definition 3.10(ii)), so they are pairwise disjoint. $\square$

Theorem 6.19 (Theorem MIX-C: the assembly with $\mathcal{V} = \emptyset$, deterministic form). *Fix the following data:*

- a valid $\text{HB}^{*\tau+}$ run on G with $n \geq N_0$ and $d_1 \geq D_*$, and a designation δ ;
- any stage-1 outcome;
- stage-3 outcomes in the good events of Lemma 4.2 (every $Z \in \text{Std}$), of Lemma 4.3 through Lemma 5.6 (every non-demoted light part), and of Theorem 4.4 through Lemma 5.8 (every demoted light part);
- any J -consumer.

Run Construction 6.17. Then:

- (a) (Order.) At step (1) of round l , $\text{Lent}(Z)$ is final for every $Z \in \text{Std}_l$. JS-LC at round l uses only classes $\mathcal{L}_{Y,l}^{\text{JS}}$, and the J -consumer only classes $\mathcal{L}_{Y,l}^{\text{JV}}$, of ancestors Y of rounds $\leq l - 2$, which are processed later. The hypotheses of Lemma 6.13 hold at every round $l \geq 3$, and every device is applied within its hypotheses.

(b) (Coverage.) The output is a decomposition of $E(G)$: every edge of G lies in exactly one output object. In detail:

- for a light Y of round r , $E_r(Y) = \text{Lent}^U(Y) \sqcup \text{Lent}^{\text{JS}}(Y) \sqcup \text{Lent}^{\text{JV}}(Y) \sqcup H_0(Y)$ if Y is not demoted, and $\text{Lent}(Y) = \emptyset$ if Y is demoted;
- for $Z \in \text{Std}_l$, $E_l(Z) = \text{Lent}(Z) \sqcup E^V(Z) \sqcup (E^Q(Z) \setminus J_l) \sqcup (J_l \cap E_l(Z))$, where the third set consists of the beads of Z ;
- every lent edge is used by at most one object;
- lent edges that are unused or returned are covered by the device of their owner, and every device tolerates such junk;
- the bound (6.3) on $|J_l|$ is computed on the sets $E_l(Z)$ with lent edges included, so lending never increases it (that bound is not used later; see the remark after it).

(c) (Cost.) The number of output objects is at most

$$\left(\frac{D_*}{2} + 745 + 338\right)n + \varepsilon_M(D_*)n + \left[80 \text{dem} + 369 \text{lp} + 169 \sum_l |\text{Lost}_l|\right] + 1.5 \sum_{(Y,l)} m_{Y,l} + \sum_l |\mathcal{O}_l|,$$

where $\varepsilon_M(D_*) := \varepsilon_K(D_*) + 169\varepsilon_A + 252/D_*$, with $\varepsilon_A = 31\varepsilon/(C' \log \log D_*)$ as defined in Lemma 2.17(e). Thus ε_M is a function of D_* alone; it depends neither on the run nor on n . Moreover $1.5 \sum_{(Y,l)} m_{Y,l} \leq 1.5 \varepsilon_{\text{CONC}}(D_*)n$ by Theorem 6.10(iv).

The itemization of (c) is:

- *K-RED*: $(D_*/2 + 745 + \varepsilon_K)n + 80 \text{dem} + 369 \text{lp}$, including Cyc and E_0 ;
- *TPV*: $169 \sum (|A_Z| + |F_Z| + |\text{Lost}_Z|) \leq 169(2 + \varepsilon_A)n + 169 \sum_l |\text{Lost}_l|$;
- *JS-LC cycles*: $\sum_l 126n/M_l + 1.5 \sum m \leq 252n/D_* + 1.5 \sum m$;
- *J-consumer*: $\sum_l |\mathcal{O}_l|$. The set J_l is handed entirely to the J -consumer.

Dependencies: 6.17, 4.2, 5.9, 5.8, 6.13, 6.15, 6.16, 6.18, 6.10, 2.11, 2.20, 2.13, 2.17, 3.10, 3.13, 4.3, 5.6, 4.4, 2.7, (T3), 6.8, 6.11

Proof. (a) Order.

- *Finality.* By Lemma 6.18(i), $\text{Lent}(Z)$ for $Z \in \text{Std}_l$ depends only on steps of rounds $\geq l + 2$. So it is final at step (1) of round l . The same holds for $H_0(Y)$ of every light part Y of round l at step (2).
- *Classes used.* At round l , JS-LC routes only in classes $\mathcal{L}_{Y,l,j}^{\text{JS}}$ with Y the class of a non-lost port of round l , so $Y \in \text{anc}_l(u)$ has round $\leq l - 2$. The J -consumer uses only classes $\mathcal{L}_{Y,l}^{\text{JV}}$ with $r(Y) \leq l - 2$, by (JC1). Such Y are processed at round $r(Y) < l$, i.e. later.
- *Hypotheses of Lemma 6.13.* Take $B_Z := E^Q(Z) \subseteq E^{\text{rem}}(Z) \subseteq E_l(Z)$. By (R5) every edge of $E_l(Z)$ lies inside $Z^0 = \text{Ret}_Z \sqcup Q_Z^*$. By (T1) of Lemma 4.2, every edge of $E^{\text{rem}}(Z)$ with both ends in Ret_Z lies in $E^V(Z)$. So every edge of B_Z has an end in Q_Z^* . By $\text{COL}(g)$ of Definition 3.10, the only consumer of $\mathcal{L}_{Y,l,j}^{\text{JS}}$ is the JS-LC system of (Y, l) at step (3) of round l . By the argument in the proof of Lemma 6.18(i), no other step up to that moment uses an edge of $E(H_Y)$. So no edge of these classes has been used.

- *Lemma 4.2 at step (1).*

- $|Z^0| \geq P_l \geq N_0$ by (T3).
- If Z is lend-good, then $O_Z = \text{Own}_Z$. By Lemma 3.13(a) it is a spanning $(2^{-5}, s_l/4)$ -expander on Z^0 , and by Lemma 3.13(e) we have $s_l/4 \geq 2^{150} L_Z^{42}$.
- If Z is lend-bad, $O_Z = X_Z^0$ is a spanning $(2^{-5}, s_l)$ -expander on Z^0 (Proposition 2.11(i)), and $s_l \geq s_l/4 \geq 2^{150} L_Z^{42}$: the inequality of Lemma 3.13(e) involves only s_l and L_Z , not the colouring.
- $E = E^{\text{rem}}(Z) \supseteq E(O_Z)$ (Lemma 6.18(iii)), and all its edges lie inside Z^0 .
- The good event of Lemma 4.2 is determined by (Z^0, O_Z, Ret_Z) and the stage-3 labels, all fixed before step (1). On it, the conclusion holds for every such E . So TPV tolerates the junk (unused and returned lent edges) left in $E^{\text{rem}}(Z)$.

- *Light parts at step (2).* Let Y be a light part of round l . If Y is not demoted, then $\text{Own}_Y \subseteq H_0(Y) \subseteq E_l(Y)$ by Lemma 6.18(iii). This is what Lemma 5.6 needs, and the child vortex covers every such $H_0(Y)$. If Y is demoted, then $\text{Lent}(Y) = \emptyset$ by Lemma 6.18(ii). So Lemma 5.8 applies to $E := E_l(Y) \supseteq E(X_Y)$, which may contain arbitrary extra edges.

- *Step (4).* The J-consumer receives J_l , which has the properties of Lemma 6.15.

(b) *Coverage.* Split the output objects into four families.

$\mathcal{O}_K$: the cycles of all Cyc_l , the edges of E_0 , and the objects of step (2) at all rounds;

$\mathcal{O}_T$: the objects of step (1);

$\mathcal{O}_S$: the cycles of step (3);

$\mathcal{O}_J$: $\bigsqcup_l \mathcal{O}_l$.

$\mathcal{O}_K$ is the output of Lemma 5.9. By Lemma 6.18(iv), the final family $(\text{Lent}^{\text{ext}}(Y))_Y$ satisfies the hypothesis of Lemma 5.9, and its $H_0(Y)$ are those of the construction. That lemma processes rounds $R, \dots, 1$. Its step at round l depends only on $H_0(Y)$ for light parts Y of round l and on its own earlier steps. At round l the sets $H_0(Y)$ of the construction are already final (part (a)). So the objects of step (2) over all rounds, together with Cyc and E_0 , are exactly the output of Lemma 5.9 for this family. Hence $\mathcal{O}_K$ decomposes

$$E_K := E(G) \setminus K^{\text{std}} \setminus \bigsqcup_{Y \text{ light}} \text{Lent}^{\text{ext}}(Y) = \bigsqcup_l \text{Cyc}_l \sqcup E_0 \sqcup \bigsqcup_{Y \text{ light}} (E_r(Y) \setminus \text{Lent}^{\text{ext}}(Y)),$$

where the equality is Proposition 2.11(iii).

The other three families.

- By (T3) and (T4), $\mathcal{O}_T$ decomposes $\bigsqcup_Z E^V(Z)$.
- By Lemma 6.13, at each round $l \geq 3$ the cycles of step (3) decompose $(\bigsqcup_{Z \in \text{Std}_l} E^Q(Z) \setminus J_l) \sqcup \text{Lent}_l^{\text{JS}}$.
- By (JC2), $\mathcal{O}_l$ decomposes $J_l \sqcup \text{Lent}_l^{\text{JV}}$. These two sets are disjoint, because $J_l \subseteq \bigcup_Z E_l(Z)$ and $\text{Lent}_l^{\text{JV}} \subseteq \bigcup_Y E(H_Y)$ with $r(Y) \leq l - 2$ (Lemma 6.15(iii)).

Every edge of $\text{Lent}_l^{\text{JS}}$ lies in exactly one class $\mathcal{L}_{Y,l,j}^{\text{JS}}$. By (JC1) and Lemma 6.15(iii), every edge of $\text{Lent}_l^{\text{JV}}$ lies in exactly one $E(H_Y)$. Distinct l use distinct classes. Hence

$$\bigsqcup_l \text{Lent}_l^{\text{JS}} = \bigsqcup_Y \text{Lent}^{\text{JS}}(Y), \quad \bigsqcup_l \text{Lent}_l^{\text{JV}} = \bigsqcup_Y \text{Lent}^{\text{JV}}(Y),$$

with all unions disjoint.

Putting the pieces together. Since $E^{\text{rem}}(Z) = E^{\text{V}}(Z) \sqcup E^{\text{Q}}(Z)$ and $J_l \cap E_l(Z) \subseteq E^{\text{Q}}(Z)$,

$$E^{\text{V}}(Z) \sqcup (E^{\text{Q}}(Z) \setminus J_l) \sqcup (J_l \cap E_l(Z)) = E^{\text{rem}}(Z) = E_l(Z) \setminus \text{Lent}(Z),$$

and $\text{Lent}(Z) = \text{Lent}^{\text{JS}}(Z) \sqcup \text{Lent}^{\text{JV}}(Z)$ for standalone Z . For light non-demoted Y ,

$$E_r(Y) \setminus \text{Lent}^{\text{ext}}(Y) = \text{Lent}^{\text{U}}(Y) \sqcup H_0(Y) \quad (\text{Lemma 6.18(iv)}),$$

and for demoted Y , $\text{Lent}^{\text{ext}}(Y) = \emptyset$. Therefore the edge sets of the four families are pairwise disjoint, and their union is

$$E_K \sqcup \bigsqcup_{Z \in \text{Std}} E_l(Z) \sqcup \bigsqcup_{Y \text{ light}} \text{Lent}^{\text{ext}}(Y) = E(G),$$

again by Proposition 2.11(iii).

Each family decomposes its own edge set, so together they decompose $E(G)$. In particular every lent edge lies in at most one object. A lent edge that is not used (not in $\text{Lent}(Y)$), including every returned edge of JS-LC, stays in $E^{\text{rem}}(Y)$ or $H_0(Y)$ and is covered by the owner's device, which tolerates it by part (a). The last item of (b) is the remark after (6.3). Its right side involves only $E_l(Z)$, δ and Lost_Z , not Lent , the TPV split or earlier rounds.

(c) *Cost.*

- $\mathcal{O}_K$. By Lemma 5.9, $|\mathcal{O}_K| \leq (D_*/2 + 745 + \varepsilon_K(D_*))n + 80 \text{ dem} + 369 \text{ lp}$.
- $\mathcal{O}_T$. By (T3), $|\mathcal{O}_T| \leq \sum_l \sum_{Z \in \text{Std}_l} 169 |\text{Ret}_Z| = 169 \sum_{l,Z} (|A_Z| + |F_Z| + |\text{Lost}_Z|)$. Here $\sum_{l,Z} |A_Z| \leq \sum_l 15.2 \varepsilon n / \log P_l \leq \varepsilon_A n$ by Proposition 2.13(K2) and Lemma 2.17(e). Also $\sum_{l,Z} |F_Z| \leq 2n$ by Proposition 2.20(i). Finally $\sum_{Z \in \text{Std}_l} |\text{Lost}_Z| = |\text{Lost}_l|$, because the Lost_Z of one round lie in the pairwise disjoint port sets U_Z (Proposition 2.11(iv)). So $|\mathcal{O}_T| \leq (338 + 169 \varepsilon_A)n + 169 \sum_l |\text{Lost}_l|$, with $338 = 169 \cdot 2$.
- $\mathcal{O}_S$. By Lemma 6.13 and $\sum_l 1/M_l \leq 2/D_*$ (Lemma 2.17(b)),

$$|\mathcal{O}_S| \leq \sum_{l \geq 3} \left(126 \frac{n}{M_l} + 1.5 \sum_{(Y,l) \text{ giant}} m_{Y,l} \right) \leq \frac{252n}{D_*} + 1.5 \sum_{(Y,l)} m_{Y,l}.$$

- $\mathcal{O}_J$. $|\mathcal{O}_J| = \sum_l |\mathcal{O}_l|$.

Adding the four bounds gives (c), since $\varepsilon_K + 169 \varepsilon_A + 252/D_* = \varepsilon_M(D_*)$. The final inequality is Theorem 6.10(iv), which holds for every valid run and every designation. $\square$

7 The junction-vertex quotient, ultra hubs, and the induction

This section supplies the J-consumer of Definition 6.16, bounds the size of the quotient graphs it produces, and closes the argument by the layered quotient induction.

Setting of this section. Throughout, D_* satisfies $(\Gamma 1)$ – $(\Gamma 4)$, G is a graph with $n \geq N_0$ vertices and $d_1 \geq D_*$, a valid $\text{HB}^{*\tau+}$ run on G is fixed (Proposition 2.15), a designation δ is fixed (Definition 6.8), and there are no VX-parts ($\mathcal{V} = \emptyset$), so that every standalone pre-part is a MIX-part. The rounds of the run are $1, \dots, R$; the round step below acts at the rounds $3 \leq l \leq R$ (rounds 1 and 2 have no classed ports). For a classed port u of round l (that is, $u \in Q_Z$ for some $Z \in \text{Std}_l$) we write $Y(u)$ for its class and $r(u)$ for the round of $Y(u)$; since $Y(u) \in \text{anc}_l(u)$ we have $r(u) \leq l - 2$ and $u \in V(Y(u))$. For an ancestor Y of round r , H_Y is its expander (X_Y if Y is light, X_Y^0 if Y is standalone), s_Y is $s_r/2$ or s_r respectively, and $p_Y = 1/(2k_{\text{lend}}(Y))$ (Definition 3.10). The functions ε_1 and ε_2 defined below contain the function $\varepsilon_A = 31\varepsilon/(C' \log \log D_*)$ of Lemma 2.17(e), which depends on D_* only and satisfies $\sum_{l \leq R} 15.2\varepsilon/\log P_l \leq \varepsilon_A$; the function ε_X does not contain ε_A .

7.1 Round-split pools and candidate junctions

Definition 7.1 (Round-split pool labels). Every vertex v of G independently draws a *pool label*

$$\text{pl}(v) \in \{\perp\} \cup \{(l, r) : 3 \leq l \leq R, 1 \leq r \leq l - 2\}, \quad \mathbb{P}(\text{pl}(v) = (l, r)) = q_l \pi_{l,r},$$

where $q_l := M_l^{-2}$ and $\pi_{l,r} := 2^{-(l-1-r)}$, and $\mathbb{P}(\text{pl}(v) = \perp) := 1 - \sum_{l,r} q_l \pi_{l,r}$. This is a probability distribution: for each l we have $\sum_{r=1}^{l-2} \pi_{l,r} = 1 - 2^{-(l-2)} < 1$, and $\sum_l q_l \leq \sum_l M_l^{-1} \leq 2/D_* < 1$ by Lemma 2.17(b). Put

$$\text{Pool}_{l,r} := \text{pl}^{-1}(l, r), \quad \text{Pool}_l := \bigcup_{r=1}^{l-2} \text{Pool}_{l,r} \quad (\text{a disjoint union}),$$

so that $\mathbb{P}(v \in \text{Pool}_l) = q_l(1 - 2^{-(l-2)}) \leq q_l$ for every v . The sets Pool_l ($3 \leq l \leq R$) are pairwise disjoint, because every vertex has one label. For $w \in \text{Pool}_l$ let $r(w)$ denote the unique r with $w \in \text{Pool}_{l,r}$. The pool labels are stage (1d) of the randomness schedule. They are independent of all other stage-1 data, in particular of the colourings and labels of Definition 3.10.

Dependencies: 2.17, 3.10

Definition 7.2 (Candidates and JV-bad ports). Let u be a classed port of round $l \geq 3$, with class $Y := Y(u)$ of round $r := r(u) \leq l - 2$. Its *candidate set* is

$$\text{Cand}_l(u) := \{w \in \text{Pool}_{l,r(u)} : uw \in \mathcal{L}_{Y,l}^{\text{JV}}\}.$$

Since $\mathcal{L}_{Y,l}^{\text{JV}} \subseteq \text{Lend}_Y \subseteq E(H_Y)$ and H_Y is a graph on $V(Y)$, automatically $\text{Cand}_l(u) \subseteq N_{H_Y}(u)$ and $N_{H_Y}(u) \subseteq V(Y)$. The candidate sets are always taken in the class graph $H_{Y(u)}$ itself; this is used in Lemma MULT below. Put

$$H_l^{\text{cd}} := \frac{\lambda_{l-2}^{95}}{8M_l^2}.$$

The port u is *JV-bad* if

$$|\text{Cand}_l(u)| < \frac{1}{2} \mathbb{E}|\text{Cand}_l(u)|,$$

and *JV-good* otherwise. Here $\mathbb{E}$ is the unconditional expectation over stage 1, which is a number determined by the run and δ . The definition applies to every classed port, whether or not it lies in Lost_l . JV-badness is a deterministic function of stage 1; it is one of the stage-2 statuses. The number H_l^{cd} is not a threshold of the definition: it is a lower bound for the threshold, since $\frac{1}{2} \mathbb{E}|\text{Cand}_l(u)| \geq H_l^{\text{cd}}$ for every classed port u of round l (Lemma 7.3(ii) below). Consequently every JV-good port has $|\text{Cand}_l(u)| \geq H_l^{\text{cd}}$ (Lemma 7.3(iv)).

Dependencies: 7.1, 3.10, 6.8

Lemma 7.3 (Candidate counts). *Let u be a classed port of round $l \geq 3$ with class Y of round r . Then:*

- (i) $|\text{Cand}_l(u)|$ is a sum of independent Bernoulli variables, one for each $w \in N_{H_Y}(u)$, each with success probability $q_l \pi_{l,r} p_Y$;
- (ii) its mean satisfies

$$\mathbb{E}|\text{Cand}_l(u)| = q_l \pi_{l,r} p_Y \deg_{H_Y}(u) \geq \frac{\lambda_r^{96} 2^{-(l-r)}}{M_l^2} \geq 2H_l^{\text{cd}};$$

- (iii) $\mathbb{P}(u \text{ is JV-bad}) \leq \exp(-H_l^{\text{cd}}/4)$;
- (iv) if u is JV-good then $|\text{Cand}_l(u)| \geq H_l^{\text{cd}} \geq 2^{10} M_l^{10}$.

Dependencies: 7.2, 7.1, 3.10, 3.11, 2.11, 2.17, 1.25

Proof. (i) By definition,

$$|\text{Cand}_l(u)| = \sum_{w \in N_{H_Y}(u)} \mathbf{1}[\text{pl}(w) = (l, r)] \cdot \mathbf{1}[uw \in \mathcal{L}_{Y,l}^{\text{JV}}].$$

The pool labels are independent across vertices. The colours of the edges of H_Y are independent across edges (Definition 3.10), and the edges uw for distinct w are distinct. The two families are independent of each other (Definition 7.1). So the summands for distinct w are functions of disjoint sets of independent variables, hence independent. For a fixed w the two indicators are independent, with $\mathbb{P}(\text{pl}(w) = (l, r)) = q_l \pi_{l,r}$. Moreover $\mathbb{P}(uw \in \mathcal{L}_{Y,l}^{\text{JV}}) = p_Y$: the edge $uw \in E(H_Y)$ lies in Lend_Y with probability $1/2$, and it then receives the colour $l \in \mathcal{I}_{\text{JV}}$ with probability $1/k_{\text{lend}}(Y)$; note that $r + 2 \leq l \leq R$, so l is an index of $\mathcal{I}_{\text{JV}}$.

(ii) The first equality follows from (i). Every H_Y has minimum degree larger than $s_Y \geq s_r/2$, and $s_r \geq \lambda_r^{100}$ (Proposition 2.11(i)). Also $p_Y \geq \lambda_r^{-4}$ (Lemma 3.11). Hence

$$\mathbb{E}|\text{Cand}_l(u)| \geq M_l^{-2} 2^{-(l-1-r)} \lambda_r^{-4} \frac{\lambda_r^{100}}{2} = \frac{\lambda_r^{96} 2^{-(l-r)}}{M_l^2}.$$

Since $2H_l^{\text{cd}} = \lambda_{l-2}^{95}/(4M_l^2)$, the second inequality is equivalent to

$$\lambda_r^{96} \geq 2^{l-r-2} \lambda_{l-2}^{95}. \quad (7.1)$$

By Lemma 2.17(a), $\lambda_r \geq \lambda_{l-2}$ because $r \leq l-2$. By Lemma 2.17(d), $\lambda_r \geq 2^{2 \log^* d_r + 2} \geq 2^{R-r} \geq 2^{l-r} \geq 2^{l-r-2}$. Multiplying $\lambda_r^{95} \geq \lambda_{l-2}^{95}$ by $\lambda_r \geq 2^{l-r-2}$ gives (7.1). (The chain of exponents is $2 \log^* d_r + 2 \geq R - r \geq l - r$; only the exponent $l - r - 2$ is needed.)

(iii) Put $\mu := \mathbb{E}|\text{Cand}_l(u)|$. By definition, u is JV-bad exactly when $|\text{Cand}_l(u)| < \mu/2$. By (i) and the lower-tail Chernoff bound for sums of independent indicators, with deviation parameter $1/2$ (Cited result 1.25), $\mathbb{P}(|\text{Cand}_l(u)| < \mu/2) \leq \exp(-\mu/8) \leq \exp(-H_l^{\text{cd}}/4)$, where the last step uses $\mu/2 \geq H_l^{\text{cd}}$ from (ii).

(iv) If u is JV-good then $|\text{Cand}_l(u)| \geq \frac{1}{2} \mathbb{E}|\text{Cand}_l(u)| \geq H_l^{\text{cd}}$, by the definition of JV-good and (ii); this is the first inequality. The second is the line “ $H_l^{\text{cd}} \geq 2^{10} M_l^{10}$ ” of the table of Lemma 3.11. It also follows directly from $M_l \leq \lambda_{l-2}^{1.6}$ (Lemma 2.17(b)): then $8M_l^2 \cdot 2^{10} M_l^{10} = 2^{13} M_l^{12} \leq 2^{13} \lambda_{l-2}^{19.2} \leq \lambda_{l-2}^{95}$, because $\lambda_{l-2} \geq M_l^{1/1.6} \geq 2^{25}$. $\square$

7.2 The randomness schedule and the round step

Definition 7.4 (The randomness schedule). All random choices of the construction are made in the following order, summarized in Table 6.

- *Stage 1* consists of four mutually independent families: (1a) the COL-JV colourings of Definition 3.10(i)–(iii); (1b) the vertex choices and sublabels of Definition 5.1; (1c) the JS labels of Definition 3.10(iv); (1d) the pool labels of Definition 7.1.
- *Selection* (1e), which is not a random family: a stage-1 outcome is then fixed in the event $\{X' \leq 3\mathbb{E}X'\}$. Here $X' \geq 0$ is the stage-1 functional introduced later in this section, a deterministic function of the run, δ and stage 1. After (1e), stage 1 is a fixed outcome: no later step of the construction uses the law of stage 1 or conditions on the event of (1e); the numbers $\mathbb{E}X'$ and $\mathbb{E}|\text{Cand}_l(u)|$ that appear later are unconditional expectations, fixed in advance.
- *Stage 2* is deterministic given stage 1: demotion, parent-bad and Bad (Definition 5.3); lend-bad, Lost, Ret and Q^* (Definition 6.11); JV-bad (Definition 7.2).
- *Stage 3* is drawn part by part: the TPV labels of every standalone pre-part, the P-vortex labels of every non-demoted light part, and the VX^+ labels of every demoted light part. Given stage 1, the label families of distinct parts are independent. Each part's labels are fixed inside that part's good event.
- *Rounds* $l = R, R-1, \dots, 3$. The round randomness ξ_l consists of the following mutually independent variables: a uniformly random bijection $\eta_h : [4M_l] \rightarrow [4M_l]$ for every vertex h ; a uniformly random 3-subset $\zeta_{h,u} \subseteq [4M_l]$ for every ordered pair (h, u) of distinct vertices; and a uniformly random linear order $\prec_u$ of $V(G)$ for every vertex u . The dependence of $\eta_h, \zeta_{h,u}, \prec_u$ on l is suppressed. The draw ξ_l is fresh: it is independent of stage 1, of stage 3 and of all $\xi_{l'}$ with $l' > l$.

The *past* $\mathfrak{F}_l^-$ at round l is the fixed outcome of stage 1, of stage 3 and of the draws $\xi_{l'}$ ($l' > l$), together with everything these determine: in particular the decompositions fixed at the rounds $l' > l$ and the set J_l . It is a fixed outcome, not a random σ -algebra. We write $\mathbb{E}[\cdot \mid \mathfrak{F}_l^-]$ and $\mathbb{P}(\cdot \mid \mathfrak{F}_l^-)$ for expectation and probability over ξ_l with $\mathfrak{F}_l^-$ fixed. The fixed rules of Construction 7.5 read ξ_l only as stated there. At each round, ξ_l is chosen in the event of step (f) of the round step. All draws in stage 1, stage 3 and the rounds are mutually independent. The outcomes actually used are chosen inside events (stage 1 in $\{X' \leq 3\mathbb{E}X'\}$, stage 3 in the good events, each ξ_l in the event of step (f)); after each choice only deterministic properties of the chosen outcome are used, and every later probability is over fresh variables with the past fixed.

Dependencies: 3.10, 5.1, 7.1, 6.11, 5.3, 7.2, 4.2, 4.3, 4.4, 5.6, 5.8

Construction 7.5 (The JV^{+*} round step at round $l \geq 3$; the J-consumer). Fix $3 \leq l \leq R$ and the past $\mathfrak{F}_l^-$. The input is the set $J_l = J_l^{\text{lost}} \cup J_l^{\text{hub}} \cup J_l^{\text{fr}} \cup J_l^{\text{par}}$ produced by JS-LC at round l , with the properties (J1)–(J3) and (i)–(v) of Lemma 6.15.

Fixed rules. The rules and orders called fixed in steps (b)–(e), (g) and (h) below are deterministic functions, chosen once and for all before any randomness is drawn, with the following arguments.

- *Functions of $\mathfrak{F}_l^-$ alone:* the pairing of the live J^{fr} items at a fresh centre and the choice of its unpaired fresh leg, and the order in which the PAR objects are coloured, in (b); the split of the live items of a non-ultra hub into groups, together with the numbering of the groups, in (c). In particular these rules do not read ξ_l .

Stage	Drawn or decided	Depends on	How it is fixed
1a	COL-JV colourings of every ancestor	independent per ancestor and per edge	part of the stage-1 outcome
1b	vertex choices and sublabels (zones)	independent per vertex	part of the stage-1 outcome
1c	JS labels $\text{lab}_{Y,l}(y)$	independent	part of the stage-1 outcome
1d	pool labels $\text{pl}(v)$	independent per vertex	part of the stage-1 outcome
1e	choice of the stage-1 outcome	the functional X'	$X' \leq 3EX'$ (probability $\geq 2/3$)
2	demotion, parent-bad, lend-bad, Lost, Ret, Q^* , JV-bad	deterministic in stage 1	determined
3	TPV, P-vortex and VX^+ labels	independent per part	inside each part's good event (probability $\geq 1/2 - o(1)$)
$l = R, \dots, 3$	$\xi_l = (\eta_h, \zeta_{h,u}, \prec_u)$: HUB lists and injections	fresh draw, $\mathfrak{F}_l^-$ fixed	event of step (f) (probability $\geq 1/2$)

Table 6: The randomness schedule (Definition 7.4).

- *Functions of $\mathfrak{F}_l^-$ and of the lists*, that is, of the variables η_h and $\zeta_{h,u}$ of ξ_l : the choice of the system of distinct representatives in (d); the processing order and the order of $V(G)$ in (e1); and the order $e_1, \dots, e_q$ of $E'(u)$ in (e2).
- None of the rules of steps (b)–(e) reads the orders $\prec_u$ of ξ_l ; within steps (a)–(f), the orders $\prec_u$ of ξ_l enter only through the assignment $e_i \mapsto w_i$ of (e2) (and through what is computed from it). Through $\mathfrak{F}_l^-$ the rules may depend on the draws $\xi_{l'}$ of the rounds $l' > l$, orders included, and in general they do, since J_l does.
- The rank order in (g) and the rule choosing $\mathcal{D}_l$ in (h) are not restricted: they are arbitrary deterministic functions of $\mathfrak{F}_l^-$ and ξ_l . They necessarily depend on ξ_l , since the layers and Q_l do.

Rules with these arguments exist, for instance lexicographically first choices for fixed orders of the finite sets involved (the greedy colouring of (b) succeeds for every order, by part (i) of the next lemma, and a split into groups as in (c) always exists). Every statement of this section about the round step holds for every choice of the fixed rules that obeys these restrictions. The restrictions are used in the claims of steps (c) and (e2), in Lemma 7.6(iii) (the grouping does not read η_h), and in Lemmas 7.10(i), 7.11(i) and 7.13(c) (nothing before the assignment $e_i \mapsto w_i$ reads the orders $\prec_u$), as well as in the opening paragraph of the subsection “Quotient size and payments” and in the proof of Lemma 7.16(ii) (the copies of non-ultra hubs are deterministic given the lists).

Items. The *items* are the edges of $J_l \setminus J_l^{\text{lost}}$. The *vertices* of an item are its port end or ends, which lie in Q_Z^* , and its centre. The centre is a hub $h \in D_l$ for a J^{hub} item and a fresh centre $x \in F_Z$ for a J^{fr} item. A J^{par} item $u-v$ has two port ends and no centre. A J^{hub} item is written (h, u) , with h its hub and u its port.

(a) *Payments bounded by stage-1 weights.* The following are *paid*, that is, output as single edges: (a1) every edge of J_l^{lost} ; (a2) every item with a vertex in Pool_l ; (a3) every item with a JV-bad port end. The remaining items are *live*. A vertex is *live* if it is a vertex of a live item, and Live denotes the set of live vertices. By (a2), $\text{Live} \cap \text{Pool}_l = \emptyset$. The sets paid in (a1)–(a3) depend on J_l , hence on

the past; only their sizes are bounded by stage-1 functionals (this is proved in the payment lemma below).

(b) *PAR objects*. Every live J^{par} item $u-v$ is a *PAR object* with *ends* at u and at v ; here $u, v \in Q_Z^*$ and $Y(u) \neq Y(v)$. At each fresh centre x , the live J^{fr} items at x are paired, by a fixed rule, into *cherries* $u-x-u'$. Here $u \neq u'$ because G is simple, so distinct items at x have distinct port ends. A cherry is a PAR object with ends at u and u' and *middle* x . If the number of live J^{fr} items at x is odd, one of them, chosen by a fixed rule, is paid: it is the *unpaired fresh leg* of x . The PAR objects are then coloured greedily, in a fixed order, with colours from $[3M_l]$, so that two PAR objects sharing a port, or sharing a middle, receive different colours.

(c) *HUB lists*. For a hub $h \in D_l$ let c_h^{live} be its number of live J^{hub} items. Put

$$\theta_l^{\text{ult}} := \lfloor M_l H_l^{\text{cd}} / 7 \rfloor, \quad K_l^{\text{HUB}} := 4M_l.$$

The hub h is *ultra* if $c_h^{\text{live}} > \theta_l^{\text{ult}}$, and *non-ultra* otherwise. For a non-ultra h with $c_h^{\text{live}} \geq 1$, its live items are split, by a fixed rule, into

$$k_h := \lceil 8c_h^{\text{live}} / H_l^{\text{cd}} \rceil$$

groups of at most $\lceil H_l^{\text{cd}} / 8 \rceil$ items each. This is possible because $k_h \lceil H_l^{\text{cd}} / 8 \rceil \geq c_h^{\text{live}}$. The i -th group receives the *list* $\{\eta_h(3i-2), \eta_h(3i-1), \eta_h(3i)\} \subseteq [4M_l]$; this needs $3k_h \leq 4M_l$, which is part (ii) of the next lemma. The lists of the groups of h are pairwise disjoint, and, since the groups and their numbering are functions of $\mathfrak{F}_l^-$ while η_h is independent of $\mathfrak{F}_l^-$, they form a uniformly random sequence of k_h pairwise disjoint 3-subsets of $[4M_l]$. Lists of distinct hubs are independent. For an ultra h , every live item (h, u) is its own group, with list $\zeta_{h,u}$.

(d) *SDR*. Every port u carrying at least one live hub item chooses, by a fixed rule, pairwise distinct colours for its live hub items, each item's colour taken from the list of its group (a system of distinct representatives). If no such choice exists, all live hub items of u are paid (an *SDR failure* at u). The items that receive a colour are the *coloured hub items*. In particular, every port carries at most one coloured hub item of each HUB colour.

(e) *Junctions*. Initially $\text{Used}(u) := \emptyset$ for every port u and $\text{Used}_\kappa(h) := \emptyset$ for every hub h and every $\kappa \in [4M_l]$.

(e1) The coloured hub items of non-ultra hubs are processed in a fixed order. The item (h, u) of colour κ receives as its *junction* the first vertex w , in a fixed order of $V(G)$, of $\text{Cand}_l(u) \setminus \text{Used}(u) \setminus \text{Used}_\kappa(h)$. Then w is added to $\text{Used}(u)$ and to $\text{Used}_\kappa(h)$.

(e2) For every port u let $E'(u)$ be the set of the ends at u of the PAR objects and of the coloured hub items (h, u) with h ultra, listed in a fixed order $e_1, \dots, e_q$. Let $w_1 \prec_u w_2 \prec_u \dots$ be the elements of $\text{Cand}_l(u) \setminus \text{Used}(u)$ in the order $\prec_u$; the end e_i receives the junction w_i . This is well defined by part (v) of the next lemma. With $\mathfrak{F}_l^-$ and the lists fixed, the map $e_i \mapsto w_i$ is a uniformly random injection of $E'(u)$ into $\text{Cand}_l(u) \setminus \text{Used}(u)$, and these injections are independent over ports: by the restrictions on the fixed rules, $\text{Used}(u)$, the set $E'(u)$ and its order $e_1, \dots, e_q$ are determined by $\mathfrak{F}_l^-$ and the lists, and the orders $\prec_u$ are independent of both. No payment is made for coincidences of junctions at ultra hubs.

(e3) *Loops*. A PAR object whose two ends received the same junction is paid. A looped J^{par} item costs one single edge; a looped cherry costs its two edges.

Every end of a coloured hub item or of an unpaid PAR object now has a junction w . It is a candidate of the end's port u , so $w \in \text{Cand}_l(u) \subseteq \text{Pool}_l$ and $uw \in \mathcal{L}_{Y(u),l}^{\text{JV}}$. The edge uw is the *junction edge* of that end.

(f) *Markov choice.* Let pay_l^{rd} be the number of edges paid in (d) and (e3), and let $\text{copies}_l := |V(Q_l)|$, with Q_l as in (g). Both are functions of $\mathfrak{F}_l^-$ and ξ_l . The round randomness ξ_l is chosen in the event

$$\{\text{copies}_l \leq 4 \mathbb{E}[\text{copies}_l \mid \mathfrak{F}_l^-]\} \cap \{\text{pay}_l^{\text{rd}} \leq 4 \mathbb{E}[\text{pay}_l^{\text{rd}} \mid \mathfrak{F}_l^-]\}.$$

By Markov's inequality in the form (b) of Cited result 1.26, taken conditionally as in its item (c) (ξ_l is independent of the variables whose outcome $\mathfrak{F}_l^-$ fixes, Definition 7.4), this event has probability at least $1/2$ given $\mathfrak{F}_l^-$ (each of the two events fails with probability at most $1/4$).

(g) *Layers, sub-layers and the quotient.* For $\kappa \in [3M_l]$ the *PAR layer* B_κ^{P} is the multigraph on the vertex set $\{[w] : w \in \text{Pool}_l\}$ of formal copies. It has one edge $[w_1][w_2]$ for every unpaid PAR object of colour κ whose ends have the junctions w_1 and w_2 ; here $w_1 \neq w_2$ by (e3). For $\kappa \in [4M_l]$ the *HUB layer* B_κ^{H} is the bipartite multigraph with sides D_l and $\{[w] : w \in \text{Pool}_l\}$. It has one edge $h[w]$ for every coloured hub item (h, u) of colour κ whose junction is w . *Rank split:* in every layer, the edges joining the same two vertices are listed in a fixed order, and the i -th edge of such a list has *rank* i . For a layer B and $\iota \geq 1$, the *sub-layer* $B^{(\iota)}$ is the spanning subgraph of B formed by its edges of rank ι . The *quotient* Q_l is the vertex-disjoint union of all sub-layers $B_\kappa^{\text{P},(\iota)}$ ($\kappa \in [3M_l]$, $\iota \geq 1$) and $B_\kappa^{\text{H},(\iota)}$ ($\kappa \in [4M_l]$, $\iota \geq 1$), with the isolated vertices of each sub-layer deleted. Vertices of distinct sub-layers are distinct vertices of Q_l ; hence every cycle of Q_l lies in one sub-layer.

(h) *Decompose and lift.* Let $\mathcal{D}_l$ be a decomposition of $E(Q_l)$ into $f(Q_l)$ objects, chosen by a fixed deterministic rule (for instance the lexicographically first one for a fixed order of $E(Q_l)$). Each member of $\mathcal{D}_l$ is *lifted* to objects of G as follows.

- A single edge of a PAR sub-layer is a PAR object. It is output as its one (J^{par}) or two (cherry) J-edges, each as a single edge. A single edge $h[w]$ of a HUB sub-layer is a coloured hub item (h, u) , and it is output as the single edge hu .
- A cycle $[w_0][w_1] \cdots [w_{m-1}]$ of a PAR sub-layer, with $m \geq 3$: indices are taken mod m , and the edge $[w_i][w_{i+1}]$ is a PAR object o_i . Let p_i be the end of o_i with junction w_i and q_i its end with junction w_{i+1} ; they are well defined because $w_i \neq w_{i+1}$. The lift is the closed walk

$$w_0 p_0 (x_0) q_0 w_1 p_1 (x_1) q_1 w_2 \cdots q_{m-1} w_0,$$

where x_i is the middle of o_i if o_i is a cherry, and is omitted if o_i is a J^{par} item.

- A cycle $h_0[w_0]h_1[w_1] \cdots h_{m-1}[w_{m-1}]h_0$ of a HUB sub-layer, with $m \geq 2$: the edge $h_i[w_i]$ is a coloured hub item (h_i, u_i) with junction w_i , and the edge $[w_i]h_{i+1}$ is a coloured hub item (h_{i+1}, u'_i) with junction w_i (indices mod m). The lift is the closed walk

$$h_0 u_0 w_0 u'_0 h_1 u_1 w_1 u'_1 h_2 \cdots u'_{m-1} h_0.$$

Put $\mathcal{O}_l :=$ (the single edges paid in (a), (b), (d) and (e3)) together with (the lifted objects of all members of $\mathcal{D}_l$), and let $\text{Lent}_l^{\text{JV}}$ be the set of junction edges that lie on lifted cycles. A junction edge assigned in (e) that does not lie on a lifted cycle is not used by the round step; it stays with its owner as junk.

Dependencies: 6.15, 6.16, 7.2, 7.1, 7.4, 1.26

Lemma 7.6 (The round step is well defined). *For every past $\mathfrak{F}_l^-$ ($3 \leq l \leq R$) the following hold.*

- (i) *PAR palette.* Every PAR object shares a port or a middle with fewer than $3M_l$ other PAR objects, so the greedy colouring of (b) succeeds. Consequently every port carries at most one PAR object of each colour, and every fresh centre is the middle of at most one cherry of each colour.

- (ii) Disjoint lists. Every non-ultra hub h has $k_h \leq \lceil 8M_l/7 \rceil$ and $3k_h \leq 4M_l$.
- (iii) Cuckoo SDR. For every port u , $\mathbb{P}(\text{SDR failure at } u \mid \mathfrak{F}_l^-) \leq (K_l^{\text{HUB}})^{-4} = (4M_l)^{-4}$.
- (iv) Greedy step. Step (e1) always finds a junction. Two coloured items of the same non-ultra hub h with the same colour κ receive distinct junctions.
- (v) Injections. At every port u carrying a live item, $|\text{Cand}_l(u) \setminus \text{Used}(u)| \geq H_l^{\text{cd}} - (M_l - 1) \geq H_l^{\text{cd}}/2 \geq M_l > |E'(u)|$, so step (e2) is well defined.

Dependencies: 7.5, 6.15, 7.3, 3.11, 1.28, 2.7

Proof. Throughout, $M_l \geq 2^{40}$ by the definition of M_l , and $H_l^{\text{cd}} \geq 2^{10}M_l^{10}$ by Lemma 7.3(iv) (equivalently, by the table of Lemma 3.11). By (J2) of Lemma 6.15, every port carries at most $M_l - 1$ J-edges, hence at most $M_l - 1$ live items. By Lemma 6.15(ii), every fresh centre carries at most $M_l - 1$ J^{fr} edges, hence is the middle of at most $(M_l - 1)/2$ cherries.

(i) A PAR object has at most two ports and at most one middle. Each of its ports carries at most $M_l - 2$ further live items, and each further PAR object at that port uses one of them. Its middle, if any, is the middle of at most $(M_l - 1)/2 - 1$ further cherries. So the object conflicts with at most $2(M_l - 2) + (M_l - 1)/2 < 3M_l$ others. When it is coloured, fewer than $3M_l$ colours are therefore blocked, and a colour of $\lceil 3M_l \rceil$ remains. The consequences are the defining property of the colouring.

(ii) If h is non-ultra then $c_h^{\text{live}} \leq \theta_l^{\text{ult}} \leq M_l H_l^{\text{cd}}/7$, so $k_h = \lceil 8c_h^{\text{live}}/H_l^{\text{cd}} \rceil \leq \lceil 8M_l/7 \rceil$. Hence $3k_h \leq 3(8M_l/7 + 1) = 24M_l/7 + 3 \leq 4M_l$, because $M_l \geq 21/4$.

(iii) Fix u and let $(h_1, u), \dots, (h_m, u)$ be its live hub items, so that $m \leq M_l - 1 \leq K/4$ with $K := K_l^{\text{HUB}} = 4M_l$. These items are distinct edges at u and G is simple, so the hubs $h_1, \dots, h_m$ are distinct. With $\mathfrak{F}_l^-$ fixed, it is determined which hubs are ultra, how the items are grouped, and which group each item belongs to (the grouping rule of step (c) reads $\mathfrak{F}_l^-$ only, by the restrictions on the fixed rules in Construction 7.5). The list of (h_i, u) is a function of η_{h_i} if h_i is non-ultra, and equals $\zeta_{h_i, u}$ if h_i is ultra. Since the h_i are distinct, the m lists are independent. Each list is a uniformly random 3-subset of $[K]$; for a non-ultra hub this holds because the sequence of its lists is a uniformly random sequence of pairwise disjoint 3-subsets, so each single list is uniform. By Hall's theorem (Cited result 1.28, applied with the m items as the index set and, for each item (h_i, u) , its list as the set of that item), an SDR fails to exist only if some set of s items has at most $s - 1$ colours in the union of their lists. Since every list has 3 elements, this forces $s \geq 4$. The union is then contained in some $(s - 1)$ -subset of $[K]$. The union bound over s , over the s -sets of items and over the $(s - 1)$ -subsets of $[K]$ gives

$$\mathbb{P}(\text{SDR failure at } u \mid \mathfrak{F}_l^-) \leq \sum_{s=4}^m T_s, \quad T_s := \binom{m}{s} \binom{K}{s-1} \left(\binom{s-1}{3} / \binom{K}{3} \right)^s.$$

We use $\binom{m}{s} \leq (em/s)^s \leq (eK/(4s))^s$, $\binom{K}{s-1} \leq (eK/(s-1))^{s-1}$, and

$$\binom{s-1}{3} / \binom{K}{3} = \frac{(s-1)(s-2)(s-3)}{K(K-1)(K-2)} \leq \left(\frac{s-1}{K} \right)^3,$$

which holds because $(s-1-j)/(K-j) \leq (s-1)/K$ for $j = 1, 2$ and $s \leq K+1$. Multiplying, and using $(s-1)^{2s+1} \leq s^{2s+1}$,

$$T_s \leq e^{2s-1} 4^{-s} s^{-s} (s-1)^{2s+1} K^{-s-1} \leq a_s := \frac{1}{e} \cdot \frac{s}{K} \cdot \left(\frac{e^2 s}{4K} \right)^s.$$

First, $a_4 = 4e^7 K^{-5} < 4400 K^{-5}$. Second, for $s \geq 4$,

$$\frac{a_{s+1}}{a_s} = \frac{s+1}{s} \cdot \frac{e^2(s+1)}{4K} \cdot \left(\frac{s+1}{s}\right)^s \leq \frac{5}{4} \cdot \frac{e^2(s+1)}{4K} \cdot e = \frac{5e^3}{16} \cdot \frac{s+1}{K},$$

which is at most $\frac{5e^3}{16}(\frac{1}{13} + \frac{1}{K}) < 0.49$ when $4 \leq s < K/13$ and $K \geq 10^4$. Hence $a_s \leq 2^{4-s} a_4$ for $4 \leq s \leq \lceil K/13 \rceil$, and the sum of these terms is at most $2a_4 < 8800 K^{-5}$. Third, for $K/13 < s \leq m \leq K/4$ we have $s/K \leq 1/4$, so $a_s \leq \frac{1}{4e}(e^2/16)^s \leq 0.47^s$, and the sum of these terms is at most $0.47^{K/13}/0.53 \leq 2 \cdot 0.47^{K/13}$. Altogether

$$\mathbb{P}(\text{SDR failure at } u \mid \mathfrak{F}_l^-) \leq 8800 K^{-5} + 2 \cdot 0.47^{K/13} \leq K^{-4} \quad (K \geq 10^4),$$

since $8800 K^{-5} \leq 0.88 K^{-4}$ and $2 \cdot 0.47^{K/13} \leq 0.1 K^{-4}$ for $K \geq 10^4$. Here $K = 4M_l \geq 2^{42}$.

(iv) Items reach (e1) only if they are live, so their ports are JV-good and $|\text{Cand}_l(u)| \geq H_l^{\text{cd}}$ (Lemma 7.3(iv)). When the item (h, u) of colour κ is processed, $\text{Used}(u)$ contains one junction for each earlier item at u , so $|\text{Used}(u)| \leq M_l - 2$. The lists of the groups of h are pairwise disjoint, so all coloured items of h with colour κ lie in the one group whose list contains κ . That group has at most $\lceil H_l^{\text{cd}}/8 \rceil$ items, so $|\text{Used}_\kappa(h)| \leq \lceil H_l^{\text{cd}}/8 \rceil - 1 < H_l^{\text{cd}}/8$. Hence at least $H_l^{\text{cd}} - (M_l - 2) - H_l^{\text{cd}}/8 > 0$ choices remain. The second statement holds because each junction chosen for h in colour κ is added to $\text{Used}_\kappa(h)$ and is excluded afterwards.

(v) After (e1), $\text{Used}(u)$ consists of one junction for each coloured item of a non-ultra hub at u . The elements of $E'(u)$ correspond to further live items at u : a PAR end at u corresponds to the J^{par} item or the leg of the cherry at u , and an ultra hub end is itself an item. So $|\text{Used}(u)| + |E'(u)| \leq M_l - 1$. Every port carrying a live item is JV-good, so $|\text{Cand}_l(u) \setminus \text{Used}(u)| \geq H_l^{\text{cd}} - (M_l - 1) \geq H_l^{\text{cd}}/2 \geq M_l > |E'(u)|$, using $H_l^{\text{cd}} \geq 2^{10} M_l^{10} \geq 2M_l$.

Remark on numerics (not used). An exact evaluation of the union bound $K^5 \sum_s T_s$ with $m = M_l - 1$ has its maximum, about 0.0502, near $M_l = 19$; it tends to $9/256$ as $M_l \rightarrow \infty$, which is the limit of $K^5 T_4$, where $T_4 = \binom{m}{4} / \binom{K}{3}^3 \sim 9m^4/K^9$. None of these numerical values is used: only the analytic bound (iii) is used, and it holds since $M_l \geq 2^{40}$. $\square$

7.3 The quotient is simple, and every decomposition lifts

For a HUB colour κ , a hub h and $w \in \text{Pool}_l$, let $m_\kappa(h, w)$ be the number of edges $h[w]$ of B_κ^{H} , that is, the number of coloured hub items (h, u) of colour κ with junction w .

Lemma 7.7 (Lemma MULT). *Let $3 \leq l \leq R$, $\kappa \in [4M_l]$, let h be a hub and let $w \in \text{Pool}_{l,r}$. Then*

$$m_\kappa(h, w) \leq \text{mult}_r(w) \leq \mu_r(w).$$

Moreover, the number of sub-layers $B_\kappa^{\text{H},(\iota)}$ ($\iota \geq 1$) in which $[w]$ is not isolated equals $\max_h m_\kappa(h, w)$, and the number of those in which h is not isolated equals $\max_w m_\kappa(h, w)$.

Dependencies: 7.5, 6.15, 7.2, 7.1, 2.13

Proof. Let $(h, u_1), \dots, (h, u_m)$ be the coloured hub items of colour κ at h with junction w , so $m = m_\kappa(h, w)$. By step (e) of Construction 7.5, the junction of (h, u_i) lies in $\text{Cand}_l(u_i) \subseteq \text{Pool}_{l,r(u_i)}$ (Definition 7.2). Every vertex has a single pool label (Definition 7.1), and $w \in \text{Pool}_{l,r}$, so $r(u_i) = r$ for all i . Moreover $w \in \text{Cand}_l(u_i) \subseteq N_{H_{Y(u_i)}}(u_i) \subseteq V(Y(u_i))$; this is where it matters that candidates are taken in the class graph $H_{Y(u_i)}$ itself. So each $Y(u_i)$ is an ancestor of round r whose vertex set contains w . The items (h, u_i) are distinct J^{hub} edges at h of round l . By (J1) of

Lemma 6.15, for each class Y there is at most one J^{hub} edge of class Y at h in round l , aggregated over all MIX-parts. So the classes $Y(u_1), \dots, Y(u_m)$ are pairwise distinct. Hence $m \leq \text{mult}_r(w)$, the number of ancestors of round r containing w , and $\text{mult}_r(w) \leq \mu_r(w)$ by Proposition 2.13.

For the second statement: $[w]$ is not isolated in $B_\kappa^{\text{H},(\iota)}$ if and only if some edge of rank ι is incident to $[w]$. By the definition of ranks, this holds if and only if some pair $\{h, [w]\}$ carries at least ι edges, that is, $\max_h m_\kappa(h, w) \geq \iota$. The statement for h is proved in the same way. $\square$

Lemma 7.8 (Q_l is simple). *Every sub-layer of Construction 7.5(g) is a simple graph: it has no parallel edges, PAR sub-layers have no loops, and HUB sub-layers are bipartite between hubs and junction copies $[w]$, where the hub h and the junction w of any edge $h[w]$ are distinct vertices of G . Consequently Q_l is a simple graph without isolated vertices.*

Dependencies: 7.5

Proof. By the definition of ranks, a sub-layer contains at most one edge of rank ι between any two vertices, so it has no parallel edges. An edge of a PAR layer joins $[w_1]$ and $[w_2]$ with $w_1 \neq w_2$, because looped PAR objects were paid in (e3); so PAR sub-layers have no loops. An edge $h[w]$ of a HUB layer joins the side D_l to the side $\{[w]\}$. Moreover h is the hub of a coloured, hence live, item, so h is a live vertex and $h \notin \text{Pool}_l$ by (a2), while $w \in \text{Pool}_l$. So $h \neq w$ as vertices of G , and HUB sub-layers have no loops even when $[w]$ is identified with w . So every sub-layer is simple. Q_l is a vertex-disjoint union of the sub-layers with their isolated vertices deleted, so it is simple and has no isolated vertices. $\square$

Lemma 7.9 (Lemma JV-L: the lift). *Let $3 \leq l \leq R$, fix any past, any ξ_l , and let Live be the set of live vertices of round l .*

- (i) $\text{Live} \cap \text{Pool}_l = \emptyset$. *Every port carries at most one PAR object of each PAR colour and at most one coloured hub item of each HUB colour. Hence, although ports are not vertices of Q_l , at most one edge of each sub-layer is an object or item with an end at a given port. Every fresh centre is the middle of at most one cherry of each PAR colour. Hubs, fresh centres and ports are pairwise disjoint.*
- (ii) *For every decomposition $\mathcal{D}$ of $E(Q_l)$ into cycles and single edges, the lift of each cycle of $\mathcal{D}$ (Construction 7.5(h)) is a cycle of G , and the lift of each single edge of $\mathcal{D}$ consists of one or two single edges of G . The lifts of distinct members of $\mathcal{D}$ are pairwise edge-disjoint. Hence the lift of $\mathcal{D}$ consists of at most $2|\mathcal{D}|$ pairwise edge-disjoint objects of G .*
- (iii) *Every item is either paid or lies in exactly one edge of Q_l . Every junction edge uw is the junction edge of at most one item-end, and that end is at u , never at w . The edge uw lies in exactly one lent class, namely $\mathcal{L}_{Y(u),l}^{\text{JV}}$, where $Y(u)$ is a lend-good ancestor of round at most $l-2$; the only consumer of this class is the round- l step. Junction edges lie in $E_r(Y)$ for ancestors Y of rounds $r \leq l-2$, items lie in $E_l(Z)$ for $Z \in \text{Std}_l$, and these edge sets are disjoint.*
- (iv) *Consequently the round step of Construction 7.5, applied at every round $3 \leq l \leq R$, is a J -consumer: $\mathcal{O}_l$ and $\text{Lent}_l^{\text{JV}}$ satisfy (JC1)–(JC3) of Definition 6.16. The objects of $\mathcal{O}_l$ other than paid single edges number at most $2|\mathcal{D}_l| = 2f(Q_l)$.*

Dependencies: 7.5, 7.8, 7.6, 6.15, 6.16, 3.10, 2.11, 6.11

Proof. (i) The first claim is (a2) of Construction 7.5. The claims about ports and fresh centres are Lemma 7.6(i) together with the SDR of step (d). Since every edge of a sub-layer of colour κ is a PAR object or a coloured hub item of colour κ , at most one edge of each sub-layer is an object or item with an end at a given port. Disjointness of the three roles is (J3) of Lemma 6.15.

(ii) *PAR cycles.* Let $[w_0] \cdots [w_{m-1}]$ be a cycle of a PAR sub-layer of colour κ . By Lemma 7.8 the sub-layer is simple, so $m \geq 3$. Distinct vertices of the sub-layer are copies of distinct pooled vertices, so $w_0, \dots, w_{m-1}$ are distinct. The objects $o_0, \dots, o_{m-1}$ are distinct, because they are distinct edges of the cycle, and all have colour κ . By (i) a port carries at most one object of colour κ , so ports of distinct objects are distinct. The two ends of one PAR object lie at distinct ports ($u \neq v$ for a J^{par} item, $u \neq u'$ for a cherry), so $p_i \neq q_i$. Hence the $2m$ ports p_i, q_i are pairwise distinct. The middles x_i of the cherries among the o_i are distinct, because a fresh centre is the middle of at most one cherry of colour κ . Ports lie in Q_Z^* and middles in F_Z , so by (i) no port is a middle. Ports and middles are live, junctions are pooled, and $\text{Live} \cap \text{Pool}_l = \emptyset$. So all vertices of the closed walk $w_0 p_0(x_0) q_0 w_1 \cdots q_{m-1} w_0$ are pairwise distinct. Consecutive vertices are adjacent in G : $w_i p_i$ and $q_i w_{i+1}$ are junction edges (edges of $H_{Y(p_i)}$ and $H_{Y(q_i)}$), and $p_i q_i$ or $p_i x_i, x_i q_i$ are the J-edges of o_i . The walk has length at least $3m \geq 9$. A closed walk of length at least 3 with pairwise distinct vertices is a cycle of G .

HUB cycles. Let $h_0[w_0]h_1[w_1] \cdots h_{m-1}[w_{m-1}]h_0$ be a cycle of a HUB sub-layer of colour κ . The sub-layer is simple and bipartite (Lemma 7.8), so the cycle has even length at least 4, and $m \geq 2$. The hubs h_i are pairwise distinct, and so are the junctions w_i . The $2m$ items (h_i, u_i) and (h_{i+1}, u'_i) are distinct, because they are distinct edges of the cycle, and all have colour κ . A port carries at most one coloured hub item of colour κ , so the $2m$ ports u_i, u'_i are pairwise distinct; in particular $u_i \neq u'_i$. Hubs and ports are disjoint by (i), and hubs and ports are live while junctions are pooled. So all vertices of the closed walk $h_0 u_0 w_0 u'_0 h_1 \cdots u'_{m-1} h_0$ are distinct. Consecutive vertices are adjacent: $h_i u_i$ and $u'_i h_{i+1}$ are items, and $u_i w_i, w_i u'_i$ are junction edges. The length is $4m \geq 8$, so the lift is a cycle of G .

Single edges. A single edge of a PAR sub-layer is lifted to the one or two J-edges of its PAR object, and a single edge of a HUB sub-layer to one item. So every member of $\mathcal{D}$ lifts to at most two objects.

Edge-disjointness. Every live item that is not paid is a coloured hub item or belongs to exactly one unpaid PAR object, so it lies in exactly one edge of Q_l . Each edge of Q_l lies in exactly one member of $\mathcal{D}$. So the item edges used by the lifts of distinct members are distinct. A junction edge uw on a lifted cycle is the junction edge of an end at u with junction w . The ends at u have pairwise distinct junctions: in (e1) every chosen junction is added to $\text{Used}(u)$ and is excluded later, and in (e2) the map is injective into $\text{Cand}_l(u) \setminus \text{Used}(u)$. So uw determines the end at u , hence the item, the edge of Q_l and the member of $\mathcal{D}$. It cannot also be the junction edge of an end at w : w is pooled, so every item with vertex w was paid in (a2), and no end lies at w . Finally, junction edges are not items: a junction edge lies in $E(H_Y) \subseteq E_r(Y)$ for an ancestor Y of round $r \leq l-2$, an item lies in $E_l(Z)$ with $Z \in \text{Std}_l$, and these edge sets are disjoint (Proposition 2.11(iii); Lemma 6.15(iii)).

(iii) The first two claims were shown in the proof of (ii); every item that is not in an edge of Q_l was paid in (a2), (a3), (b), (d) or (e3). Let uw be a junction edge of an end at u . Then $w \in \text{Cand}_l(u)$, so $uw \in \mathcal{L}_{Y(u),l}^{\text{JV}} \subseteq E(H_{Y(u)})$. The graphs H_Y of distinct ancestors are edge-disjoint, since $E(H_Y) \subseteq E_r(Y)$ and these sets are pairwise disjoint (Proposition 2.11(iii)). Inside $\text{Lend}_{Y(u)}$ the edge has exactly one colour. So uw lies in exactly one lent class, namely $\mathcal{L}_{Y(u),l}^{\text{JV}}$. By $\text{COL}(g)$ of Definition 3.10, the only consumer of this class is the round- l J-consumer. The port u carries an end, so it is live, and hence $u \in Q_Z^* = Q_Z \setminus \text{Lost}_Z$. Since Lost_Z contains every classed port whose class is lend-bad, $Y(u)$ is lend-good. Finally $r(u) \leq l-2$. The disjointness claim was shown in (ii).

(iv) (JC1): $\text{Lent}_l^{\text{JV}}$ consists of junction edges on lifted cycles, and by (iii) each lies in $\mathcal{L}_{Y,l}^{\text{JV}}$ for a lend-good ancestor Y of round at most $l-2$. (JC2): the paid single edges are J-edges that lie in no edge of Q_l ; they are distinct from each other and from all edges used by the lifts, which consist of items in edges of Q_l and of junction edges. Together with (ii), the objects of $\mathcal{O}_l$ are pairwise edge-disjoint cycles and single edges. Their union contains every edge of J_l^{lost} (paid in (a1)) and every item: each item is paid or lies in an edge of Q_l , whose member of $\mathcal{D}_l$ lifts to objects containing all items of that edge. The only other edges used are the junction edges on lifted cycles, which form $\text{Lent}_l^{\text{JV}}$ by definition. So the union is exactly $J_l \cup \text{Lent}_l^{\text{JV}}$. (JC3) holds because no other edge is used. The bound $2|\mathcal{D}_l| = 2f(Q_l)$ is (ii) for $\mathcal{D} = \mathcal{D}_l$. Junction edges that were assigned but do not lie on a lifted cycle are not in $\text{Lent}_l^{\text{JV}}$; they stay in $E_r(Y) \setminus \text{Lent}(Y)$ as junk for their owner. $\square$

7.4 Quotient size and payments

In this subsection $3 \leq l \leq R$ and the past $\mathfrak{F}_l^-$ are fixed. We say that *the lists are fixed* when, in addition, the variables η_h and $\zeta_{h,u}$ of ξ_l are fixed. Then steps (a)–(d) and (e1) of Construction 7.5, the sets $E'(u)$ and their orders $e_1, \dots, e_q$ are determined (by the restrictions on the fixed rules), and only the orders $\prec_u$ are random. By step (e2), the junctions of the ends at distinct ports are then independent, and the junction of an end in $E'(u)$ is uniform on $\text{Cand}_l(u) \setminus \text{Used}(u)$, a set of size at least $H_l^{\text{cd}}/2$ (Lemma 7.6(v)). By (J2) of Lemma 6.15, $|J_l| \leq n(M_l - 1)$, hence $|J_l| \leq 1.37n(M_l - 1)$; in this subsection we deliberately use the looser form (it only enlarges constants). Hence the number of PAR objects and the number of hub items are each at most $1.37nM_l$. For a PAR colour κ and $w \neq w'$ in Pool_l , let $m_\kappa(w, w')$ be the number of edges $[w][w']$ of B_κ^{P} . As in Lemma 7.7, $[w]$ is not isolated in exactly $\max_{w'} m_\kappa(w, w')$ of the sub-layers $B_\kappa^{\text{P},(l)}$.

Lemma 7.10 (Lemma CC: PAR copies). *Let the past and the lists be fixed. Fix a PAR colour κ and $w \in \text{Pool}_l$. Condition further on the indicators*

$$I_x := \mathbf{1}[\text{the end at } x \text{ of the } \kappa\text{-object at } x \text{ receives the junction } w]$$

for every port x carrying a PAR object of colour κ . Let S_w be the set of PAR objects of colour κ with exactly one end whose junction is w .

(i) *The partner ports v_o of the objects $o \in S_w$ (the ports of the ends whose junction is not w) are pairwise distinct. Under the conditioning, the junctions of the partner ends are independent, and the junction of the partner end of o is uniform on $(\text{Cand}_l(v_o) \setminus \text{Used}(v_o)) \setminus \{w\}$. In particular it equals a given w' with probability at most $3/H_l^{\text{cd}}$.*

(ii) *For every integer $t \geq 1$,*

$$\mathbb{E}\left[\max_{w'} m_\kappa(w, w') \mid \mathfrak{F}_l^-, \text{lists}, (I_x)_x\right] \leq t \cdot \mathbf{1}[S_w \neq \emptyset] + \frac{6e|S_w|}{H_l^{\text{cd}}} + 4e2^{-t}|S_w|.$$

(iii) *With $t := t_l^{\text{CC}} := \lceil 2 \log_2 M_l \rceil$, the number of vertices of Q_l lying in PAR sub-layers satisfies*

$$\mathbb{E}[\#\{\text{PAR copies}\} \mid \mathfrak{F}_l^-, \text{lists}] \leq 3M_l(t_l^{\text{CC}} + 1)|\text{Pool}_l| + 44.7 \frac{nM_l}{H_l^{\text{cd}}} + 29.8 \frac{n}{M_l}.$$

These bounds hold for every past, in particular for every realized J_l , and for every outcome of the lists, the SDR and the greedy step (e1): only the orders $\prec_u$ are random.

Dependencies: 7.5, 7.6, 1.25, 6.15

Proof. The edges at $[w]$ in B_κ^P are exactly the objects in S_w . Indeed, an object with both ends at junction w is looped and was paid in (e3), and an object with no end at junction w is not incident to $[w]$. So $m_\kappa(w, w')$ is the number of $o \in S_w$ whose partner end has junction w' .

(i) A PAR object has its two ends at distinct ports, and every port carries at most one PAR object of colour κ (Lemma 7.6(i)). So every such port x carries exactly one end of colour κ , the indicator I_x is well defined, and the partner ports of distinct objects of S_w are distinct. The set S_w and the partner ports are determined by the indicators. With the past and the lists fixed, $\text{Cand}_l(x) \setminus \text{Used}(x)$, $E'(x)$ and its order are fixed for every port x (restrictions on the fixed rules in Construction 7.5), so the junction of the κ -end at a port x is a function of $\prec_x$ alone, and the orders of distinct ports are independent. So conditioning on $(I_x)_x$ keeps the partner ends independent, and it conditions the partner end at v_o only on the event $I_{v_o} = 0$. The junction of that end is uniform on $C(v_o) := \text{Cand}_l(v_o) \setminus \text{Used}(v_o)$, since it is the image of a fixed element under a uniform injection. Given $I_{v_o} = 0$ it is therefore uniform on $C(v_o) \setminus \{w\}$. By Lemma 7.6(v), $|C(v_o) \setminus \{w\}| \geq H_l^{\text{cd}}/2 - 1 \geq H_l^{\text{cd}}/3$.

(ii) If $S_w = \emptyset$ the left side is 0. Otherwise, by (i), each $m_\kappa(w, w')$ is, under the conditioning, a sum of independent indicators with mean $\mu_{w'} \leq 3|S_w|/H_l^{\text{cd}}$, and $\sum_{w'} \mu_{w'} = |S_w|$. Put $T := \max(t, \lceil 6e|S_w|/H_l^{\text{cd}} \rceil)$. For every integer $j \geq T$ we have $j \geq 2e\mu_{w'}$. By Cited result 1.25, $\mathbb{P}(m_\kappa(w, w') \geq j) \leq (e\mu_{w'}/j)^j \leq (e\mu_{w'}/j) 2^{1-j}$. Hence, writing m for $m_\kappa(w, w')$,

$$\begin{aligned} \mathbb{E}[m \mathbf{1}[m \geq T]] &= T \mathbb{P}(m \geq T) + \sum_{j>T} \mathbb{P}(m \geq j) \leq e\mu_{w'} 2^{1-T} + \sum_{j>T} \frac{e\mu_{w'}}{j} 2^{1-j} \\ &\leq e\mu_{w'} 2^{2-T} \leq e\mu_{w'} 2^{2-t}. \end{aligned}$$

Moreover

$$\max_{w'} m_\kappa(w, w') \leq (T-1) + \sum_{w'} m_\kappa(w, w') \mathbf{1}[m_\kappa(w, w') \geq T] :$$

if the maximum is below T it is at most $T-1$, and otherwise it is one of the summands. Taking conditional expectations,

$$\mathbb{E}\left[\max_{w'} m_\kappa(w, w') \mid \cdot\right] \leq T-1 + 4e 2^{-t} |S_w| \leq t + \frac{6e|S_w|}{H_l^{\text{cd}}} + 4e 2^{-t} |S_w|,$$

because $T-1 \leq \max(t-1, 6e|S_w|/H_l^{\text{cd}})$.

(iii) The number of PAR copies is $\sum_{\kappa \in [3M_l]} \sum_{w \in \text{Pool}_l} \max_{w'} m_\kappa(w, w')$. Averaging (ii) over the indicators and summing over the at most $3M_l |\text{Pool}_l|$ pairs (κ, w) gives at most $3M_l t |\text{Pool}_l| + (6e/H_l^{\text{cd}} + 4e 2^{-t}) \mathbb{E} \sum_{\kappa, w} |S_w|$. Every PAR object lies in at most two of the sets S_w (one for each end), so $\sum_{\kappa, w} |S_w| \leq 2 \cdot 1.37nM_l = 2.74nM_l$. With $t = t_l^{\text{CC}}$ we have $2^{-t} \leq M_l^{-2}$, and $6e \cdot 2.74 < 44.7$ and $4e \cdot 2.74 < 29.8$. Finally $t \leq t+1$. $\square$

Lemma 7.11 (Ultra-hub copies). *Let the past and the lists be fixed.*

(i) *Let h be an ultra hub, $\kappa \in [4M_l]$, and let $\omega_1, \dots, \omega_N$ be the junctions of the coloured hub items of h of colour κ . These items lie at N distinct ports, because distinct items at h are distinct edges hu and G is simple. The ω_i are independent, and each is uniform on a set of size at least $H_l^{\text{cd}}/2$.*

(ii) *If $N \geq 1$ then, with $\mu^* := 2N/H_l^{\text{cd}}$, $\mathbb{E}[\max_w m_\kappa(h, w) \mid \mathfrak{F}_l^-, \text{lists}] \leq \log_2 N + 2e\mu^* + 8$.*

(iii) *The number of vertices of HUB sub-layers that are copies of h satisfies $\mathbb{E}[\cdot \mid \mathfrak{F}_l^-, \text{lists}] \leq 4M_l(\log_2 c_h^{\text{live}} + 8) + 4e c_h^{\text{live}}/H_l^{\text{cd}}$.*

(iv) Summed over all ultra hubs, the expected number of hub copies of ultra hubs, given $\mathfrak{F}_l^-$ and the lists, is at most

$$320 \frac{nM_l^3 (\log_2 \theta_l^{\text{ult}} + 8)}{\lambda_{l-2}^{95}}.$$

The bound depends on the past only through deterministic counts, so it holds for every past and every realized J_l .

Dependencies: 7.5, 7.6, 6.15, 1.25, 7.7, 7.3

Proof. (i) Items of ultra hubs receive their junctions in (e2). The junction of the item (h, u_i) is the image of its end under the uniformly random injection at u_i , so it is uniform on $\text{Cand}_l(u_i) \setminus \text{Used}(u_i)$, a set of size at least $H_l^{\text{cd}}/2$ by Lemma 7.6(v). The ports u_i are distinct, and injections at distinct ports are independent, so the ω_i are independent. (Distinctness of the ports comes from the simplicity of G ; it does not use (J1).)

(ii) Let $m_w := \#\{i : \omega_i = w\}$. It is a sum of independent indicators with mean $\mu_w \leq 2N/H_l^{\text{cd}} = \mu^*$, and $\sum_w \mu_w = N$. Put $T := \max(\lceil 2e\mu^* \rceil, \lceil \log_2 N \rceil + 1)$. For $j \geq T$ we have $e\mu_w/j \leq 1/2$, so by Cited result 1.25 $\mathbb{P}(m_w \geq j) \leq (e\mu_w/j)^j \leq (e\mu_w/j)2^{1-j}$, and summing over w gives $\sum_w \mathbb{P}(m_w \geq j) \leq (eN/j)2^{1-j}$. Hence

$$\mathbb{E}[\max_w m_w] = \sum_{j \geq 1} \mathbb{P}(\max_w m_w \geq j) \leq (T-1) + \sum_{j \geq T} \frac{eN}{j} 2^{1-j} \leq (T-1) + \frac{eN}{T} 2^{2-T} \leq T-1 + \frac{2e}{T},$$

using $2^{-T} \leq 2^{-\log_2 N-1} = 1/(2N)$. Finally $T-1 \leq \max(2e\mu^*, \log_2 N + 1) \leq \log_2 N + 2e\mu^* + 1$ and $2e/T \leq 2e < 7$.

(iii) By Lemma 7.7, the number of copies of h in the sub-layers of colour κ is $\max_w m_\kappa(h, w)$. Let N_κ be the number of coloured items of h of colour κ . Then $N_\kappa \leq c_h^{\text{live}}$ and $\sum_\kappa N_\kappa \leq c_h^{\text{live}}$, and at most $4M_l$ colours have $N_\kappa \geq 1$. By (ii) the expected number of copies of h is at most $\sum_{\kappa: N_\kappa \geq 1} (\log_2 c_h^{\text{live}} + 8) + 2e \sum_\kappa 2N_\kappa/H_l^{\text{cd}} \leq 4M_l(\log_2 c_h^{\text{live}} + 8) + 4e c_h^{\text{live}}/H_l^{\text{cd}}$.

(iv) The function $g(x) := (\log_2 x + 8)/x$ is decreasing on $[1, \infty)$, since $x^2 g'(x) = 1/\ln 2 - 8 - \log_2 x < 0$ there. For an ultra hub, $c_h^{\text{live}} > \theta_l^{\text{ult}} \geq 1$, so $\log_2 c_h^{\text{live}} + 8 = c_h^{\text{live}} g(c_h^{\text{live}}) \leq c_h^{\text{live}} (\log_2 \theta_l^{\text{ult}} + 8)/\theta_l^{\text{ult}}$. By (J2) of Lemma 6.15, $\sum_h c_h^{\text{live}} \leq |J_l| \leq 1.37nM_l$. We have $M_l H_l^{\text{cd}}/7 = \lambda_{l-2}^{95}/(56M_l)$, and $M_l H_l^{\text{cd}} \geq 2^{10} M_l^{11}$ (Lemma 7.3(iv)), so $\lambda_{l-2}^{95}/M_l \geq 2^{453}$. Hence $\theta_l^{\text{ult}} \geq \lambda_{l-2}^{95}/(56M_l) - 1 \geq \lambda_{l-2}^{95}/(57M_l)$, and also $\theta_l^{\text{ult}} \geq 2^8$. Summing (iii) over ultra hubs,

$$\sum_{h \text{ ultra}} \left(4M_l(\log_2 c_h^{\text{live}} + 8) + \frac{4e c_h^{\text{live}}}{H_l^{\text{cd}}} \right) \leq 4M_l \cdot 1.37nM_l \cdot \frac{57M_l}{\lambda_{l-2}^{95}} (\log_2 \theta_l^{\text{ult}} + 8) + 4e \cdot 1.37nM_l \cdot \frac{8M_l^2}{\lambda_{l-2}^{95}}.$$

The first term is $312.36 nM_l^3 (\log_2 \theta_l^{\text{ult}} + 8)/\lambda_{l-2}^{95}$ and the second is at most $119.2 nM_l^3/\lambda_{l-2}^{95}$. Since $\log_2 \theta_l^{\text{ult}} + 8 \geq 16$, the second term is at most $7.5 nM_l^3 (\log_2 \theta_l^{\text{ult}} + 8)/\lambda_{l-2}^{95}$, and $312.36 + 7.5 < 320$. $\square$

Lemma 7.12 (The stage-1 weight X_V). *For $3 \leq l \leq R$ put*

$$X_{V,l} := 3M_l(t_l^{\text{CC}} + 1)|\text{Pool}_l| + 4M_l \sum_{w \in \text{Pool}_l} \text{mult}_{r(w)}(w), \quad X_V := \sum_{l=3}^R X_{V,l}.$$

Each $X_{V,l}$ is a deterministic function of the run and the pool labels, and

$$\mathbb{E}X_{V,l} \leq \frac{(6 \log_2 M_l + 12) n}{M_l}, \quad \mathbb{E}X_V \leq 2.1 \frac{(6 \log_2 D_* + 12) n}{D_*}.$$

Dependencies: 7.1, 2.13, 2.17

Proof. The quantities M_l , t_l^{CC} and $\text{mult}_r(w)$ are determined by the run, and Pool_l and $r(w)$ by the pool labels. Since $t_l^{\text{CC}} + 1 \leq 2 \log_2 M_l + 2$ and $\mathbb{E}|\text{Pool}_l| \leq q_l n = n/M_l^2$, the first summand has expectation at most $(6 \log_2 M_l + 6)n/M_l$. For the second,

$$\mathbb{E} \sum_{w \in \text{Pool}_l} \text{mult}_{r(w)}(w) = \sum_w \sum_{r=1}^{l-2} q_l \pi_{l,r} \text{mult}_r(w) = q_l \sum_{r=1}^{l-2} \pi_{l,r} \sum_w \text{mult}_r(w) \leq q_l \cdot 1.37n,$$

since $\sum_w \text{mult}_r(w) \leq 1.37n$ for every r (Proposition 2.13) and $\sum_r \pi_{l,r} < 1$. So the second summand has expectation at most $4M_l \cdot 1.37n/M_l^2 = 5.48n/M_l$. Adding, $\mathbb{E}X_{V,l} \leq (6 \log_2 M_l + 11.48)n/M_l$. Summing over l and using $\sum_l \psi(M_l) \leq 2.1 \psi(D_*)$ for $\psi(x) = (6 \log_2 x + 12)/x$ (Lemma 2.17(b)) gives the bound for $\mathbb{E}X_V$. $\square$

Lemma 7.13 (Payments). *For every past and every $3 \leq l \leq R$ the following hold.*

- (a1) $|J_l^{\text{lost}}| \leq (M_l - 1)|\text{Lost}_l|$.
- (a2) *The number of items with a vertex in Pool_l is at most $\sum_{v \in \text{Pool}_l} \omega_l(v)$, where $\omega_l(v) := c_{v,l}^{\text{agg}}$ if $v \in D_l$ and $\omega_l(v) := M_l$ otherwise.*
- (a3) *The number of items with a JV-bad port end is at most $M_l - 1$ times the number of JV-bad classed ports of round l .*
- (b) *Every fresh centre has at most one unpaired fresh leg per round, so the number of unpaired fresh legs over all rounds is at most $\sum_{l,Z} |F_Z| \leq 2n$.*
- (c) $\mathbb{E}[\text{SDR-failure payments} \mid \mathfrak{F}_l^-] \leq n(M_l - 1)(4M_l)^{-4} \leq n/(256M_l^3)$, and $\mathbb{E}[\text{loop-paid edges} \mid \mathfrak{F}_l^-] \leq 2 \cdot (2/H_l^{\text{cd}}) \cdot \#\{\text{PAR objects}\} \leq 5.5nM_l/H_l^{\text{cd}}$.
- (d) *If ξ_l lies in the event of Construction 7.5(f), then $\text{pay}_l^{\text{rd}} \leq 4(n/(256M_l^3) + 5.5nM_l/H_l^{\text{cd}}) \leq 2n/M_l$; if this holds at every round, then $\sum_l \text{pay}_l^{\text{rd}} \leq 9n/D_*$.*
- (e) *No other item is paid. In particular, no item is paid because of a coincidence of junctions at an ultra hub, and no item is paid for a concentrated pair.*

Dependencies: 7.5, 7.6, 6.15, 2.20, 2.17

Proof. We use the properties of Lemma 6.15.

(a1) Every J^{lost} edge joins a vertex of Lost_l to a port, and every vertex carries at most $M_l - 1$ J-edges by (J2).

(a2) Let $v \in \text{Pool}_l$. If $v \in D_l$, then v is a hub and not a port or a fresh centre, by (J3). The items with vertex v are therefore J^{hub} items (v, u) : J^{par} items have both ends in Q_Z^* , and J^{fr} items have their centre in F_Z . By (J1) there is at most one such item per class Y . Each such class has $d_{Y,l}(v) \geq 1$, because the item is an edge $vu \in E_l(Z)$ with $u \in Q_Z$ and $Y(u) = Y$. So there are at most $c_{v,l}^{\text{agg}}$ items with vertex v . If $v \notin D_l$, then v carries at most $M_l - 1 \leq M_l$ J-edges by (J2).

(a3) A port carries at most $M_l - 1$ items by (J2).

(b) The pairing at a fresh centre x leaves at most one live J^{fr} item unpaired. Every fresh centre of round l lies in F_Z for some $Z \in \text{Std}_l$, and $\sum_{l,Z} |F_Z| \leq 2n$ by Proposition 2.20(i).

(c) By (J3), every non-hub vertex has one role in one part per round, so there are at most n ports. A port u pays at most $M_l - 1$ items for an SDR failure, and it fails with probability at most $(4M_l)^{-4}$ given $\mathfrak{F}_l^-$ (Lemma 7.6(iii)). This gives the first bound. For loops, fix the lists. A PAR object has its two ends at distinct ports $p \neq q$. Their junctions are independent, and the junction

at q is uniform on a set of size at least $H_l^{\text{cd}}/2$ (Lemma 7.6(v)). So the probability that the two junctions coincide is $\sum_w \mathbb{P}(p \rightarrow w) \mathbb{P}(q \rightarrow w) \leq \max_w \mathbb{P}(q \rightarrow w) \leq 2/H_l^{\text{cd}}$. A looped object costs at most two edges (a cherry costs two, a J^{par} item one), and there are at most $1.37nM_l$ PAR objects ((J2)). So the expected number of loop-paid edges is at most $4 \cdot 1.37nM_l/H_l^{\text{cd}} \leq 5.5nM_l/H_l^{\text{cd}}$. This bound does not depend on the lists, so it also holds given $\mathfrak{F}_l^-$ alone.

(d) The first inequality is the definition of the event in (f), together with (c). Next, $4 \cdot 5.5nM_l/H_l^{\text{cd}} = 176nM_l^3/\lambda_{l-2}^{95}$. By Lemma 2.17(b), $M_l \leq \lambda_{l-2}^{1.6}$, so $\lambda_{l-2} \geq M_l^{1/1.6} \geq 2^{25}$ and $176M_l^3/\lambda_{l-2}^{95} \leq 176\lambda_{l-2}^{-90.2} \leq \lambda_{l-2}^{-1.6} \leq 1/M_l$. Also $4n/(256M_l^3) \leq n/M_l$. Hence $\text{pay}_l^{\text{rd}} \leq 2n/M_l$, and $\sum_l \text{pay}_l^{\text{rd}} \leq 2n \sum_l 1/M_l \leq 4n/D_* \leq 9n/D_*$ by Lemma 2.17(b).

(e) In Construction 7.5, single edges are paid only in (a1), (a2), (a3), (b), (d) and (e3). $\square$

Definition 7.14 (The stage-1 functional X'). With ω_l as in Lemma 7.13(a2), put

$$X_{\text{pool}} := \sum_{l=3}^R \left[\sum_{v \in \text{Pool}_l} \omega_l(v) + (M_l - 1) \cdot \#\{\text{JV-bad classed ports of round } l\} \right],$$

let X_V be as in Lemma 7.12, and let X_U be as in Definition 6.11. Then

$$X' := X_U + X_{\text{pool}} + X_V.$$

Each of the three terms is non-negative and is a deterministic function of the run, δ and the stage-1 outcome. Indeed, $\omega_l(v)$ is determined by the run and δ (through $c_{v,l}^{\text{agg}}$), Pool_l is determined by the pool labels, and JV-badness is determined by stage 1. X' is the functional used in stage (1e) of the schedule.

Dependencies: 6.11, 7.12, 7.13

Lemma 7.15 (Expectation of X'). $\mathbb{E}X' \leq \varepsilon_X(D_*)n$, where

$$\varepsilon_X(D_*) := \varepsilon_U(D_*) + \frac{10.3}{D_*} + 2.1 \frac{6 \log_2 D_* + 12}{D_*}$$

and ε_U is the function of Lemma 5.5 (through Lemma 6.12). More precisely, $\mathbb{E}X_U \leq \varepsilon_U(D_*)n + 5.5n/D_*$, $\mathbb{E}X_{\text{pool}} \leq 4.8n/D_*$ and $\mathbb{E}X_V \leq 2.1(6 \log_2 D_* + 12)n/D_*$.

Dependencies: 7.14, 6.12, 7.3, 7.12, 2.13, 2.11, 2.17, 5.5, 2.9

Proof. The bound on $\mathbb{E}X_U$ is Lemma 6.12, and the bound on $\mathbb{E}X_V$ is Lemma 7.12. For X_{pool} , fix a round l .

Pooled weights. The weights $\omega_l(v)$ are determined by the run and δ , so they are independent of the pool labels. Hence $\mathbb{E} \sum_{v \in \text{Pool}_l} \omega_l(v) \leq q_l \sum_v \omega_l(v) \leq q_l (\sum_{h \in D_l} c_{h,l}^{\text{agg}} + nM_l)$. Now $c_{h,l}^{\text{agg}} \leq \sum_Y d_{Y,l}(h)$, and $\sum_Y d_{Y,l}(h)$ counts the edges $hu \in E_l(Z)$ with $Z \in \text{Std}_l$ and $u \in Q_Z$. Summing over h counts each such edge at most once for each of its classed-port ends. So $\sum_h c_{h,l}^{\text{agg}} \leq \sum_{Z \in \text{Std}_l} \sum_{u \in Q_Z} \deg_{E_l(Z)}(u)$. Every vertex has $E_l(Z)$ -degree at most $M_l - 1$ (Lemma 2.9(ii)), and $\sum_{Z \in \text{Std}_l} |Z^0| \leq 1.37n$ by Proposition 2.13. Hence $\sum_{h \in D_l} c_{h,l}^{\text{agg}} \leq 1.37nM_l$, and the expected pooled weight is at most $2.37n/M_l$.

JV-bad ports. There are at most n classed ports of round l , because $Q_Z \subseteq U_Z$ and the port sets U_Z of distinct $Z \in \text{Std}_l$ are pairwise disjoint (Proposition 2.11(iv)); this is the count used in the proofs of (K6) of Lemma 6.12 and of (J2) of Lemma 6.15. By Lemma 7.3(iii) each is JV-bad with probability at most $\exp(-H_l^{\text{cd}}/4)$, and $\exp(-H_l^{\text{cd}}/4) \leq M_l^{-3}$ because $H_l^{\text{cd}} \geq 2^{10}M_l^{10}$ (Lemma 7.3(iv)). So the expected JV-bad weight is at most

$$(M_l - 1) \cdot n \cdot M_l^{-3} \leq n/M_l^2 \leq 0.03n/M_l.$$

The pooled-weight bound above counts the same classed ports by the deliberately looser $\sum_{Z \in \text{Std}_l} |Z^0| \leq 1.37n$; this only enlarges its constant.

Hence $\mathbb{E}X_{\text{pool}} \leq \sum_l 2.4n/M_l \leq 4.8n/D_*$, using $\sum_l 1/M_l \leq 2/D_*$ (Lemma 2.17(b)). Adding the three bounds gives $\mathbb{E}X' \leq \varepsilon_U(D_*)n + (5.5 + 4.8)n/D_* + 2.1(6\log_2 D_* + 12)n/D_* = \varepsilon_X(D_*)n$. $\square$

Lemma 7.16 (Lemma UH*-split: quotient size). *For $3 \leq l \leq R$ put*

$$\det_l := 3|D_l| + 78 \frac{nM_l}{H_l^{\text{cd}}} + 30 \frac{n}{M_l} + 320 \frac{nM_l^3(\log_2 \theta_l^{\text{ult}} + 8)}{\lambda_{l-2}^{95}},$$

a deterministic function of the run.

- (i) $X_{V,l}$ is the stage-1 weight of Lemma 7.12.
- (ii) For every past, $\mathbb{E}[\text{copies}_l \mid \mathfrak{F}_l^-] \leq X_{V,l} + \det_l$.
- (iii) If the stage-1 outcome satisfies $X' \leq 3\mathbb{E}X'$ and at every round ξ_l lies in the event of Construction 7.5(f), then

$$\sum_{l=3}^R |V(Q_l)| \leq 12\mathbb{E}X' + 4 \sum_{l=3}^R \det_l.$$

- (iv) Put $F(x) := (3184 + 30400\log_2 x)x^{-90.2}$ for $x \geq 1$. Then $\sum_l \det_l \leq 3\varepsilon_A n + 60n/D_* + 2F(\log_2 D_*)n$, and $2F(\log_2 D_*) < 2^{-1000}$. Hence, in the situation of (iii),

$$\sum_{l=3}^R |V(Q_l)| \leq \theta_Q n, \quad \theta_Q := \varepsilon_2(D_*) := 12\varepsilon_X(D_*) + 4\left(3\varepsilon_A + \frac{60}{D_*} + 2F(\log_2 D_*)\right),$$

with $\varepsilon_A = 31\varepsilon/(C' \log \log D_*)$ as in Lemma 2.17(e). So θ_Q depends on D_* only, and $\theta_Q \leq 1/4$ by condition (Γ4).

Dependencies: 7.10, 7.7, 7.11, 7.12, 7.15, 7.4, 2.13, 2.17, 2.16, (Γ4), (Γ1), 7.5, 7.6, 2.7, 6.15

Proof. (ii) Fix the past and the lists. The vertices of Q_l are of four kinds.

PAR copies. Their conditional expectation is at most $3M_l(t_l^{\text{CC}} + 1)|\text{Pool}_l| + 44.7nM_l/H_l^{\text{cd}} + 29.8n/M_l$ by Lemma 7.10(iii).

Copies of non-ultra hubs (deterministic given the lists). Let h be non-ultra. By Lemma 7.6(iv) (see step (e1) of Construction 7.5), $m_\kappa(h, w) \leq 1$ for all κ and w . So by Lemma 7.7, h has exactly one copy in each colour that one of its items uses, and none in the other colours. The colours of its items lie in the union of its k_h lists, which has $3k_h$ elements. So the number of copies of non-ultra hubs is at most $\sum_{h: c_h^{\text{live}} \geq 1} 3k_h \leq \sum_{h \in D_l} 3(1 + 8c_h^{\text{live}}/H_l^{\text{cd}}) \leq 3|D_l| + 24 \cdot 1.37nM_l/H_l^{\text{cd}}$, since $\sum_h c_h^{\text{live}} \leq 1.37nM_l$. Here $24 \cdot 1.37 = 32.88 < 32.9$.

Junction copies in HUB sub-layers (the count depends on the junctions, but the bound below is deterministic). By Lemma 7.7, in colour κ the vertex $[w]$ has $\max_h m_\kappa(h, w) \leq \text{mult}_{r(w)}(w)$ copies. Summing over the $4M_l$ HUB colours and $w \in \text{Pool}_l$ gives at most $4M_l \sum_{w \in \text{Pool}_l} \text{mult}_{r(w)}(w)$.

Copies of ultra hubs. Their conditional expectation is at most $320nM_l^3(\log_2 \theta_l^{\text{ult}} + 8)/\lambda_{l-2}^{95}$ by Lemma 7.11(iv).

The first summands of the PAR bound and of the junction bound add up to $X_{V,l}$. The remaining terms add up to at most $\det_l$, because $44.7 + 32.9 \leq 78$ and $29.8 \leq 30$. Averaging over the lists gives (ii).

(iii) By the choice in (f) and by (ii), $|V(Q_l)| = \text{copies}_l \leq 4 \mathbb{E}[\text{copies}_l \mid \mathfrak{F}_l^-] \leq 4(X_{V,l} + \det_l)$ at every round. Summing over l , $\sum_l |V(Q_l)| \leq 4X_V + 4 \sum_l \det_l \leq 4X' + 4 \sum_l \det_l \leq 12 \mathbb{E}X' + 4 \sum_l \det_l$, because $X_U, X_{\text{pool}} \geq 0$ and $X' \leq 3\mathbb{E}X'$ (stage (1e) of Definition 7.4).

(iv) By Proposition 2.13(K2), $|D_l| \leq 7.6\epsilon n / \log P_l$. So $\sum_l 3|D_l| \leq 3n \sum_l 7.6\epsilon / \log P_l \leq 1.5\epsilon_A n \leq 3\epsilon_A n$ by Lemma 2.17(e) (the factor 2 of slack is deliberate; it only enlarges ϵ_2). Next, $\sum_l 30n/M_l \leq 60n/D_*$ by Lemma 2.17(b). For the remaining terms, $78nM_l/H_l^{\text{cd}} = 624nM_l^3/\lambda_{l-2}^{95}$, and $\log_2 \theta_l^{\text{ult}} \leq \log_2(M_l H_l^{\text{cd}}/7) \leq 95 \log_2 \lambda_{l-2}$. With $M_l \leq \lambda_{l-2}^{1.6}$ (Lemma 2.17(b)), these terms are at most $nF(\lambda_{l-2})$, with F as in (iv). Here $\lambda_{l-2} \geq M_l^{1/1.6} \geq 2^{25}$. On $[2^{25}, \infty)$, F is decreasing, and $F(2x) \leq F(x)/2$, since the first factor at most doubles while $x^{-90.2}$ is multiplied by $2^{-90.2}$. By Lemma 2.16(ii), $\lambda_r \geq 2^{\lambda_{r+1}/A} \geq 2\lambda_{r+1}$ for consecutive rounds, because $\lambda_{r+1} \geq 2^{25}$. So $X_l := F(\lambda_{l-2})$ ($3 \leq l \leq R$) satisfies $X_l \leq X_{l+1}/2$. By Lemma 2.16(i), $\sum_l X_l \leq 2X_R = 2F(\lambda_{R-2})$ (if $R \leq 2$ the sum is empty). Moreover $\lambda_{R-2} \geq \lambda_R = \log_2 d_R \geq \log_2 D_*$, by Lemma 2.17(a) and because round R has $d_R \geq D_*$ by rule (R0) of Definition 2.7; and $\log_2 D_* \geq 2^{256} \geq 2^{25}$ by (Γ1)(a) at $\mu = \log_2 \log_2 D_*$. Since F is decreasing on $[2^{25}, \infty)$, we get

$$\sum_l X_l \leq 2F(\log_2 D_*) \leq 2F(2^{25}) \leq 2 \cdot 2^{20} \cdot 2^{-2255} < 2^{-1000}.$$

Together, $\sum_l \det_l \leq 3\epsilon_A n + 60n/D_* + 2F(\log_2 D_*)n$. The bound on $\sum_l |V(Q_l)|$ follows from (iii) and $\mathbb{E}X' \leq \epsilon_X(D_*)n$ (Lemma 7.15). The functions ϵ_X , ϵ_A and $F(\log_2 D_*)$ depend on D_* only, and $\theta_Q \leq 1/4$ is part of condition (Γ4). $\square$

7.5 One joint outcome, the cost, and Theorem JV⁺*

Lemma 7.17 (One joint outcome exists). *There is a joint outcome, consisting of a stage-1 outcome, stage-3 outcomes for all parts, and draws $\xi_R, \dots, \xi_3$, such that:*

- (1) $X' \leq 3\mathbb{E}X'$;
- (2) *the stage-3 labels of every part lie in that part's good event: the event of Lemma 4.2 for every standalone pre-part (with the data of step (1) of Construction 6.17); the event of Lemma 4.3 as used in Lemma 5.6 for every non-demoted light part; and the event of Theorem 4.4 as used in Lemma 5.8 for every demoted light part;*
- (3) *at every round $3 \leq l \leq R$, ξ_l lies in the event of Construction 7.5(f). Consequently $\text{copies}_l \leq 4(X_{V,l} + \det_l)$ and $\text{pay}_l^{\text{rd}} \leq 4(n/(256M_l^3) + 5.5nM_l/H_l^{\text{cd}})$.*

Such an outcome is obtained as follows. (i) Choose the stage-1 outcome in $\{X' \leq 3\mathbb{E}X'\}$; stage 2 is then determined. (ii) For every part, choose its stage-3 labels in its good event. (iii) For $l = R, R-1, \dots, 3$ in turn: run steps (1)–(3) of Construction 6.17 at round l , which are deterministic given the past; draw ξ_l and choose it in the event of Construction 7.5(f); build Q_l , fix $\mathcal{D}_l$, lift it, and record $\text{Lent}_l^{\text{JV}}$ (step (4) of Construction 6.17). (iv) At rounds 2 and 1, run steps (1)–(2) of Construction 6.17; here $J_2 = J_1 = \emptyset$.

Dependencies: 7.4, 7.5, 6.17, 4.2, 4.3, 4.4, 5.6, 5.8, 7.15, 7.13, 7.16, 1.26, (Γ3)

Proof. Step (i). X' is a non-negative random variable on a finite probability space. By Markov's inequality (Cited result 1.26), $\mathbb{P}(X' \leq 3\mathbb{E}X') \geq 2/3 > 0$, so the event is non-empty. Fix an outcome in it; this gives (1). Stage 2 is a deterministic function of stage 1 (Definition 7.4).

Step (ii). With stage 1 fixed, the good event of each part is determined by stage-1 and stage-2 data together with that part's own stage-3 labels.

- For a standalone pre-part Z , Lemma 4.2 is applied with O_Z and $P = \text{Ret}_Z$, both fixed at stage 2, and its good event is determined by (Z^0, O_Z, Ret_Z) and the labels.
- For a non-demoted light part, the good event of Lemma 4.3 depends only on the own classes, Rt , φ and the labels (Lemma 5.6).
- For a demoted light part, the good event of Theorem 4.4 depends only on $O = X_Y$ and the labels (Lemma 5.8).

Each of these events has probability at least $1/2 - o(1)$, and by condition (Γ3) every such $o(1)$ is at most $1/100$. So each event is non-empty, whatever the stage-1 outcome; only this non-emptiness is used. The events of distinct parts involve disjoint families of labels, so choosing each part's labels inside its own event gives one stage-3 outcome in all of the events at once; no union bound is needed. None of these events depends on the edge sets that the devices later receive: the conclusions of Lemma 4.2, Lemma 4.3 and Theorem 4.4 hold on their good events for *every* admissible edge set (every $E \supseteq E(O)$, every H_0 containing the own classes, every graph containing O). So the events do not depend on junk, on E^{rem} , on H_0 or on later rounds. This gives (2).

Step (iii). At round l the past $\mathfrak{F}_l^-$ is fixed, and steps (1)–(3) of Construction 6.17 are deterministic given it; in particular J_l is determined. The draw ξ_l is independent of the past, and copies_l and pay_l^{rd} are non-negative functions of ξ_l with the past fixed. By Markov's inequality (Cited result 1.26), the event of Construction 7.5(f) has probability at least $1/2$. This holds for every past and needs nothing beyond Markov's inequality. So a choice of ξ_l in the event exists, whatever happened before. On this event, the explicit bounds of (3) follow from Lemma 7.16(ii) and Lemma 7.13(c), which hold for *every* past; this is the only place where the uniformity in the past is used.

Step (iv). Rounds 1 and 2 have no classed ports, since every port there is fresh. So JS-LC produces no J-set, and there is no round step. $\square$

Proposition 7.18 (Cost on the chosen outcome). *On an outcome as in Lemma 7.17, with the round step of Construction 7.5 as the J-consumer, the decomposition of $E(G)$ given by Theorem 6.19 has at most*

$$\left(\frac{D_*}{2} + 1085 + \varepsilon_1(D_*)\right)n + 2 \sum_{l=3}^R f(Q_l)$$

objects, where

$$\varepsilon_1(D_*) := \varepsilon_K(D_*) + 169\varepsilon_A + \frac{252}{D_*} + 1.5\varepsilon_{\text{CONC}}(D_*) + 3\varepsilon_X(D_*) + \frac{9}{D_*},$$

with $\varepsilon_A = 31\varepsilon/(C' \log \log D_)$ as in Lemma 2.17(e). The terms are itemized in Table 7. The constant is $745 + 338 + 2 = 1085$, and ε_1 is counted once. Consequently $C_0 = D_*/2 + 1085 + \varepsilon_1(D_*) \leq D_*/2 + 1091$ under (Γ4). No term involves c_{EG} . The only Markov multipliers are the fixed 3 (stage 1) and 4 (per round). There is no term for VX-parts ($\mathcal{V} = \emptyset$).*

Dependencies: 6.19, 6.10, 7.9, 7.13, 7.15, 7.17, 5.9, 2.20, 2.13, 7.5, 2.17, (Γ4), 6.11, 7.14

Proof. By Lemma 7.9(iv), the round step is a J-consumer. The outcome of Lemma 7.17 has stage-3 labels in all good events. So Theorem 6.19 applies to the run, δ , the chosen stage-1 and stage-3

Item	Bound on the number of objects	Source
Long cycles, E_0 and light parts (K-RED)	$(D_*/2 + 745 + \varepsilon_K)n + [80 \text{ dem} + 369 \text{ lp}]$	Lemma 5.9
TPV in standalone pre-parts	$169(2 + \varepsilon_A)n + [169 \sum_l \text{Lost}_l]$	Thm. 6.19(c); Props. 2.20, 2.13
JS-LC cycles	$252 n/D_* + 1.5 \varepsilon_{\text{CONC}}(D_*) n$	Thms. 6.19(c), 6.10
J^{lost} items	$[\sum_l (M_l - 1) \text{Lost}_l]$	Lemma 7.13(a1)
Pooled and JV-bad items	$[X_{\text{pool}}]$	Lemma 7.13(a2),(a3)
Unpaired fresh legs	$2n$	Lemma 7.13(b)
SDR failures and loops	$9n/D_*$	Lemma 7.13(d)
Lift of the quotients	$2 \sum_l \mathcal{D}_l = 2 \sum_l f(Q_l)$	Lemma 7.9
All bracketed terms together	$\leq X_U + X_{\text{pool}} \leq X' \leq 3 \mathbb{E}X' \leq 3 \varepsilon_X(D_*) n$	Lemmas 7.15, 7.17

Table 7: The cost line on the chosen outcome (Proposition 7.18). Summing the rows gives $(D_*/2 + 745 + 338 + 2)n + \varepsilon_1(D_*)n + 2 \sum_l f(Q_l)$.

outcomes and this J-consumer, and it yields a decomposition of $E(G)$. By Theorem 6.19(c), its number of objects is at most

$$\left(\frac{D_*}{2} + 745 + 338\right)n + \varepsilon_M(D_*)n + \left[80 \text{ dem} + 369 \text{ lp} + 169 \sum_l |\text{Lost}_l|\right] + 1.5 \sum_{(Y,l)} m_{Y,l} + \sum_l |\mathcal{O}_l|,$$

with $\varepsilon_M(D_*) = \varepsilon_K(D_*) + 169\varepsilon_A + 252/D_*$. This itemization combines the K-RED bound of Lemma 5.9; the TPV bound

$$169 \sum_{l,Z} (|A_Z| + |F_Z| + |\text{Lost}_Z|),$$

with $\sum |F_Z| \leq 2n$ (Proposition 2.20(i)) and $\sum |A_Z| \leq \varepsilon_A n$ (Proposition 2.13(K2) and Lemma 2.17(e)); and the JS-LC bound. By Theorem 6.10(iv), $1.5 \sum_{(Y,l)} m_{Y,l} \leq 1.5 \varepsilon_{\text{CONC}}(D_*) n$.

The objects of $\mathcal{O}_l$ are the paid single edges and the lifted objects. By Lemma 7.13, the paid single edges number at most: $(M_l - 1)|\text{Lost}_l|$ in (a1); the round- l summand of X_{pool} in (a2) and (a3); at most $2n$ unpaired fresh legs over all rounds; and pay_l^{rd} SDR-failure and loop edges, with $\sum_l \text{pay}_l^{\text{rd}} \leq 9n/D_*$ by Lemma 7.13(d), since the outcome satisfies Lemma 7.17(3). By Lemma 7.9(iv), the lifted objects number at most $2f(Q_l)$. Hence

$$\sum_l |\mathcal{O}_l| \leq \sum_l (M_l - 1)|\text{Lost}_l| + X_{\text{pool}} + 2n + \frac{9n}{D_*} + 2 \sum_l f(Q_l).$$

The bracketed terms and the terms $\sum_l (M_l - 1)|\text{Lost}_l|$ and X_{pool} together equal $80 \text{ dem} + 369 \text{ lp} + \sum_l (169 + M_l - 1)|\text{Lost}_l| + X_{\text{pool}} \leq X_U + X_{\text{pool}} \leq X'$. By Lemma 7.17(1) and Lemma 7.15, $X' \leq 3 \mathbb{E}X' \leq 3 \varepsilon_X(D_*)n$. Adding everything gives at most

$$\left(\frac{D_*}{2} + 745 + 338 + 2\right)n + \left(\varepsilon_K + 169\varepsilon_A + \frac{252}{D_*} + 1.5\varepsilon_{\text{CONC}} + 3\varepsilon_X + \frac{9}{D_*}\right)n + 2 \sum_l f(Q_l),$$

which is the claim, since $745 + 338 + 2 = 1085$. Every term of ε_1 is a function of D_* alone, and $\varepsilon_1(D_*) \leq 6$ is part of (Γ4), which gives $C_0 \leq D_*/2 + 1091$. $\square$

Theorem 7.19 (Theorem JV⁺). *Assume that D_* satisfies (Γ1)–(Γ4). Let G be a graph with $n \geq N_0$ vertices and $d_1 \geq D_*$. Take any valid HB^{*+} run on G (Proposition 2.15), any designation δ , and no VX-parts ($\mathcal{V} = \emptyset$). Then there are simple graphs $Q_3, \dots, Q_R$ without isolated vertices such that*

(1) $f(G) \leq C_0 n + 2 \sum_{l=3}^R f(Q_l)$, where $C_0 := D_*/2 + 1085 + \varepsilon_1(D_*)$;

(2) $\sum_{l=3}^R |V(Q_l)| \leq \theta_Q n$, where $\theta_Q = \varepsilon_2(D_*) \leq 1/4$.

Here ε_1 and ε_2 are the explicit functions of D_* defined in Proposition 7.18 and Lemma 7.16. Items at ultra hubs cost no collision payments. They pay only the standard costs (pooled vertices, JV-bad ports, SDR failures), and they add quotient vertices only through Lemma 7.16.

Dependencies: 7.18, 7.16, 7.8, 7.9, 7.17, 6.19, 2.15, 7.13, 7.11, 1.33

Proof. Take the outcome of Lemma 7.17 and let $Q_3, \dots, Q_R$ be the quotients built on it. By Lemma 7.8 each Q_l is simple and has no isolated vertices. By Lemma 7.9(iv), the round step is a J-consumer, so Theorem 6.19 gives a decomposition of $E(G)$. By Proposition 7.18 it has at most $C_0 n + 2 \sum_l f(Q_l)$ objects, which proves (1), since $f(G)$ is at most the number of objects of any decomposition. The outcome satisfies the hypotheses of Lemma 7.16(iii): $X' \leq 3EX'$, and the event of (f) holds at every round. So Lemma 7.16(iv) gives (2). The statement about ultra hubs is Lemma 7.13(e) together with Lemma 7.11. $\square$

7.6 The layered quotient induction and the proof of the Main Theorem

Theorem 7.20 (Theorem HI'' (layered quotient induction)). *Let $C \geq D_*/2$ and $\vartheta \in [0, 1/2)$ be constants with the following property. Every graph G without isolated vertices, with $n \geq N_0$ vertices and $d_1 = 2|E(G)|/n \geq D_*$, admits finitely many simple graphs $Q_1, \dots, Q_k$ ($k \geq 0$) such that*

$$f(G) \leq Cn + 2 \sum_{i=1}^k f(Q_i) \quad \text{and} \quad \sum_{i=1}^k |V(Q_i)| \leq \vartheta n.$$

Then $f(G) \leq c|V(G)|$ for every graph G , where $c := \max(C/(1 - 2\vartheta), N_0/2)$.

Dependencies: 1.5, 1.4, 1.31

Proof. Suppose the conclusion fails, and let G be a counterexample with $n := |V(G)|$ minimal, so that $f(G) > cn$. The graph without vertices has $f = 0$, so $n \geq 1$.

(1) G has no isolated vertex. If v were isolated, then $f(G - v) = f(G)$ by Fact 1.5(d), so $f(G - v) > cn \geq c(n - 1) = c|V(G - v)|$, and $G - v$ would be a smaller counterexample.

(2) $n \geq N_0$. Otherwise $n - 1 < N_0$, and by Fact 1.5(a), $f(G) \leq |E(G)| \leq n(n - 1)/2 < (N_0/2)n \leq cn$.

(3) $d_1 \geq D_*$. Otherwise, by Fact 1.5(a), $f(G) \leq |E(G)| = d_1 n/2 < (D_*/2)n \leq Cn \leq cn$, because $c \geq C/(1 - 2\vartheta) \geq C$. This is the trivial bound $f(G) \leq |E(G)|$; no other estimate is used for graphs of small average degree.

(4) By (1)–(3), the hypothesis applies to G and gives simple graphs $Q_1, \dots, Q_k$ with $f(G) \leq Cn + 2 \sum_i f(Q_i)$ and $\sum_i |V(Q_i)| \leq \vartheta n < n$. Each Q_i has fewer vertices than G , so by the minimality of n , $f(Q_i) \leq c|V(Q_i)|$.

(5) Hence $f(G) \leq Cn + 2c \sum_i |V(Q_i)| \leq Cn + 2c\vartheta n \leq c(1 - 2\vartheta)n + 2c\vartheta n = cn$, where we used $C \leq c(1 - 2\vartheta)$. This contradicts $f(G) > cn$.

The constants C and ϑ do not depend on c , and minimality is applied only to the graphs Q_i , which have strictly fewer vertices than G . $\square$

Lemma 7.21 (The galactic conditions are satisfiable). *Let N_0 be as in Definition 1.31(ii). Then:*

(i) each item (a)–(f) of $(\Gamma 1)$ (Condition [\(F1\)](#)) holds for all sufficiently large real μ ; hence there is $\mu_1 \geq 1$ such that all of them hold for every $\mu \geq \mu_1$, and $(\Gamma 1)$ holds for every $D_* > 2$ with $\log_2 \log_2 D_* \geq \mu_1$;

(ii) $\varepsilon_1(D_*) \rightarrow 0$ and $\varepsilon_2(D_*) \rightarrow 0$ as $D_* \rightarrow \infty$;

(iii) there is D_0 such that every $D_* \geq D_0$ satisfies $(\Gamma 1)$ – $(\Gamma 4)$ simultaneously.

In particular a constant D_* as in Definition [1.31](#)(iii) exists. Only this existence is used: neither μ_1 nor D_0 is computed.

Dependencies: [1.33](#), [7.18](#), [7.16](#), [5.5](#), [5.9](#), [2.17](#), [7.15](#), [6.10](#), [1.31](#), [3.13](#), [5.4](#), [3.12](#)

Proof. (i) We use inequalities of type (E) as defined in Lemma [3.12](#), which has no hypothesis: by part (i) of that lemma, every inequality of type (E) holds for all sufficiently large μ . For $\mu \geq 1$ we have $0 \leq \log_2 \mu \leq \log_2(A\mu)$, and: item (a) holds for every $\mu \geq 2^8$; item (b), that is $\mu - 14 - \log_2 A - 3 \log_2 \mu \geq 0$, follows from the inequality $\mu - (14 + \log_2 A) - 3 \log_2(A\mu) \geq 0$ of type (E); item (c) is the inequality $1.6\mu - 2A \log_2(A\mu) \geq 0$ of type (E); item (d), that is $\mu - 8 - 2 \log_2 \mu \geq 0$, follows from the inequality $\mu - 8 - 2 \log_2(A\mu) \geq 0$ of type (E); item (e), which is (G^*) , is (after taking $\log_2$ of both sides) the inequality $36\mu - 240 - 46A \log_2(A\mu) \geq 0$ of type (E); and every inequality of item (f) holds for all sufficiently large μ by Lemma [3.12](#)(iii). So each of the finitely many inequalities of $(\Gamma 1)$ holds for all μ at least some threshold; let $\mu_1 \geq 1$ be at least all of these thresholds. If $D_* > 2$ and $\log_2 \log_2 D_* \geq \mu_1$, every $\mu \geq \log_2 \log_2 D_*$ satisfies $\mu \geq \mu_1$, so $(\Gamma 1)$ holds.

(ii) By Proposition [7.18](#) and Lemmas [7.15](#) and [7.16](#), ε_1 and ε_2 are finite sums of non-negative explicit functions of D_* with fixed coefficients: $1/D_*$; $(6 \log_2 D_* + 12)/D_*$; $\varepsilon_A = 31\varepsilon/(C' \log \log D_*)$ (Lemma [2.17](#)(e)); $\varepsilon_{\text{CONC}}(D_*)$; $\varepsilon_U(D_*)$; $\varepsilon_K(D_*)$; and, in ε_2 only, $F(\log_2 D_*)$ with $F(x) = (3184 + 30400 \log_2 x) x^{-90.2}$ as in Lemma [7.16](#)(iv), which tends to 0 because $F(x) \rightarrow 0$ as $x \rightarrow \infty$. The function $\varepsilon_{\text{CONC}}$ is the explicit expression of Theorem [6.10](#)(iv). It is a combination of $\log^* D_*/\log D_*$, $\log^* D_*/\log \log D_*$ and $(2 \log^* D_* + 2)/(\log D_*)^{1/2}$, each of which tends to 0. The functions ε_U (Lemma [5.5](#)) and ε_K (Lemma [5.9](#)) are, by their defining statements, explicit functions of D_* that tend to 0 as $D_* \rightarrow \infty$. Every other listed function tends to 0. Hence $\varepsilon_1(D_*) \rightarrow 0$ and $\varepsilon_2(D_*) \rightarrow 0$.

(iii) Put $D_1 := 2^{2^{\mu_1}}$; then $D_1 \geq 4 > 2$, as $\mu_1 \geq 1$. By (i), $(\Gamma 1)$ holds for every $D_* \geq D_1$, since then $\log_2 \log_2 D_* \geq \mu_1$. $(\Gamma 2)$ (a) holds for every $D_* \geq 2^{117}$, and $(\Gamma 2)$ (b),(c) are implied by $(\Gamma 1)$ (Condition [1.33](#)). $(\Gamma 3)$ holds for every $D_* \geq 2^{(2N_0)^{1/C'}}$; here N_0 is fixed before D_* . By (ii) there is D'' with $\varepsilon_1(D_*) \leq 6$ and $\varepsilon_2(D_*) \leq 1/4$ for all $D_* \geq D''$, which is $(\Gamma 4)$. Take $D_0 := \max(D_1, 2^{117}, 2^{(2N_0)^{1/C'}}, D'')$. $\square$

Corollary 7.22 (Corollary JV⁺-EG; proof of the Main Theorem). *Theorem [1.1](#) holds.*

Dependencies: [7.20](#), [7.19](#), [2.15](#), [7.21](#), [6.8](#), [1.33](#), [1.31](#)

Proof. Let D_* be as in Theorem [1.1](#), that is, a constant satisfying $(\Gamma 1)$ – $(\Gamma 4)$. Such constants exist by Lemma [7.21](#), and the argument below applies to each of them. Let $C_0 = D_*/2 + 1085 + \varepsilon_1(D_*)$, $\theta_Q = \varepsilon_2(D_*)$ and $c_{\text{EG}} = \max(C_0/(1 - 2\theta_Q), N_0/2)$, as in Theorem [1.1](#). We apply Theorem [7.20](#) with $C := C_0$ and $\vartheta := \theta_Q$. Its assumptions on the constants hold: $C_0 \geq D_*/2$, since $\varepsilon_1 \geq 0$; and $0 \leq \theta_Q \leq 1/4 < 1/2$.

Let G be a graph without isolated vertices, with $n \geq N_0$ vertices and $d_1 \geq D_*$. By Proposition [2.15](#), G has a valid HB^{* τ +} run. A designation exists: for every standalone pre-part Z of a round $l \geq 3$ and every $u \in Q_Z$, the set $\text{anc}_l(u)$ is non-empty by the definition of classed ports, and $Y(u)$ can be chosen in it by a fixed rule (Definition [6.8](#)). With no VX-parts, Theorem [7.19](#) gives simple graphs $Q_3, \dots, Q_R$ with $f(G) \leq C_0 n + 2 \sum_l f(Q_l)$ and $\sum_l |V(Q_l)| \leq \theta_Q n$. This is

the hypothesis of Theorem 7.20, which therefore gives $f(G) \leq c_{\text{EG}}|V(G)|$ for every graph G . In particular $f(n) \leq c_{\text{EG}}n$ for every n , so $f(n) = O(n)$, where c_{EG} depends only on $\sigma, C', A, \varepsilon, N_0$ and D_* .

This completes the proof of Theorem 1.1. $\square$

Remark 7.23 (Why the argument is not circular). (1) C_0 and θ_Q are fixed functions of D_* (Proposition 7.18, Lemma 7.16), and D_* is fixed before any graph is considered. The constant c_{EG} enters only as the constant c of Theorem 7.20 (throughout its statement and proof); it plays no other role in any construction or bound of Sections 2–7. (2) Minimality is applied only to the quotients Q_l . They are simple and have at most $\theta_Q n \leq n/4 < n$ vertices in total, and the induction uses about them only the inequality $f(Q_l) \leq c_{\text{EG}}|V(Q_l)|$. (3) The quotient Q_l depends on the decompositions $\mathcal{D}_{l'}$ ($l' > l$), through J_l and through junk. This is harmless. Each $\mathcal{D}_{l'}$ is fixed at round l' by a deterministic rule. The Markov choice at round l succeeds with probability at least $1/2$ whatever the past. Every bound of Lemmas 7.10, 7.11 and 7.13 holds for every past. (4) No other link calls itself. Lemma TPV, Lemma PV and Theorem VX^+ finish their runs with the elementary long-cycle bound of Fact 1.6, which is proved directly and does not use the statement being proved; all three are unconditional.

Dependencies: 7.20, 7.19, 7.18, 7.16, 7.10, 7.11, 7.13, 7.8, 7.5, 4.2, 4.3, 4.4, (Γ4), 1.6

8 Formal verification

Theorem 1.1, in the form $f(n) = O(n)$, has been formalized and proved in the Lean 4 proof assistant [5] with its mathematical library Mathlib [13]. The formal target is not a statement of our own: it is the theorem `Erdos184.erdos_184` of the formal-conjectures repository [12] (file `FormalConjectures/ErdosProblems/184.lean`, commit 2424bb48). Up to typesetting it reads:

```
open scoped Classical in
theorem erdos_184 :
  ∃ f : ℕ → ℝ,
    (f = 0[atTop] fun n : ℕ ↦ (n : ℝ)) ∧
    ∀ {V : Type*} [Fintype V] [DecidableEq V] (G : SimpleGraph V),
      ∃ (D : Finset G.Subgraph),
        (∀ H ∈ D, IsCycleOrEdge H.coe) ∧
        IsDecomposition G D ∧
        (D.card : ℝ) ≤ f (Fintype.card V)
```

(in namespace `Erdos184`, under open `Filter SimpleGraph`). Here `IsCycleOrEdge H` means that the finite simple graph H is connected and 2-regular, or has exactly one edge; `IsDecomposition G D` means that the edge sets of the members of D are pairwise disjoint with union $E(G)$. A connected 2-regular finite simple graph is a cycle of length at least 3, so the statement says that some $f(n) = O(n)$ bounds, for every finite simple graph on n vertices, the size of a partition of its edge set into cycles and single edges: this is the “in particular” part of Theorem 1.1 (take $f(n) = c_{\text{EG}}n$).

The Lean proof of `Erdos184.erdos_184` depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. No published result is assumed: the cited inputs of Section 1 that the proof uses are proved in Lean, except Hall’s theorem (Cited result 1.28), which is taken from Mathlib. The proof has been checked by the Lean kernel and by the independent checker *comparator* [10] in release mode, which verifies that the proved theorem has exactly the statement of formal-conjectures, that no axioms other than the three above are used, and that the proof is accepted on a

kernel replay. The code is available at <https://github.com/steelwheel01/erdos-gallai-lean> [3] (release tag `release-2026-10-01`, commit `a11bb45847101e0d1595089bc390c59f82ad1135`; verification run `36795612102` of its GitHub Actions release workflow). The formalization follows this paper section by section; a table matching every numbered statement of Sections 1–7 with its Lean counterpart, where there is one, is given in the file `CORRESPONDENCE.md` of that repository.

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